

Canonical GHZ configurations - cluster P3

species: electron
polarization cluster: P3
source cluster configurations: 87
canonical configurations collected: 53

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction $\geq 2.0\%$
normalized amplitude display: bar length = $|A_{\text{tilde}}|$, bar color/text = phase

cluster retained-component count distribution:

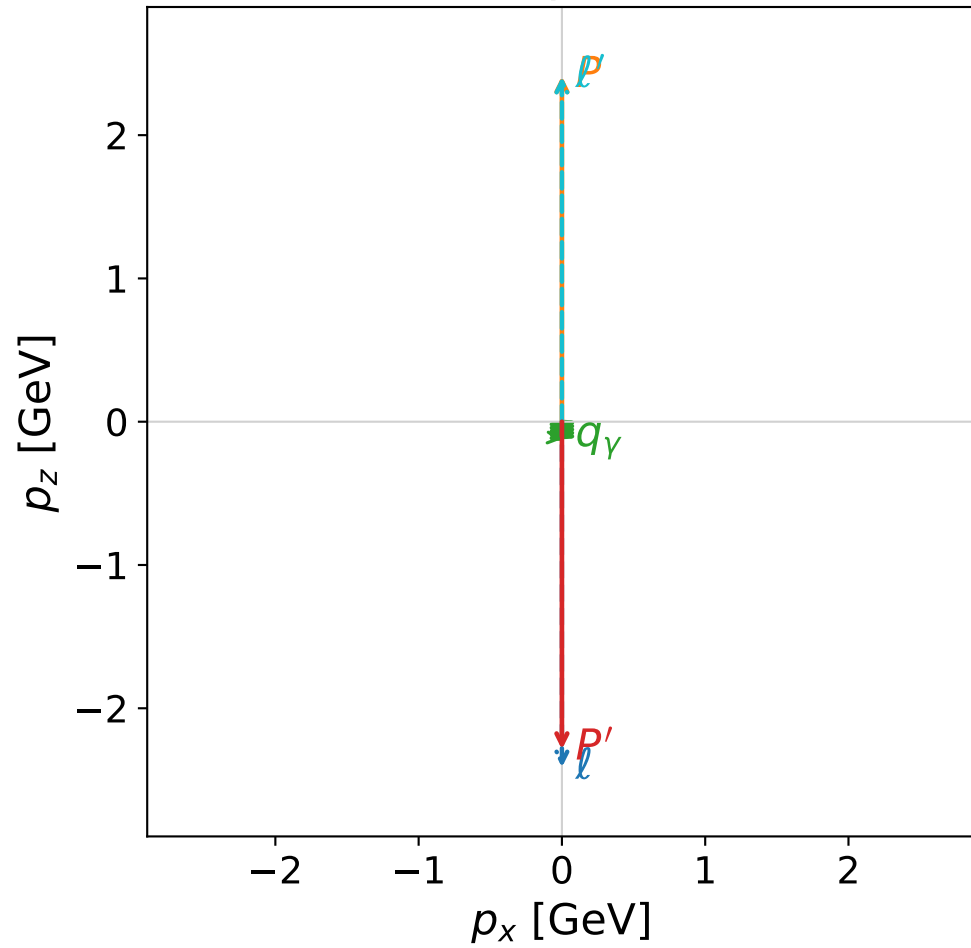
2 components:	53 configurations
4 components:	8 configurations
5 components:	4 configurations
6 components:	19 configurations
7 components:	3 configurations

CSV index:
Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_03\combined\
canonical_dghz_polarization_cluster_03_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

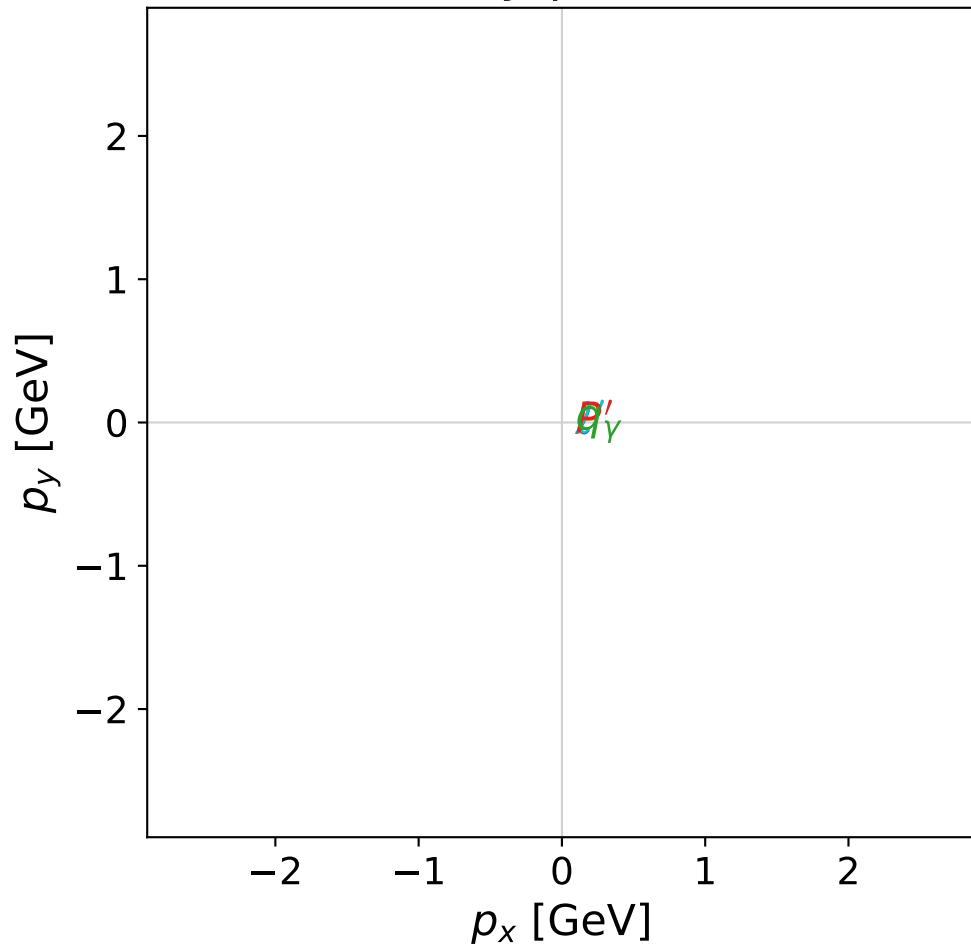


dghz_mixing_angles_polarization_03_local_minimum_1
 $d_{\{\rm GHZ\}}=2.28069\text{e-}08$
 region: polarization_cluster_03_local_minimum_1
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0001
 lepton mixing angle: $\alpha_e=2.4389177$ rad
 incoming lepton: $-0.763116|+\rangle +0.646261|-\rangle$
 proton mixing angle: $\alpha_p=0.70266188$ rad
 incoming proton: $0.763125|+\rangle +0.646251|-\rangle$

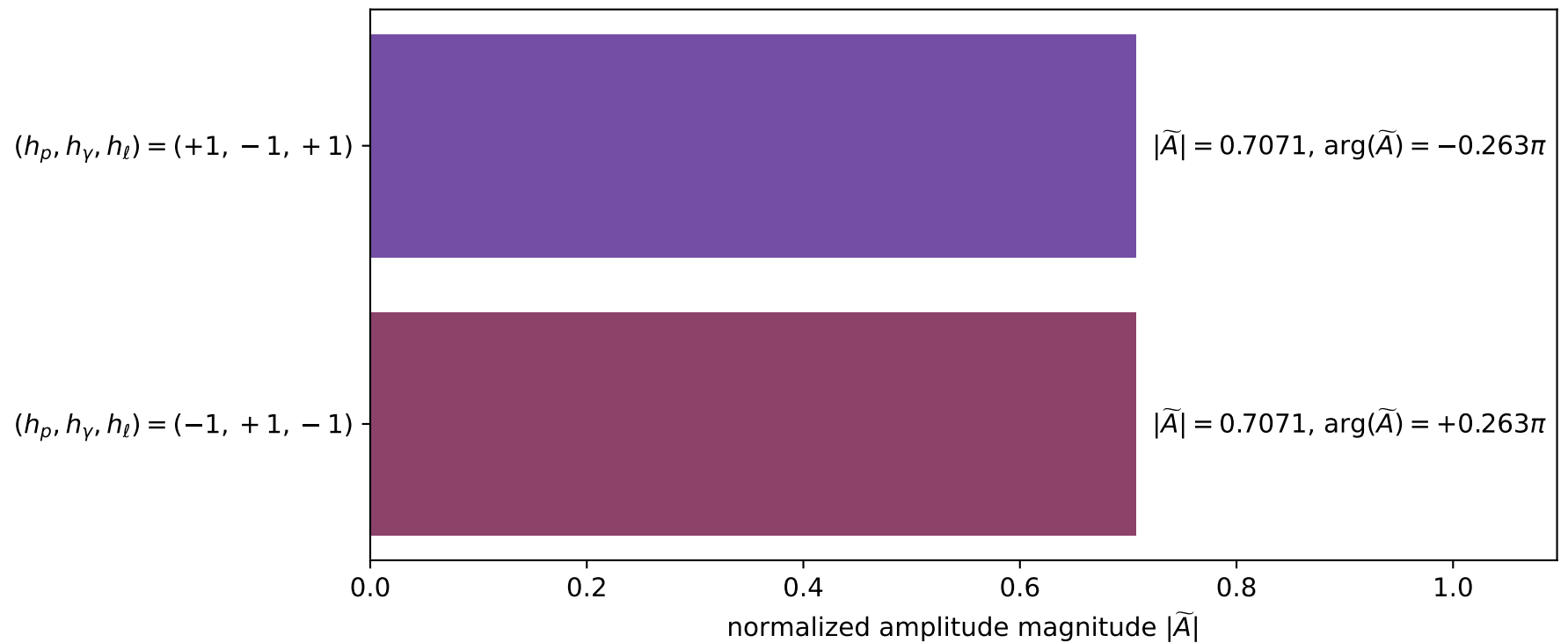
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.28696$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=8.73394\text{e-}09$, $\phi_\gamma=2.31425$
 $E_\gamma=0.120598$, $\theta_{l'}=0$, $\phi_{l'}=3.18183$
 $Q^2=23.2283$, $x_B=0.998084$, $t=-22.0669$, $W^2=0.924438$, $y=0.964871$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.4076$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.4076$
 P' $E= 2.4718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2870$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.1206$

x--y plane

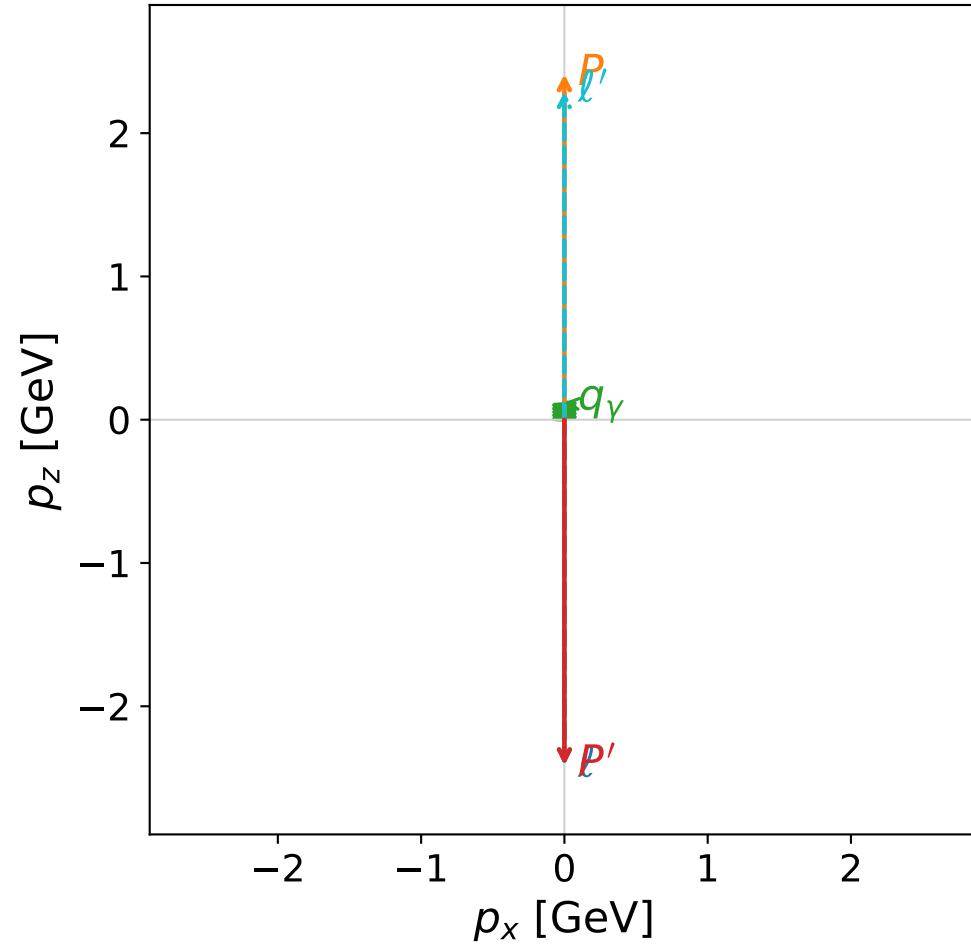


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

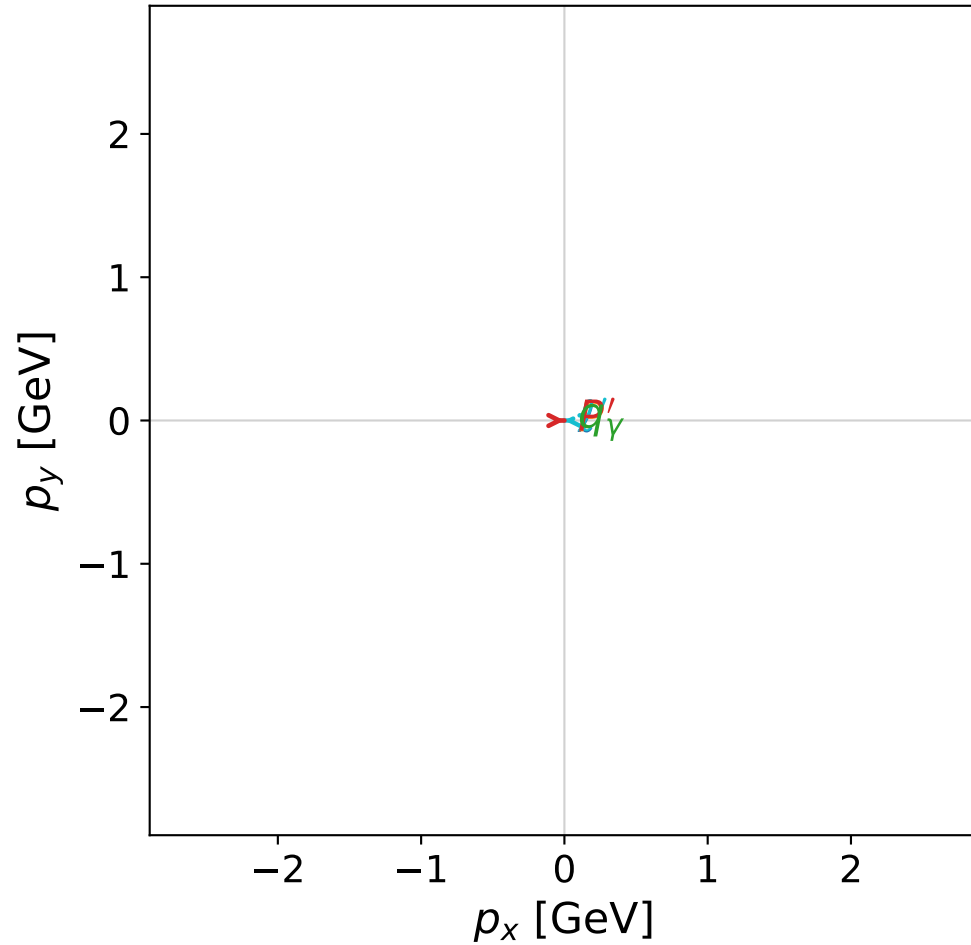


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

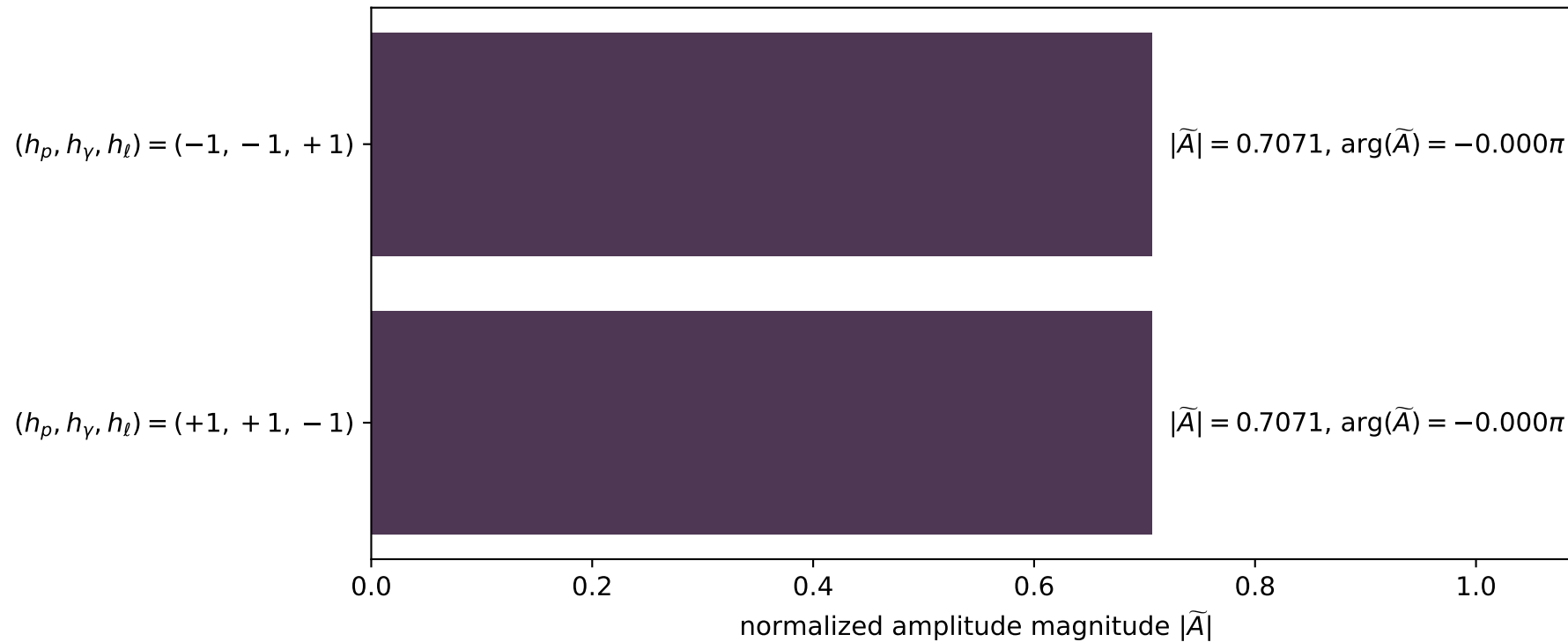


dghz_mixing_angles_polarization_03_local_minimum_5
 $d_{\text{GHZ}}=2.39007\text{e-}08$
 region: polarization_cluster_03_local_minimum_5
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0005
 lepton mixing angle: $\alpha_e=2.2809494$ rad
 incoming lepton: $-0.65195|+\rangle +0.758262|-\rangle$
 proton mixing angle: $\alpha_p=0.71016018$ rad
 incoming proton: $0.758257|+\rangle +0.651955|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=6.28319$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

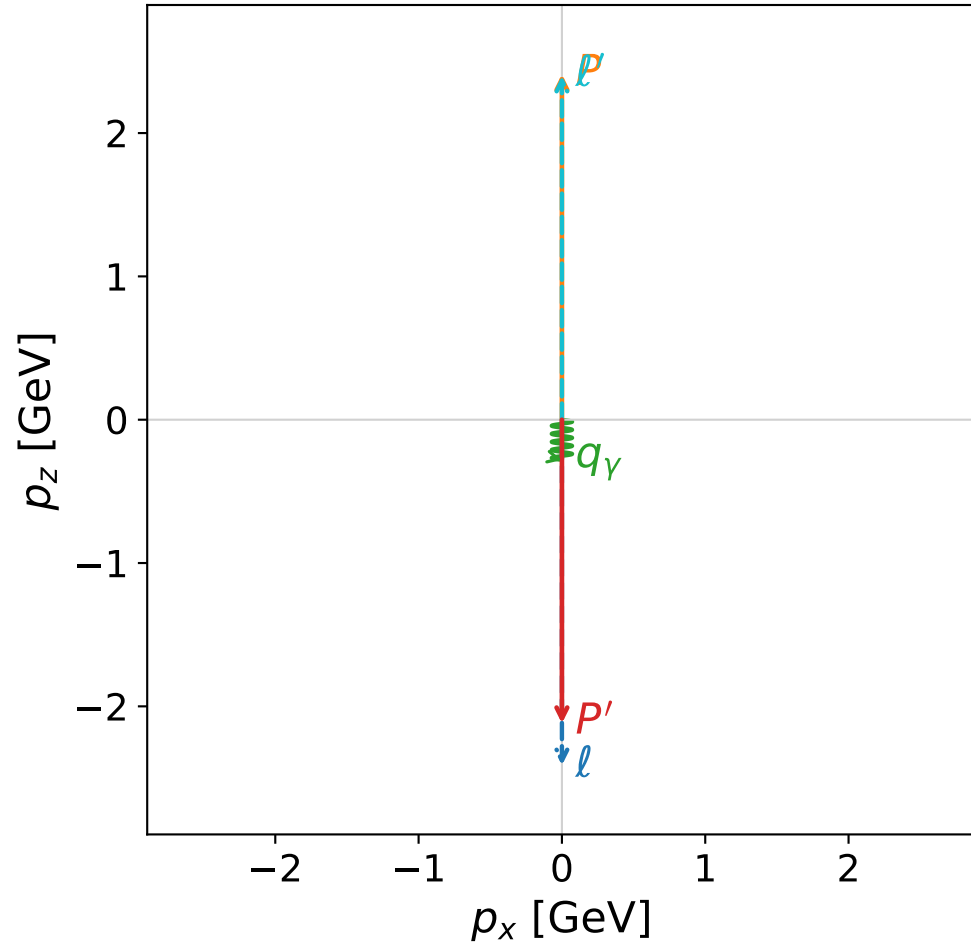
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

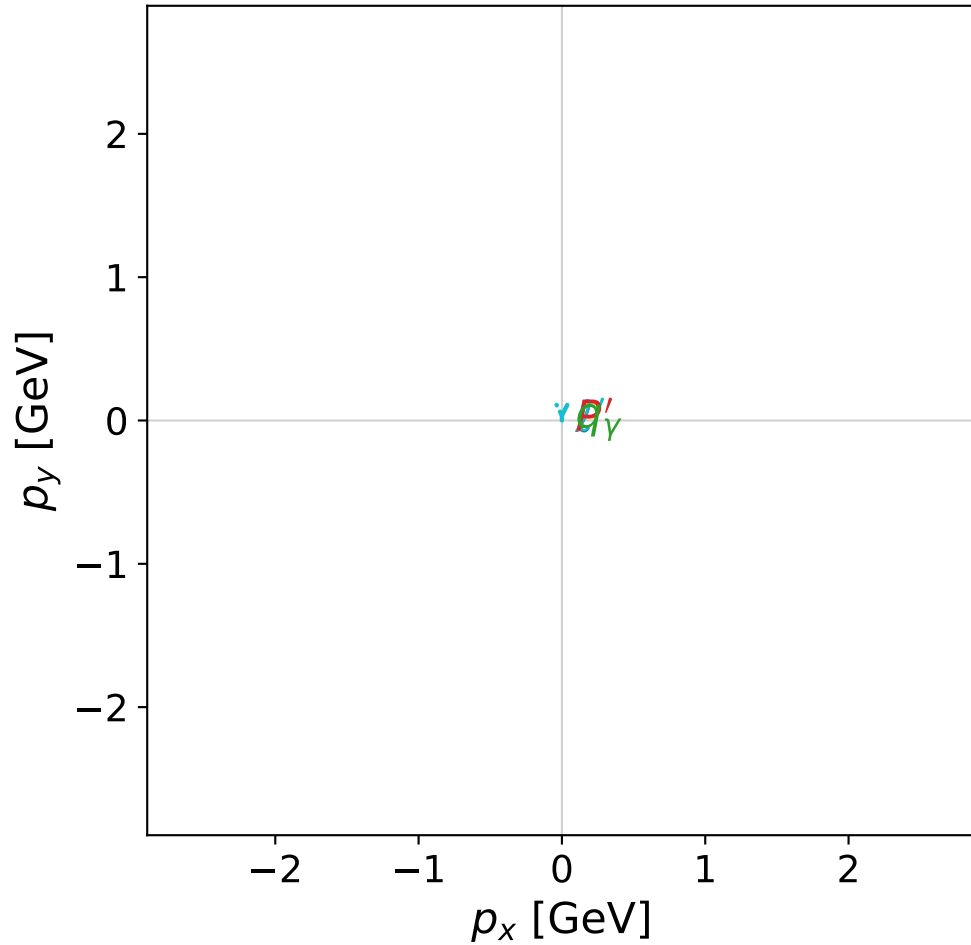


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_12

$d_{\{\rm GHZ\}}=2.46147\text{e-}08$

region: polarization_cluster_03_local_minimum_12

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0012

lepton mixing angle: $\alpha_e=2.1502889$ rad

incoming lepton: $-0.547599|+\rangle +0.836741|-\rangle$

proton mixing angle: $\alpha_p=0.99129539$ rad

incoming proton: $0.547606|+\rangle +0.836736|-\rangle$

$s=24.9869$, $\sqrt{s}=4.99869$, $|\vec{P}|=2.41134$, $|\vec{P}'|=2.11911$

$\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.71126$, $\phi_\gamma=1.54366$

$E_\gamma=0.281071$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.83329$

$Q^2=23.1506$, $x_B=0.995208$, $t=-20.4521$, $W^2=0.991326$, $y=0.964951$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.4113$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4113$

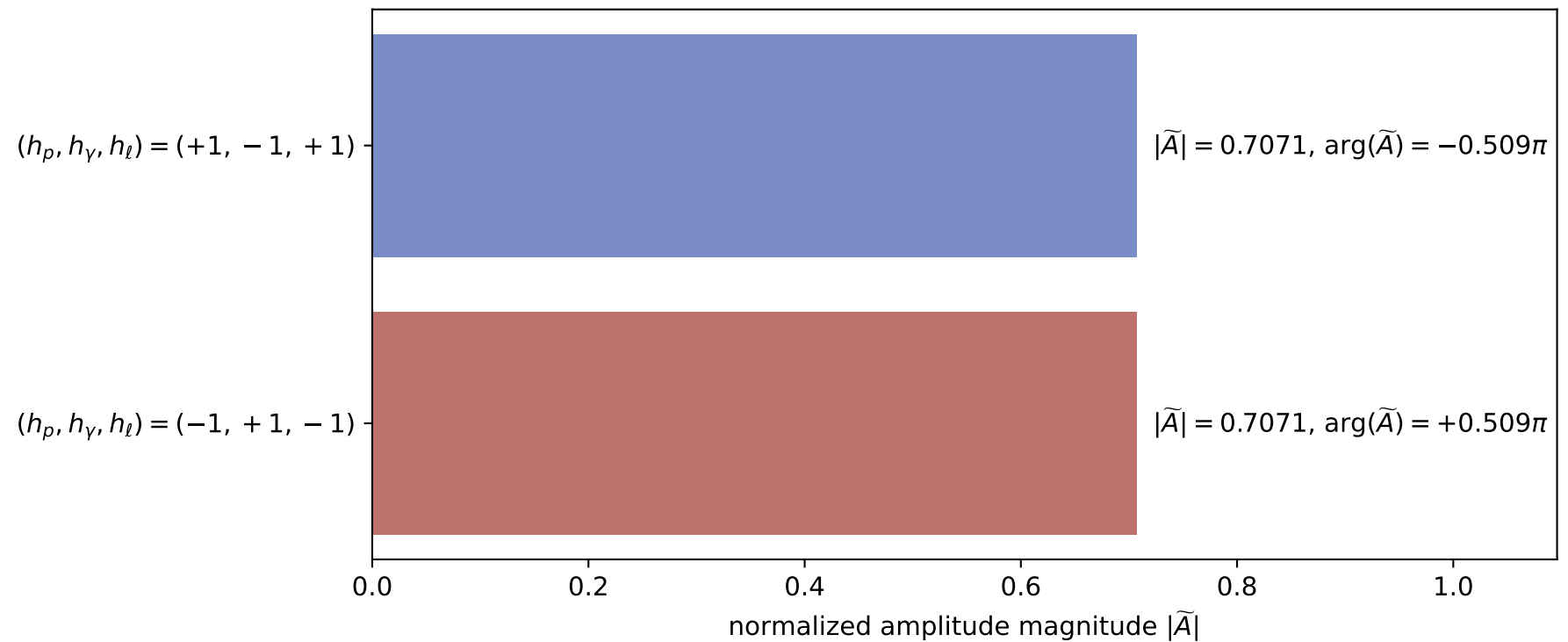
P $E= 2.5874$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4113$

ℓ' $E= 2.4002$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.4002$

P' $E= 2.3174$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.1191$

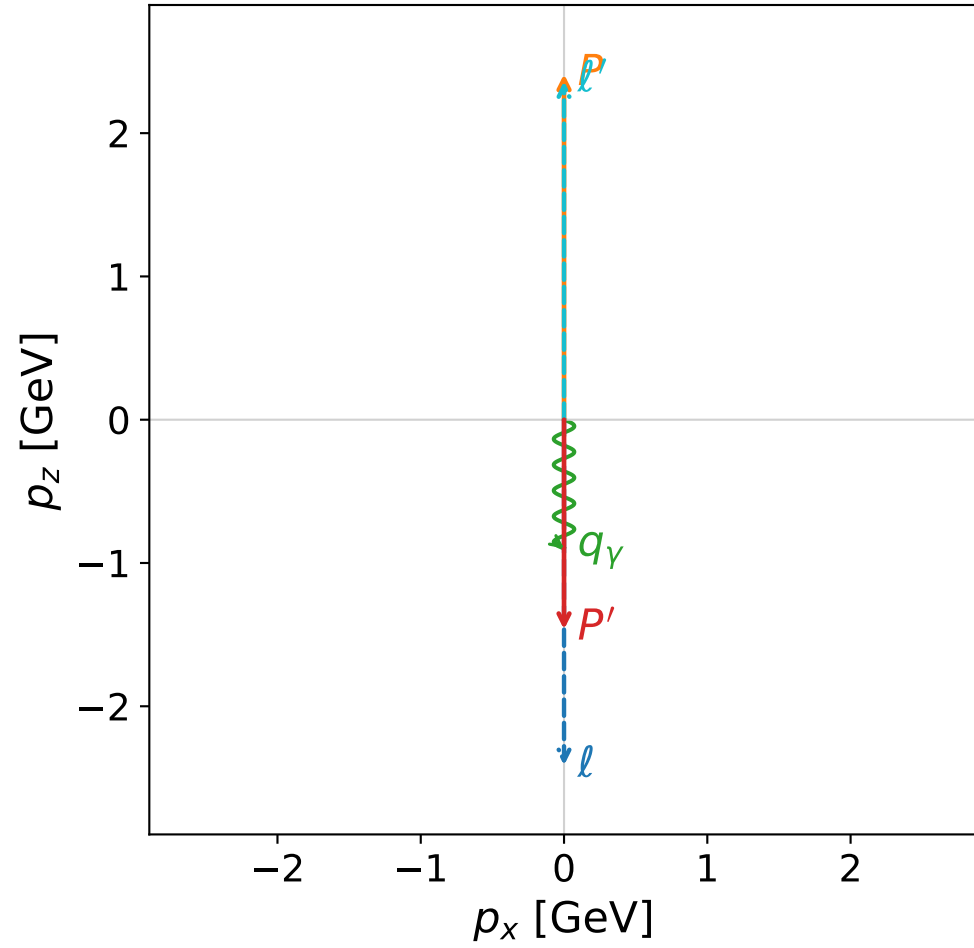
q_γ $E= 0.2811$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2811$

Leading normalized final-state amplitudes
(N=2, retained=100.0%)

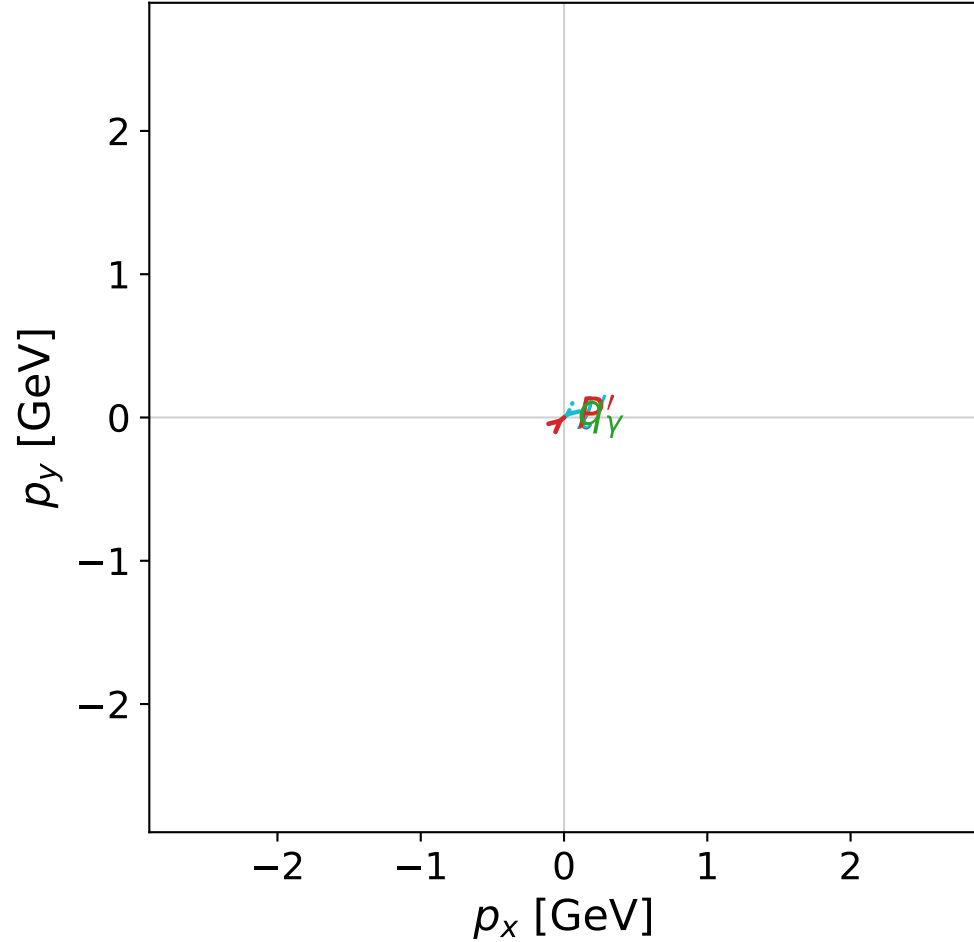


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



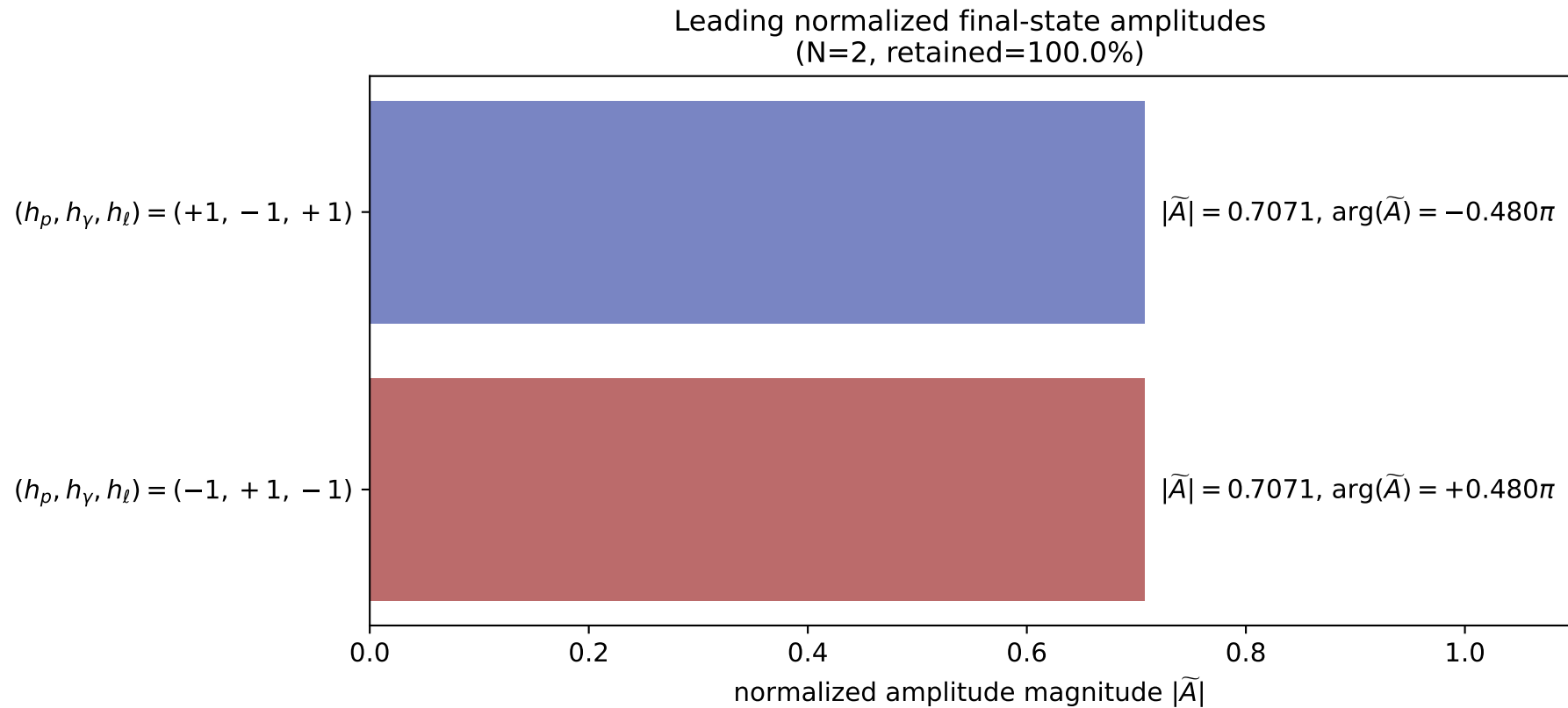
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_13
 $d_{\{\rm GHZ\}}=2.46222\text{e-}08$
 region: polarization_cluster_03_local_minimum_13
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0013
 lepton mixing angle: $\alpha_e=2.1691893$ rad
 incoming lepton: $-0.563315|+\rangle +0.826242|-\rangle$
 proton mixing angle: $\alpha_p=0.97239039$ rad
 incoming proton: $0.563326|+\rangle +0.826235|-\rangle$

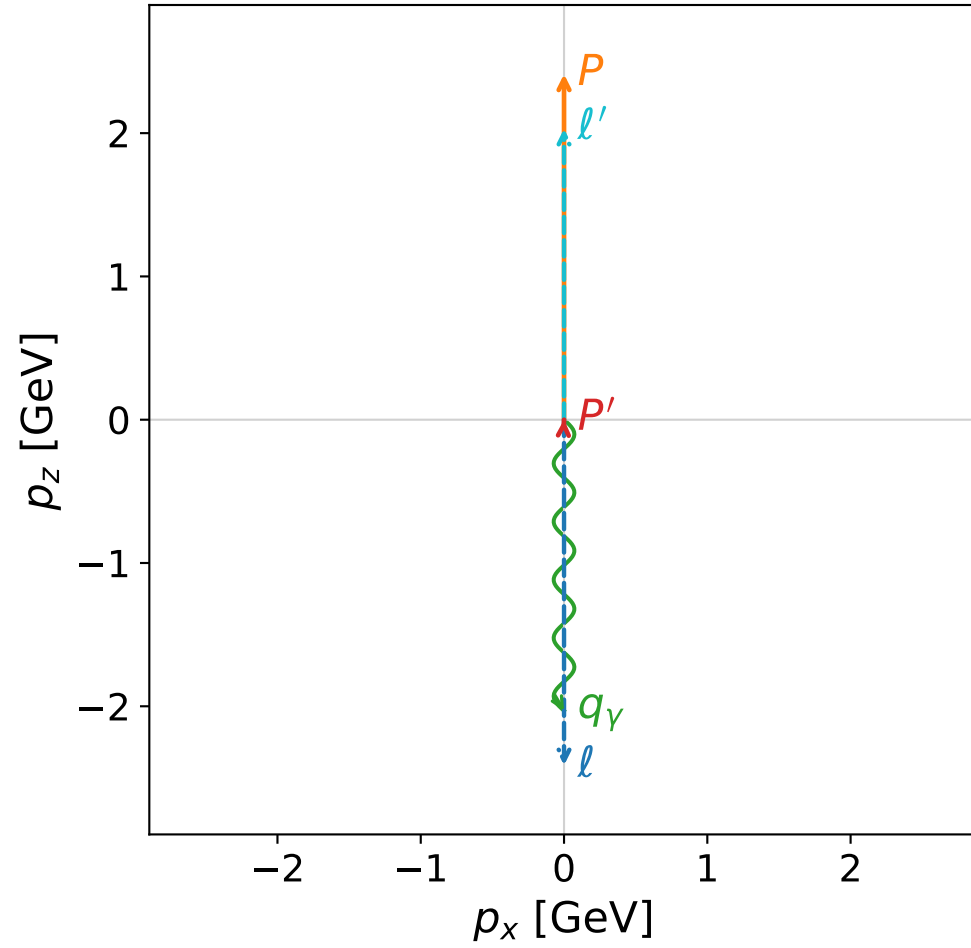
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.46356$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.711355$, $\phi_\gamma=2.34502$
 $E_\gamma=0.899044$, $\theta_{\ell'}=4.02607\text{e-}07$, $\phi_{\ell'}=3.85295$
 $Q^2=22.7946$, $x_B=0.978784$, $t=-14.2982$, $W^2=1.37394$, $y=0.965527$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3626$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.3626$
 P' $E= 1.7383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.4636$
 q_γ $E= 0.8990$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.8990$

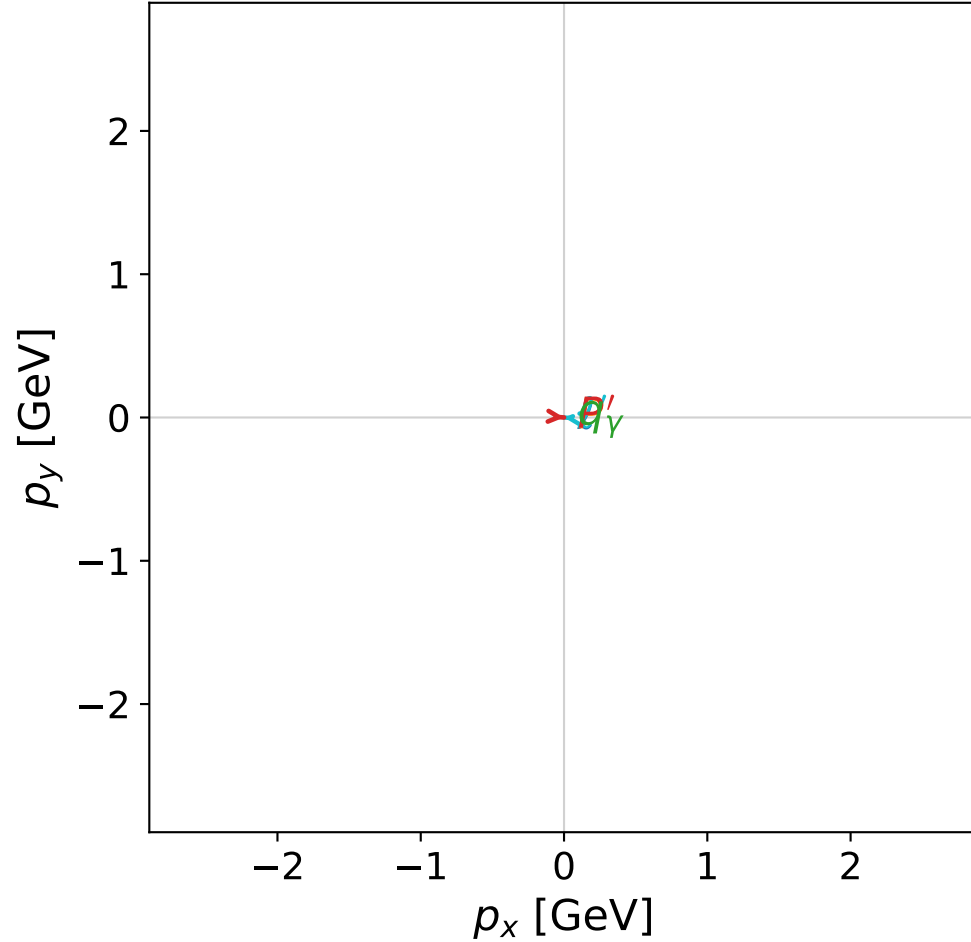


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



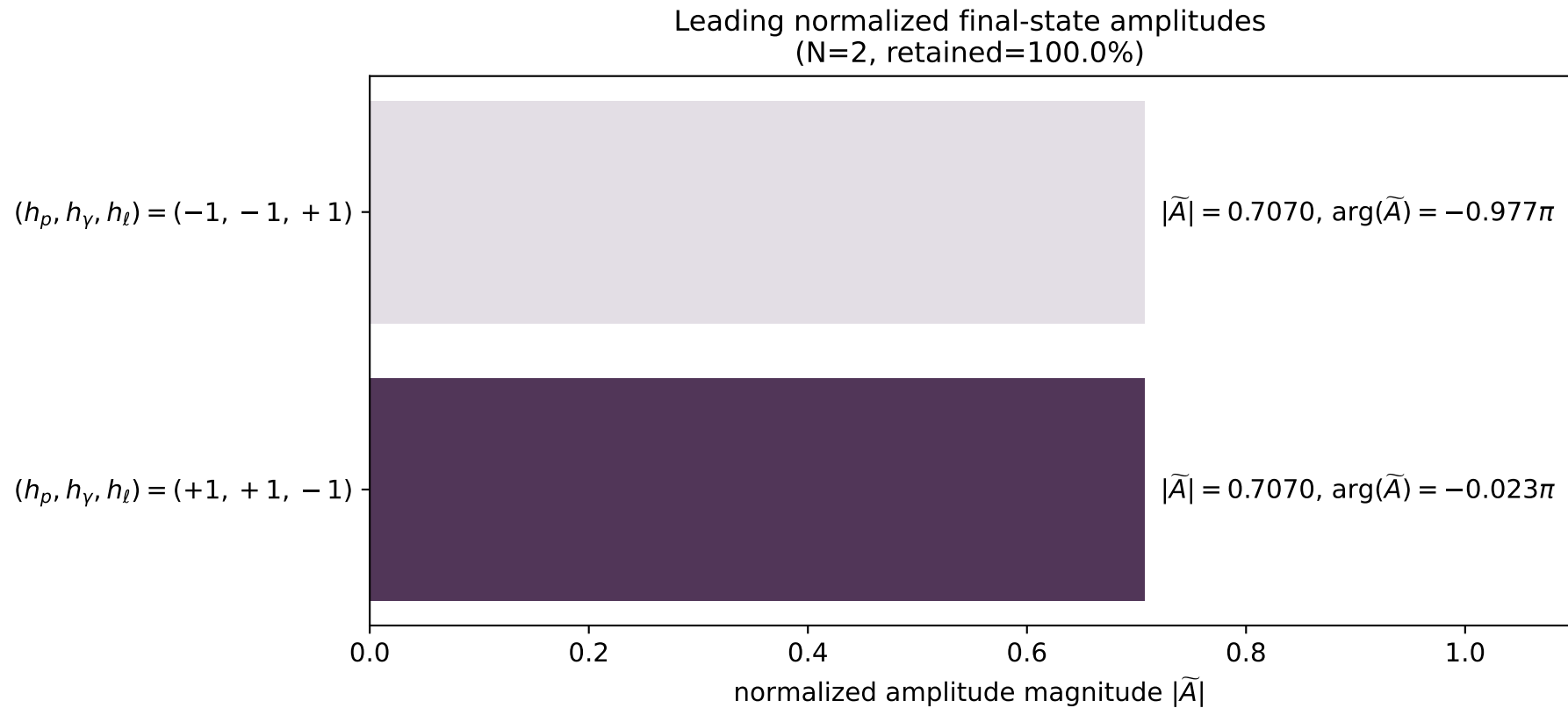
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_15
 $d_{\text{GHZ}}=2.47594\text{e-}08$
 region: polarization_cluster_03_local_minimum_15
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0015
 lepton mixing angle: $\alpha_e=2.4020754$ rad
 incoming lepton: $-0.738794|+\rangle +0.673931|-\rangle$
 proton mixing angle: $\alpha_p=0.73950355$ rad
 incoming proton: $0.738803|+\rangle +0.673921|-\rangle$

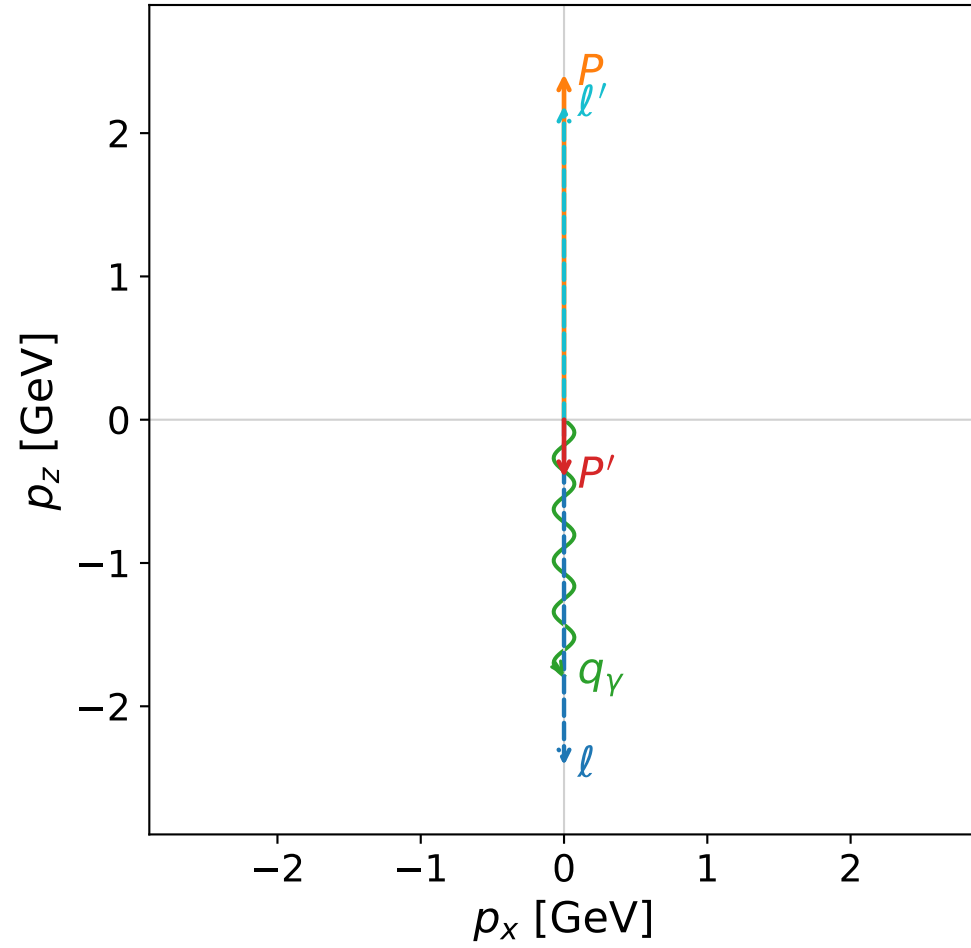
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.00056453$
 $\theta_{P'}=0.0393096$, $\theta_\gamma=3.14159$, $\phi_{P'}=6.20497$, $\phi_\gamma=0.0735511$
 $E_\gamma=2.03121$, $\theta_{\ell'}=1.09255\text{e-}05$, $\phi_{\ell'}=3.06338$
 $Q^2=19.5912$, $x_B=0.837086$, $t=-3.09252$, $W^2=4.69268$, $y=0.970368$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0306$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0306$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0006$
 q_γ $E= 2.0312$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0312$

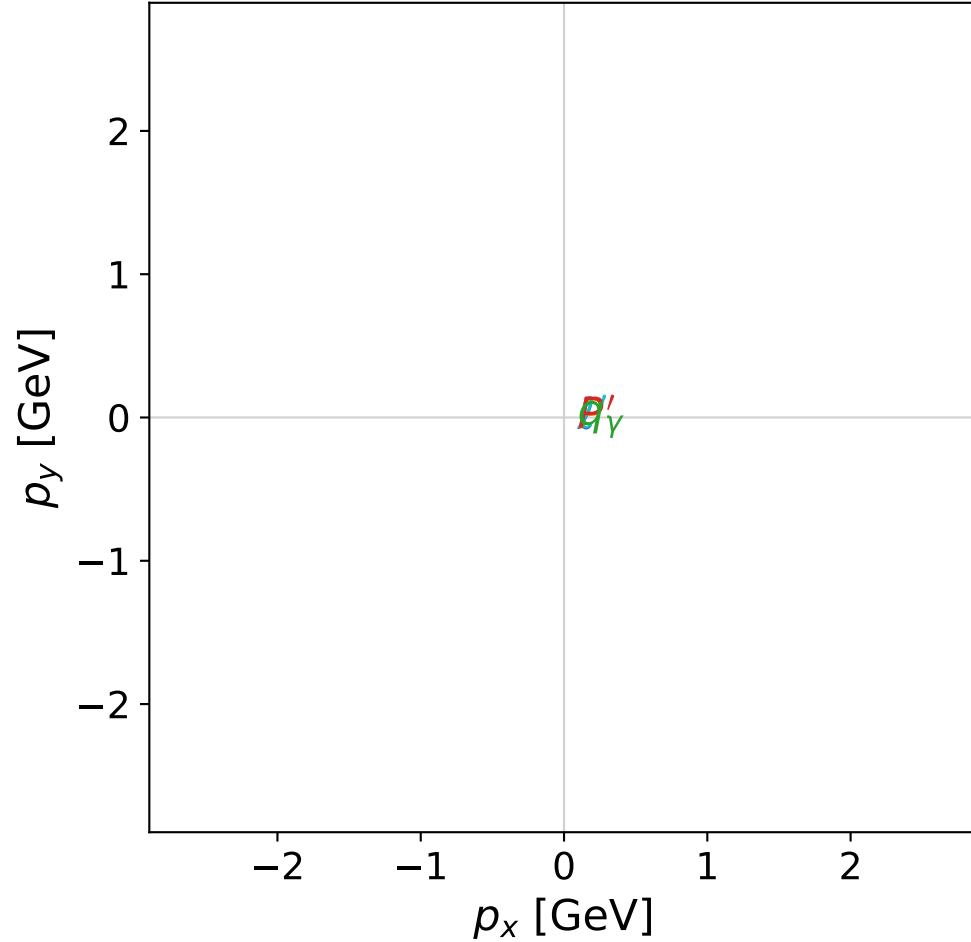


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

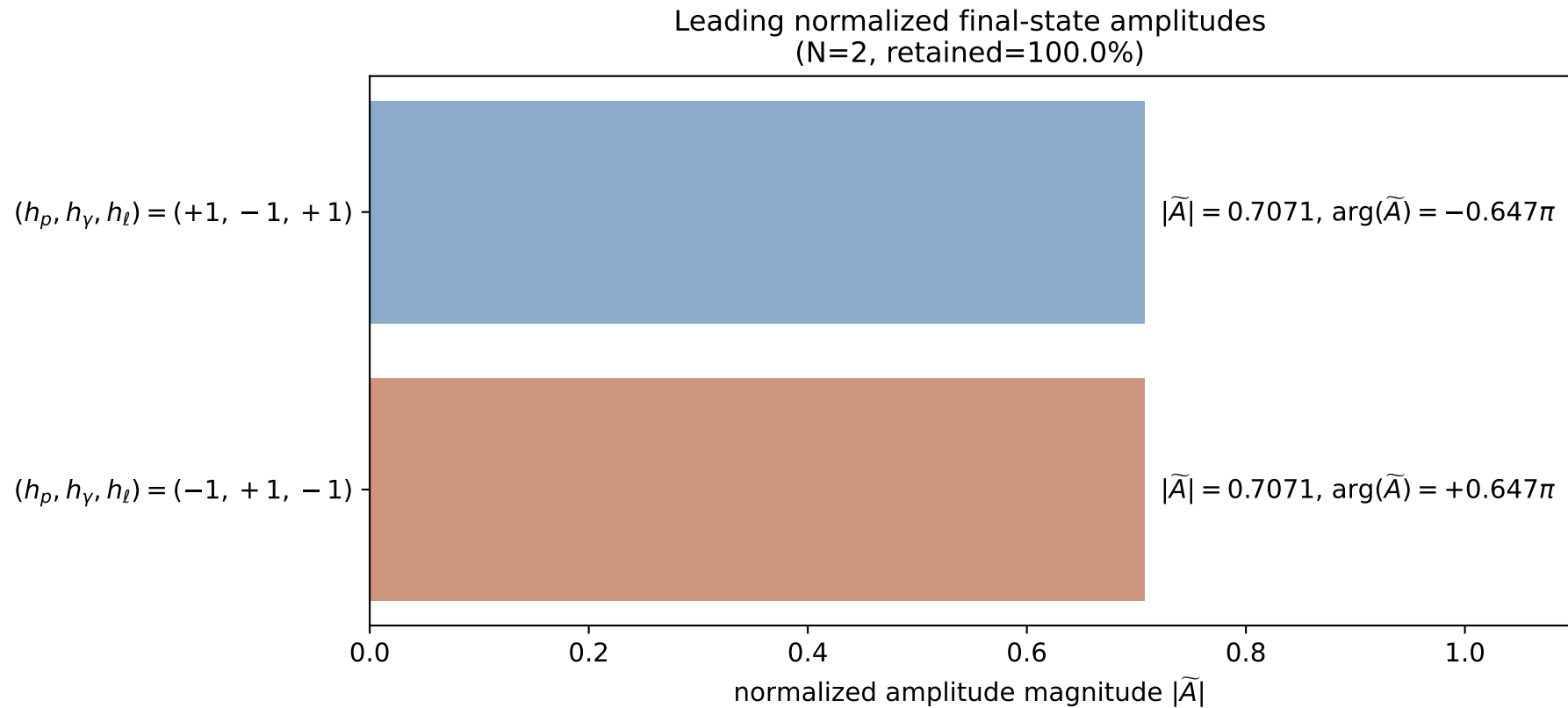


dghz_mixing_angles_polarization_03_local_minimum_17
 $d_{\{\rm GHZ\}}=2.4853\text{e-}08$
 region: polarization_cluster_03_local_minimum_17
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0017
 lepton mixing angle: $\alpha_e=2.4106629$ rad
 incoming lepton: $-0.744554|+\rangle +0.667562|-\rangle$
 proton mixing angle: $\alpha_p=0.73092649$ rad
 incoming proton: $0.744556|+\rangle +0.66756|-\rangle$

$s=24.9975$, $\sqrt{s}=4.99975$, $|\vec{P}|=2.41189$, $|\vec{P}'|=0.403477$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.09629$, $\phi_\gamma=1.10925$
 $E_\gamma=1.78759$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.47422$
 $Q^2=21.1384$, $x_B=0.905421$, $t=-5.47151$, $W^2=3.08794$, $y=0.968025$

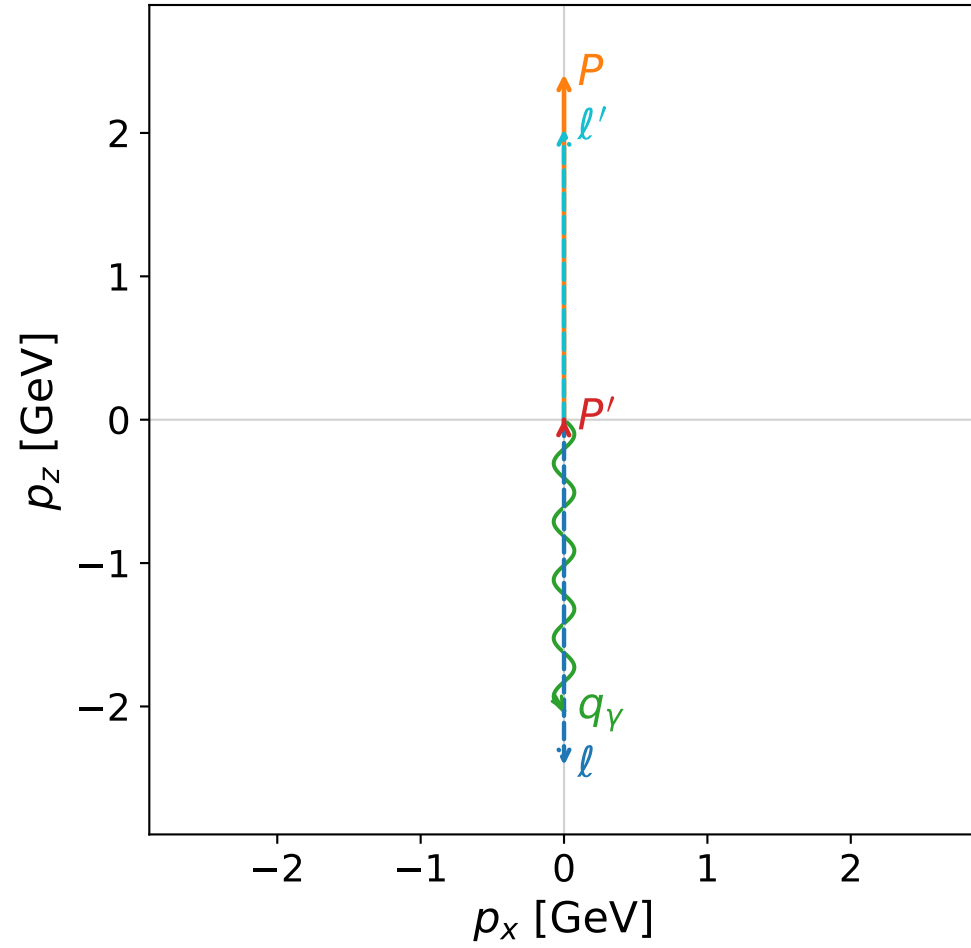
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.4119	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-2.4119
P	$E=$	2.5879	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	2.4119
ℓ'	$E=$	2.1911	$p_x=$	-0.0000	$p_y=$	-0.0000	$p_z=$	2.1911
P'	$E=$	1.0211	$p_x=$	-0.0000	$p_y=$	0.0000	$p_z=$	-0.4035
q_γ	$E=$	1.7876	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-1.7876

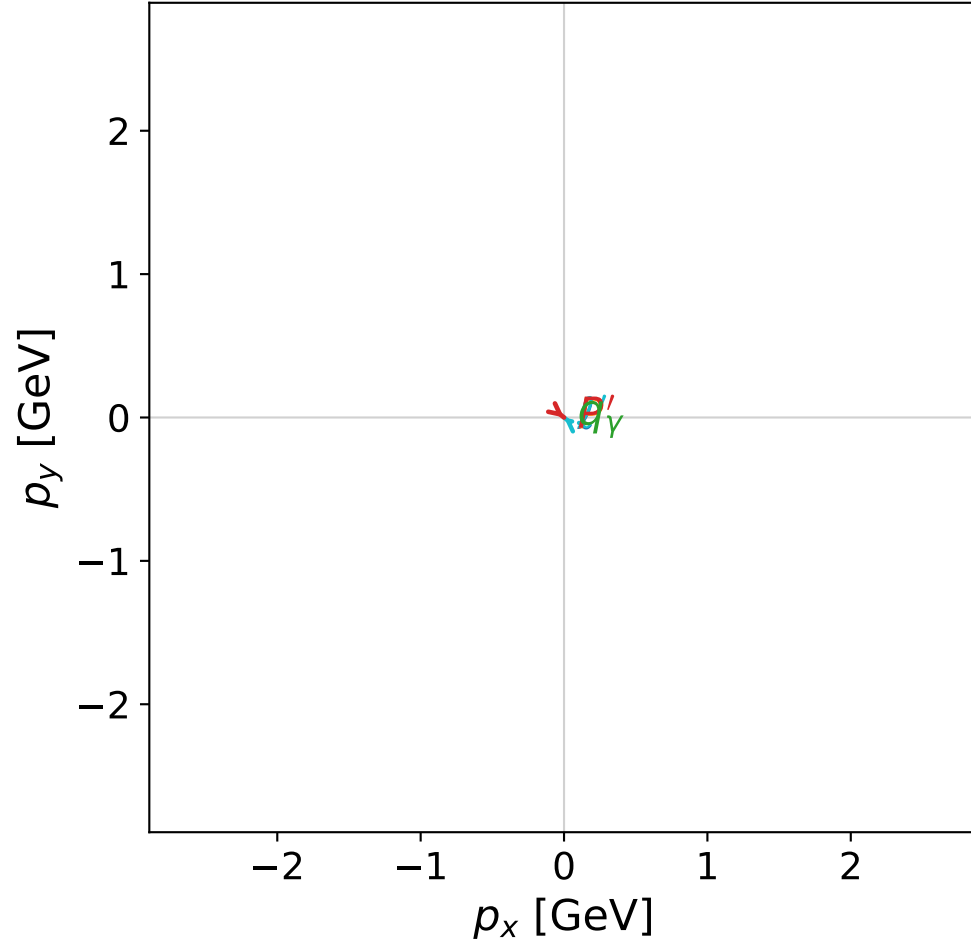


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



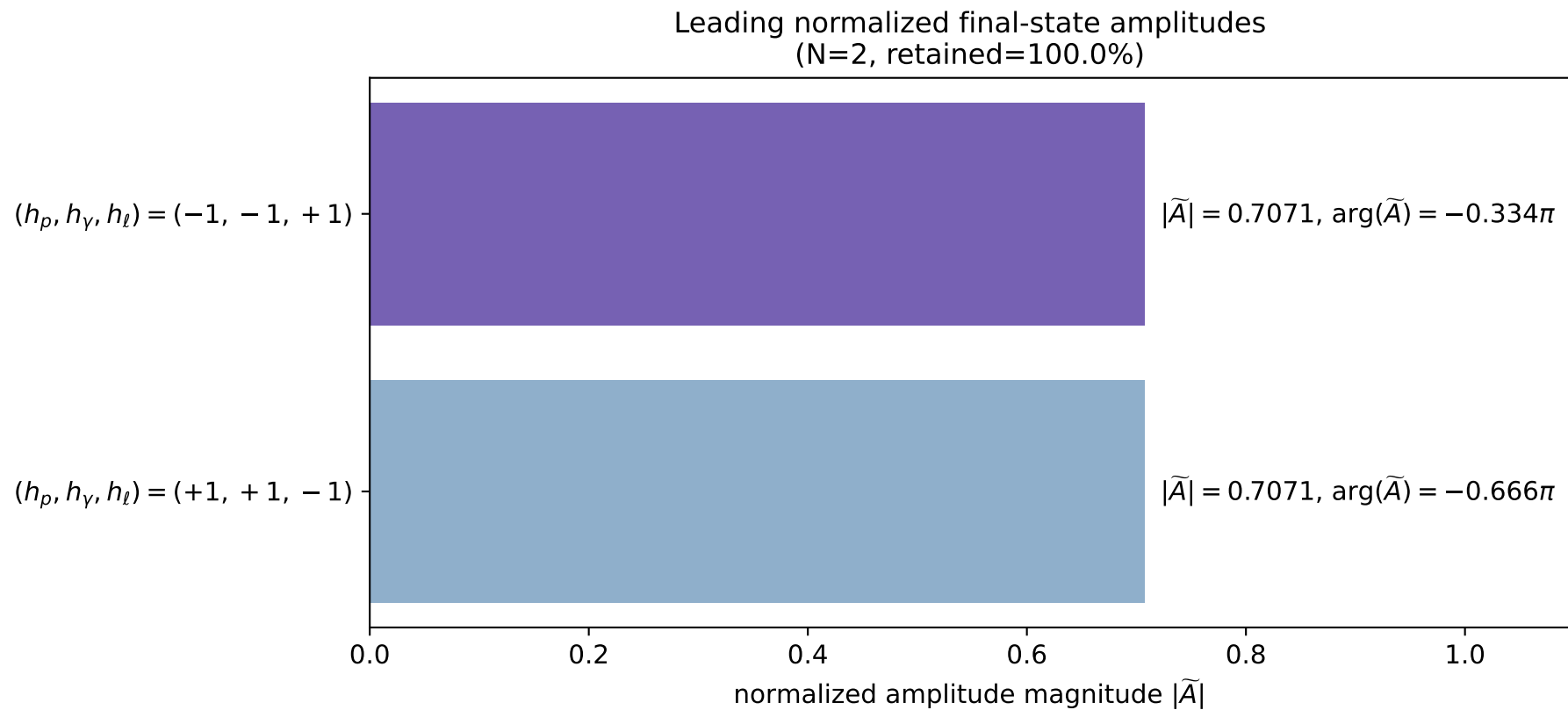
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_20
 $d_{\{\rm GHZ\}}=2.54044\text{e-}08$
 region: polarization_cluster_03_local_minimum_20
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0020
 lepton mixing angle: $\alpha_e=2.3449091$ rad
 incoming lepton: $-0.699082|+\rangle +0.715042|-\rangle$
 proton mixing angle: $\alpha_p=0.79667666$ rad
 incoming proton: $0.699087|+\rangle +0.715037|-\rangle$

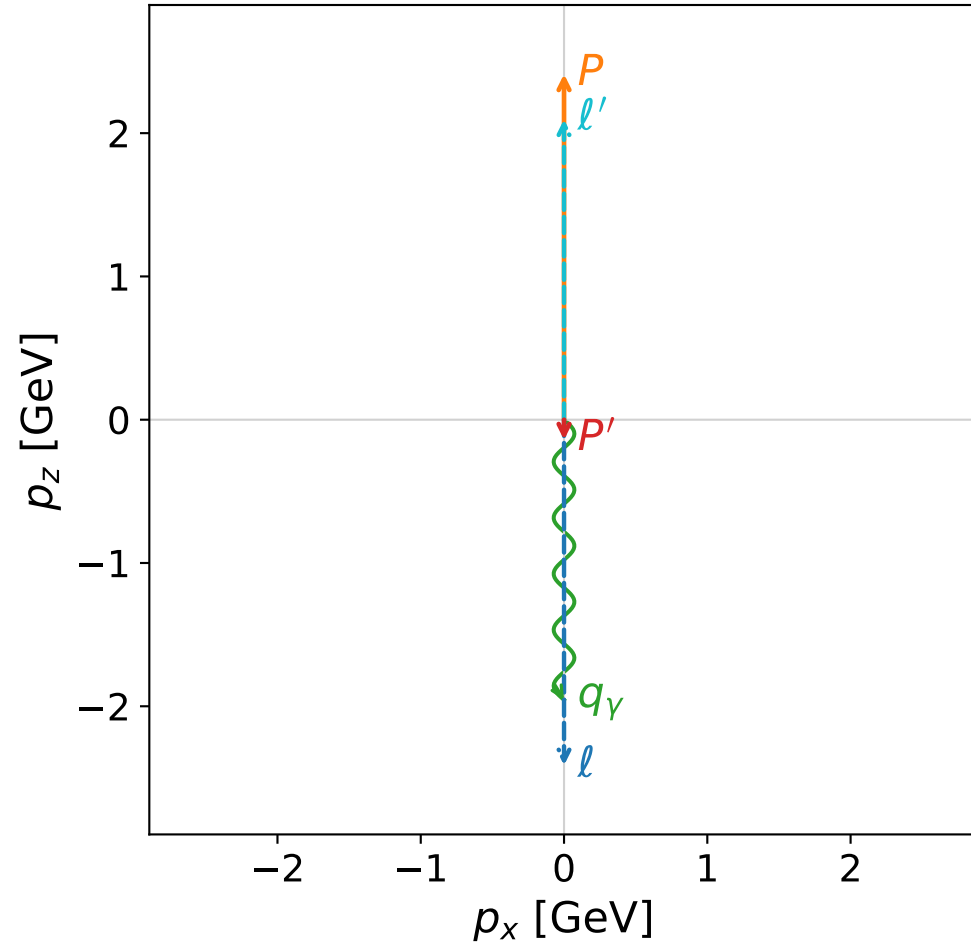
$s=24.9686$, $\sqrt{s}=4.99686$, $|\vec{P}|=2.41039$, $|\vec{P}'|=0.0036789$
 $\theta_{P'}=0.019635$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.61047$, $\phi_\gamma=2.09314$
 $E_\gamma=2.03127$, $\theta_{l'}=3.56238\text{e-}05$, $\phi_{l'}=2.46887$
 $Q^2=19.5491$, $x_B=0.836335$, $t=-3.07484$, $W^2=4.70547$, $y=0.970358$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4104$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4104$
 P $E= 2.5865$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4104$
 l' $E= 2.0276$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= 0.0037$
 q_γ $E= 2.0313$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0313$

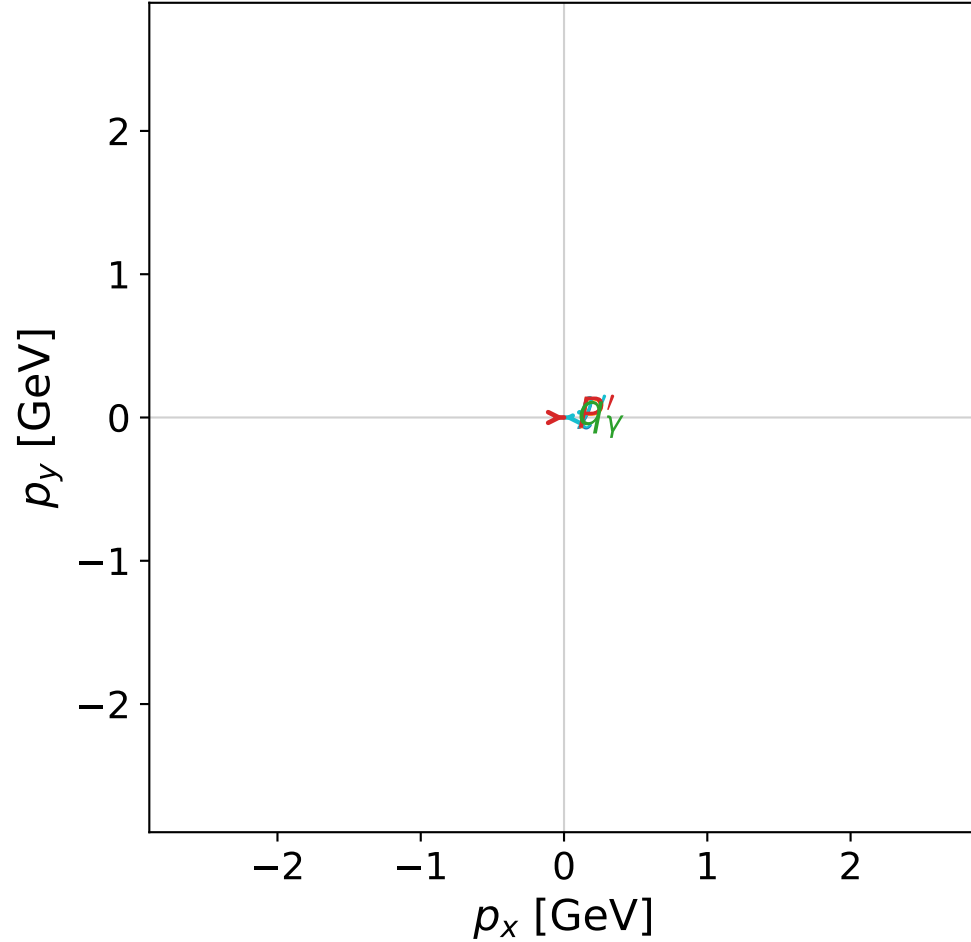


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



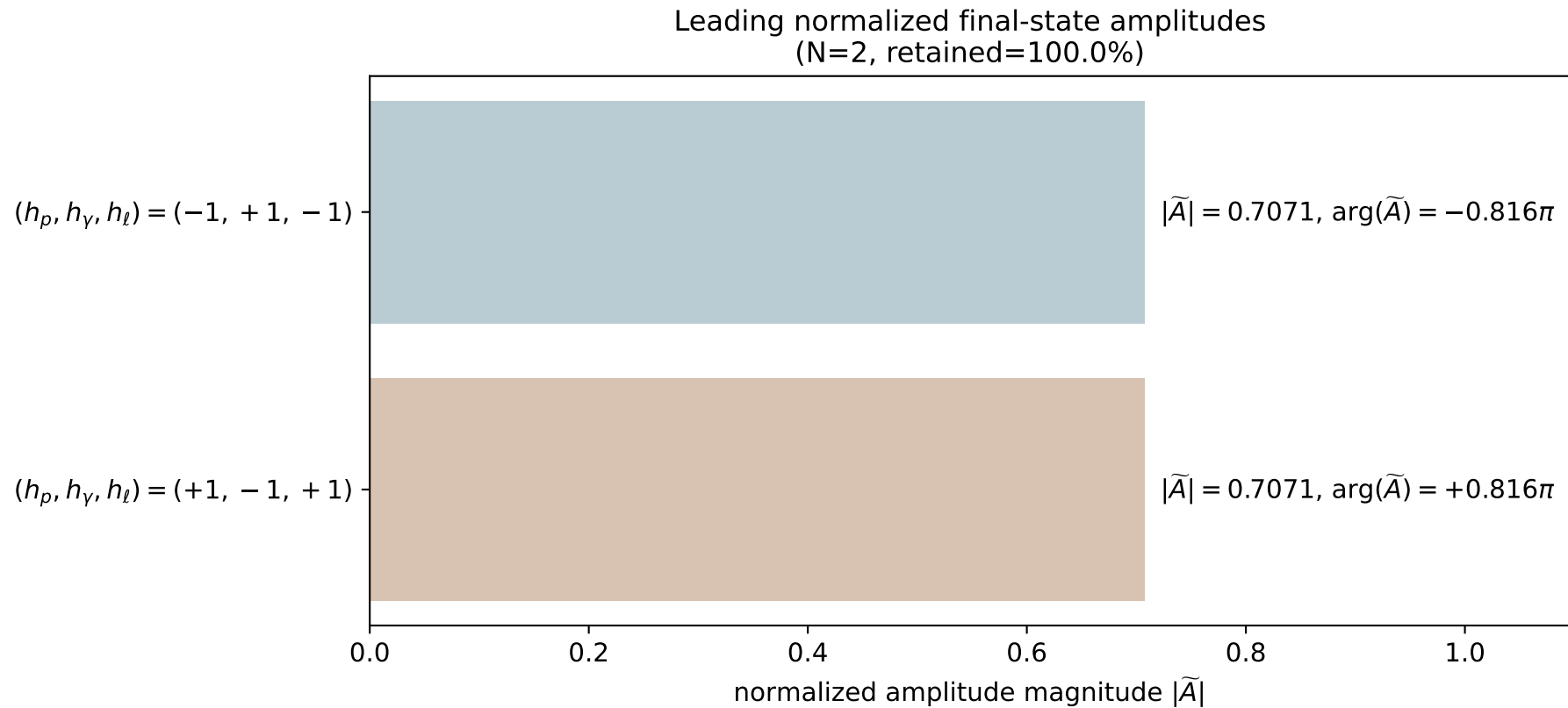
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_38
 $d_{\text{GHZ}}=2.64248\text{e-}08$
 region: polarization_cluster_03_local_minimum_38
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0038
 lepton mixing angle: $\alpha_e=2.4362483$ rad
 incoming lepton: $-0.761388|+\rangle +0.648296|-\rangle$
 proton mixing angle: $\alpha_p=0.70535267$ rad
 incoming proton: $0.761383|+\rangle +0.648302|-\rangle$

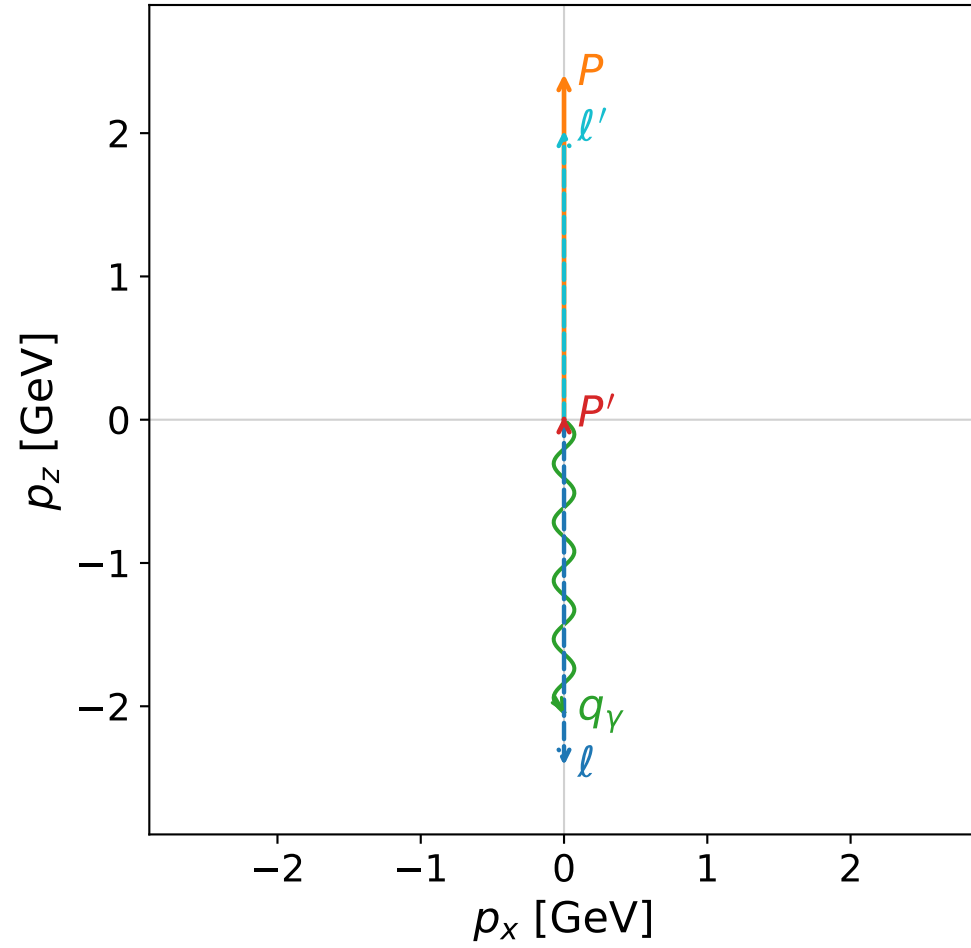
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.140649$
 $\theta_{p'}=3.14094$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.000316874$, $\phi_\gamma=5.70698$
 $E_\gamma=1.95543$, $\theta_{\ell'}=4.39858\text{e-}05$, $\phi_{\ell'}=3.14191$
 $Q^2=20.2231$, $x_B=0.864884$, $t=-3.82814$, $W^2=4.03919$, $y=0.969416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0961$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0961$
 P' $E= 0.9485$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= -0.1406$
 q_γ $E= 1.9554$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.9554$

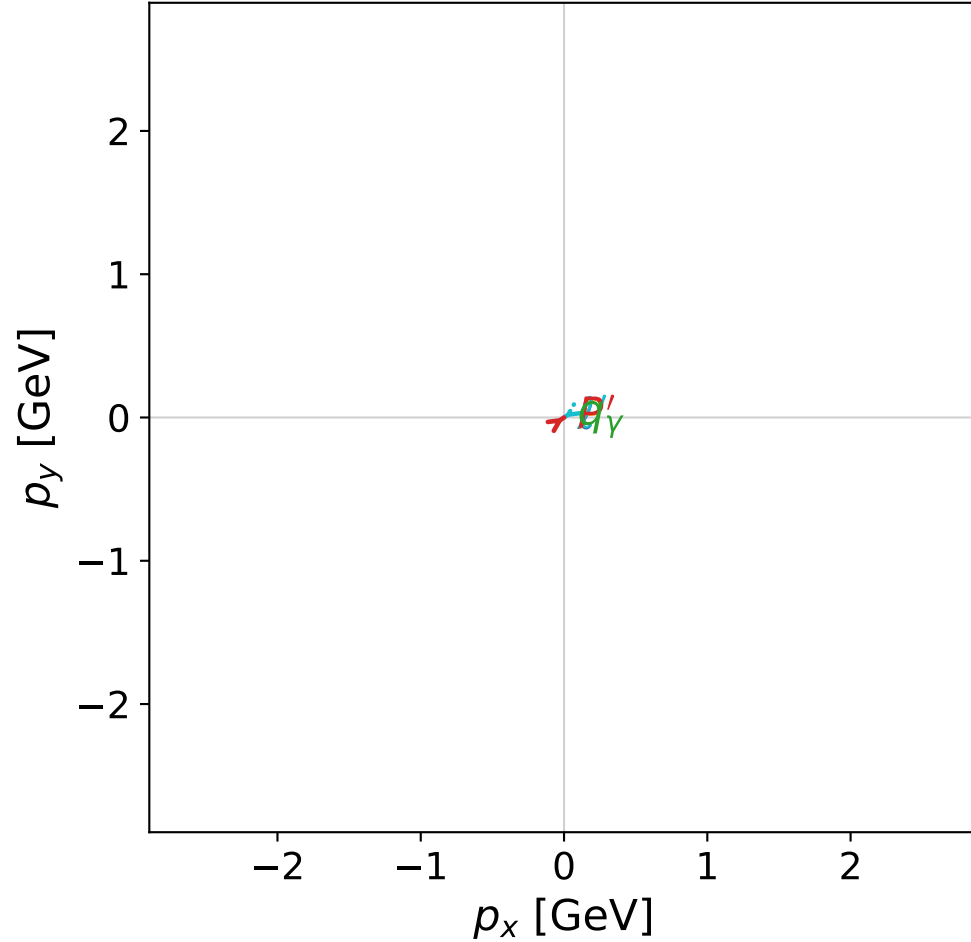


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



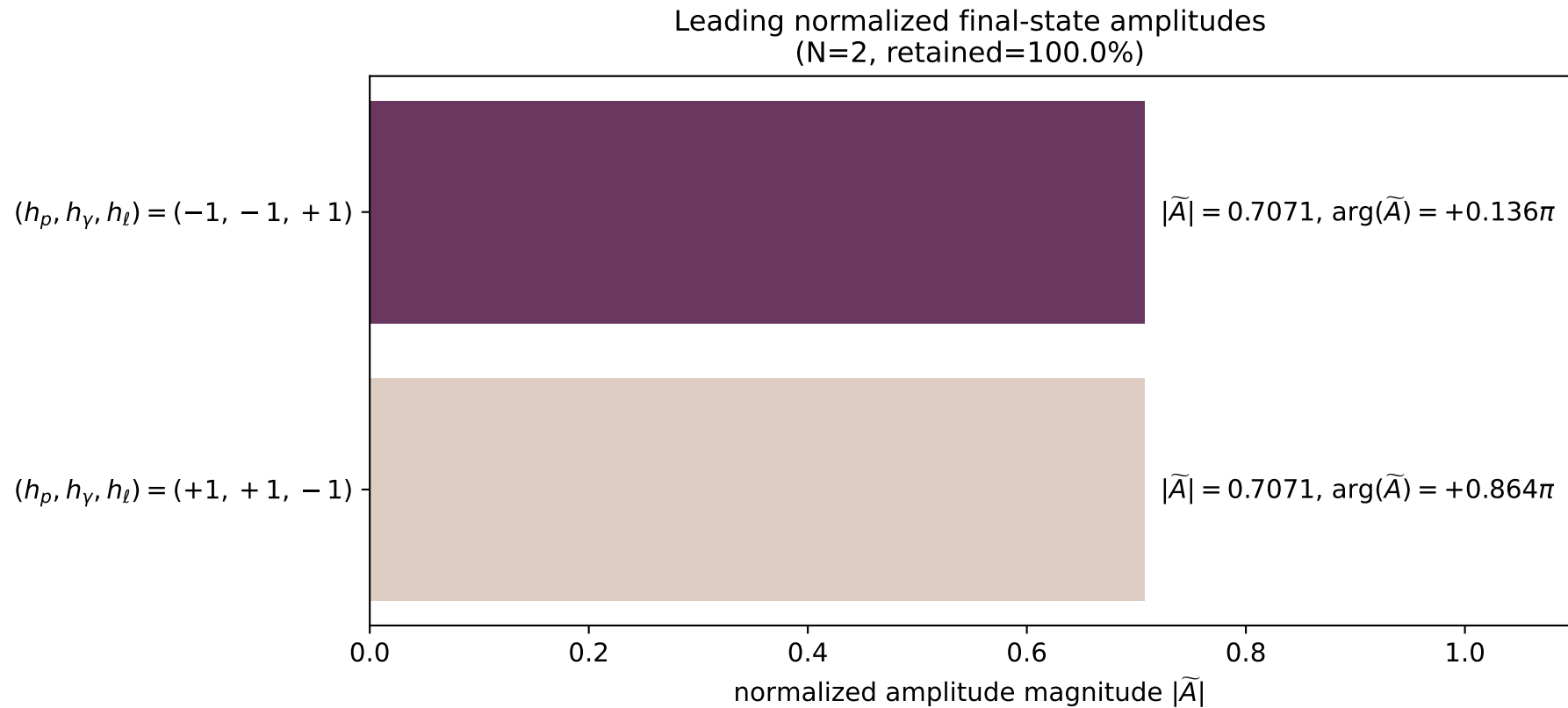
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_43
 $d_{\{\text{rm GHZ}\}}=2.6533\text{e-}08$
 region: polarization_cluster_03_local_minimum_43
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0043
 lepton mixing angle: $\alpha_e=2.3049352$ rad
 incoming lepton: $-0.669948|+\rangle +0.742408|-\rangle$
 proton mixing angle: $\alpha_p=0.83665018$ rad
 incoming proton: $0.669953|+\rangle +0.742403|-\rangle$

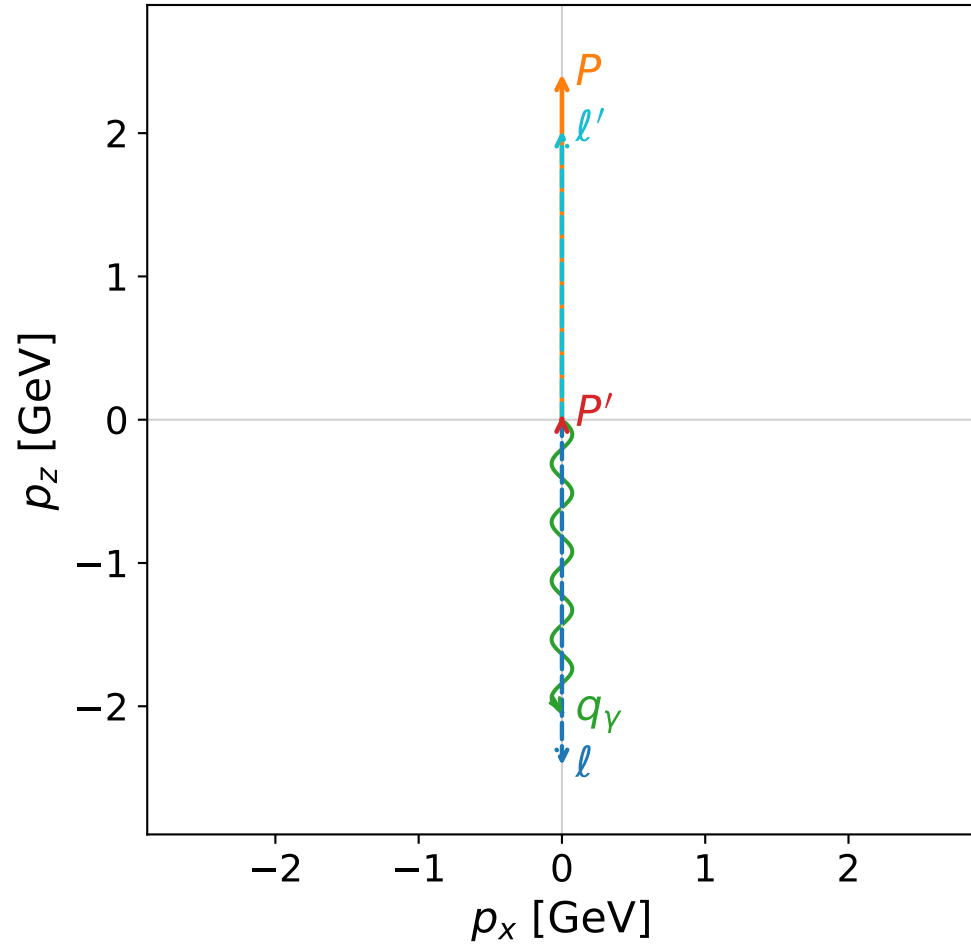
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.023742$
 $\theta_{p'}=3.55227\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.592674$, $\phi_\gamma=3.5688$
 $E_\gamma=2.04272$, $\theta_{\ell'}=4.1803\text{e-}07$, $\phi_{\ell'}=3.73427$
 $Q^2=19.4792$, $x_B=0.832104$, $t=-2.98239$, $W^2=4.81021$, $y=0.970541$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0190$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0190$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0237$
 q_γ $E= 2.0427$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0427$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

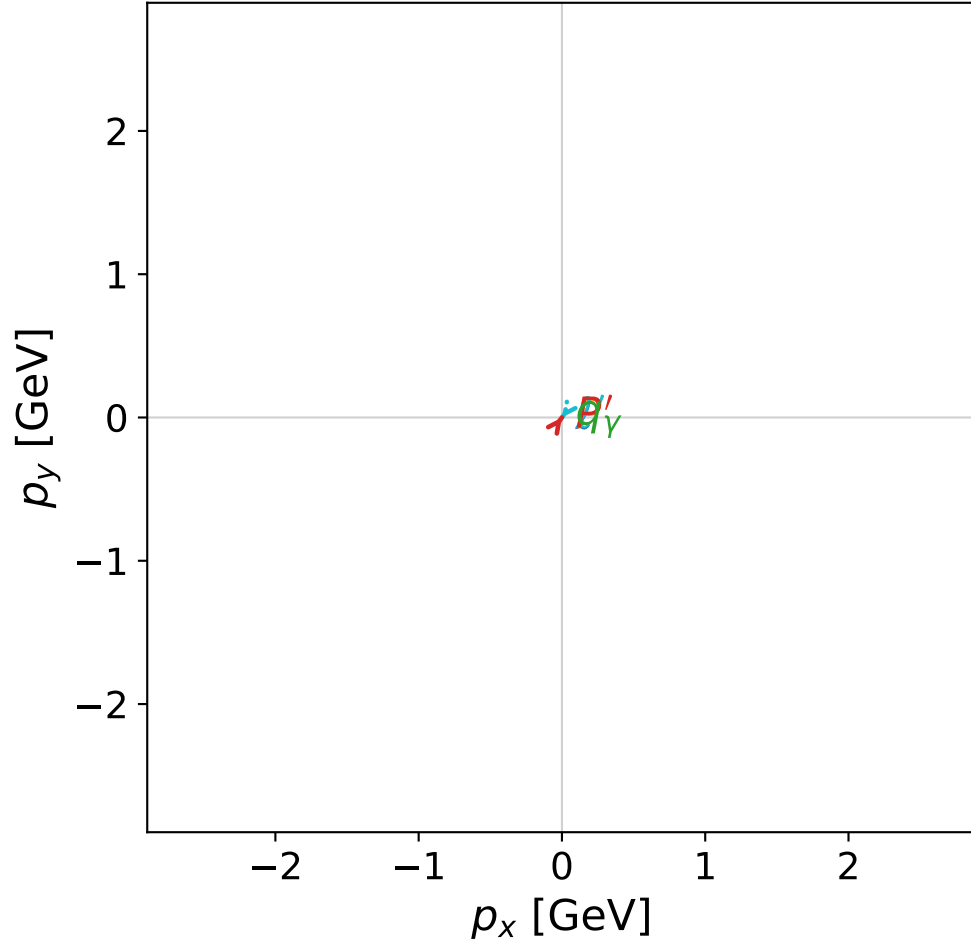


dghz_mixing_angles_polarization_03_local_minimum_45
 $d_{\text{GHZ}}=2.6597\text{e-}08$
 region: polarization_cluster_03_local_minimum_45
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0045
 lepton mixing angle: $\alpha_e=2.3367051$ rad
 incoming lepton: $-0.693192|+\rangle +0.720753|-\rangle$
 proton mixing angle: $\alpha_p=0.80488895$ rad
 incoming proton: $0.693191|+\rangle +0.720754|-\rangle$

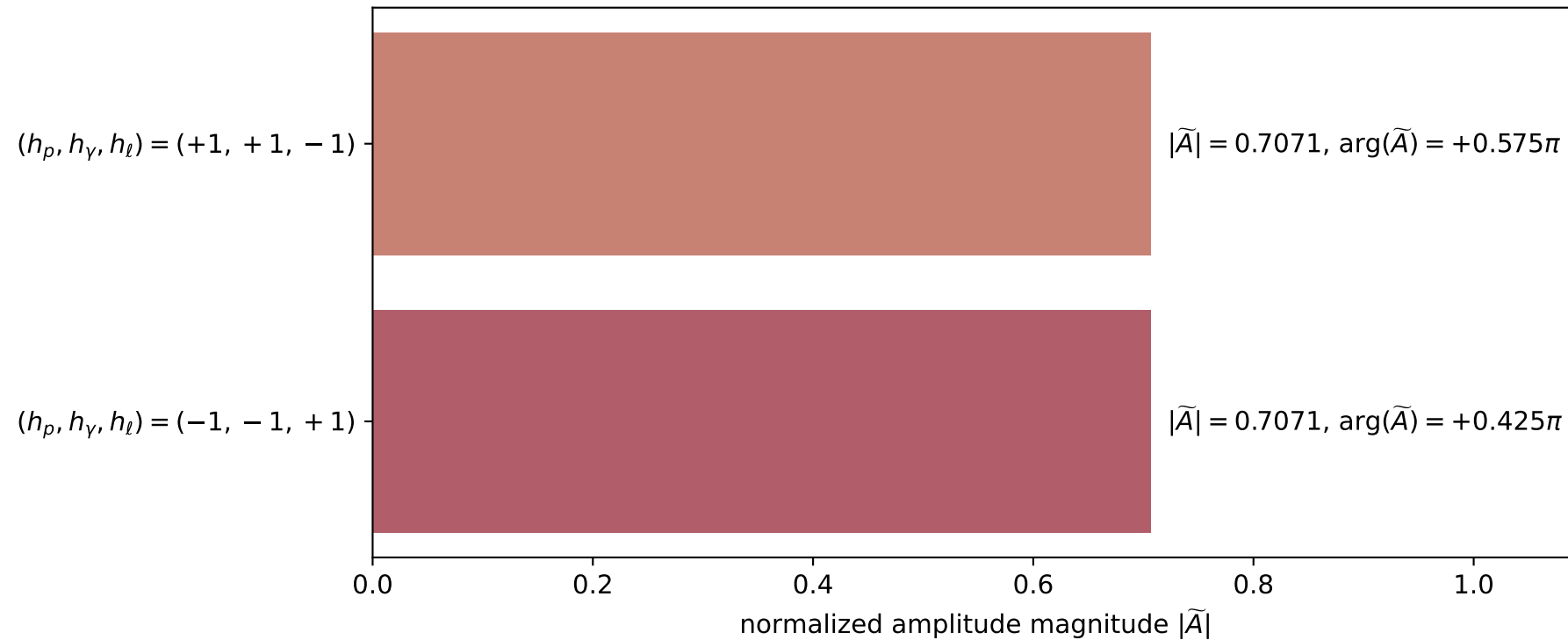
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0277887$
 $\theta_{P'}=0.00245437$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.937225$, $\phi_\gamma=4.47598$
 $E_\gamma=2.04469$, $\theta_{\ell'}=3.38161\text{e-}05$, $\phi_{\ell'}=4.07882$
 $Q^2=19.4592$, $x_B=0.831222$, $t=-2.96345$, $W^2=4.831$, $y=0.970571$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0169$ $p_x= -0.0000$ $p_y= -0.0001$ $p_z= 2.0169$
 P' $E= 0.9384$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0278$
 q_γ $E= 2.0447$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0447$

x--y plane

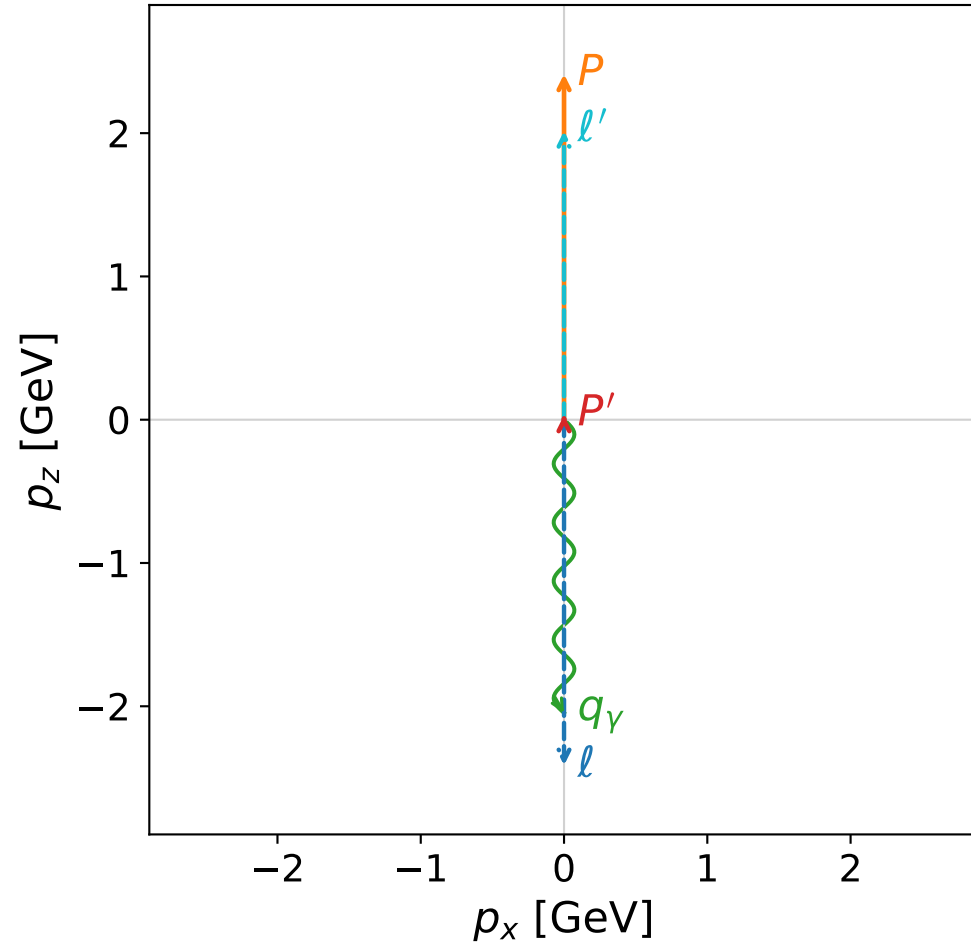


Leading normalized final-state amplitudes (N=2, retained=100.0%)

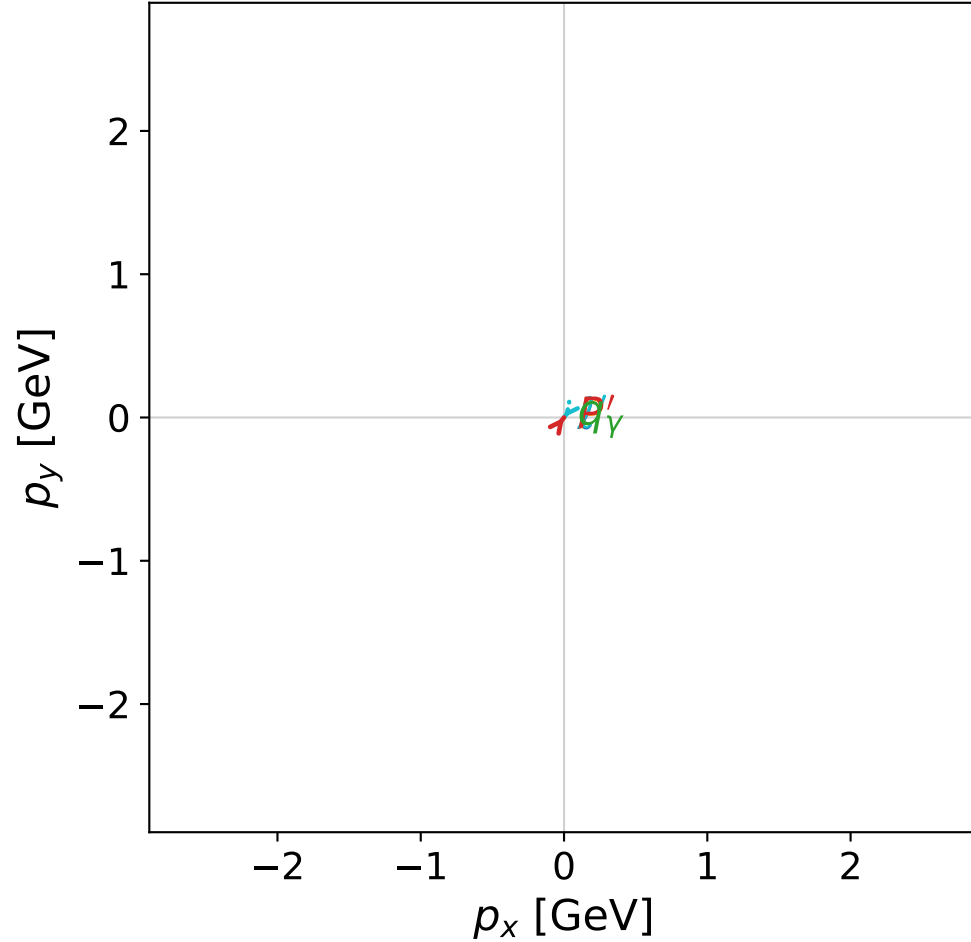


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



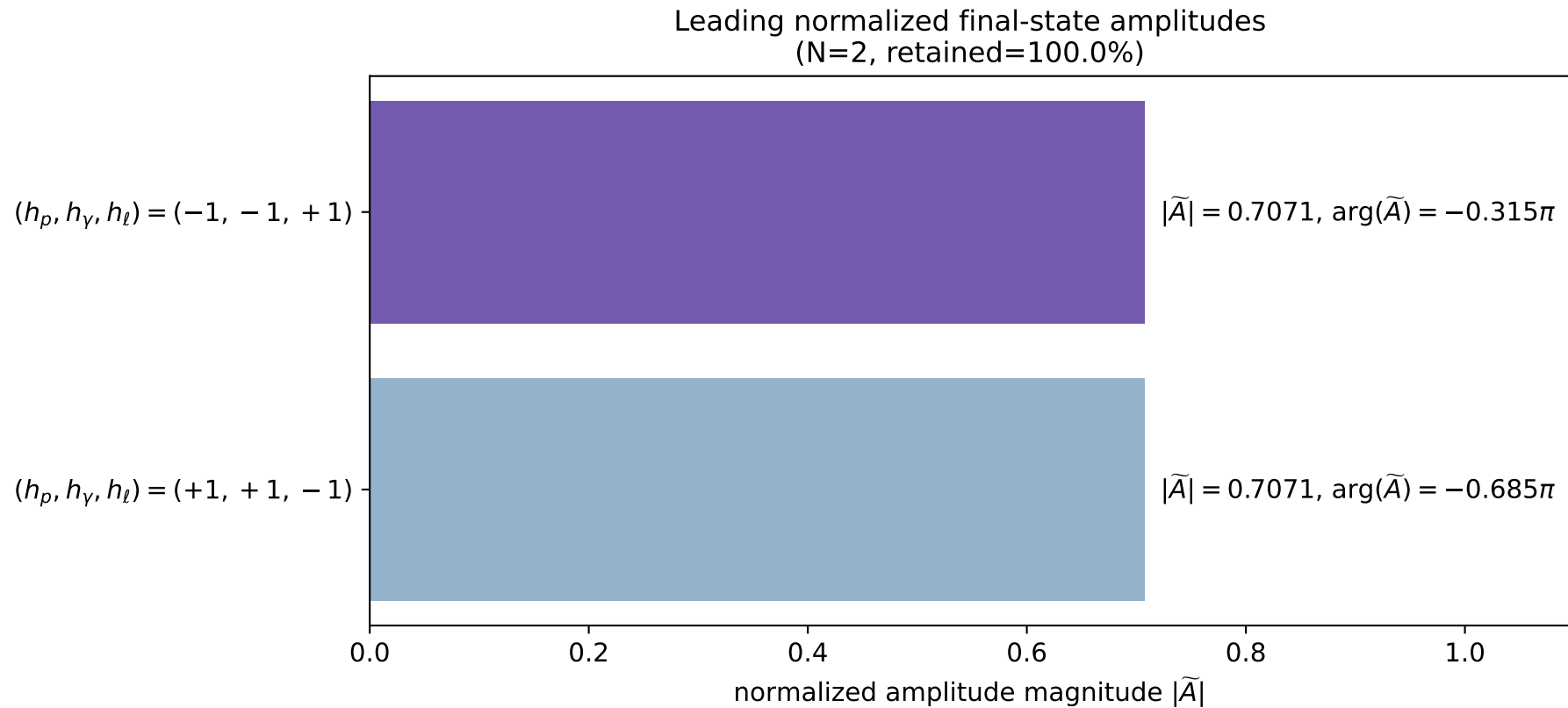
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_48
 $d_{\{\rm GHZ\}}=2.67266\text{e-}08$
 region: polarization_cluster_03_local_minimum_48
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0048
 lepton mixing angle: $\alpha_e=2.3692897$ rad
 incoming lepton: $-0.716306|+\rangle +0.697787|-\rangle$
 proton mixing angle: $\alpha_p=0.772294$ rad
 incoming proton: $0.716312|+\rangle +0.69778|-\rangle$

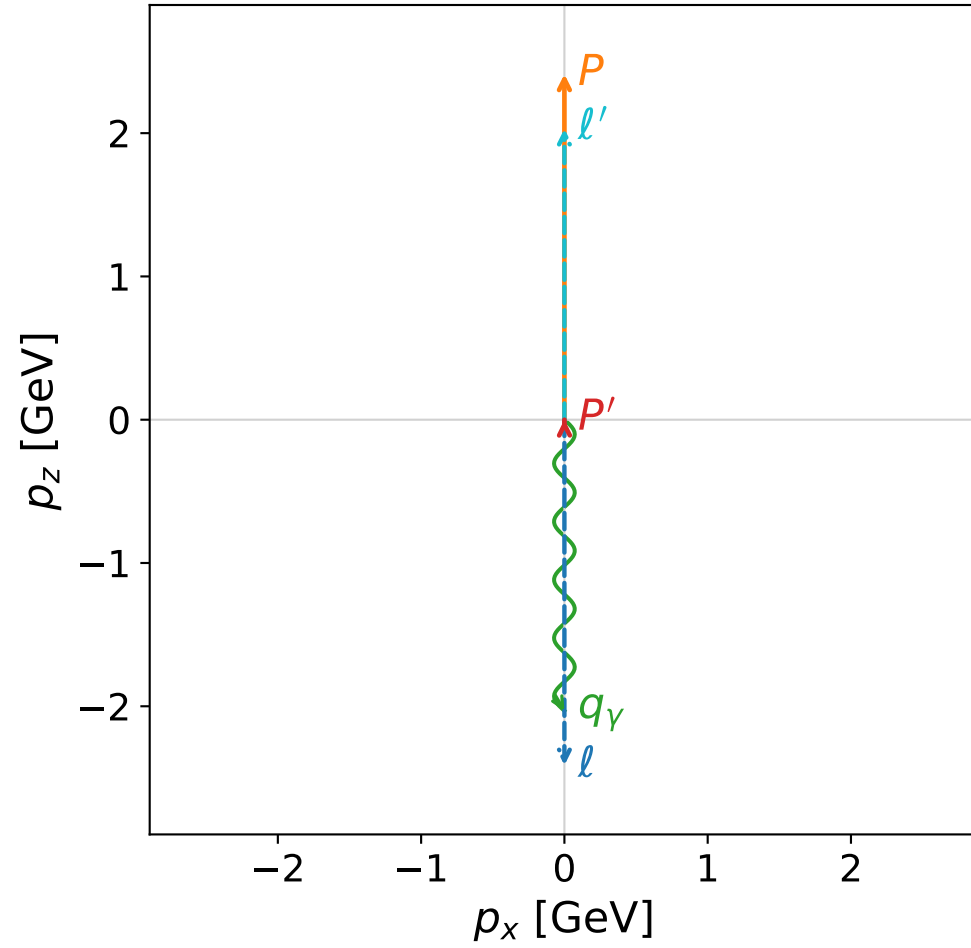
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0314813$
 $\theta_{p'}=0.000613592$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.92167$, $\phi_\gamma=2.15105$
 $E_\gamma=2.04648$, $\theta_{\ell'}=9.58645\text{e-}06$, $\phi_{\ell'}=4.06326$
 $Q^2=19.4408$, $x_B=0.830413$, $t=-2.94624$, $W^2=4.85005$, $y=0.970599$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0150$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.0150$
 P' $E= 0.9385$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0315$
 q_γ $E= 2.0465$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0465$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

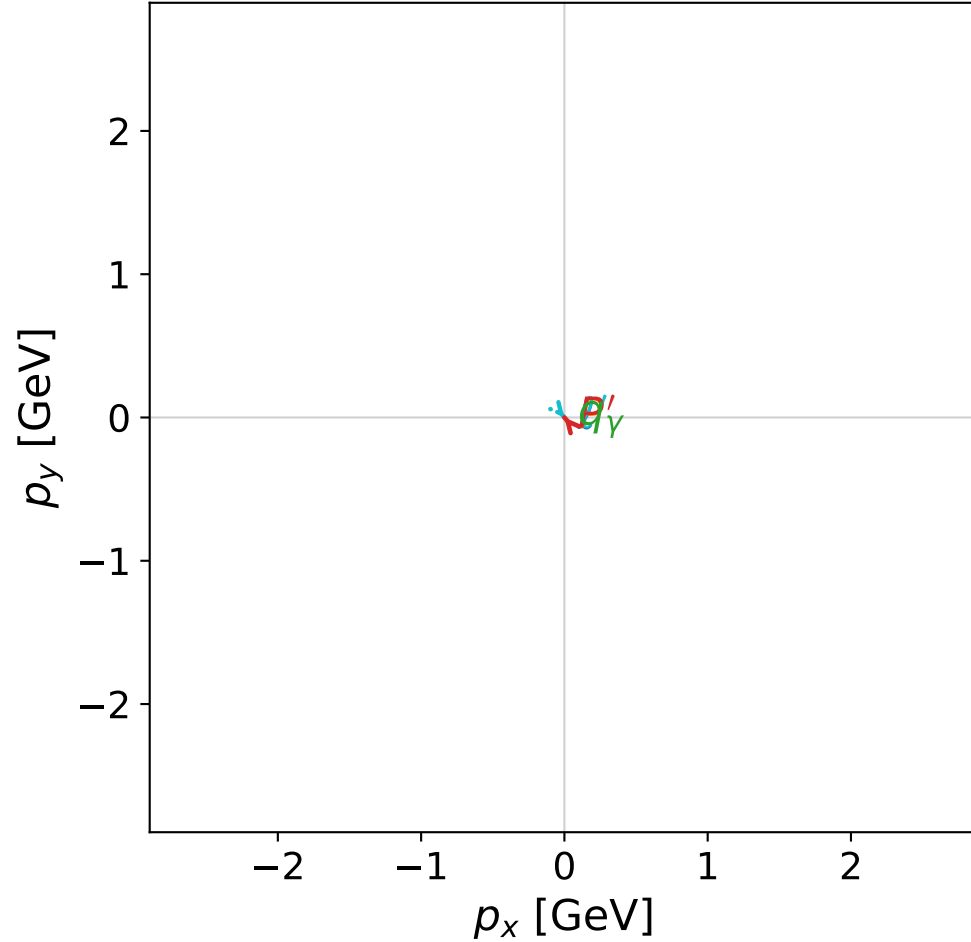


dghz_mixing_angles_polarization_03_local_minimum_51
 $d_{\text{GHZ}}=2.68417\text{e-}08$
 region: polarization_cluster_03_local_minimum_51
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0051
 lepton mixing angle: $\alpha_e=2.2220181$ rad
 incoming lepton: $-0.606159|+\rangle +0.795344|-\rangle$
 proton mixing angle: $\alpha_p=0.91958237$ rad
 incoming proton: $0.606152|+\rangle +0.795349|-\rangle$

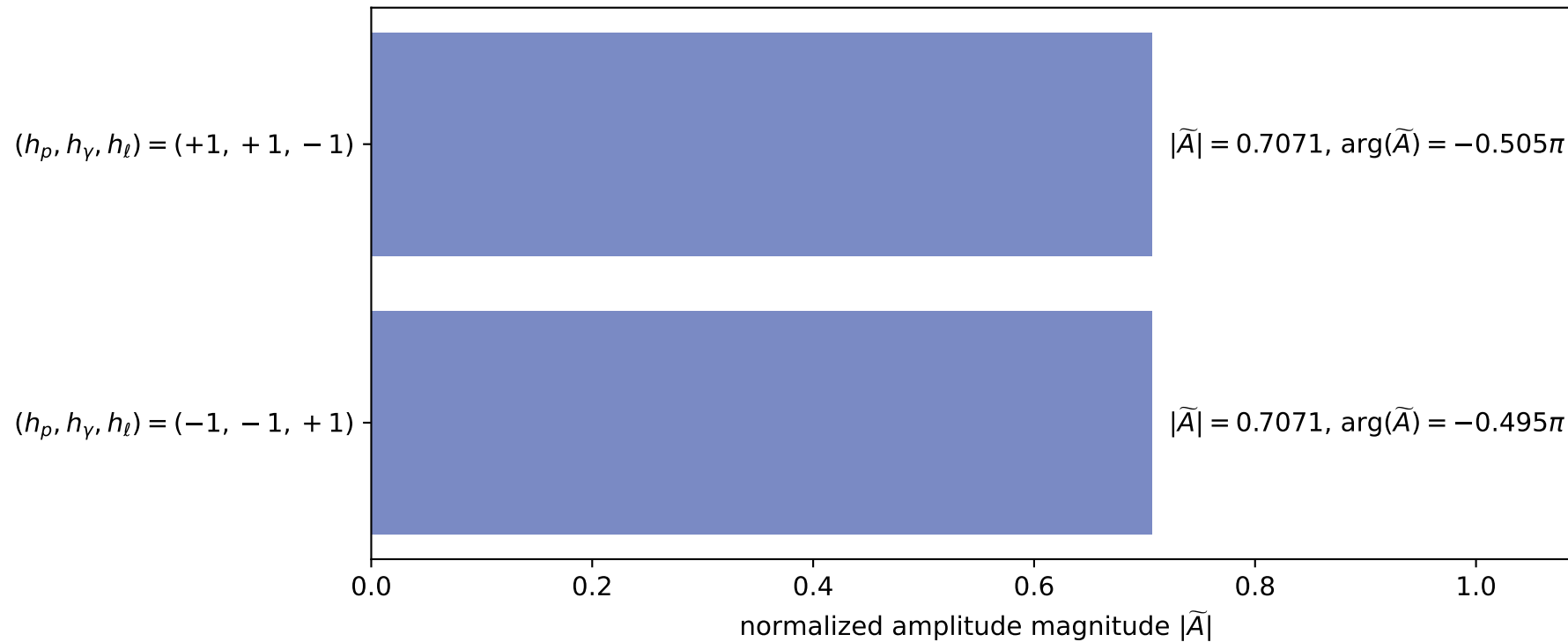
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.001359$
 $\theta_{P'}=0.00981748$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.27173$, $\phi_\gamma=1.5862$
 $E_\gamma=2.03168$, $\theta_{\ell'}=6.57126\text{e-}06$, $\phi_{\ell'}=5.41332$
 $Q^2=19.5887$, $x_B=0.836921$, $t=-3.08882$, $W^2=4.6968$, $y=0.970376$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0303$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0303$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0014$
 q_γ $E= 2.0317$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0317$

x--y plane

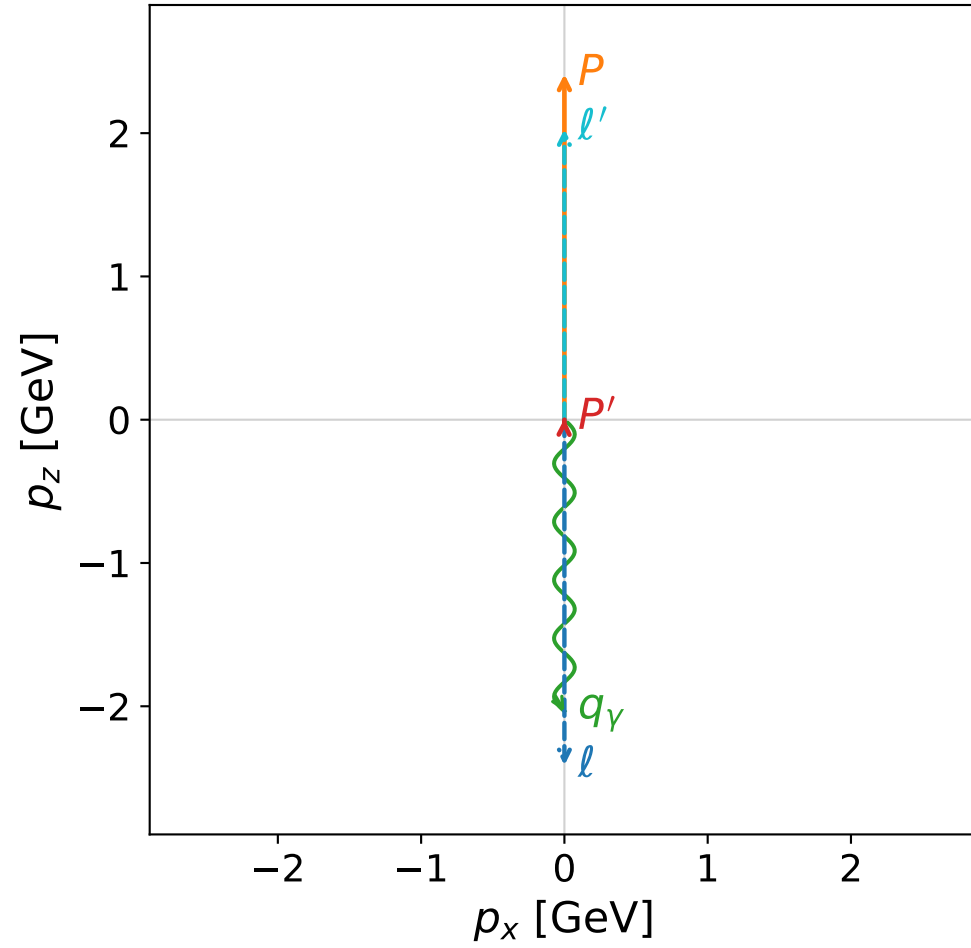


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

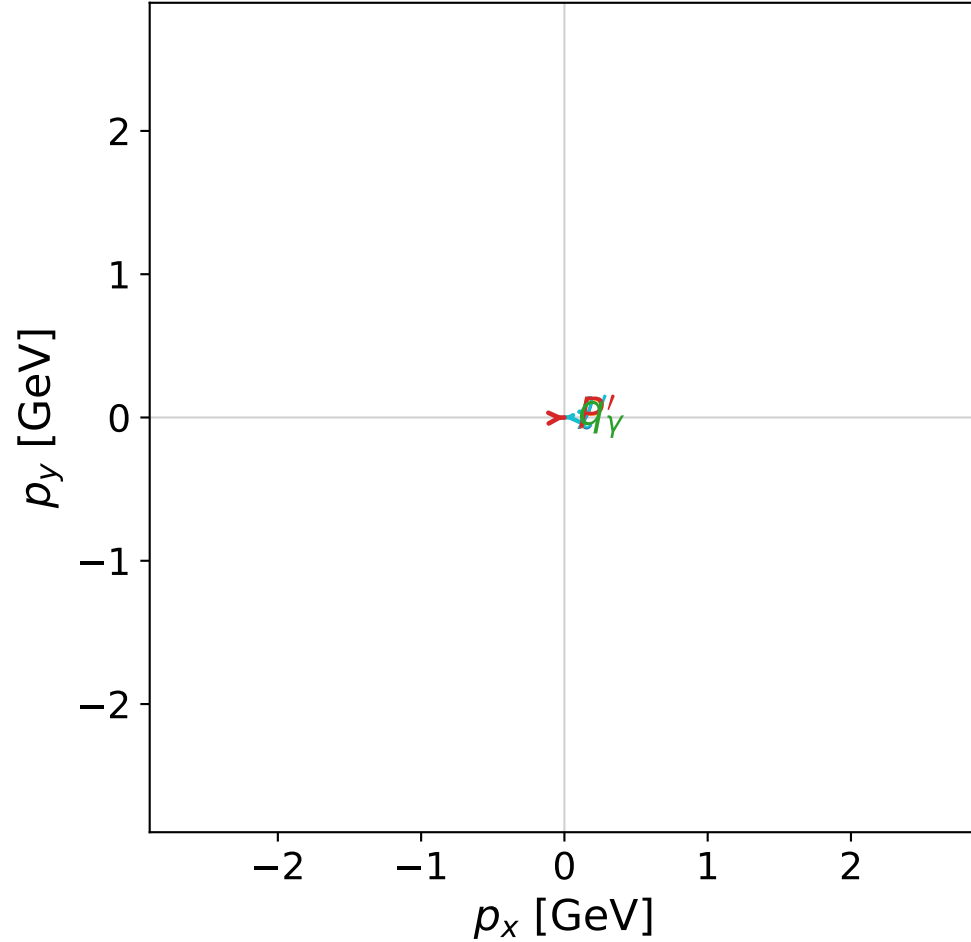


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



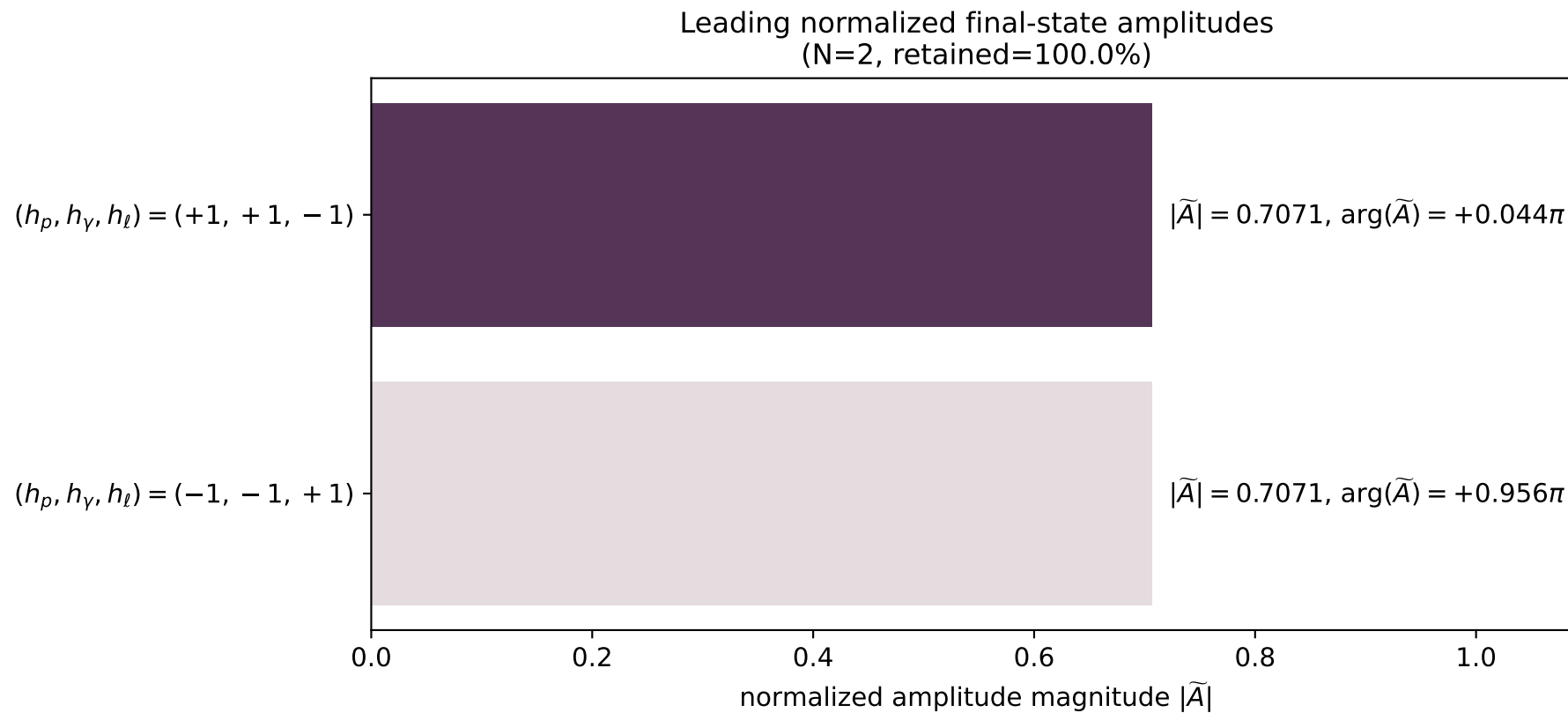
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_52
 $d_{\text{GHZ}}=2.69014\text{e-}08$
 region: polarization_cluster_03_local_minimum_52
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0052
 lepton mixing angle: $\alpha_e=2.4223145$ rad
 incoming lepton: $-0.752281|+\rangle +0.658842|-\rangle$
 proton mixing angle: $\alpha_p=0.71927442$ rad
 incoming proton: $0.752284|+\rangle +0.658839|-\rangle$

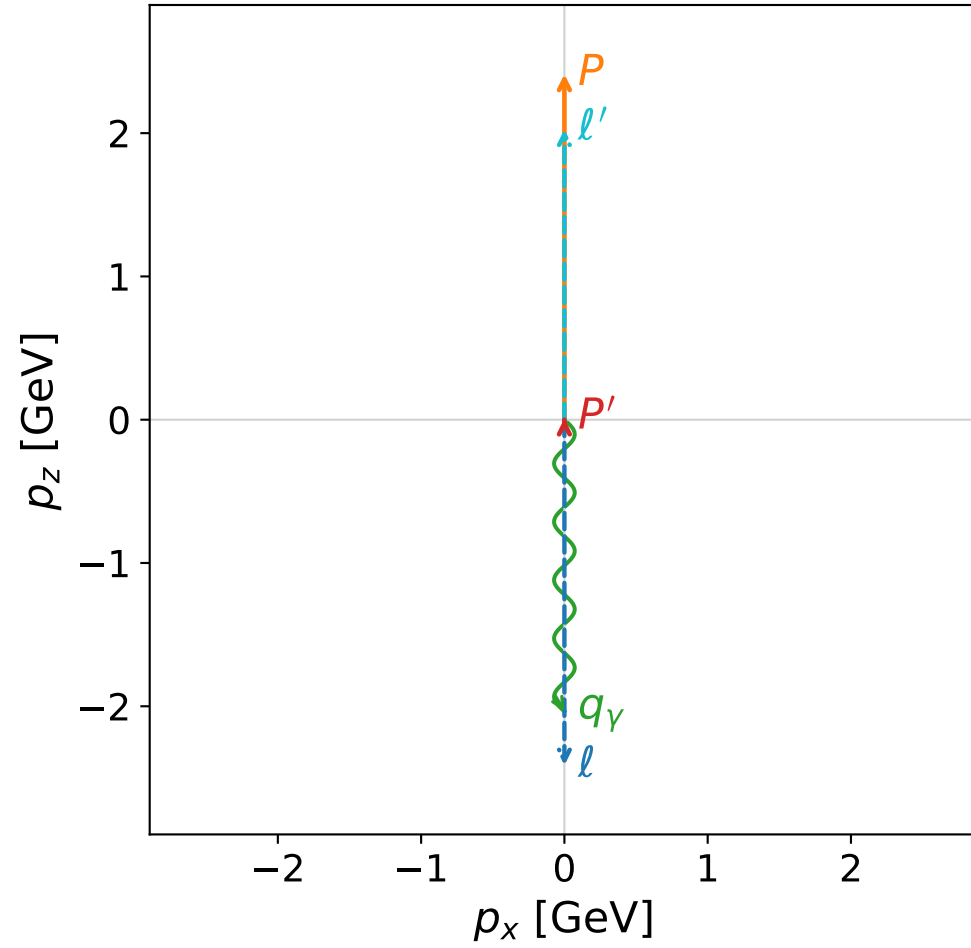
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.00684209$
 $\theta_{P'}=0.0196328$, $\theta_Y=3.14159$, $\phi_{P'}=0.0386213$, $\phi_Y=6.14533$
 $E_Y=2.03441$, $\theta_{\ell'}=6.62473\text{e-}05$, $\phi_{\ell'}=3.18021$
 $Q^2=19.5621$, $x_B=0.835752$, $t=-3.0625$, $W^2=4.72433$, $y=0.970416$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0276$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0276$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0068$
 q_Y $E= 2.0344$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0344$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

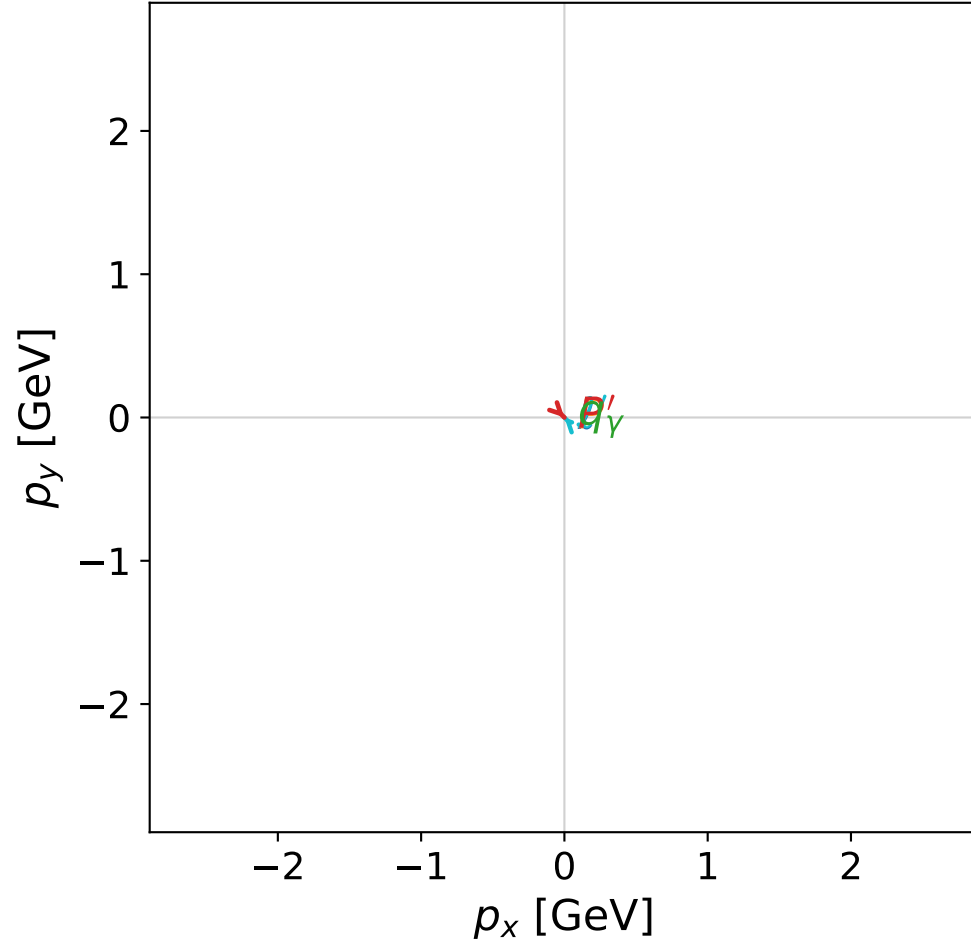


dghz_mixing_angles_polarization_03_local_minimum_53
 $d_{\text{GHZ}}=2.69649\text{e-}08$
 region: polarization_cluster_03_local_minimum_53
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0053
 lepton mixing angle: $\alpha_e=2.2615246$ rad
 incoming lepton: $-0.637099|+\rangle +0.770782|-\rangle$
 proton mixing angle: $\alpha_p=0.8800608$ rad
 incoming proton: $0.637104|+\rangle +0.770778|-\rangle$

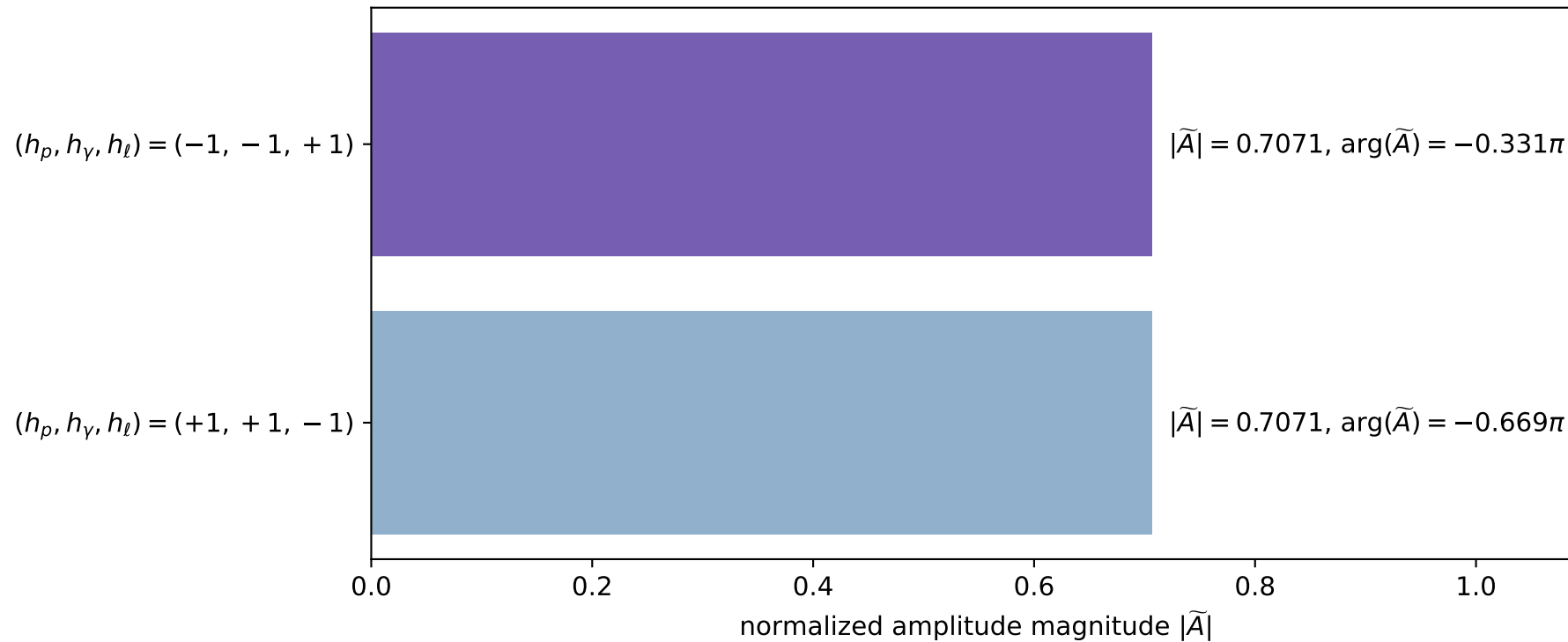
$s=24.9999$, $\sqrt{s}=4.99999$, $|\vec{P}|=2.41201$, $|\vec{P}'|=0.00908808$
 $\theta_{P'}=0.000472367$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.49656$, $\phi_\gamma=2.10193$
 $E_\gamma=2.03552$, $\theta_{\ell'}=2.11843\text{e-}06$, $\phi_{\ell'}=2.35497$
 $Q^2=19.5511$, $x_B=0.83527$, $t=-3.05175$, $W^2=4.73565$, $y=0.970432$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0264$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0264$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0091$
 q_γ $E= 2.0355$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0355$

x--y plane

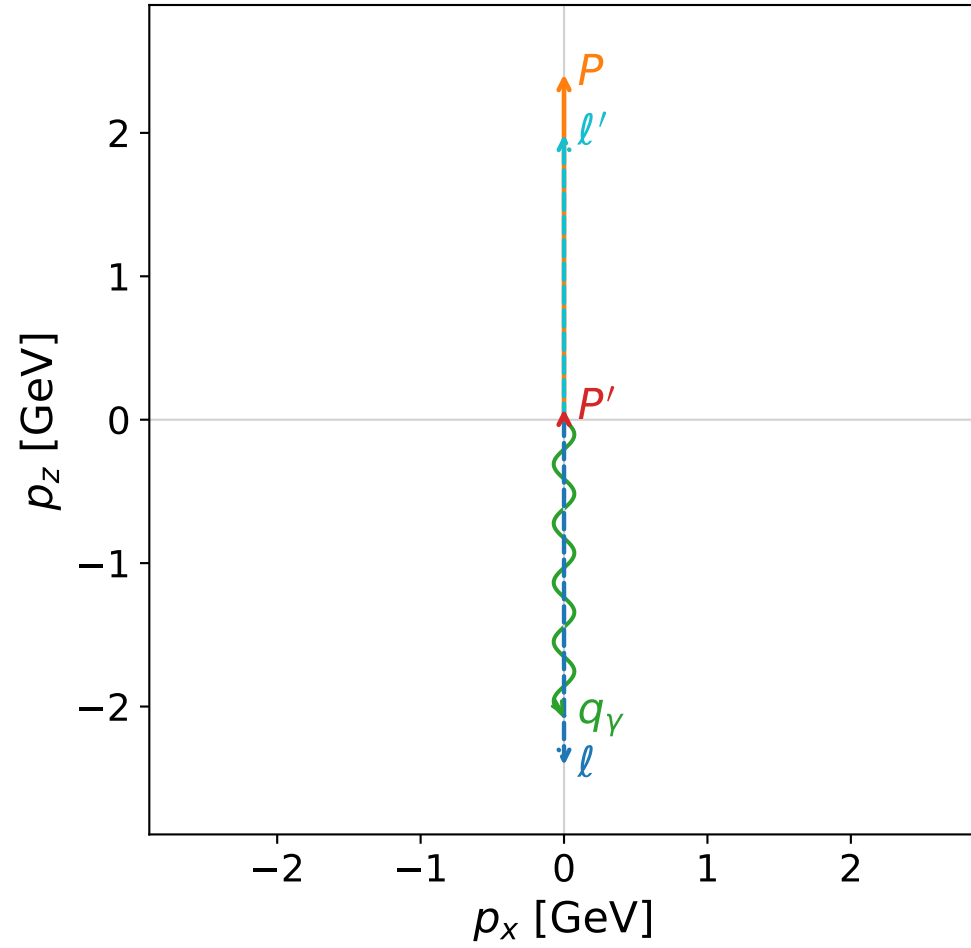


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

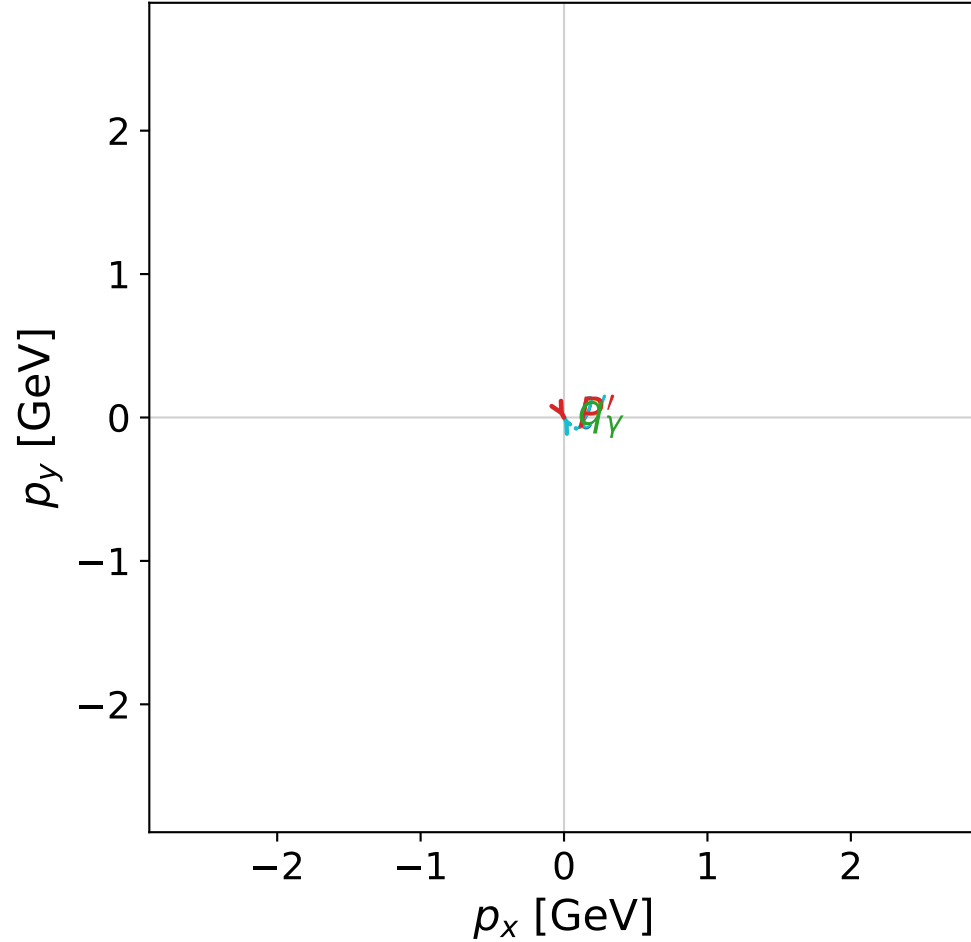


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



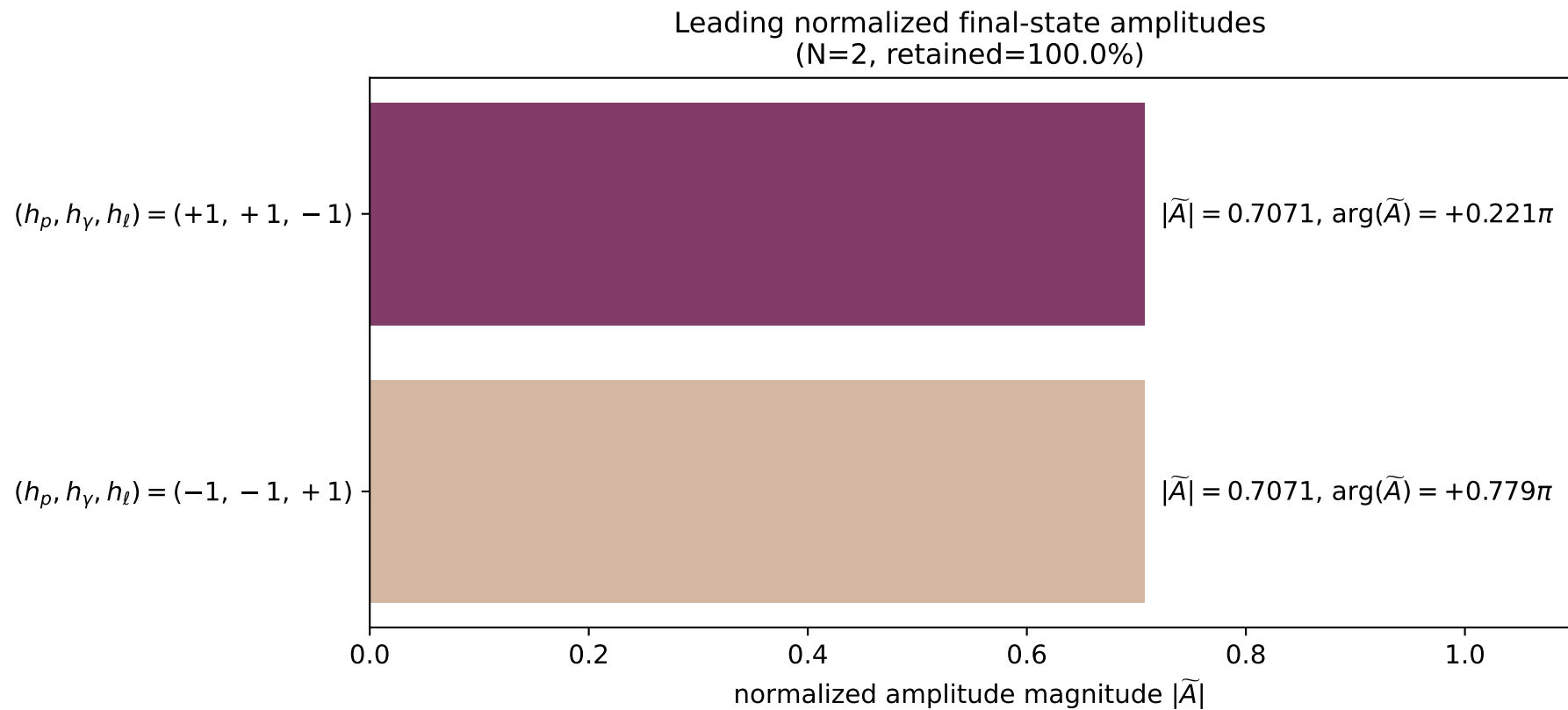
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_55
 $d_{\text{GHZ}}=2.70614\text{e-}08$
 region: polarization_cluster_03_local_minimum_55
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0055
 lepton mixing angle: $\alpha_e=2.360551$ rad
 incoming lepton: $-0.710181|+\rangle +0.70402|-\rangle$
 proton mixing angle: $\alpha_p=0.78104501$ rad
 incoming proton: $0.710178|+\rangle +0.704022|-\rangle$

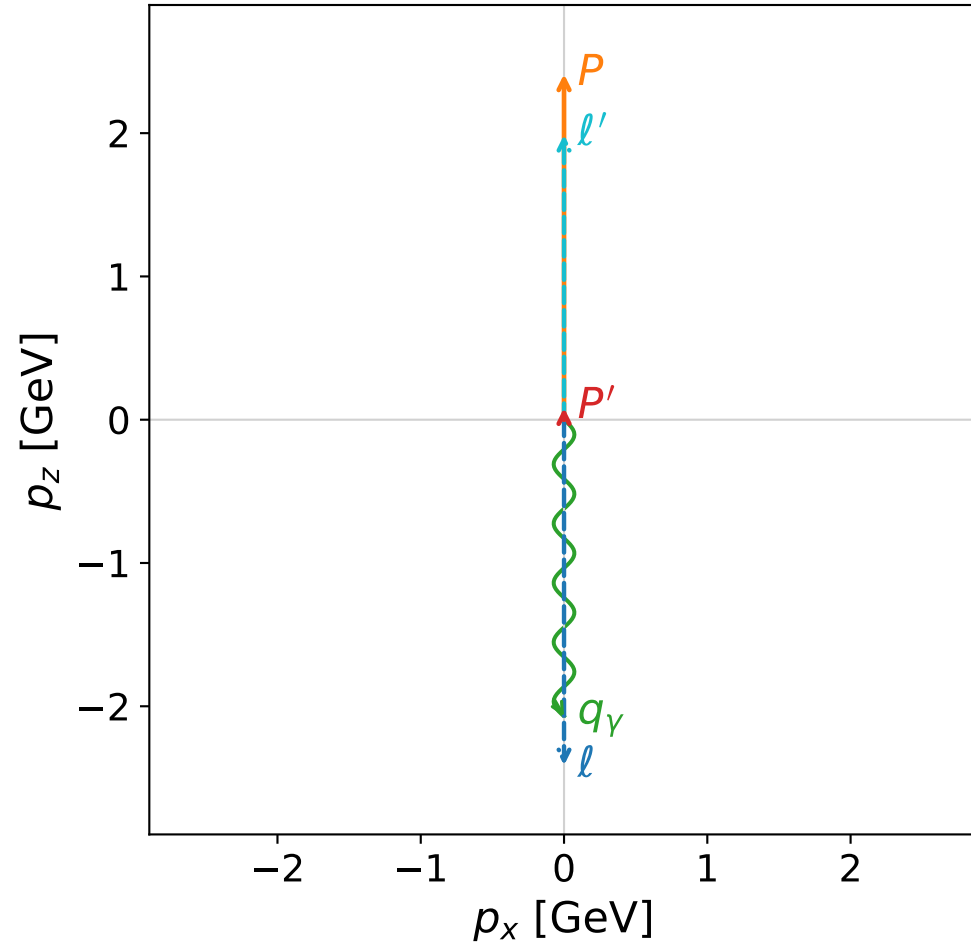
$s=24.9555$, $\sqrt{s}=4.99555$, $|\vec{P}|=2.40971$, $|\vec{P}'|=0.0757091$
 $\theta_{P'}=7.31897\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.21638$, $\phi_\gamma=5.58978$
 $E_\gamma=2.0651$, $\theta_{\ell'}=2.78532\text{e-}06$, $\phi_{\ell'}=2.07479$
 $Q^2=19.1755$, $x_B=0.820345$, $t=-2.74225$, $W^2=5.07927$, $y=0.970893$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4097$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4097$
 P $E= 2.5858$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4097$
 ℓ' $E= 1.9894$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9894$
 P' $E= 0.9411$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0757$
 q_γ $E= 2.0651$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0651$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

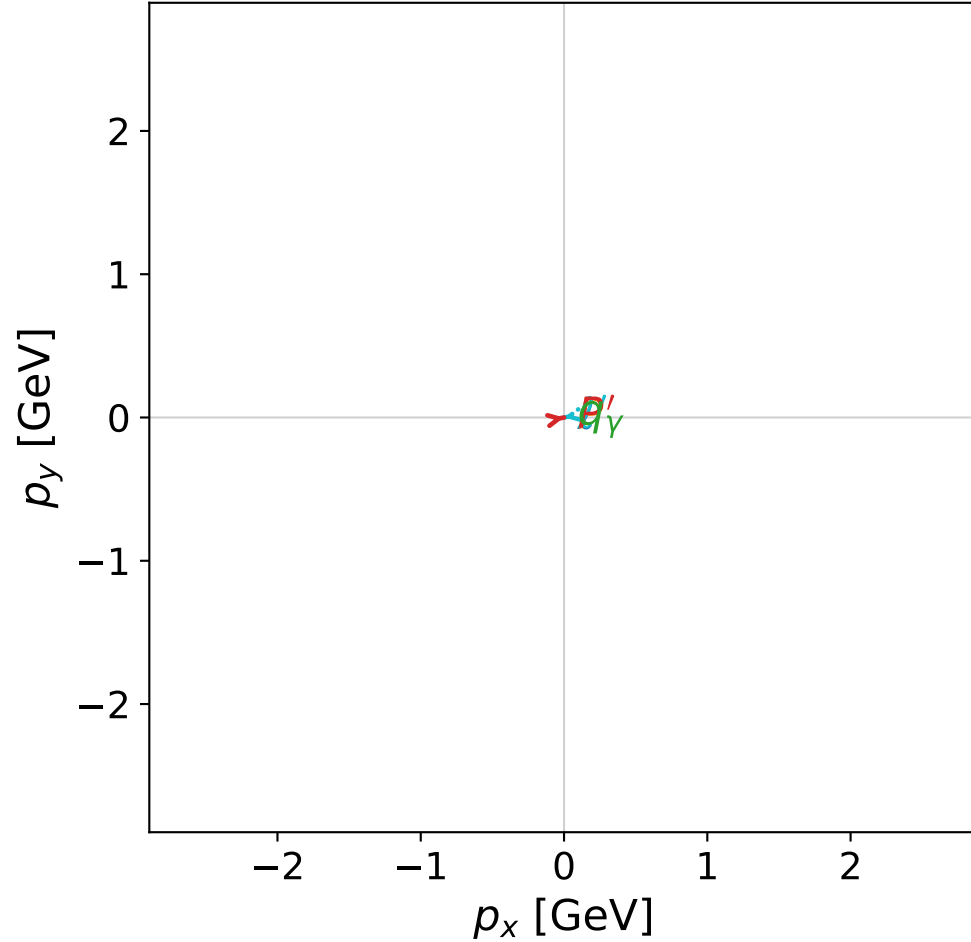


dghz_mixing_angles_polarization_03_local_minimum_56
 $d_{\{\rm GHZ\}}=2.70735\text{e-}08$
 region: polarization_cluster_03_local_minimum_56
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0056
 lepton mixing angle: $\alpha_e=2.3601707$ rad
 incoming lepton: $-0.709913|+\rangle +0.70429|-\rangle$
 proton mixing angle: $\alpha_p=0.78143038$ rad
 incoming proton: $0.709907|+\rangle +0.704296|-\rangle$

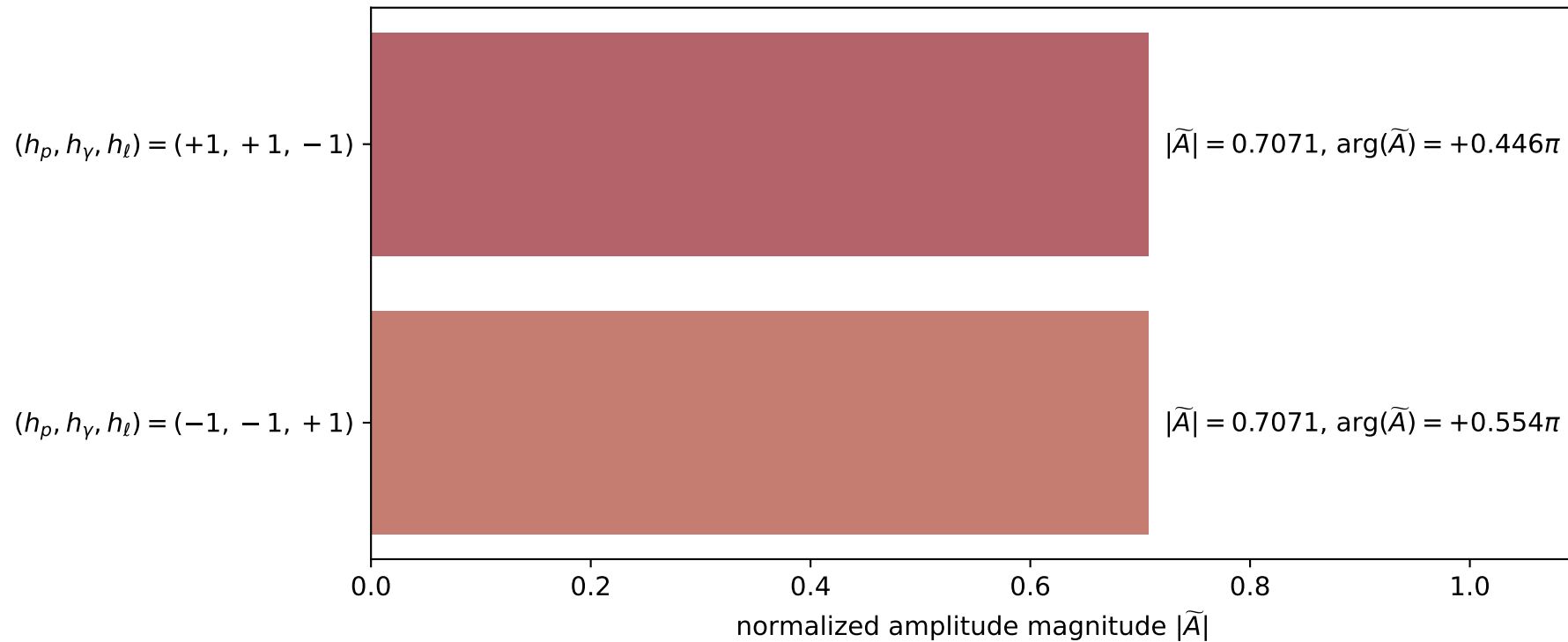
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.0826338$
 $\theta_{P'}=7.6699\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.192921$, $\phi_\gamma=4.88356$
 $E_\gamma=2.0705$, $\theta_{\ell'}=3.18833\text{e-}06$, $\phi_{\ell'}=3.33451$
 $Q^2=19.1791$, $x_B=0.818899$, $t=-2.71555$, $W^2=5.12133$, $y=0.970995$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9879$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9879$
 P' $E= 0.9416$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0826$
 q_γ $E= 2.0705$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.0705$

x--y plane

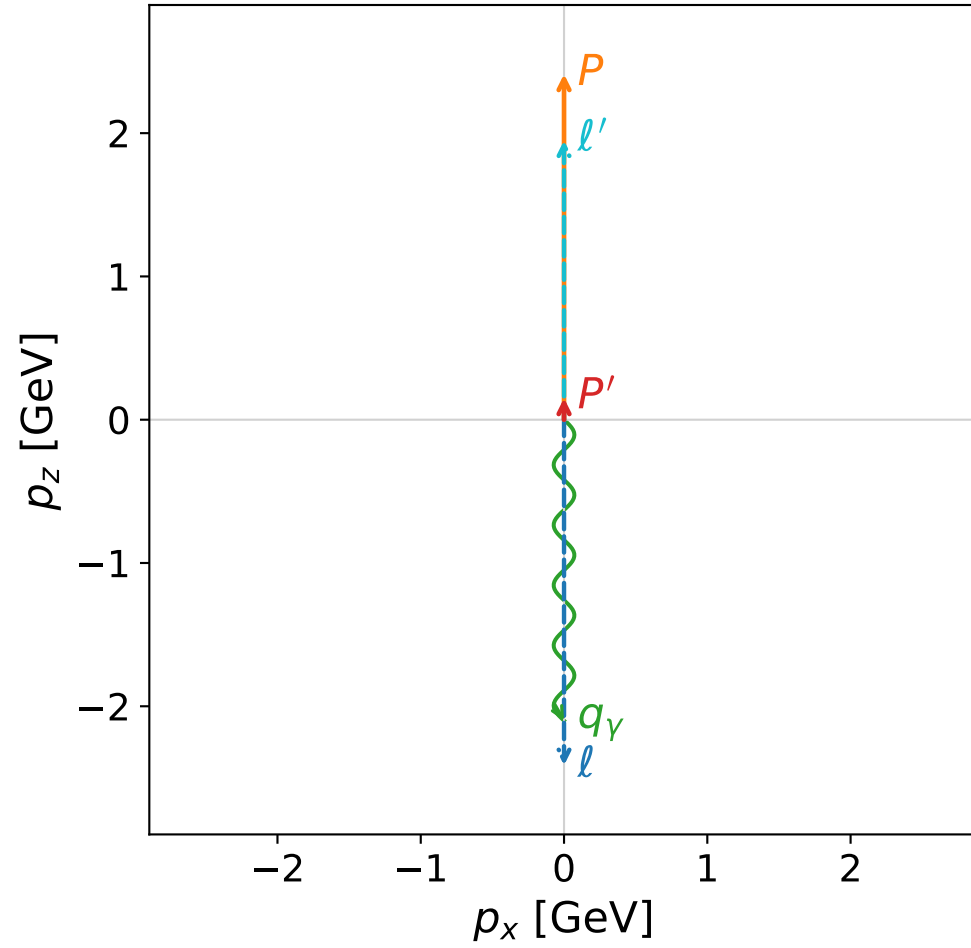


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

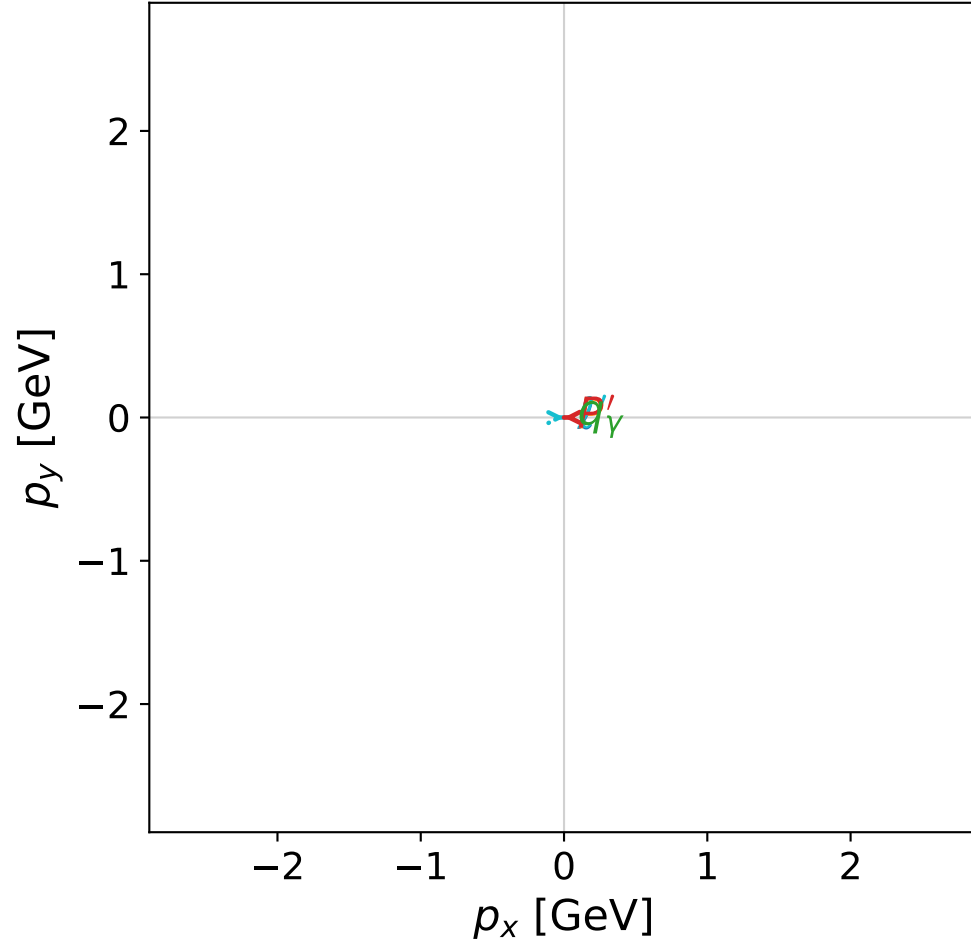


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



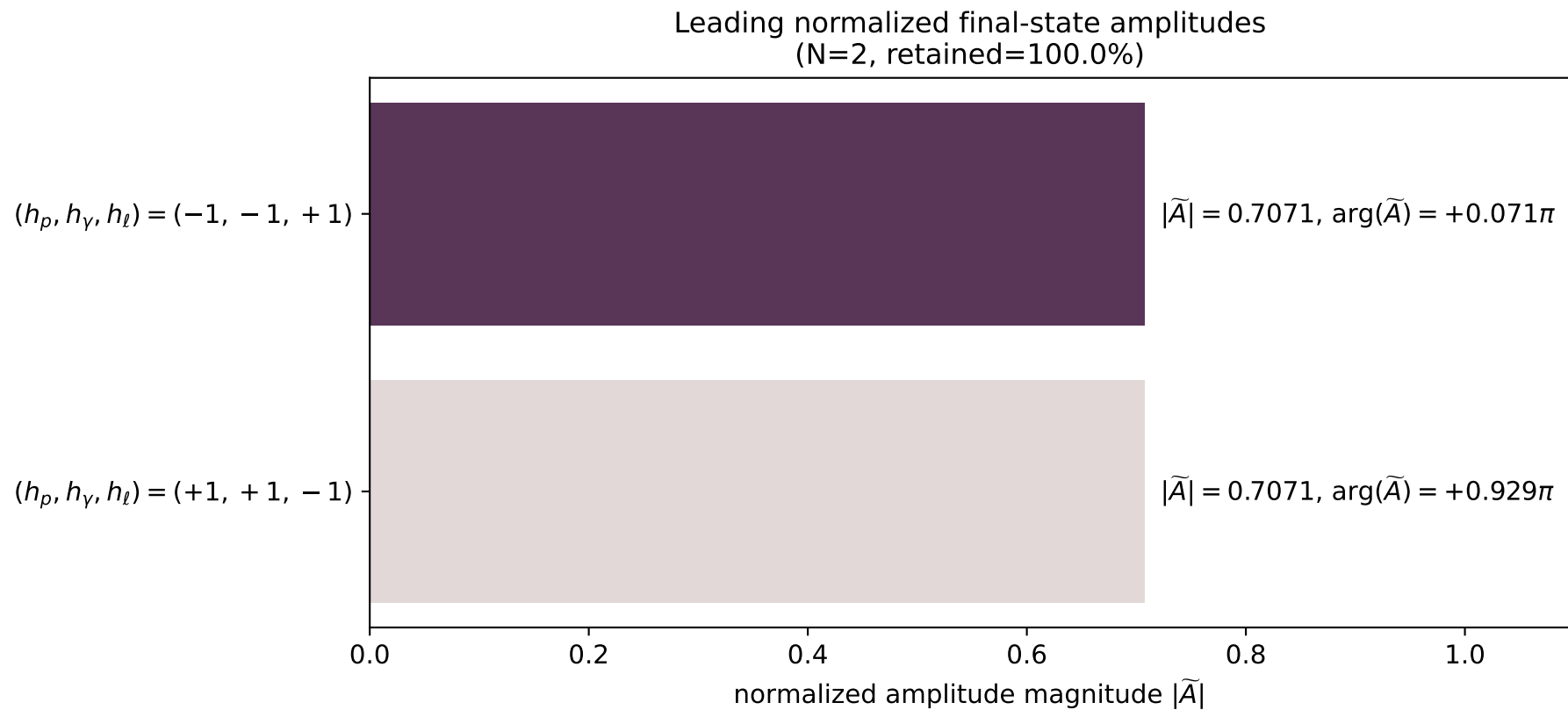
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_60
 $d_{\{\rm GHZ\}}=2.74796\text{e-}08$
 region: polarization_cluster_03_local_minimum_60
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0060
 lepton mixing angle: $\alpha_e=2.3243577$ rad
 incoming lepton: $-0.68424|+\rangle +0.729257|-\rangle$
 proton mixing angle: $\alpha_p=0.8172232$ rad
 incoming proton: $0.684249|+\rangle +0.729249|-\rangle$

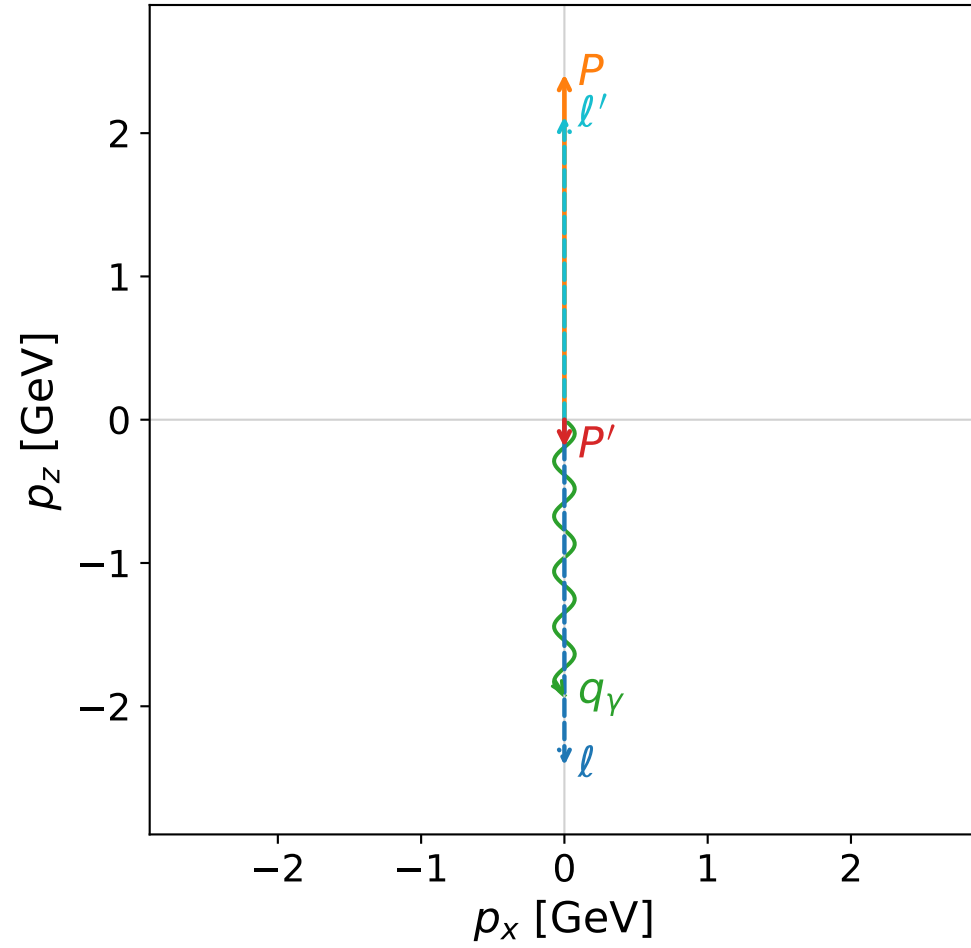
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.150619$
 $\theta_{p'}=5.43472\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.14933$, $\phi_\gamma=3.36397$
 $E_\gamma=2.1003$, $\theta_{\ell'}=4.19848\text{e-}06$, $\phi_{\ell'}=0.00773789$
 $Q^2=18.8107$, $x_B=0.802708$, $t=-2.43097$, $W^2=5.50317$, $y=0.971552$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.9497$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9497$
 P' $E= 0.9500$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1506$
 q_γ $E= 2.1003$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1003$

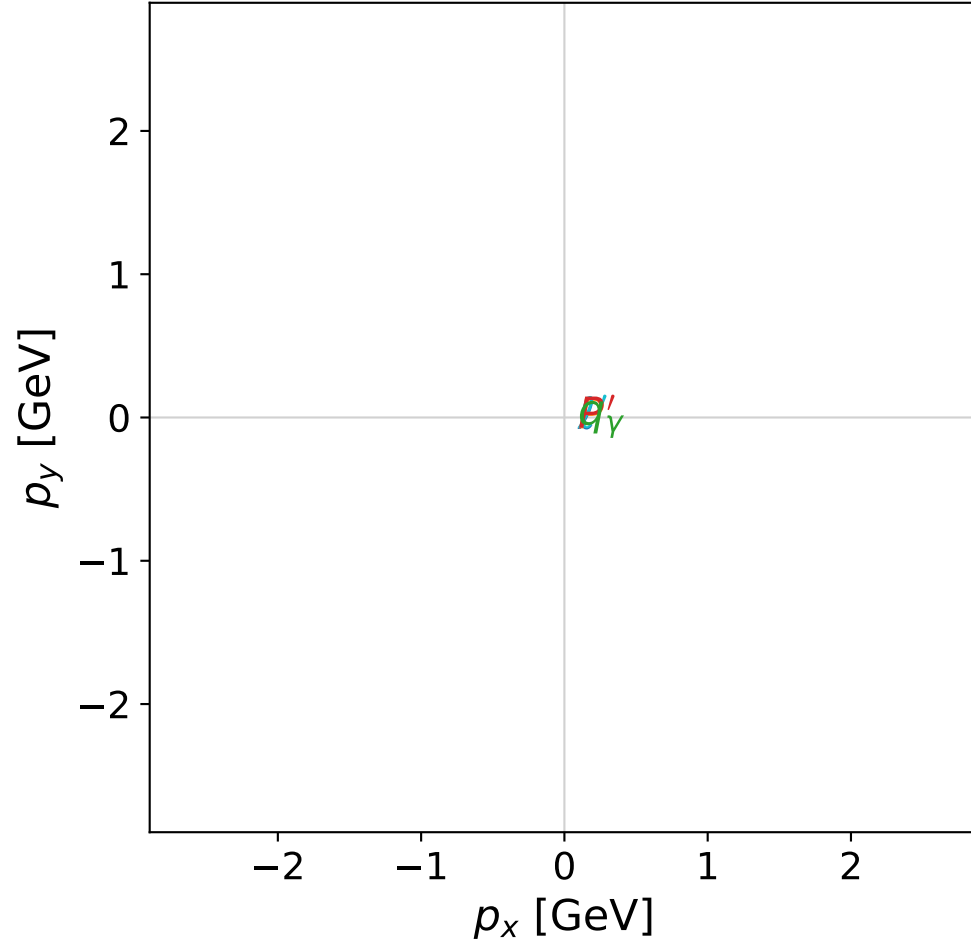


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

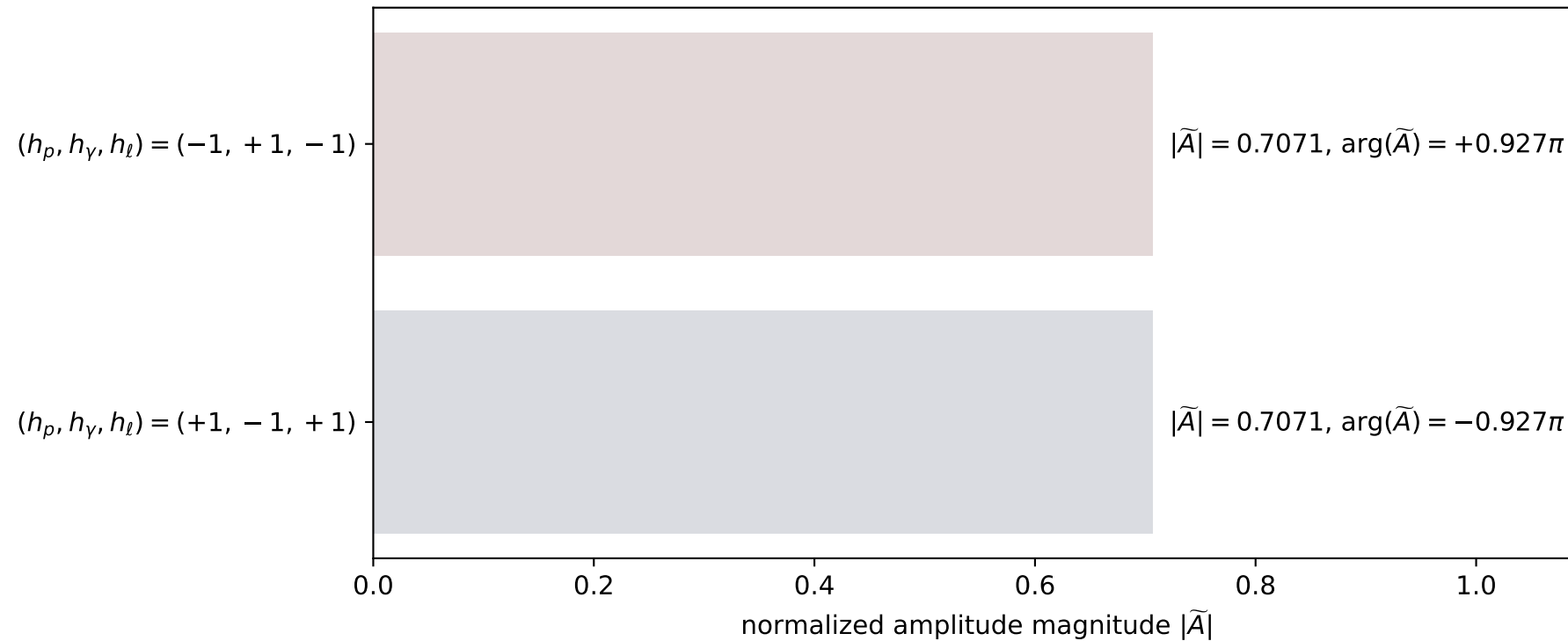


dghz_mixing_angles_polarization_03_local_minimum_66
 $d_{\text{GHZ}}=2.78662\text{e-}08$
 region: polarization_cluster_03_local_minimum_66
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0066
 lepton mixing angle: $\alpha_e=2.5600686$ rad
 incoming lepton: $-0.835626|+\rangle +0.549298|-\rangle$
 proton mixing angle: $\alpha_p=0.58153866$ rad
 incoming proton: $0.835618|+\rangle +0.54931|-\rangle$

$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.192212$
 $\theta_{P'}=3.14159$, $\theta_Y=3.14159$, $\phi_{P'}=3.14281$, $\phi_Y=0.229158$
 $E_Y=1.92515$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.39574$
 $Q^2=20.4284$, $x_B=0.873944$, $t=-4.12349$, $W^2=3.8264$, $y=0.969106$

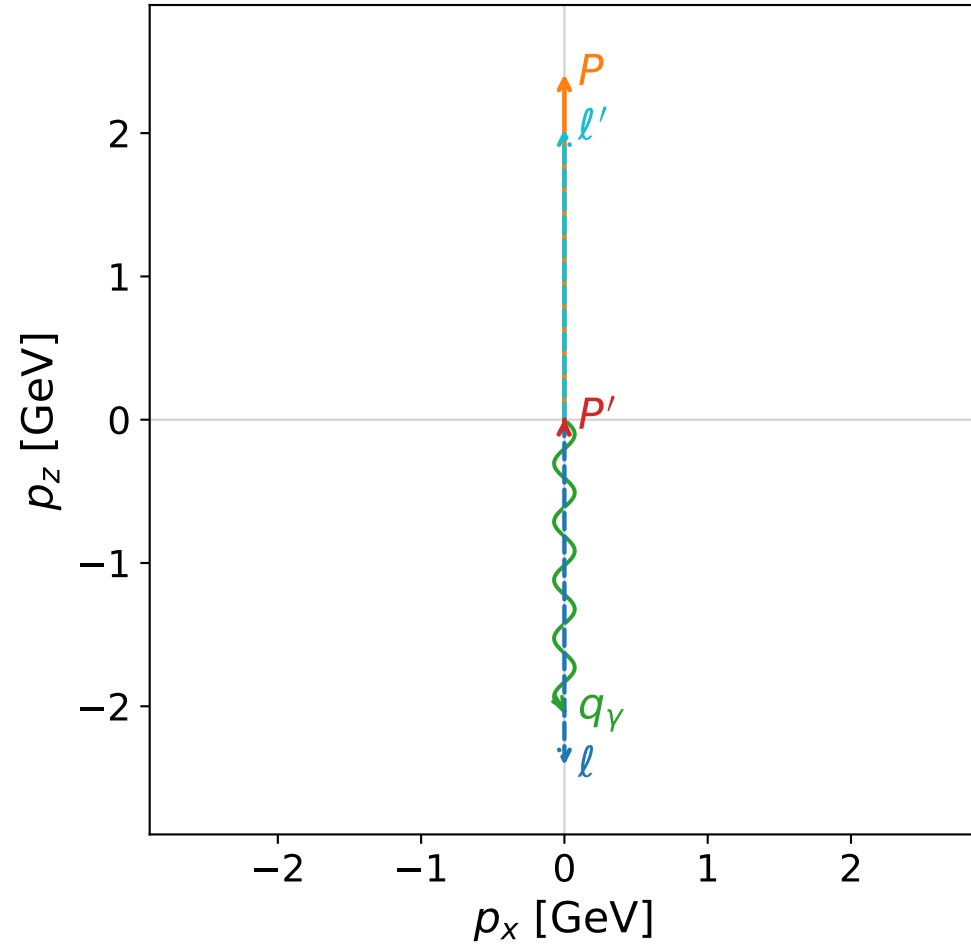
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.1174$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.1174$
 P' $E= 0.9575$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1922$
 q_Y $E= 1.9251$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9251$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

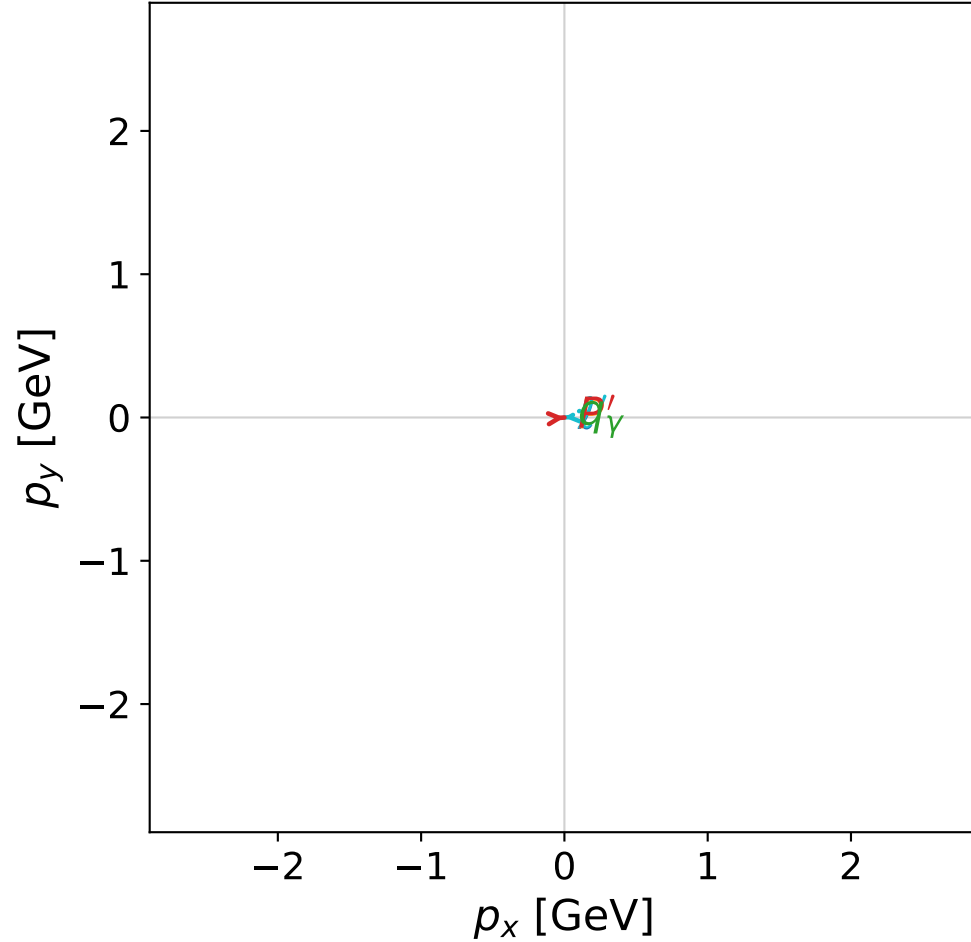


dghz_mixing_angles_polarization_03_local_minimum_67
 $d_{\text{GHZ}}=2.80689\text{e-}08$
 region: polarization_cluster_03_local_minimum_67
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0067
 lepton mixing angle: $\alpha_e=2.235145$ rad
 incoming lepton: $-0.616546|+\rangle +0.787319|-\rangle$
 proton mixing angle: $\alpha_p=0.90642644$ rad
 incoming proton: $0.616563|+\rangle +0.787305|-\rangle$

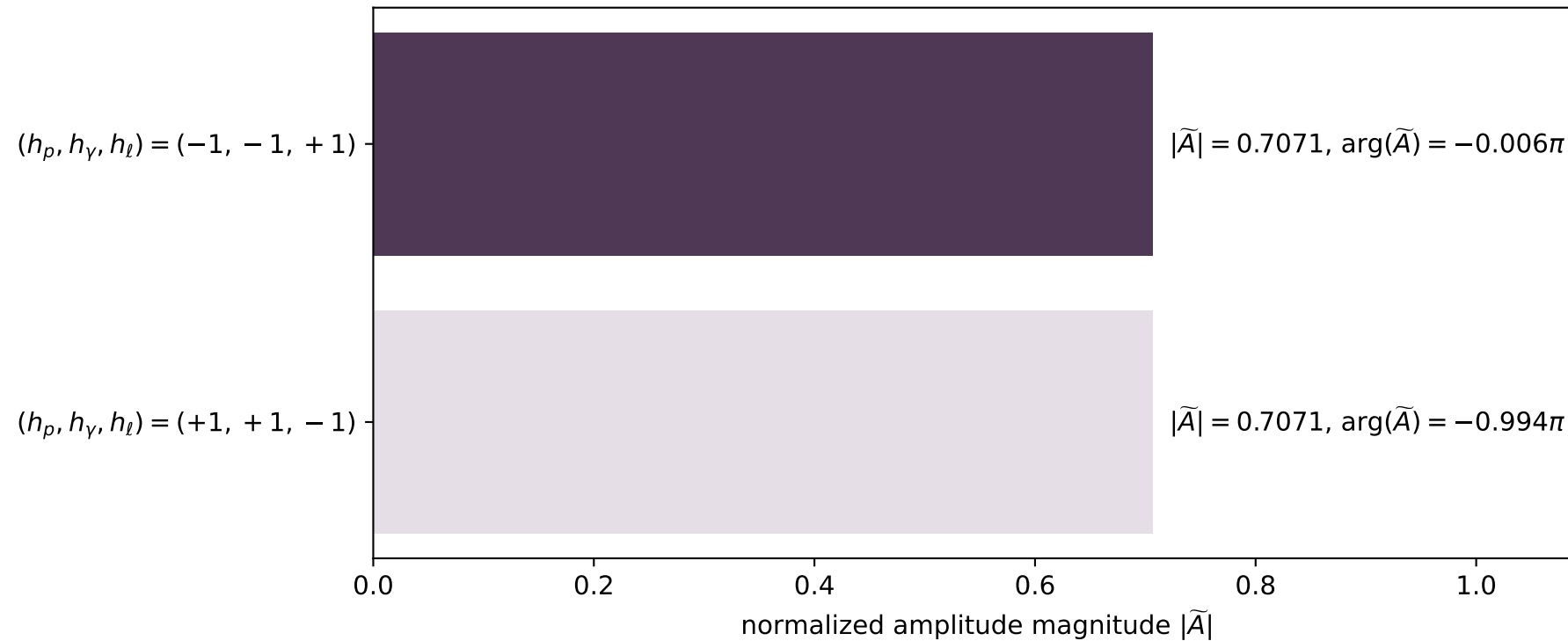
$s=24.9985$, $\sqrt{s}=4.99985$, $|\vec{P}|=2.41194$, $|\vec{P}'|=0.0078299$
 $\theta_{P'}=0.0184851$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0794118$, $\phi_\gamma=3.1225$
 $E_\gamma=2.03482$, $\theta_{\ell'}=7.14003\text{e-}05$, $\phi_{\ell'}=3.221$
 $Q^2=19.556$, $x_B=0.835536$, $t=-3.05764$, $W^2=4.72917$, $y=0.970421$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4119$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4119$
 P $E= 2.5879$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4119$
 ℓ' $E= 2.0270$ $p_x= -0.0001$ $p_y= -0.0000$ $p_z= 2.0270$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= 0.0000$ $p_z= 0.0078$
 q_γ $E= 2.0348$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.0348$

x--y plane

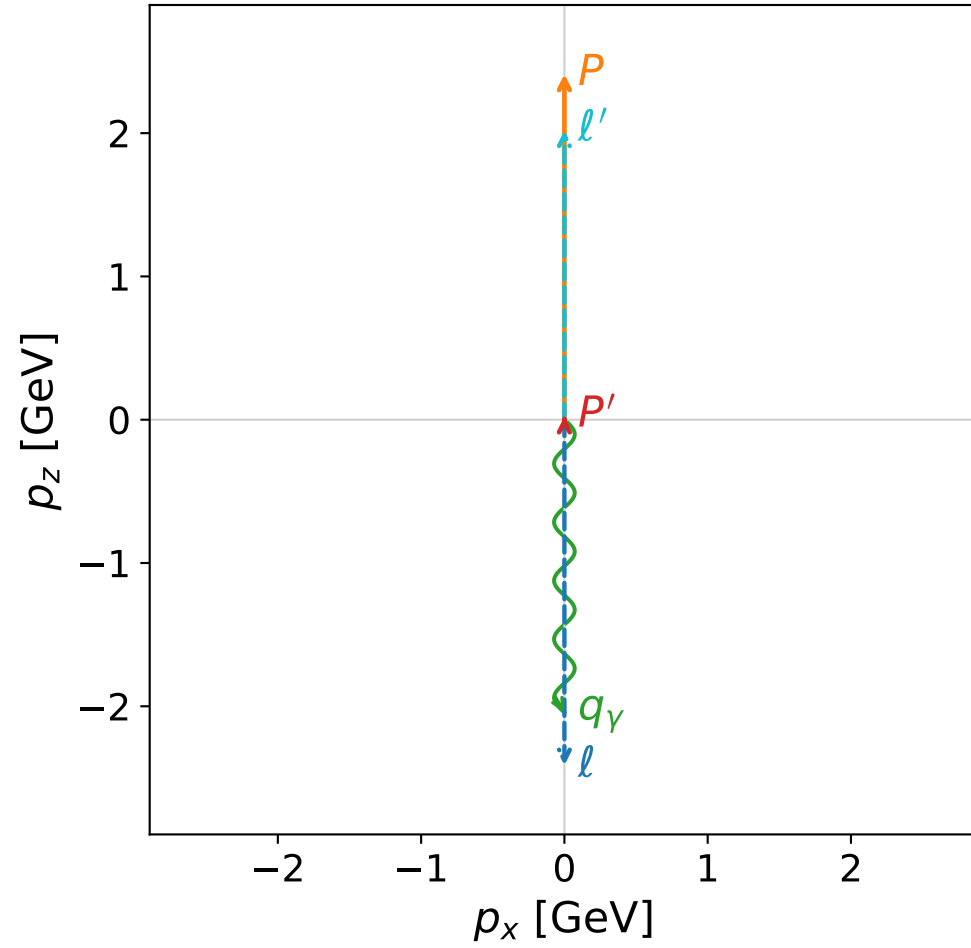


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

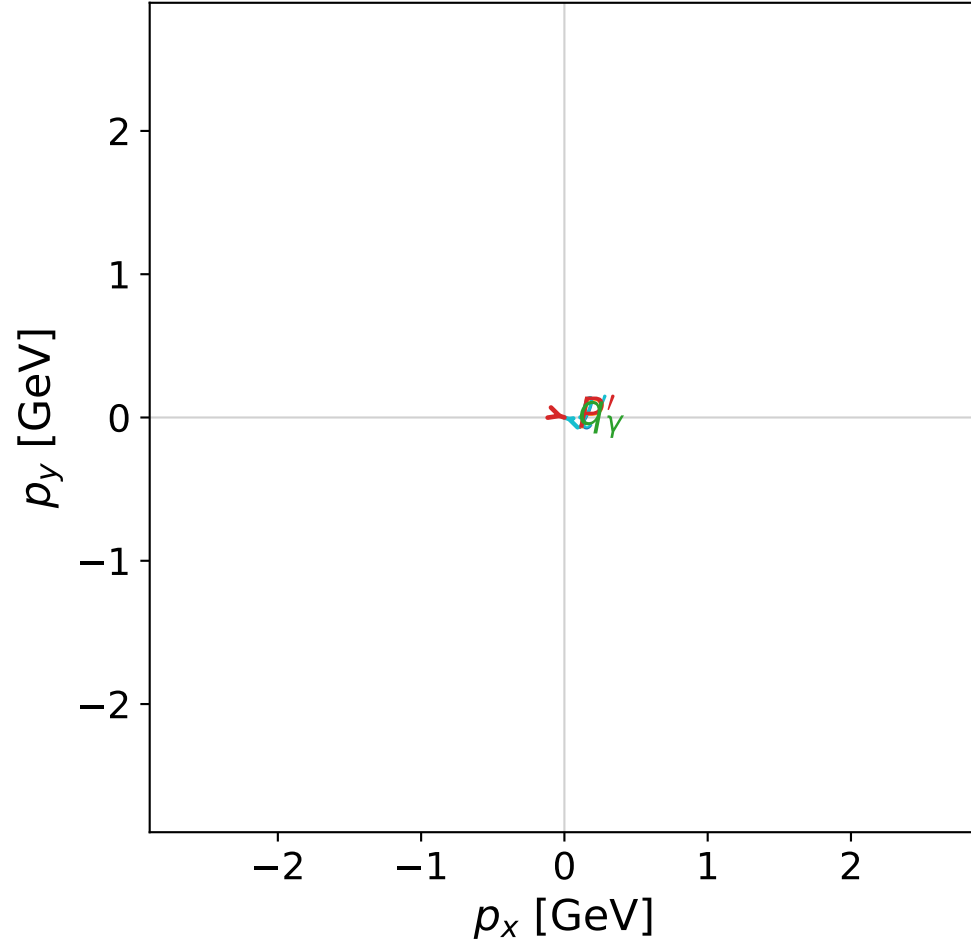


dghz_mixing_angles_polarization_03_local_minimum_73
 $d_{\text{GHZ}}=2.91539\text{e-}08$
 region: polarization_cluster_03_local_minimum_73
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0073
 lepton mixing angle: $\alpha_e=2.1495745$ rad
 incoming lepton: $-0.547002|+\rangle +0.837132|-\rangle$
 proton mixing angle: $\alpha_p=0.99200753$ rad
 incoming proton: $0.54701|+\rangle +0.837126|-\rangle$

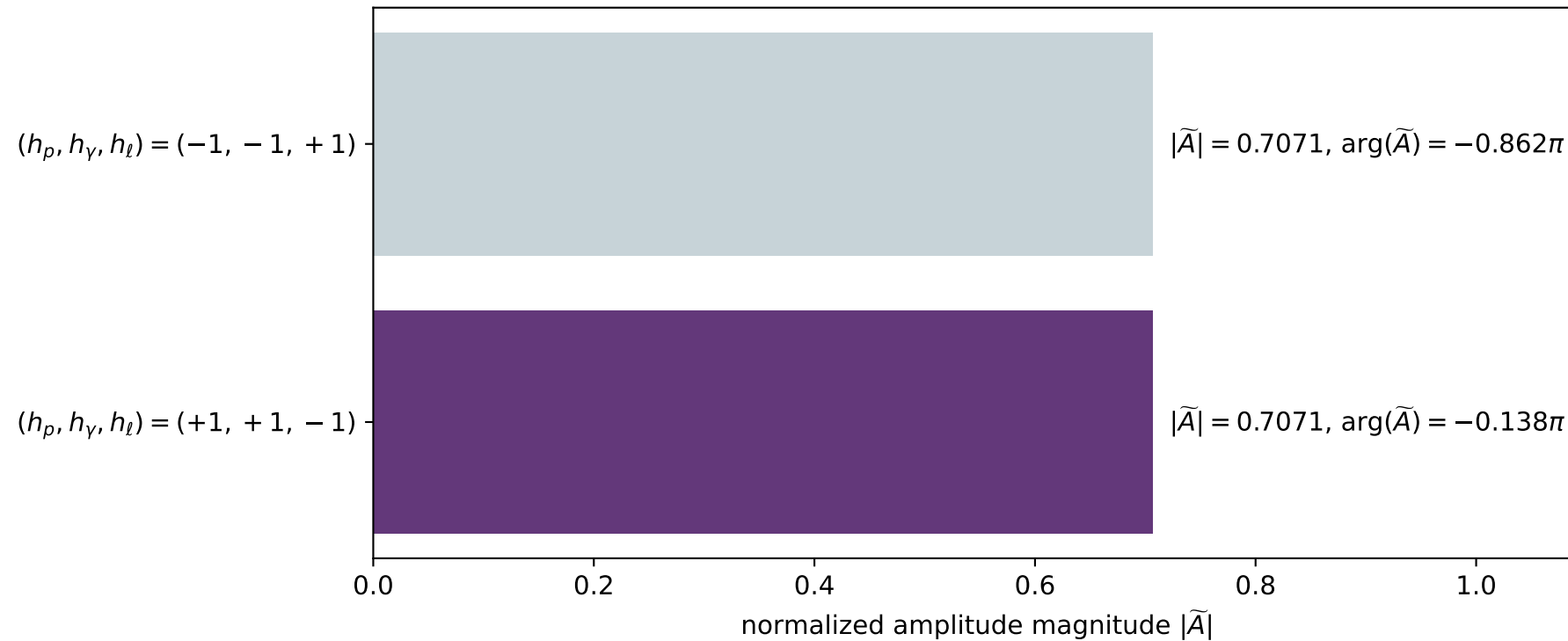
$s=24.9997$, $\sqrt{s}=4.99997$, $|\vec{P}|=2.412$, $|\vec{P}'|=0.0232778$
 $\theta_{P'}=0.000120512$, $\theta_\gamma=3.14159$, $\phi_{P'}=5.96763$, $\phi_\gamma=0.433102$
 $E_\gamma=2.04248$, $\theta_{\ell'}=1.38933\text{e-}06$, $\phi_{\ell'}=2.82604$
 $Q^2=19.4813$, $x_B=0.832205$, $t=-2.98455$, $W^2=4.8078$, $y=0.970537$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0192$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 2.0192$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0233$
 q_γ $E= 2.0425$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0425$

x--y plane

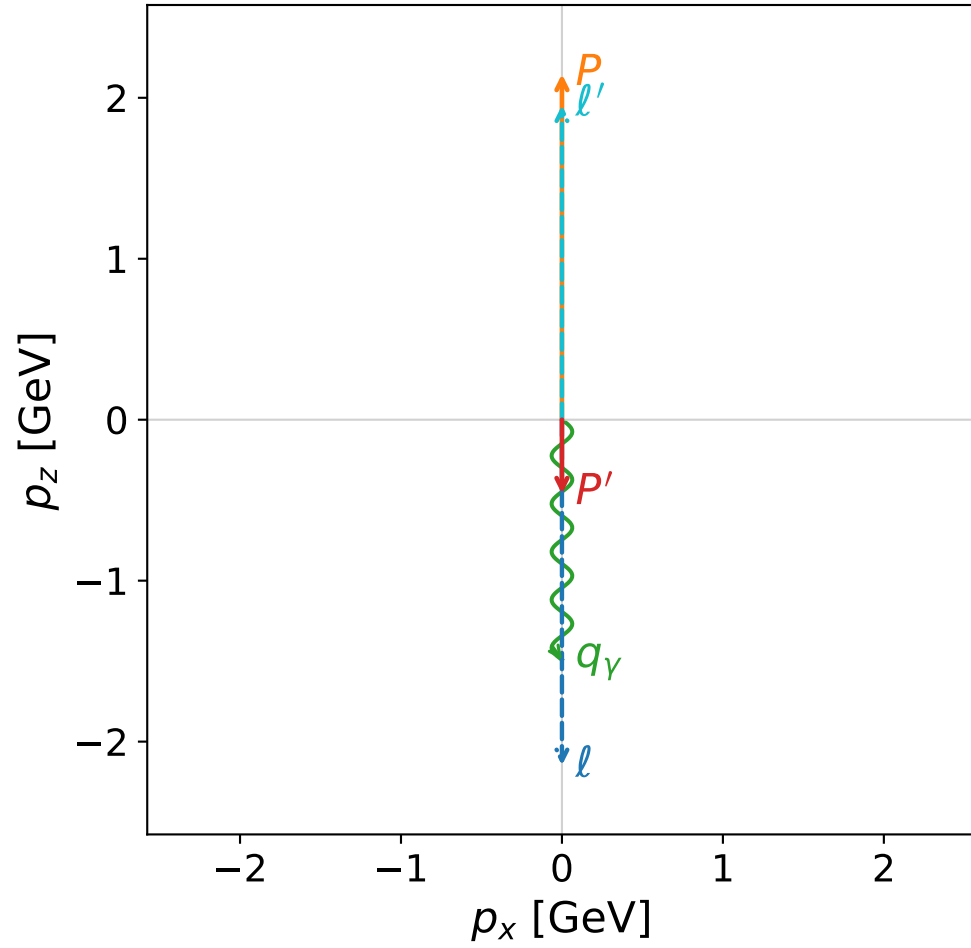


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

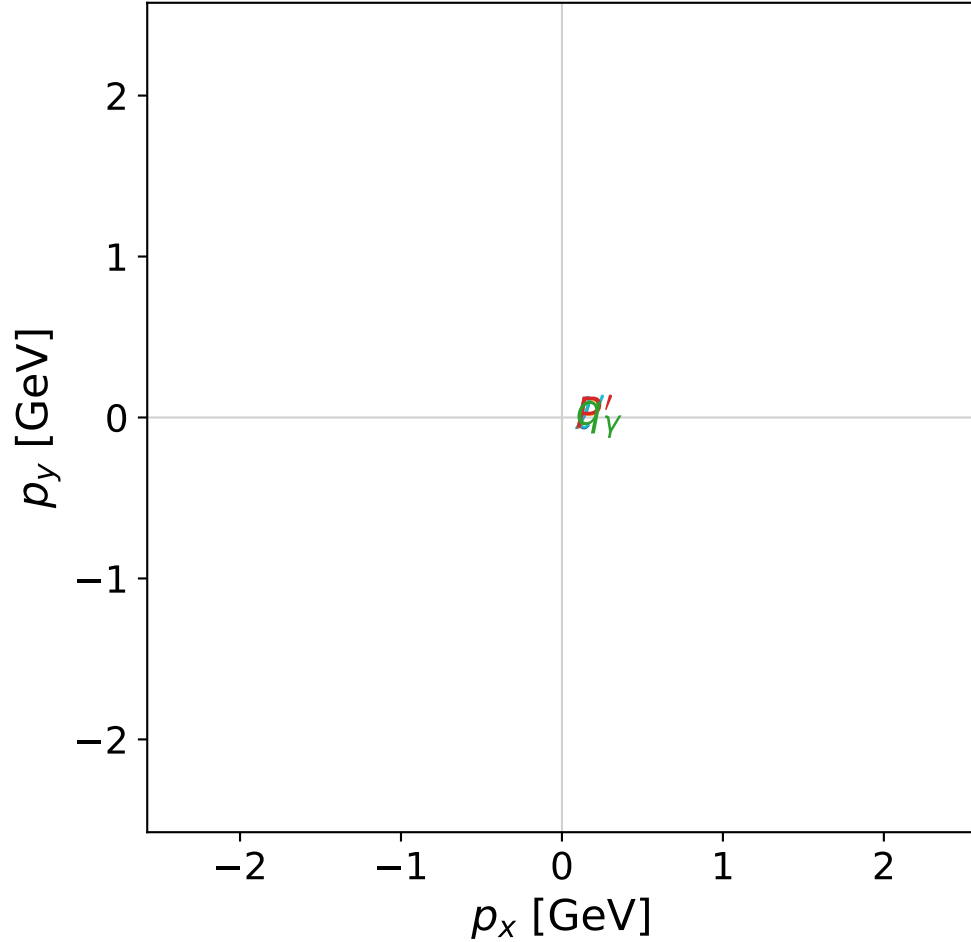


dghz_mixing_angles_polarization_03_local_minimum_87
 $d_{\text{GHZ}}=3.19716\text{e-}08$
 region: polarization_cluster_03_local_minimum_87
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0087
 lepton mixing angle: $\alpha_e=2.4687294$ rad
 incoming lepton: $-0.78204|+\rangle +0.623228|-\rangle$
 proton mixing angle: $\alpha_p=0.67288163$ rad
 incoming proton: $0.782029|+\rangle +0.623242|-\rangle$

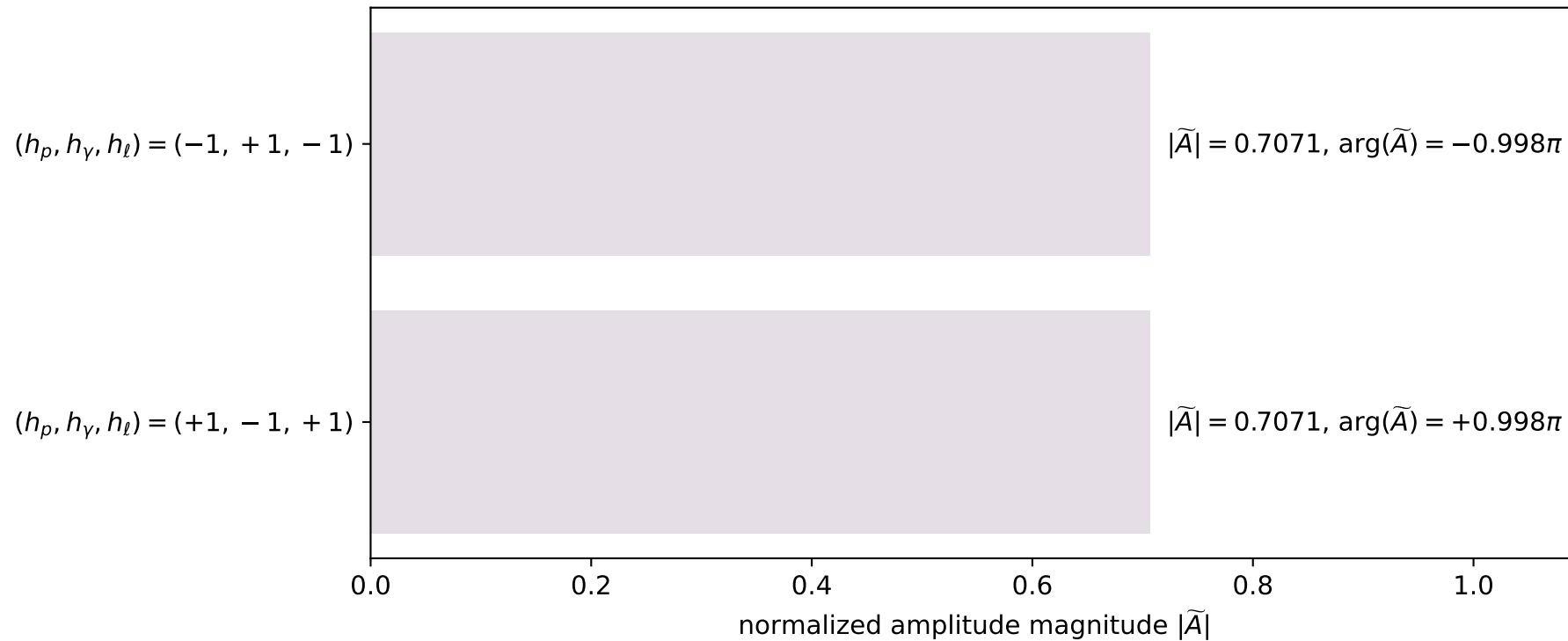
$s=20.1618$, $\sqrt{s}=4.49019$, $|\vec{P}|=2.14712$, $|\vec{P}'|=0.461231$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.72593$, $\phi_\gamma=6.27676$
 $E_\gamma=1.49185$, $\theta_{l'}=0$, $\phi_{l'}=2.90749$
 $Q^2=16.774$, $x_B=0.905891$, $t=-5.1192$, $W^2=2.62242$, $y=0.960305$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1471$
 P $E= 2.3431$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1471$
 l' $E= 1.9531$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.9531$
 P' $E= 1.0453$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.4612$
 q_γ $E= 1.4918$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4918$

x--y plane

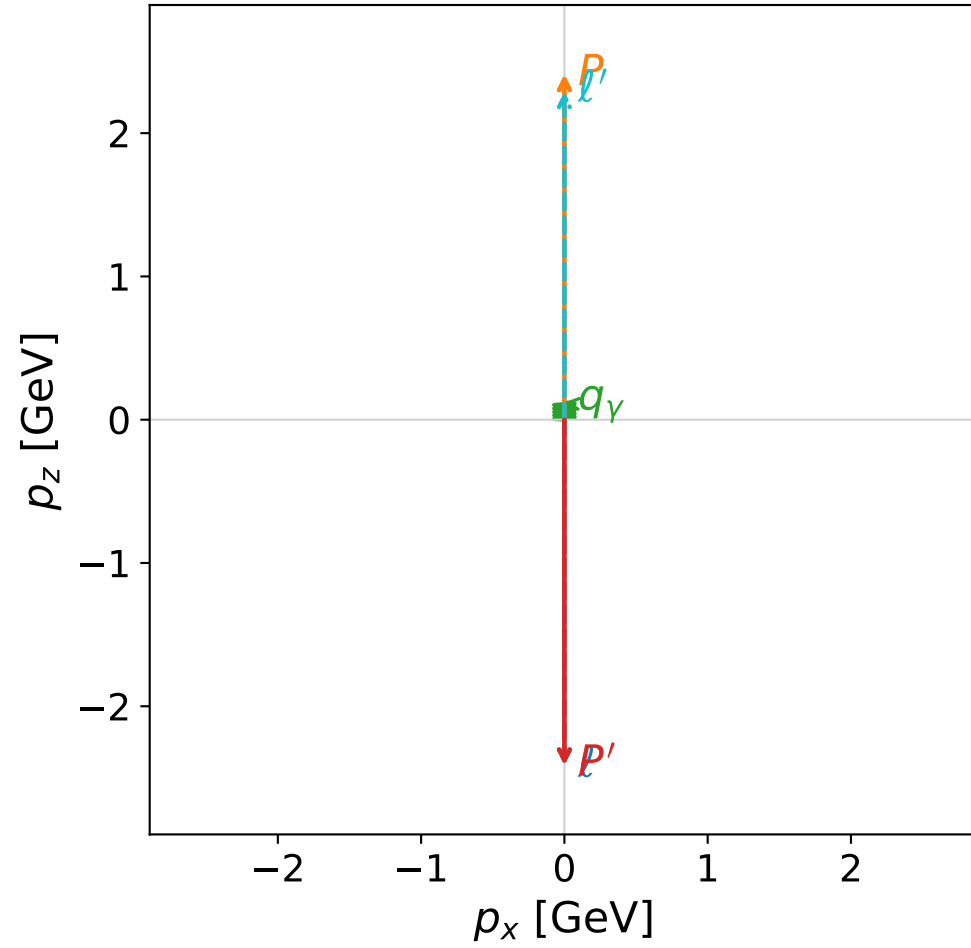


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

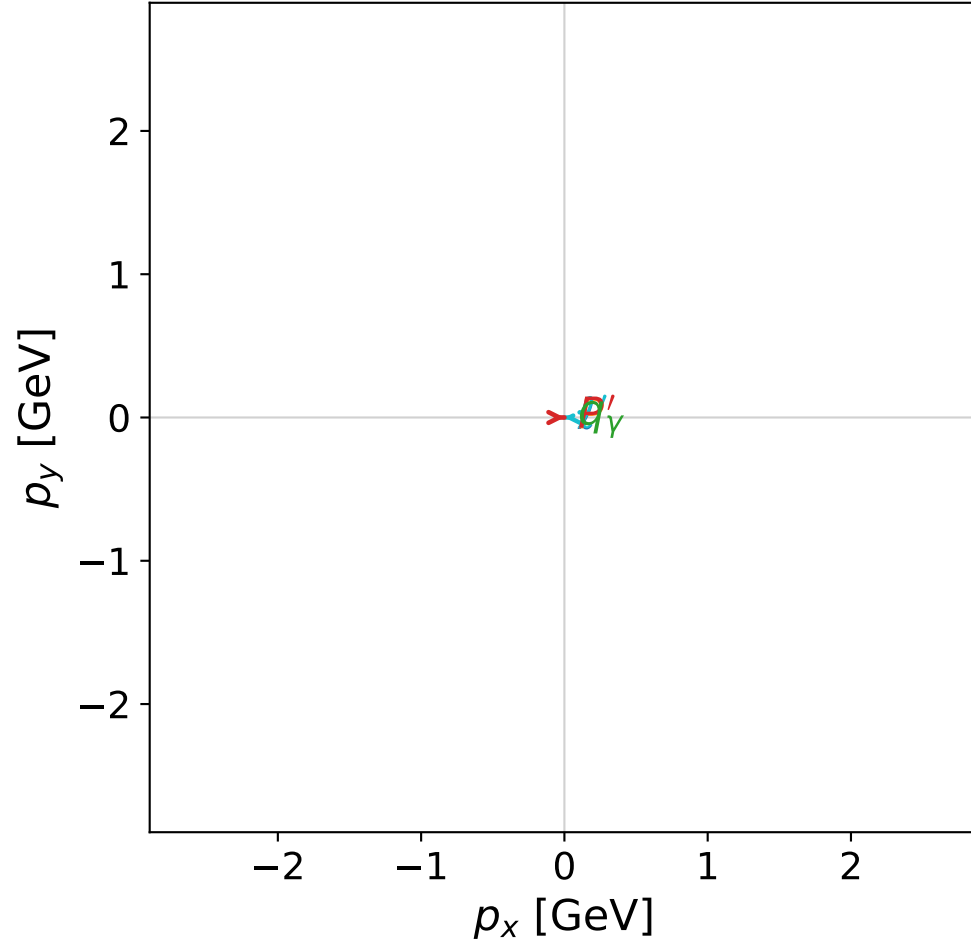


dghz_mixing_angles_polarization_03_local_minimum_89
 $d_{\text{GHZ}}=3.21673\text{e-}08$
 region: polarization_cluster_03_local_minimum_89
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0089
 lepton mixing angle: $\alpha_e=1.9834494$ rad
 incoming lepton: $-0.401041|+\rangle +0.91606|-\rangle$
 proton mixing angle: $\alpha_p=0.41264296$ rad
 incoming proton: $0.916064|+\rangle +0.401032|-\rangle$

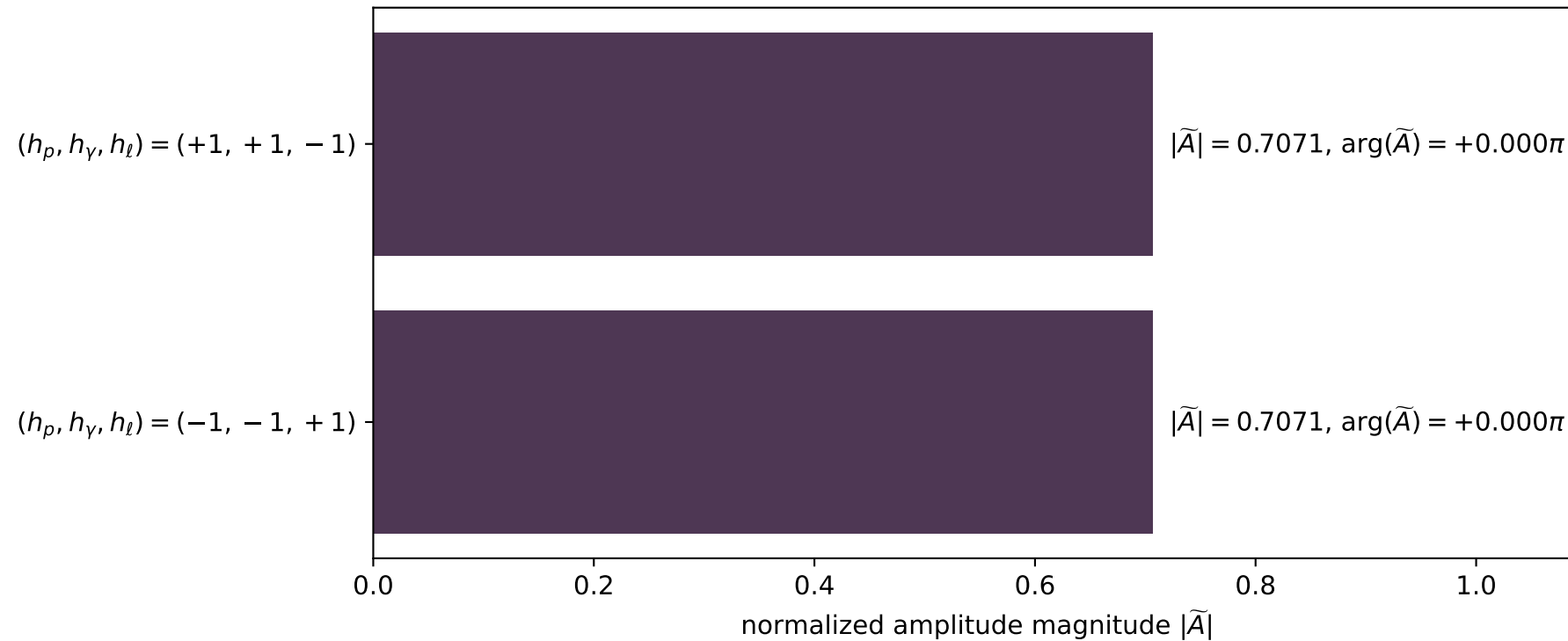
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0$, $\phi_\gamma=0$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=3.14159$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$

x--y plane

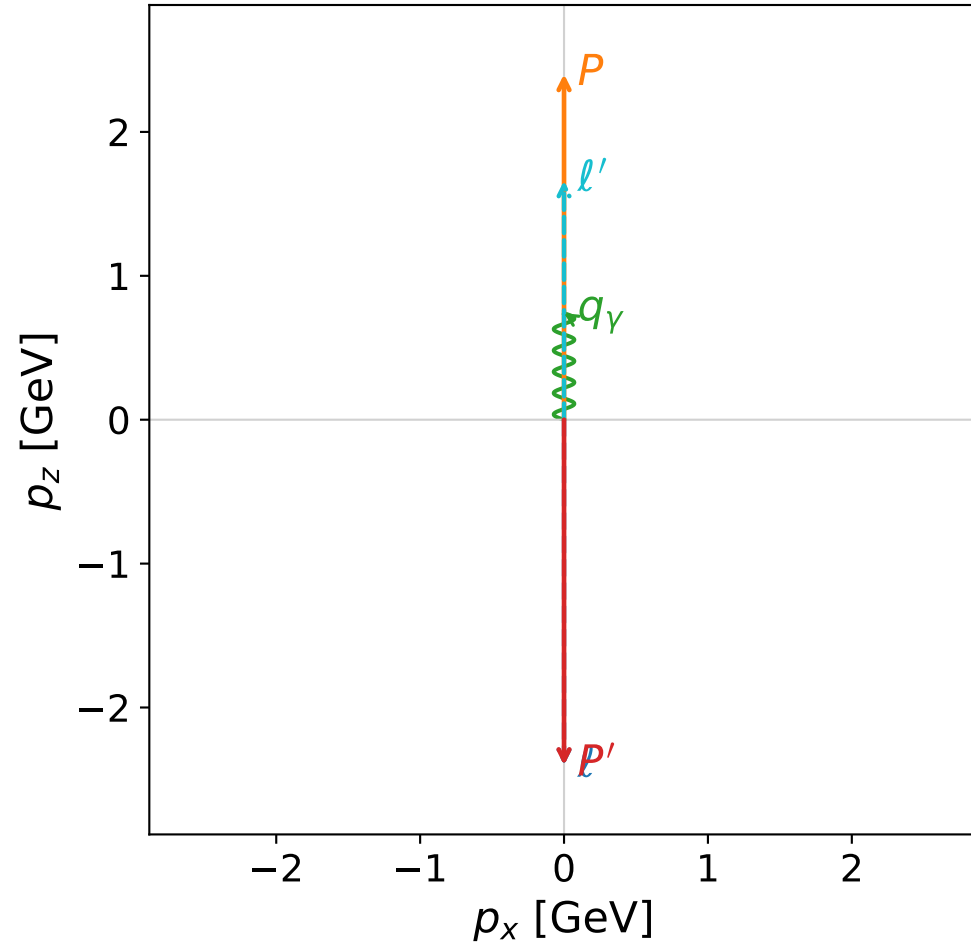


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

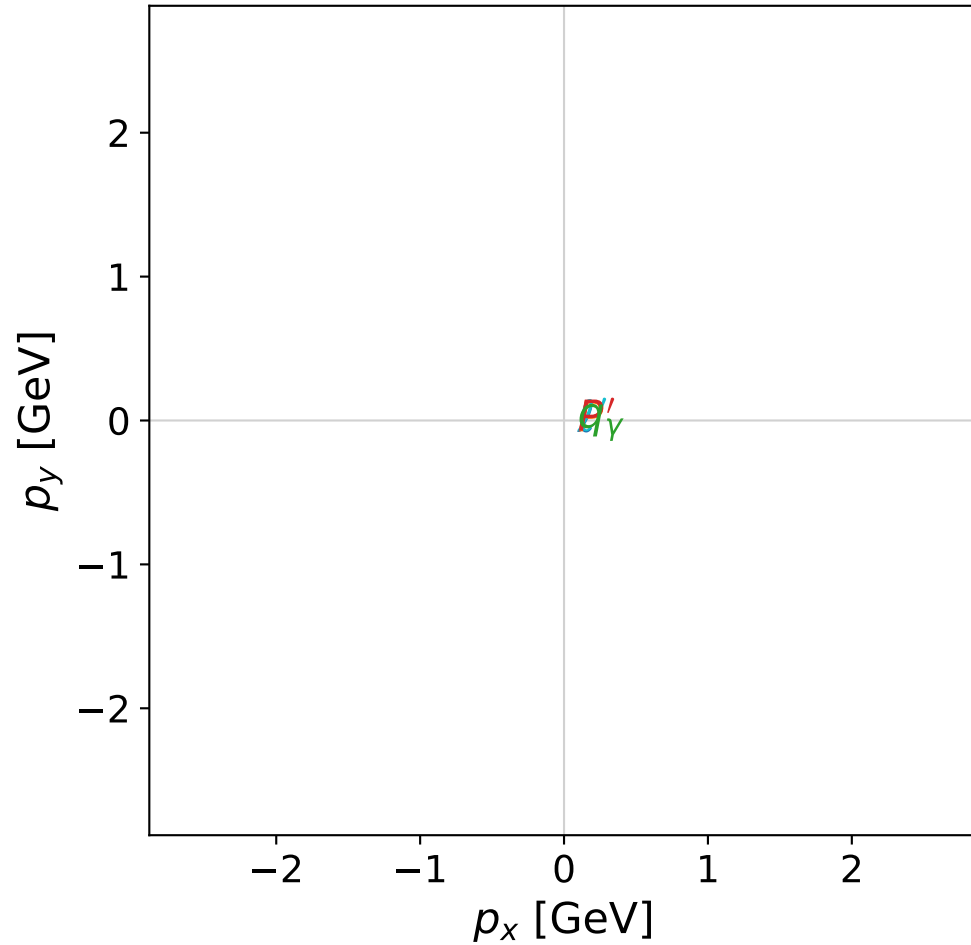


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

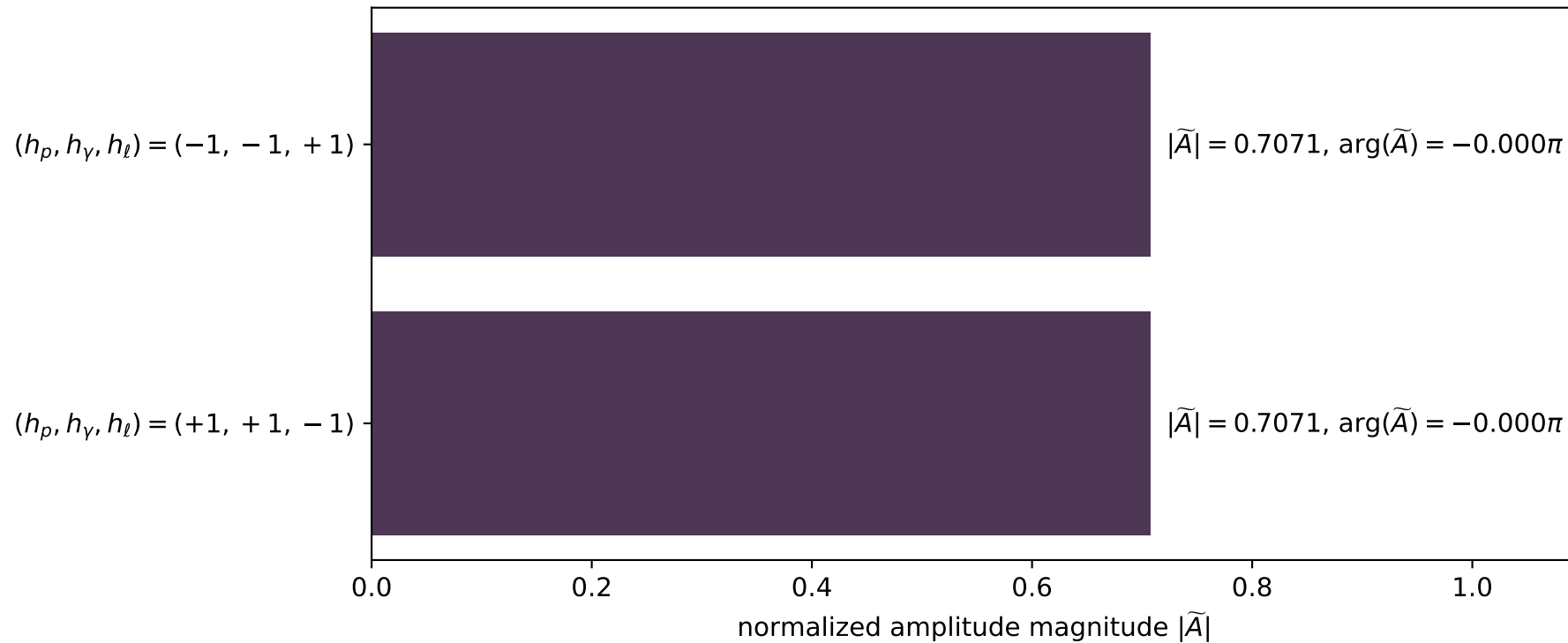


dghz_mixing_angles_polarization_03_local_minimum_92
 $d_{\{\text{rm GHZ}\}}=3.31027\text{e-}08$
 region: polarization_cluster_03_local_minimum_92
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0092
 lepton mixing angle: $\alpha_e=2.4281603$ rad
 incoming lepton: $-0.75612|+\rangle +0.654433|-\rangle$
 proton mixing angle: $\alpha_p=0.85738407$ rad
 incoming proton: $0.654418|+\rangle +0.756133|-\rangle$

$s=24.8013$, $\sqrt{s}=4.98009$, $|\vec{P}|=2.40171$, $|\vec{P}'|=2.40171$
 $\theta_{p'}=3.14159$, $\theta_Y=0$, $\phi_{p'}=5.23216$, $\phi_Y=5.8557$
 $E_Y=0.740299$, $\theta_{l'}=0$, $\phi_{l'}=2.09057$
 $Q^2=15.9609$, $x_B=0.684007$, $t=-23.0728$, $W^2=8.25336$, $y=0.975459$

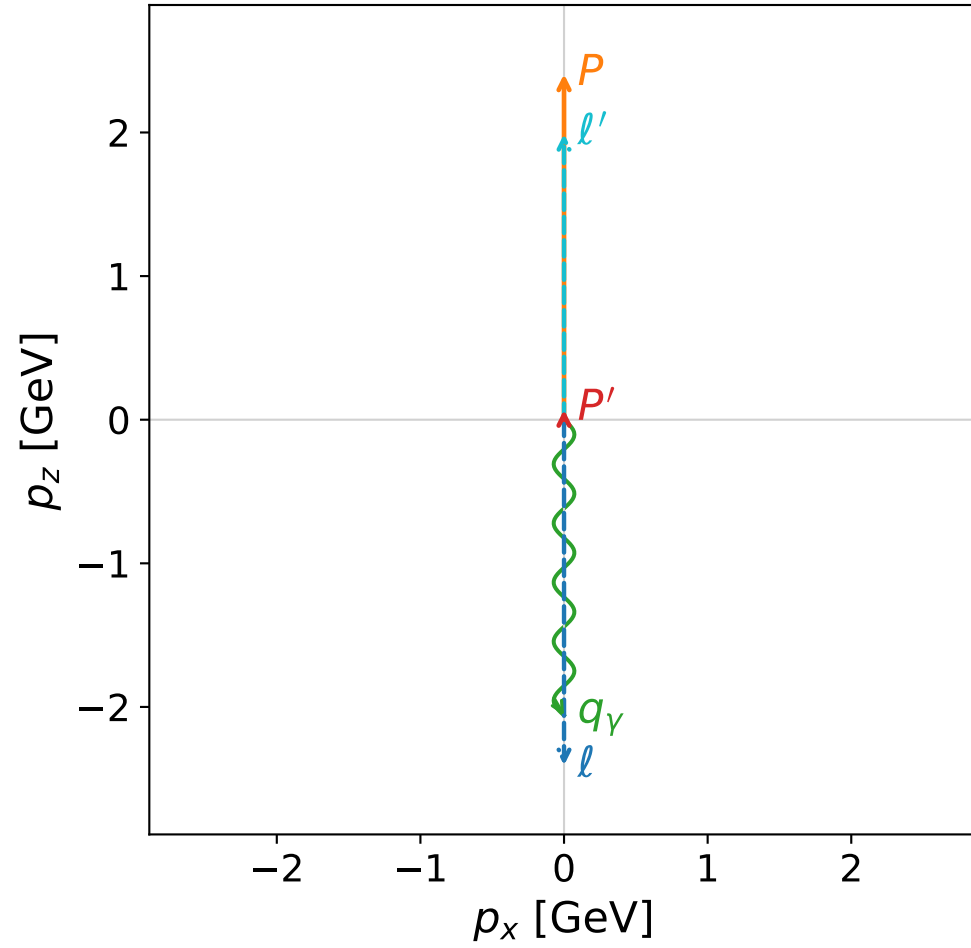
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4017$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4017$
 P $E= 2.5784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4017$
 l' $E= 1.6614$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.6614$
 P' $E= 2.5784$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4017$
 q_Y $E= 0.7403$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7403$

Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

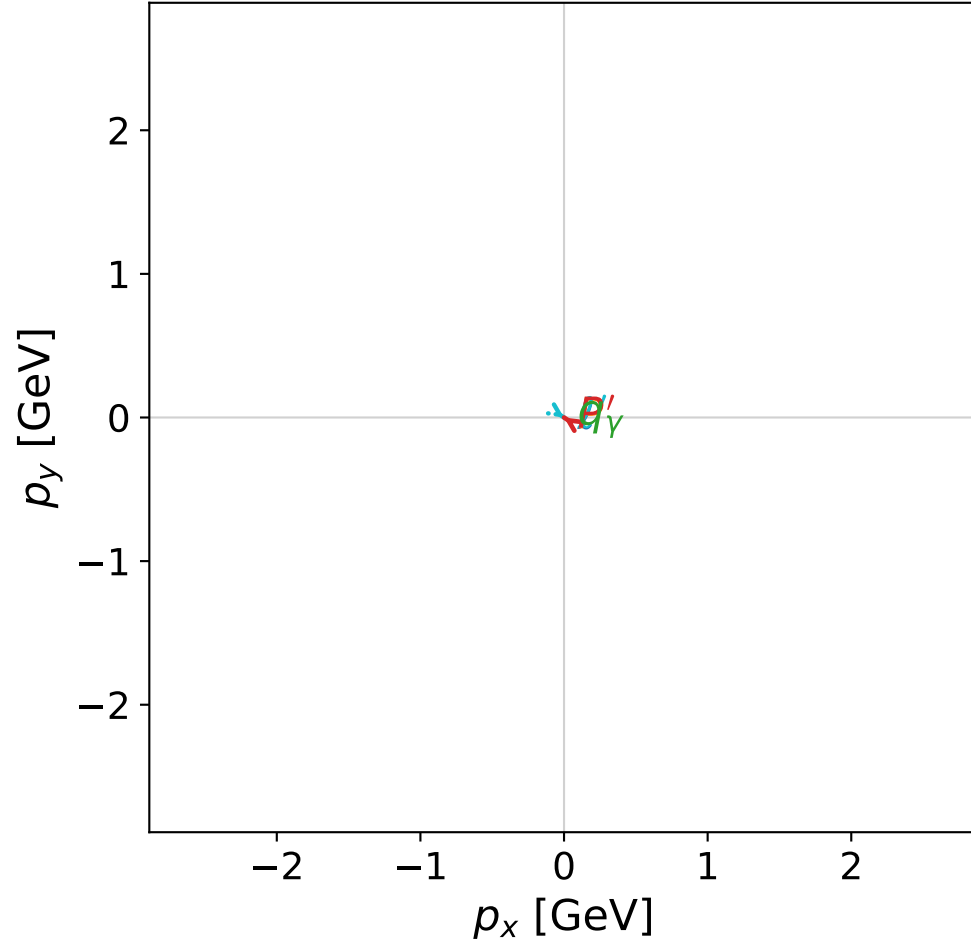


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



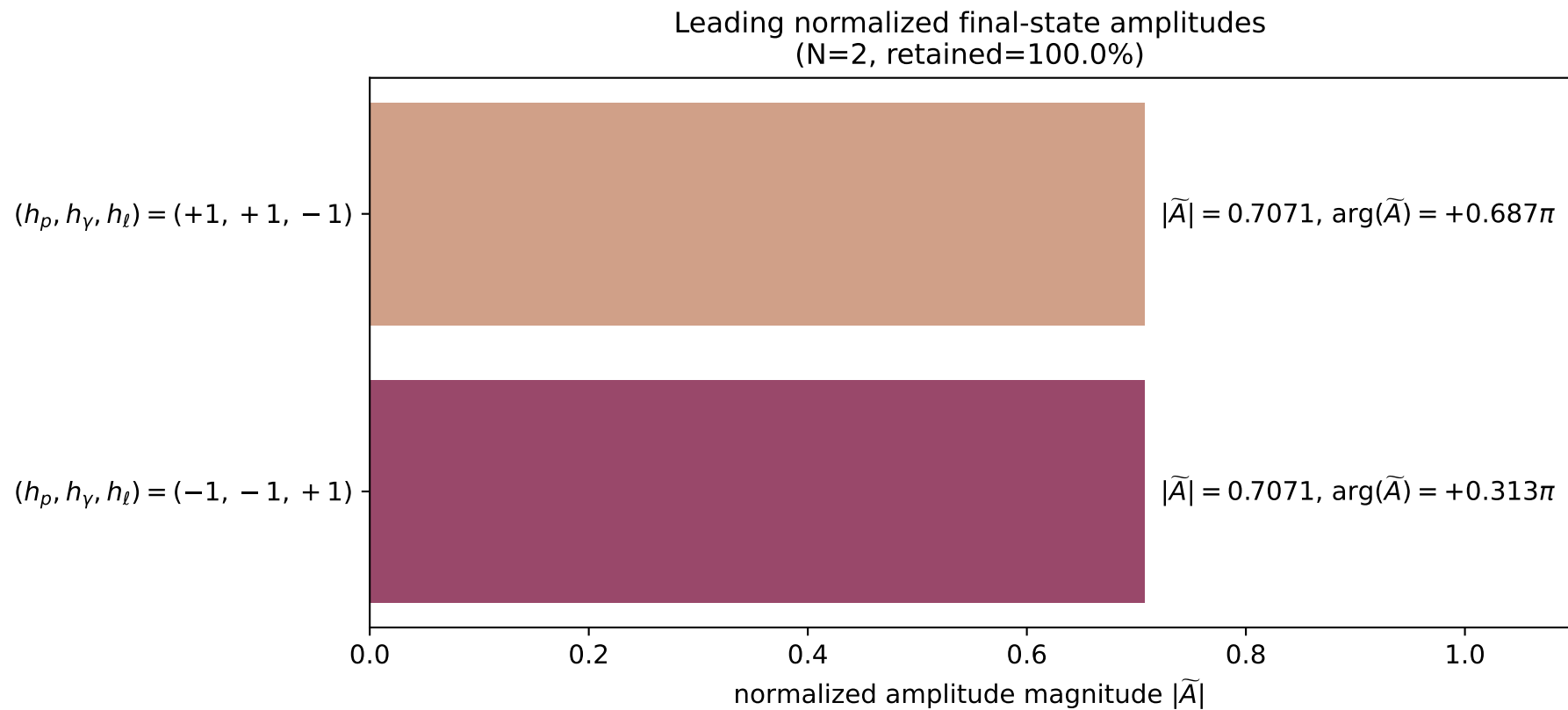
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_93
 $d_{\text{GHZ}}=3.31851\text{e-}08$
 region: polarization_cluster_03_local_minimum_93
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0093
 lepton mixing angle: $\alpha_e=2.6709187$ rad
 incoming lepton: $-0.891263|+\rangle +0.453487|-\rangle$
 proton mixing angle: $\alpha_p=0.47067803$ rad
 incoming proton: $0.891261|+\rangle +0.453491|-\rangle$

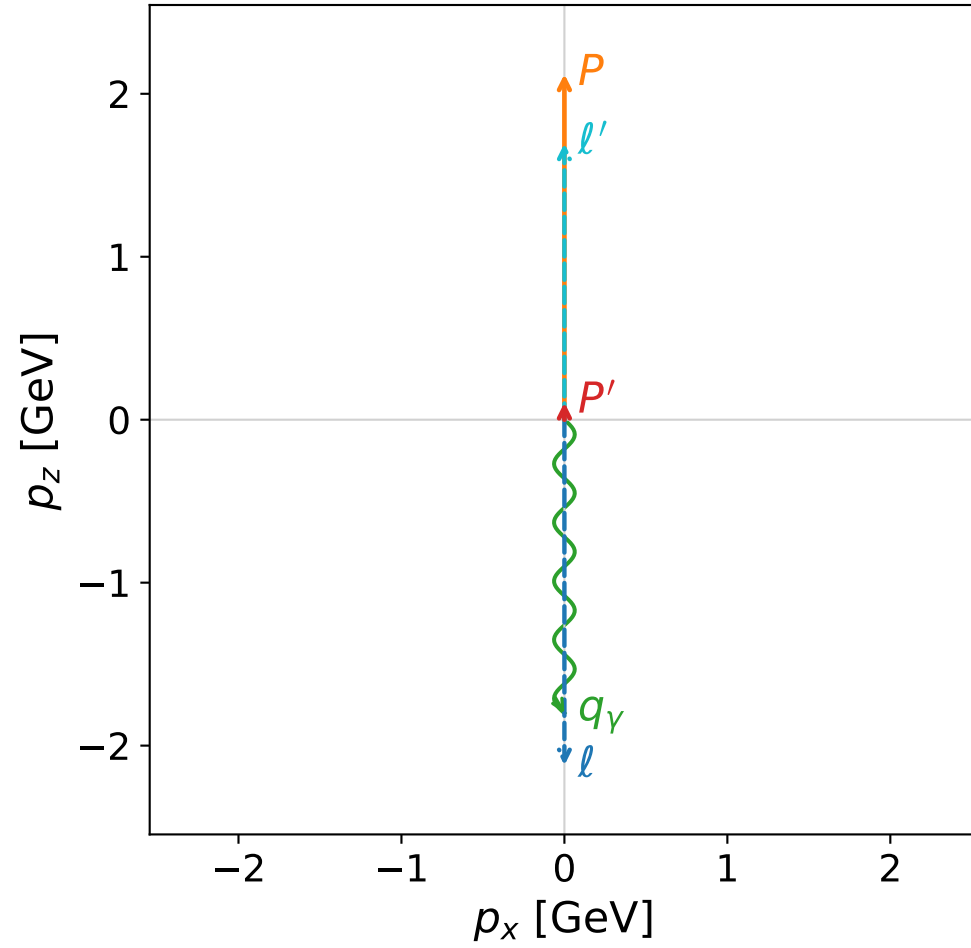
$s=24.8907$, $\sqrt{s}=4.98906$, $|\vec{P}|=2.40635$, $|\vec{P}'|=0.0697271$
 $\theta_{P'}=1.43494\text{e-}05$, $\theta_\gamma=3.14159$, $\phi_{P'}=2.55848$, $\phi_\gamma=4.12637$
 $E_\gamma=2.0591$, $\theta_{\ell'}=5.029\text{e-}07$, $\phi_{\ell'}=5.70007$
 $Q^2=19.1485$, $x_B=0.821501$, $t=-2.76326$, $W^2=5.04052$, $y=0.970777$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4064$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4064$
 P $E= 2.5827$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4064$
 ℓ' $E= 1.9894$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9894$
 P' $E= 0.9406$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0697$
 q_γ $E= 2.0591$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0591$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

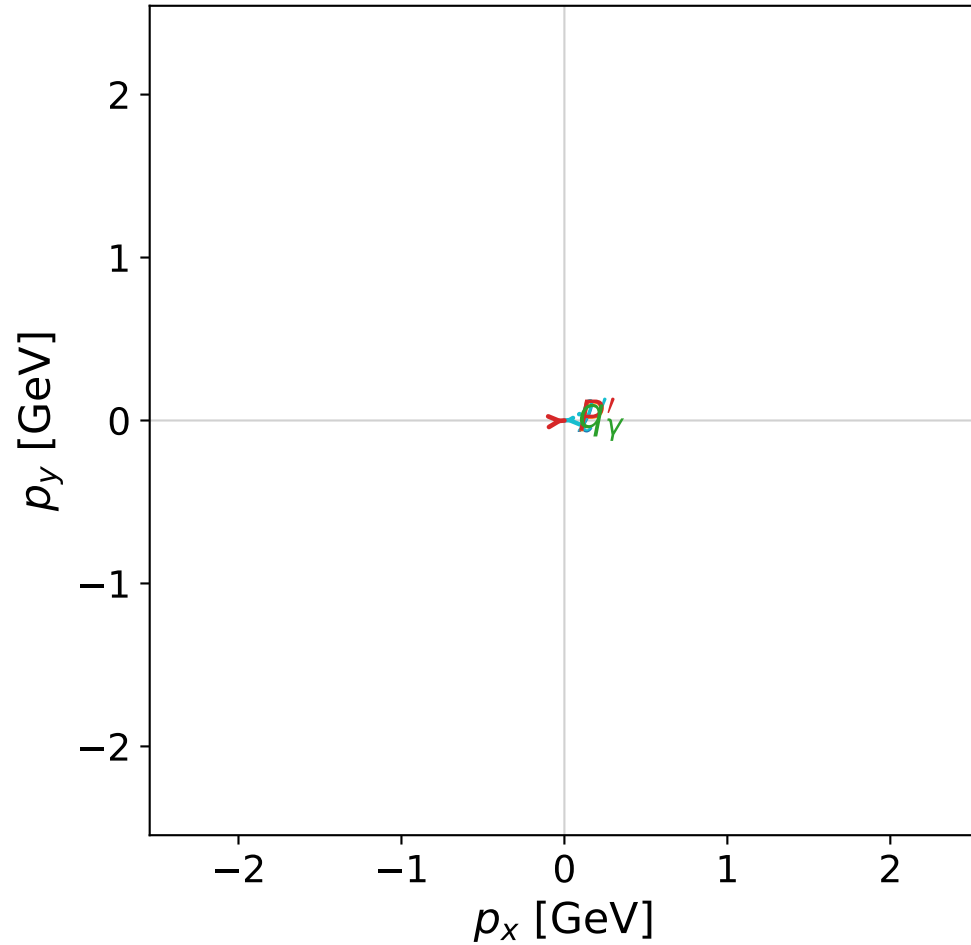


dghz_mixing_angles_polarization_03_local_minimum_97
 $d_{\text{GHZ}}=3.45768\text{e-}08$
 region: polarization_cluster_03_local_minimum_97
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0097
 lepton mixing angle: $\alpha_e=2.363027$ rad
 incoming lepton: $-0.711922|+\rangle +0.702259|-\rangle$
 proton mixing angle: $\alpha_p=0.77853959$ rad
 incoming proton: $0.71194|+\rangle +0.70224|-\rangle$

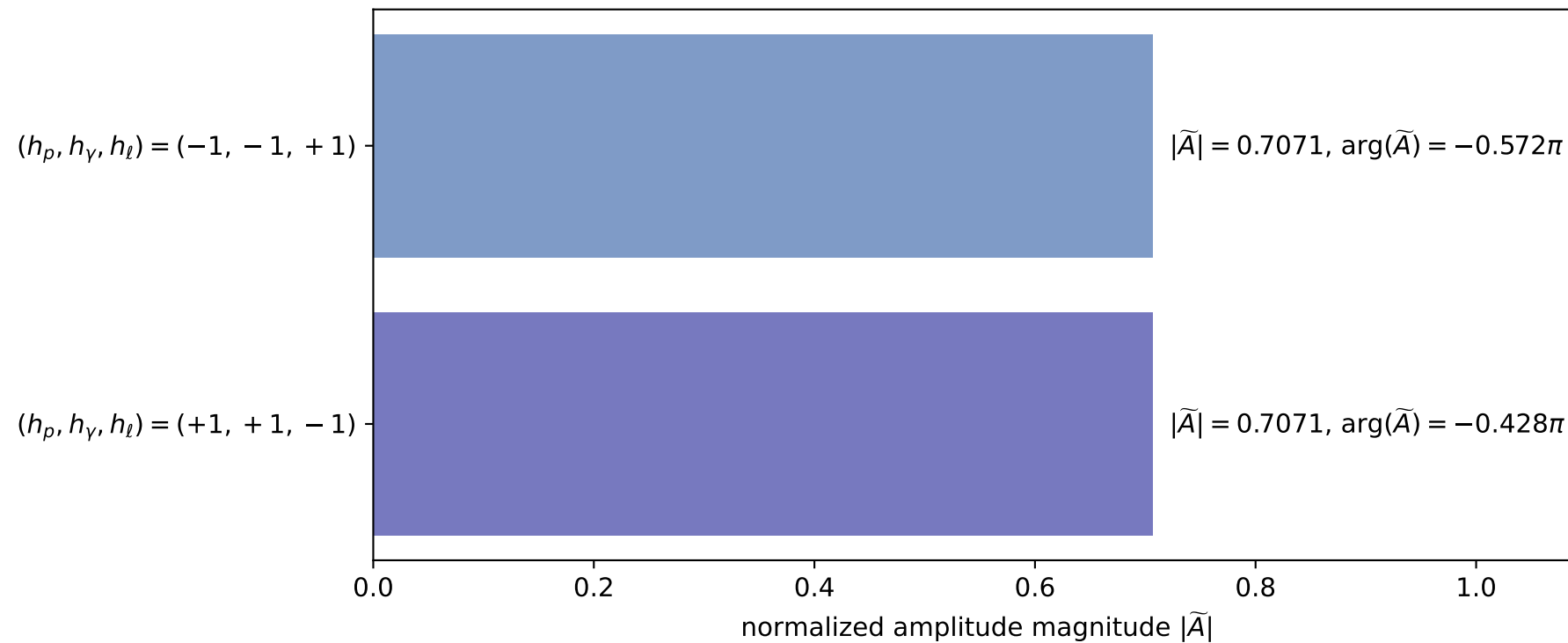
$s=19.711$, $\sqrt{s}=4.4397$, $|\vec{P}|=2.12076$, $|\vec{P}'|=0.105882$
 $\theta_{P'}=3.23647\text{e-}06$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.0751304$, $\phi_\gamma=1.34534$
 $E_\gamma=1.80081$, $\theta_{\ell'}=2.02129\text{e-}07$, $\phi_{\ell'}=3.21672$
 $Q^2=14.3782$, $x_B=0.79178$, $t=-2.16917$, $W^2=4.66097$, $y=0.964325$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1208$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1208$
 P $E= 2.3189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1208$
 ℓ' $E= 1.6949$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.6949$
 P' $E= 0.9440$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1059$
 q_γ $E= 1.8008$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8008$

x--y plane

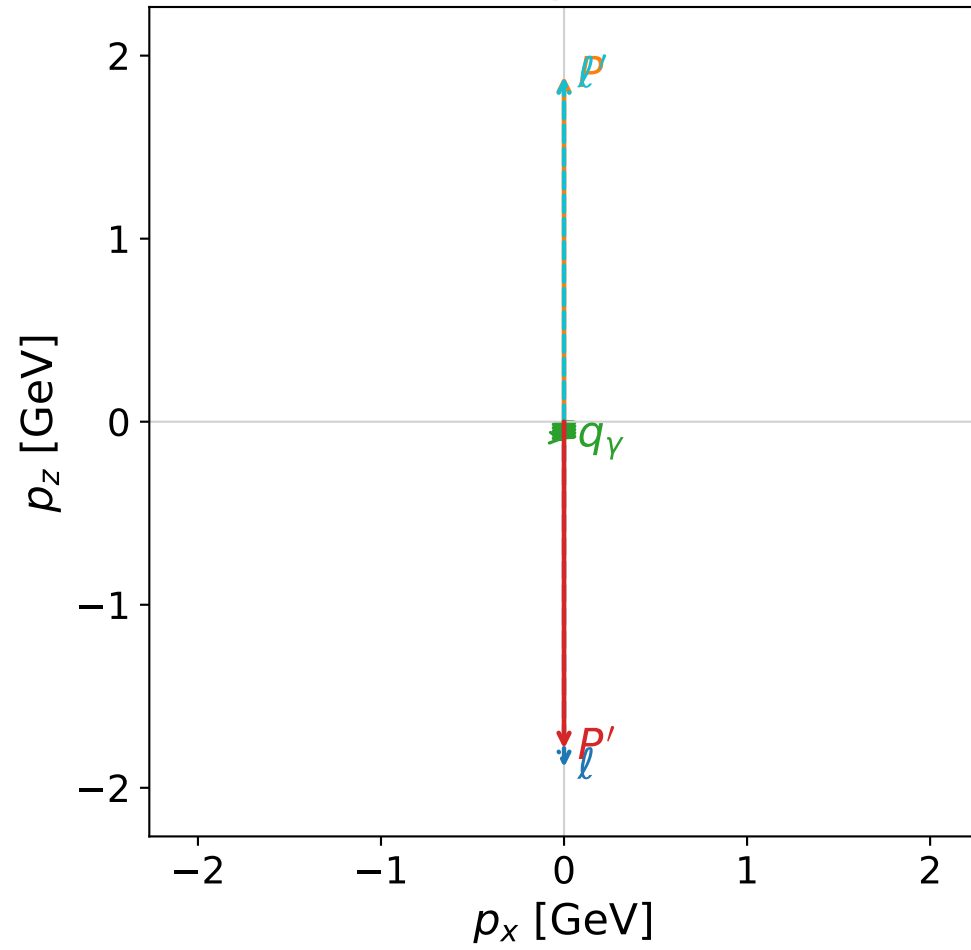


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

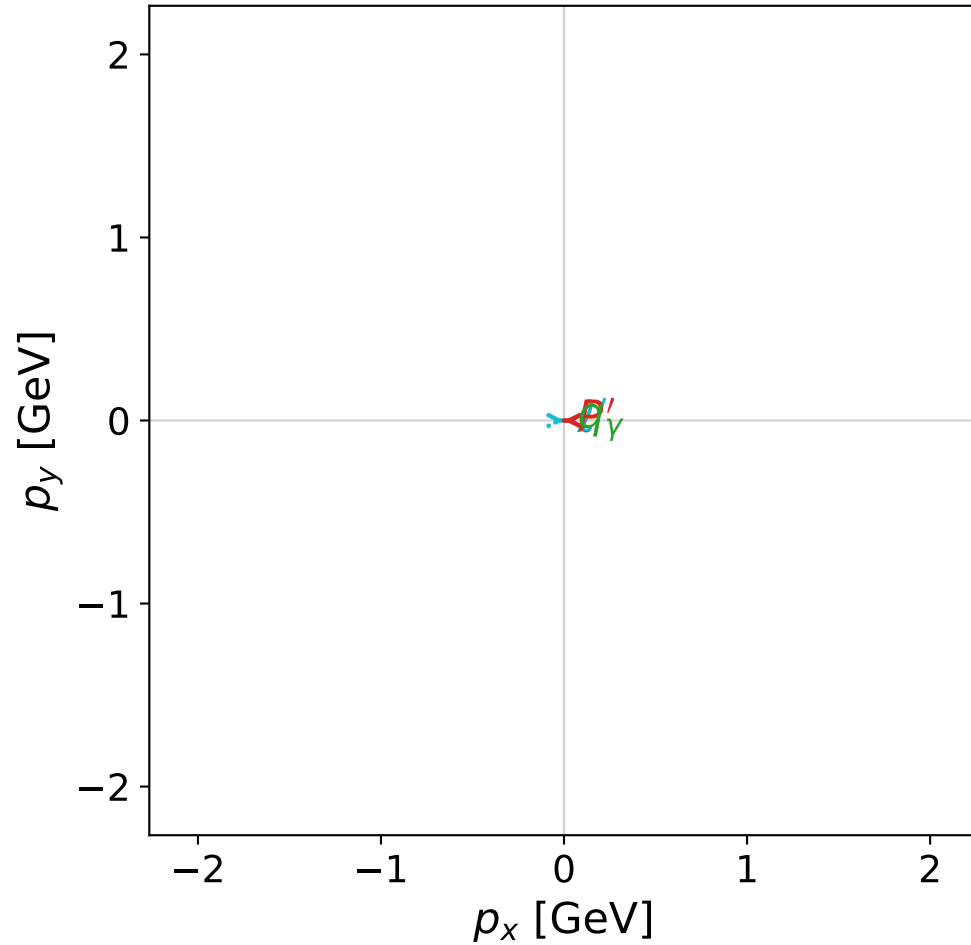


dghz_mixing_angles_polarization_03_local_minimum_102
 $d_{\{\text{rm GHZ}\}}=3.6745\text{e-}08$
 region: polarization_cluster_03_local_minimum_102
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0102
 lepton mixing angle: $\alpha_e=2.3754496$ rad
 incoming lepton: $-0.72059|+\rangle +0.693361|-\rangle$
 proton mixing angle: $\alpha_p=0.7661545$ rad
 incoming proton: $0.720582|+\rangle +0.693369|-\rangle$

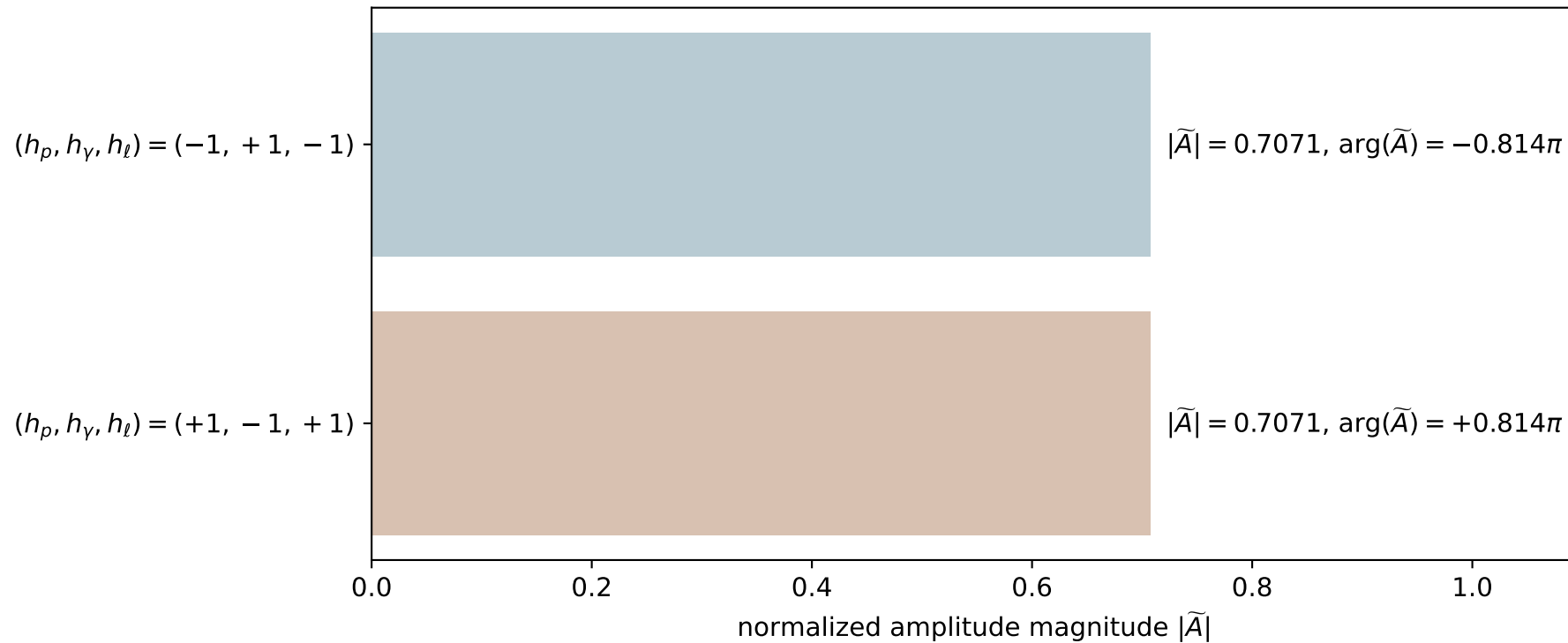
$s=15.9699$, $\sqrt{s}=3.99624$, $|\vec{P}|=1.88803$, $|\vec{P}'|=1.78706$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.12959$, $\phi_\gamma=5.69987$
 $E_\gamma=0.0954542$, $\theta_{l'}=0$, $\phi_{l'}=0.0182279$
 $Q^2=14.217$, $x_B=0.996905$, $t=-13.4982$, $W^2=0.923985$, $y=0.945067$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.8880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8880$
 P $E= 2.1082$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8880$
 l' $E= 1.8825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8825$
 P' $E= 2.0183$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.7871$
 q_γ $E= 0.0955$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0955$

x--y plane

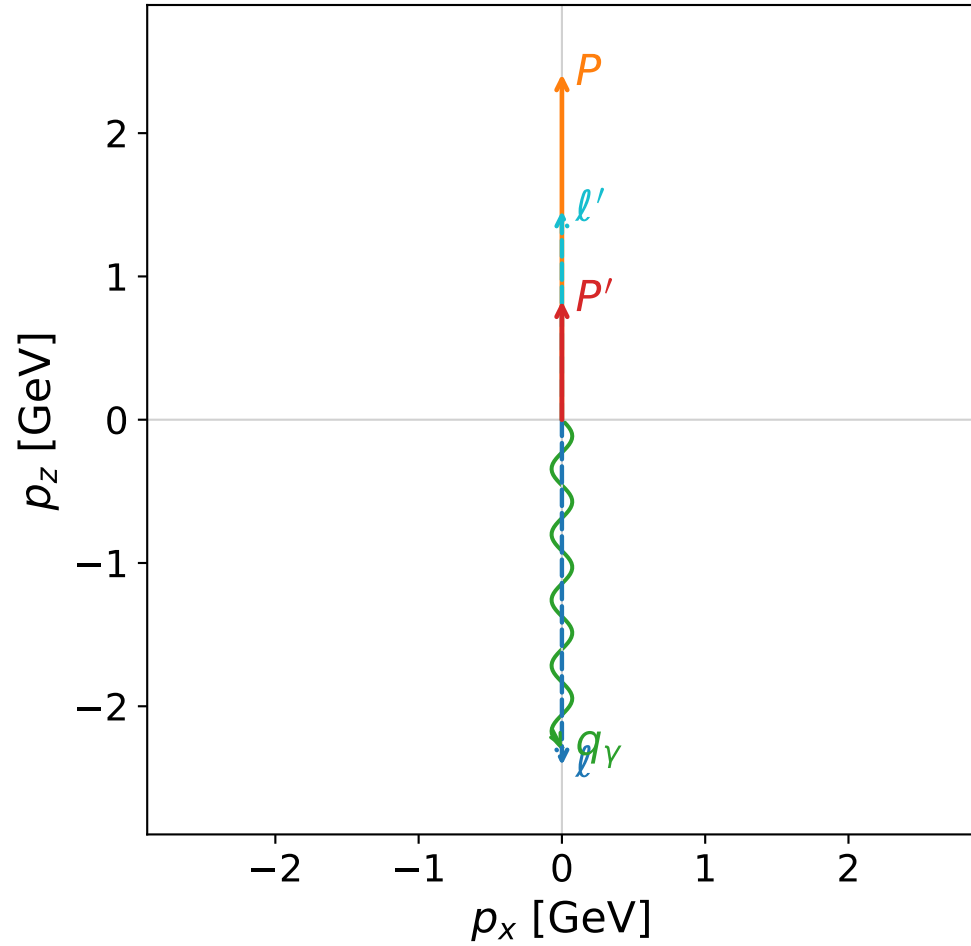


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)

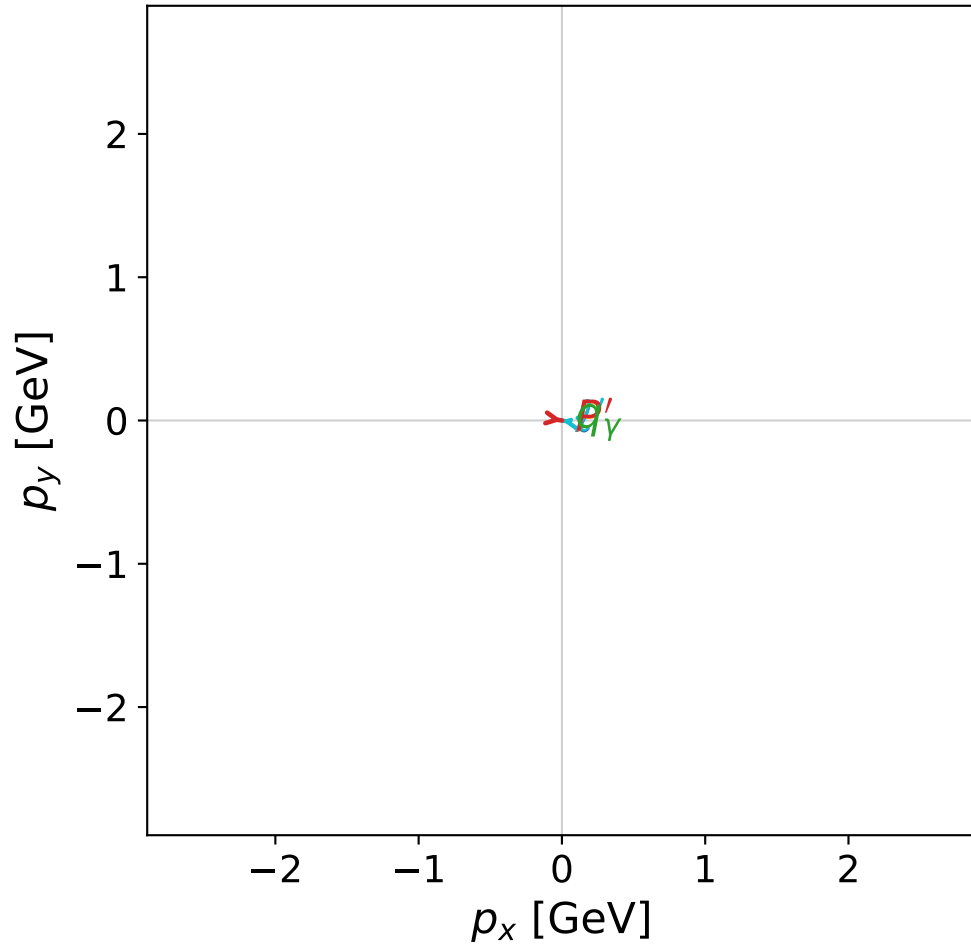


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



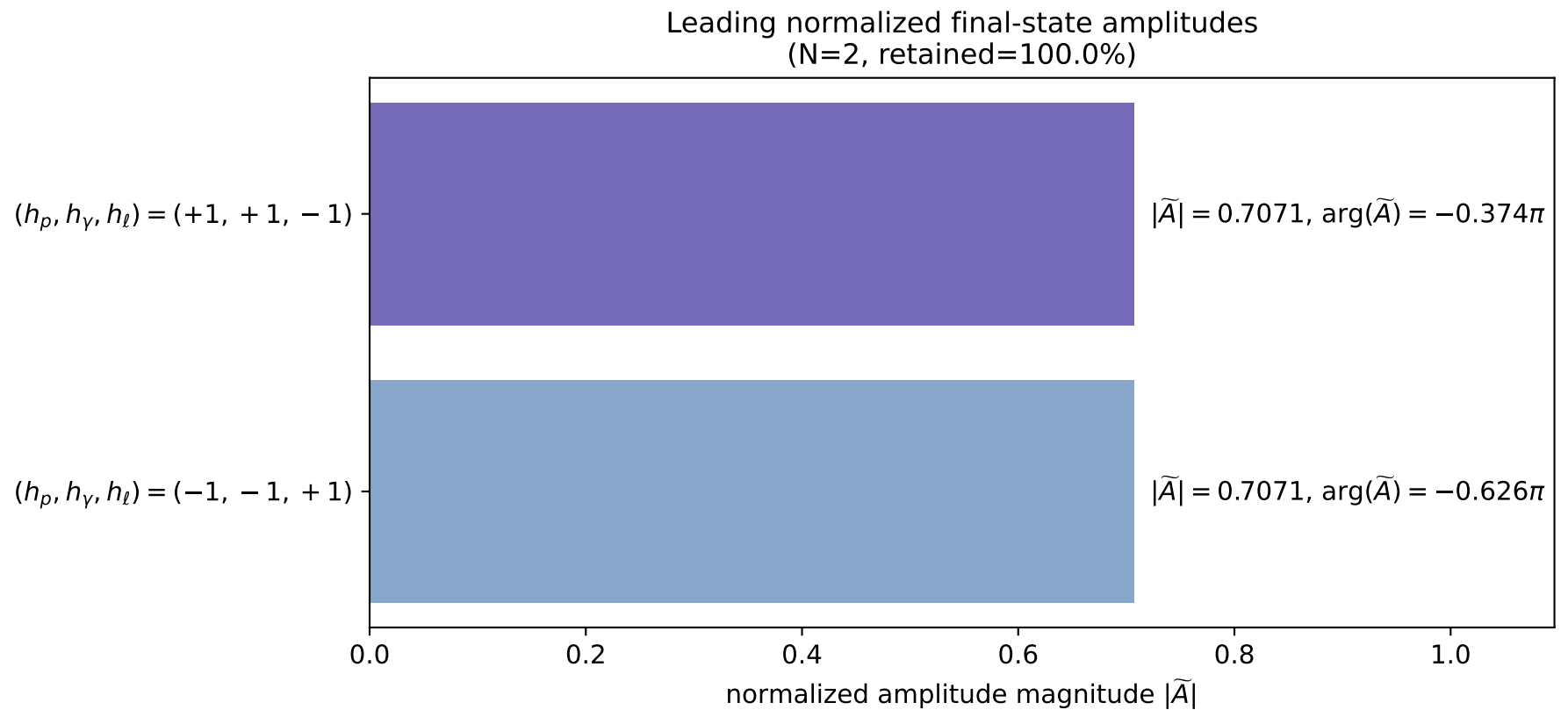
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_103
 $d_{\{\rm GHZ\}}=3.69193\text{e-}08$
 region: polarization_cluster_03_local_minimum_103
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0103
 lepton mixing angle: $\alpha_e=2.3646323$ rad
 incoming lepton: $-0.713048|+\rangle +0.701115|-\rangle$
 proton mixing angle: $\alpha_p=0.77696696$ rad
 incoming proton: $0.713043|+\rangle +0.70112|-\rangle$

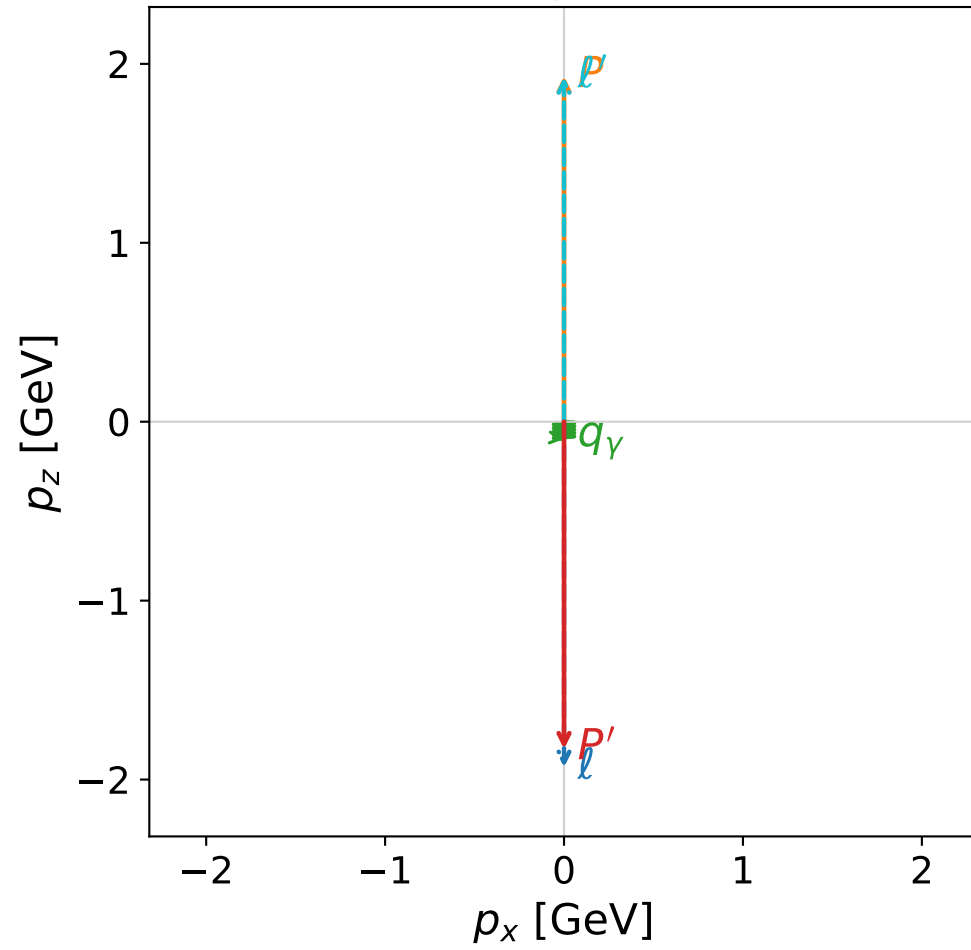
$s=24.9995$, $\sqrt{s}=4.99995$, $|\vec{P}|=2.41199$, $|\vec{P}'|=0.829186$
 $\theta_{p'}=3.04918\text{e-}05$, $\theta_Y=3.14159$, $\phi_{p'}=6.11999$, $\phi_Y=1.17541$
 $E_Y=2.28859$, $\theta_{\ell'}=1.73244\text{e-}05$, $\phi_{\ell'}=2.9784$
 $Q^2=14.0803$, $x_B=0.596469$, $t=-0.720361$, $W^2=10.4056$, $y=0.978705$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 1.4594$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.4594$
 P' $E= 1.2520$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8292$
 q_Y $E= 2.2886$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2886$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

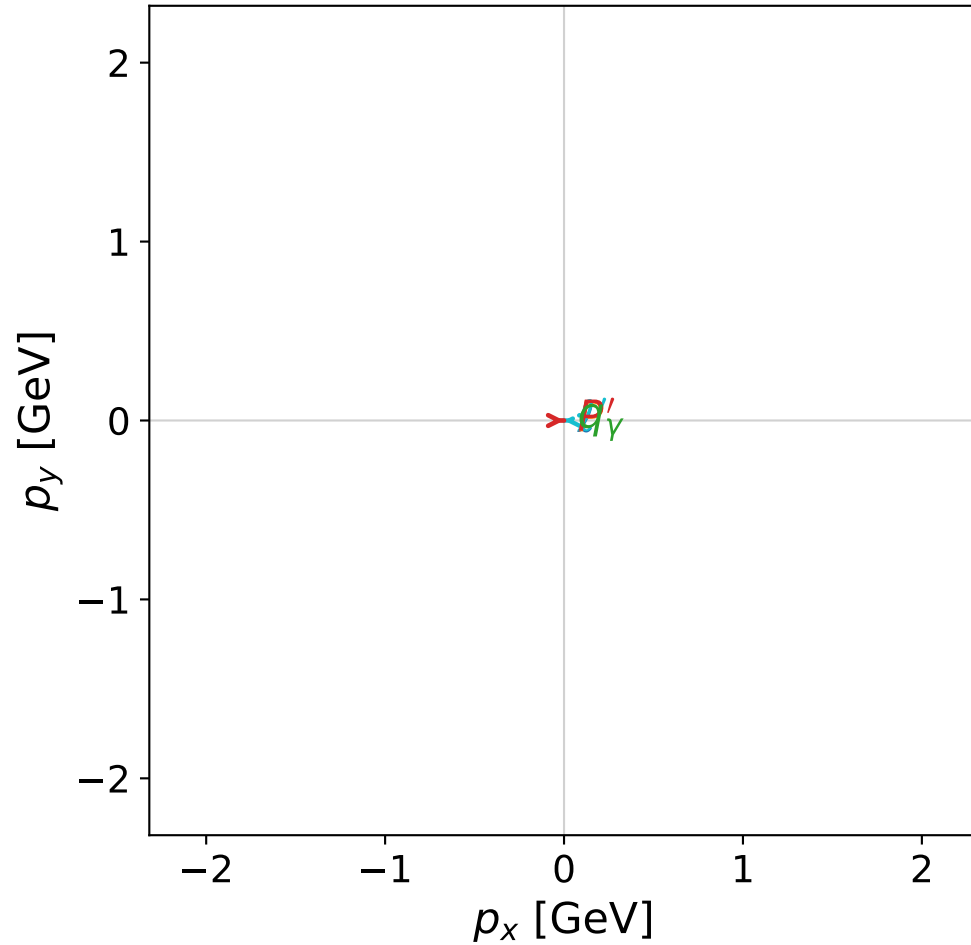


dghz_mixing_angles_polarization_03_local_minimum_106
 $d_{\{\rm GHZ\}}=3.85907\text{e-}08$
 region: polarization_cluster_03_local_minimum_106
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0106
 lepton mixing angle: $\alpha_e=2.1441764$ rad
 incoming lepton: $-0.542475|+\rangle +0.840072|-\rangle$
 proton mixing angle: $\alpha_p=0.99739265$ rad
 incoming proton: $0.542494|+\rangle +0.840059|-\rangle$

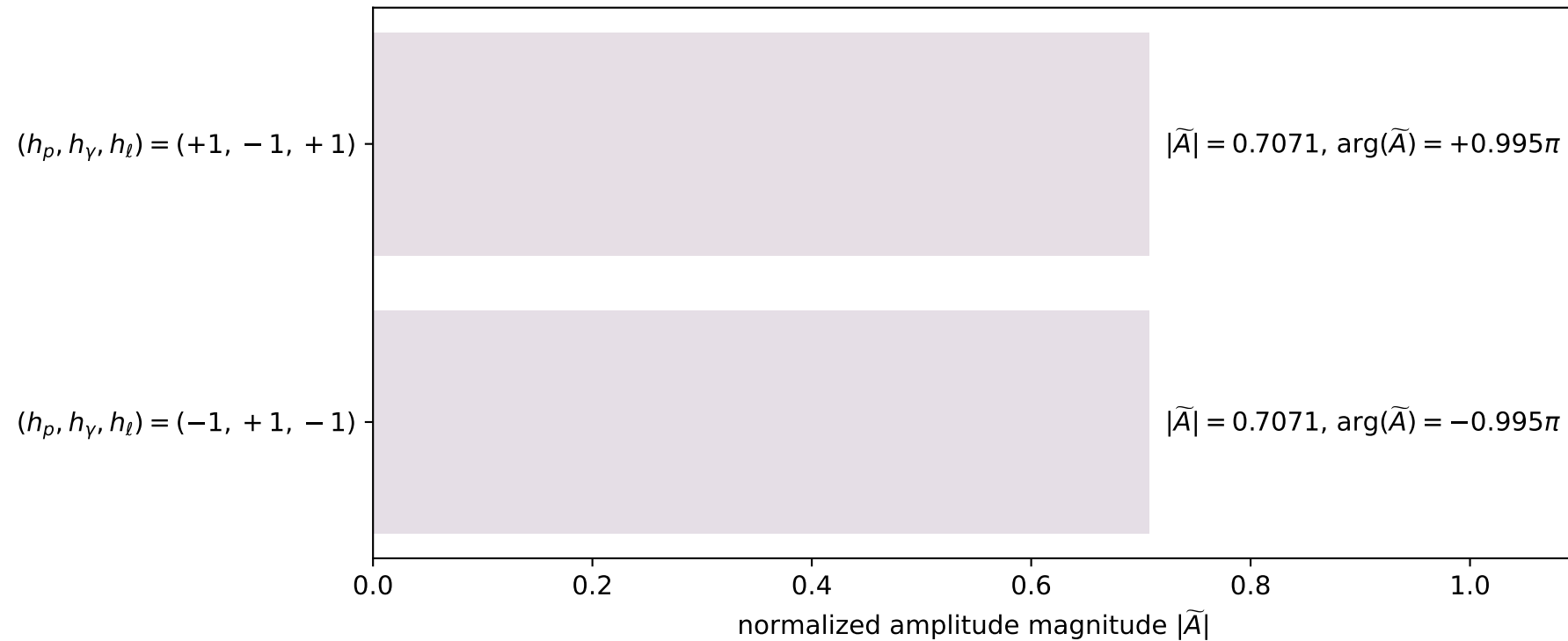
$s=16.6364$, $\sqrt{s}=4.07877$, $|\vec{P}|=1.93153$, $|\vec{P}'|=1.82957$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=0.0149987$, $\phi_\gamma=6.26819$
 $E_\gamma=0.0966003$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.15509$
 $Q^2=14.8818$, $x_B=0.997069$, $t=-14.1375$, $W^2=0.923592$, $y=0.94726$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9315$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9315$
 P $E= 2.1472$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9315$
 ℓ' $E= 1.9262$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.9262$
 P' $E= 2.0560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8296$
 q_γ $E= 0.0966$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0966$

x--y plane

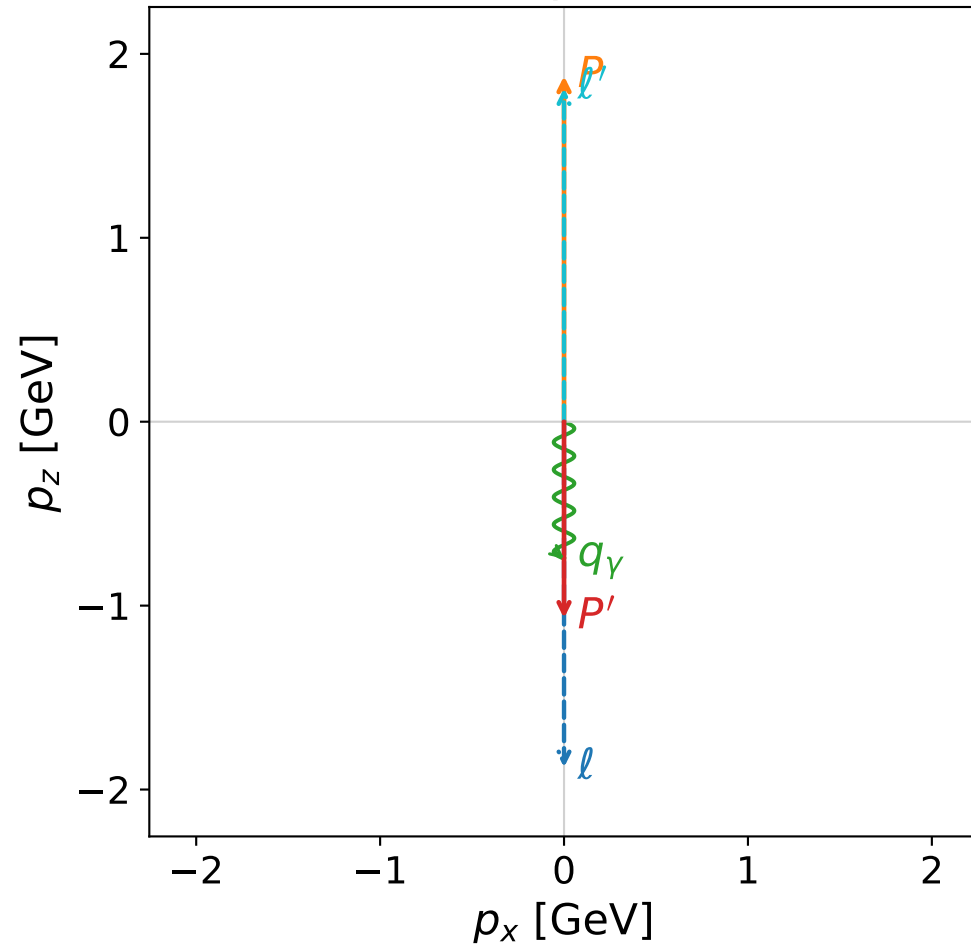


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

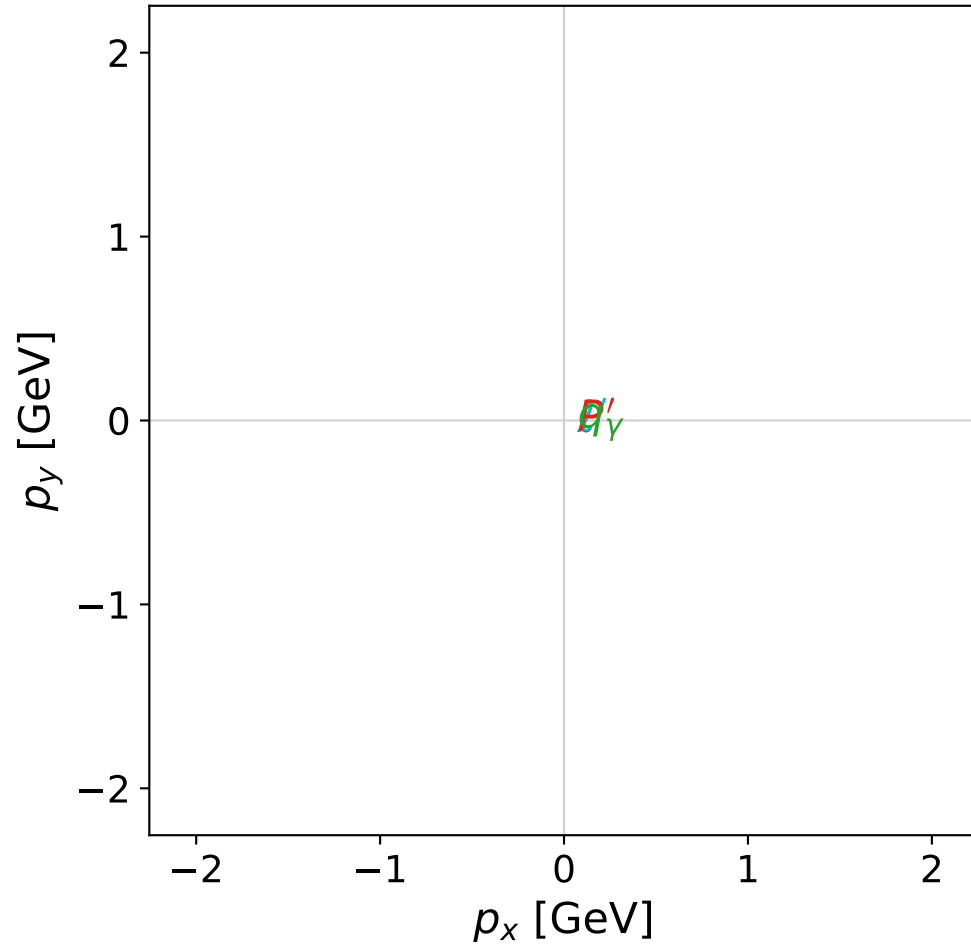


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



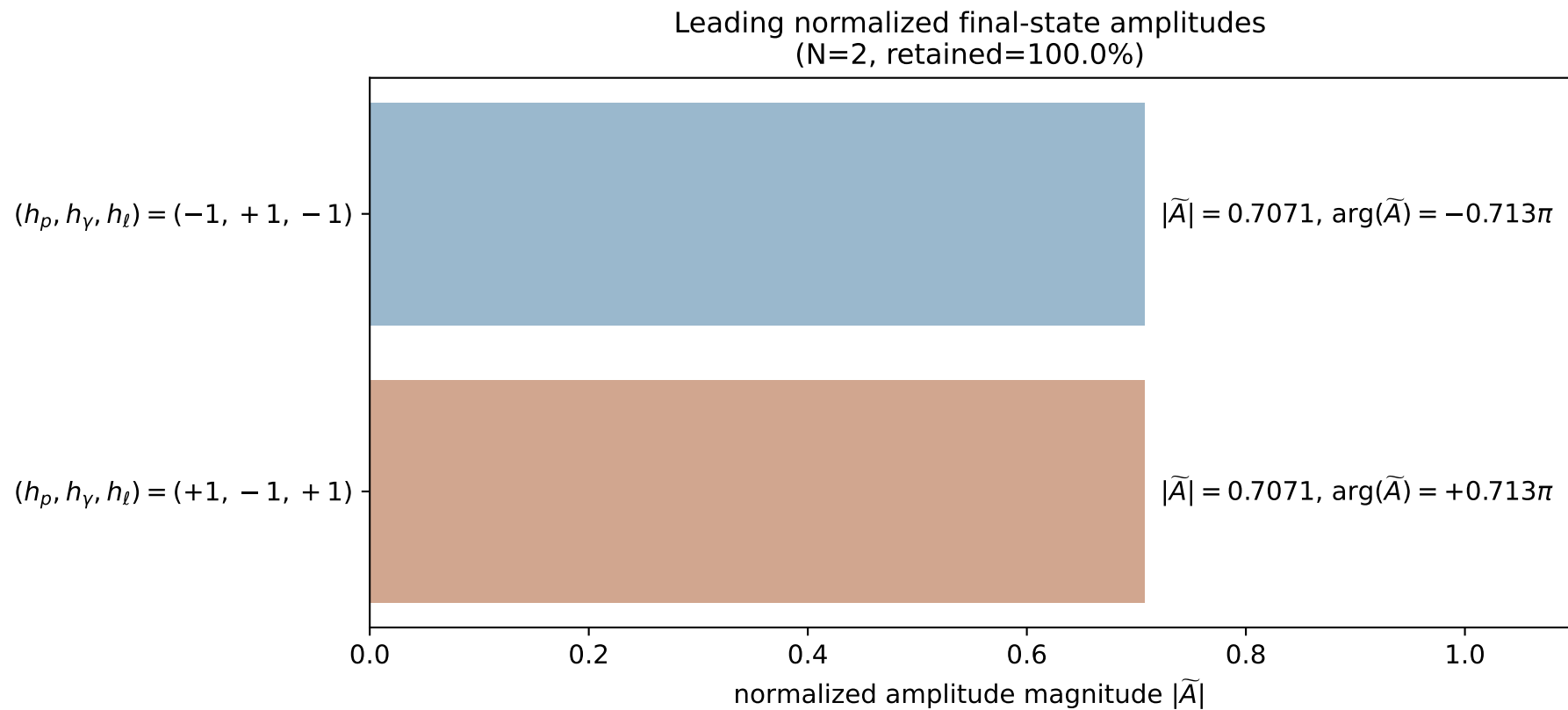
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_107
 $d_{\text{GHZ}}=3.86512\text{e-}08$
 region: polarization_cluster_03_local_minimum_107
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0107
 lepton mixing angle: $\alpha_e=2.4205789$ rad
 incoming lepton: $-0.751137|+\rangle +0.660146|-\rangle$
 proton mixing angle: $\alpha_p=0.72102631$ rad
 incoming proton: $0.751129|+\rangle +0.660156|-\rangle$

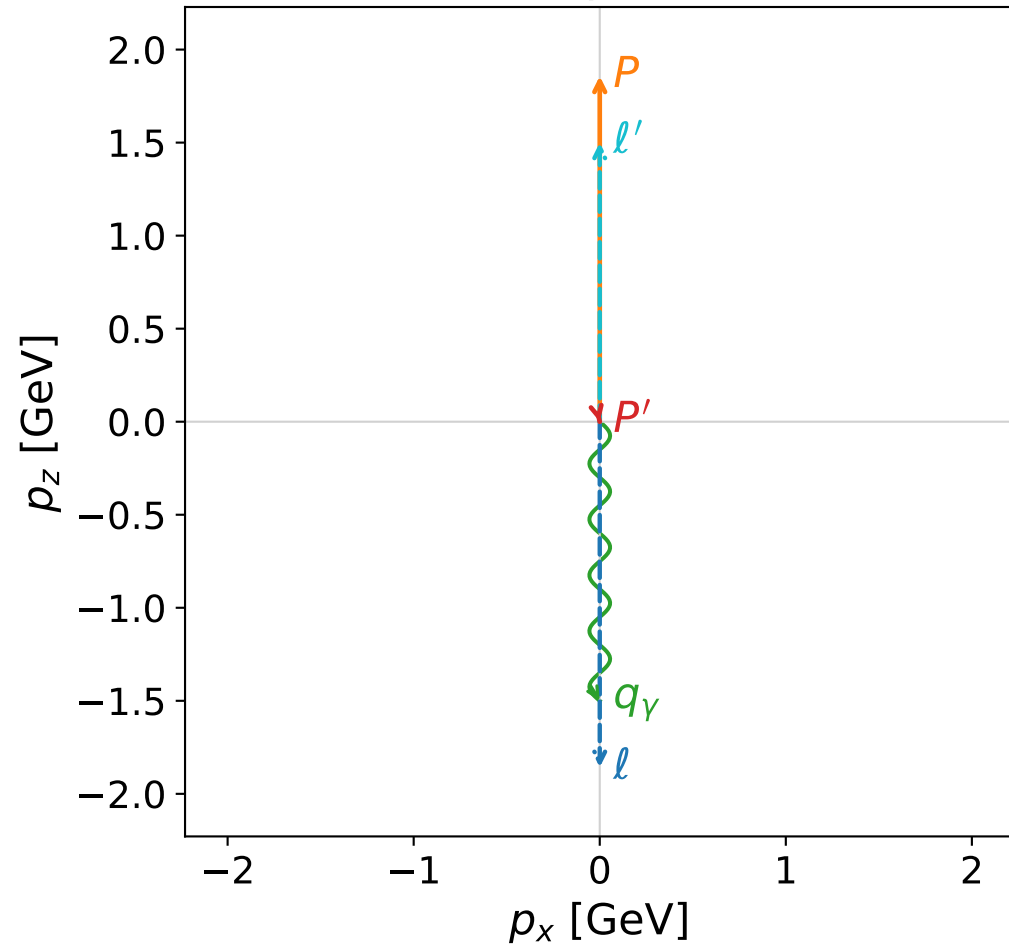
$s=15.8347$, $\sqrt{s}=3.97928$, $|\vec{P}|=1.87909$, $|\vec{P}'|=1.06808$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=1.05624$, $\phi_\gamma=5.38261$
 $E_\gamma=0.744851$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.47839$
 $Q^2=13.6266$, $x_B=0.962801$, $t=-8.22517$, $W^2=1.40632$, $y=0.946392$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8791$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8791$
 P $E= 2.1002$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8791$
 ℓ' $E= 1.8129$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.8129$
 P' $E= 1.4215$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.0681$
 q_γ $E= 0.7449$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.7449$

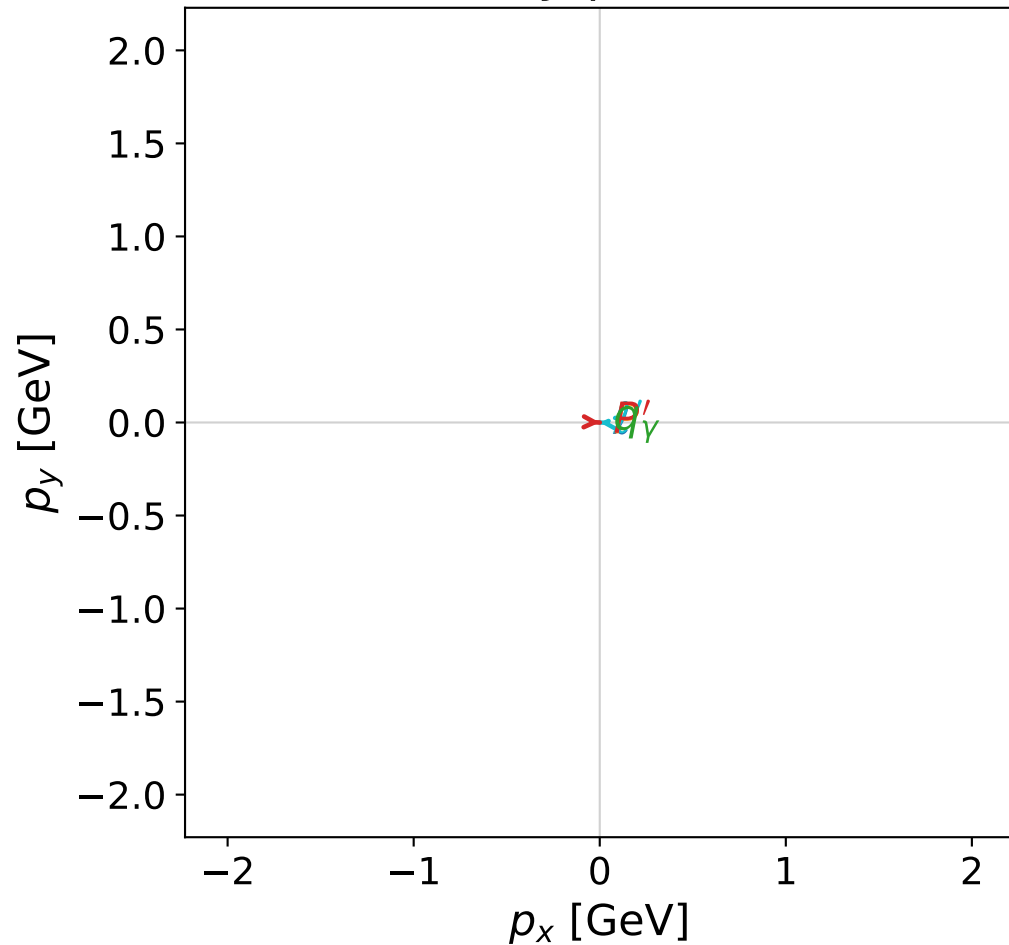


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



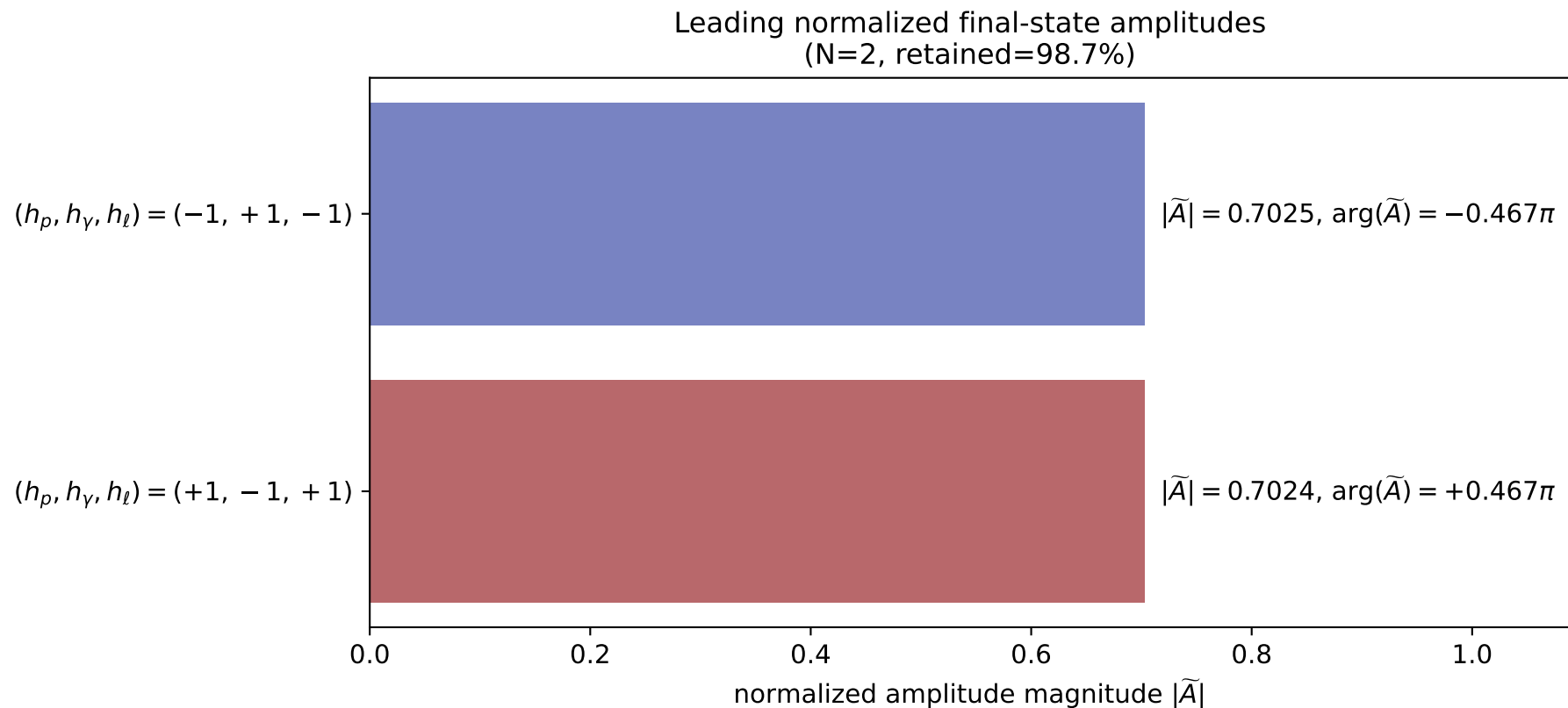
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_118
 $d_{\{\rm GHZ\}}=4.43159\text{e-}08$
 region: polarization_cluster_03_local_minimum_118
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0118
 lepton mixing angle: $\alpha_e=2.3993315$ rad
 incoming lepton: $-0.736942|+\rangle +0.675956|-\rangle$
 proton mixing angle: $\alpha_p=0.74228136$ rad
 incoming proton: $0.736928|+\rangle +0.675971|-\rangle$

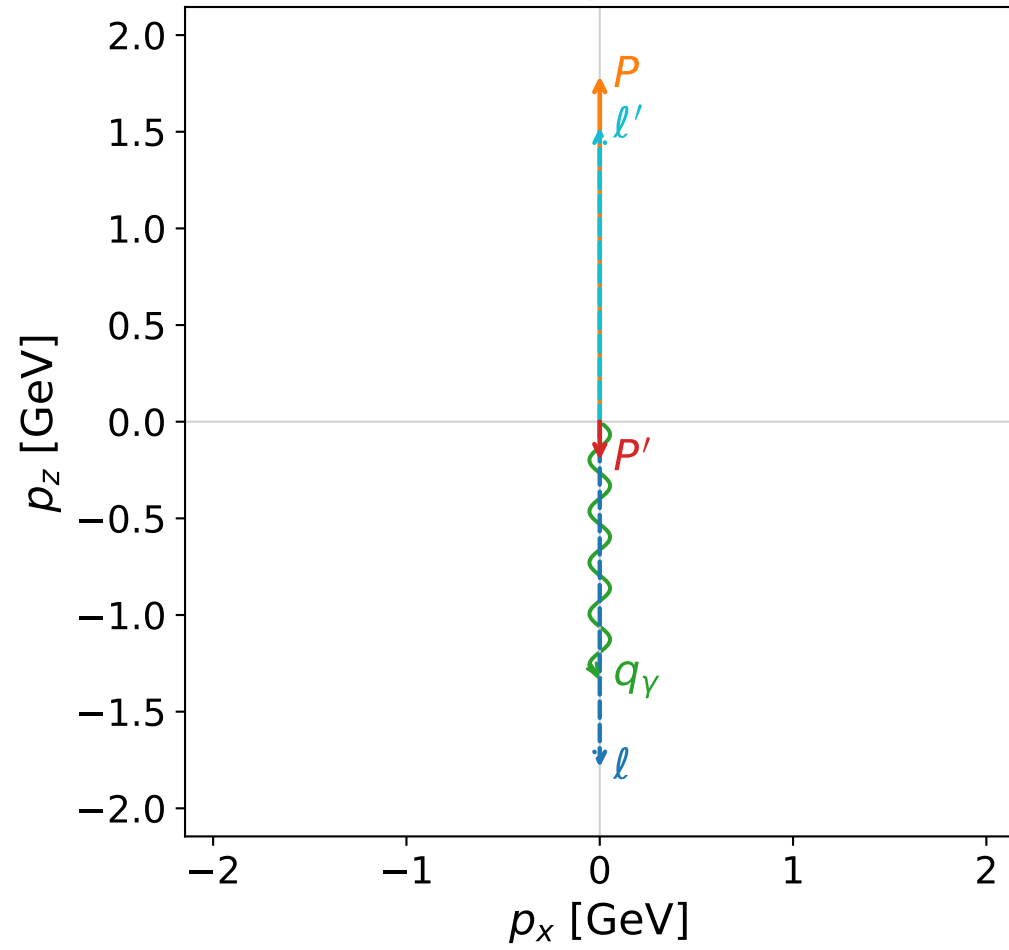
$s=15.5089$, $\sqrt{s}=3.93814$, $|\vec{P}|=1.85736$, $|\vec{P}'|=0.000247174$
 $\theta_{p'}=2.91175$, $\theta_\gamma=3.14159$, $\phi_{p'}=6.24252$, $\phi_\gamma=4.56954$
 $E_\gamma=1.49995$, $\theta_{\ell'}=3.7536\text{e-}05$, $\phi_{\ell'}=3.10092$
 $Q^2=11.1456$, $x_B=0.798465$, $t=-2.14474$, $W^2=3.69303$, $y=0.954178$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8574$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8574$
 P $E= 2.0808$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8574$
 ℓ' $E= 1.5002$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 1.5002$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0002$
 q_γ $E= 1.4999$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.4999$

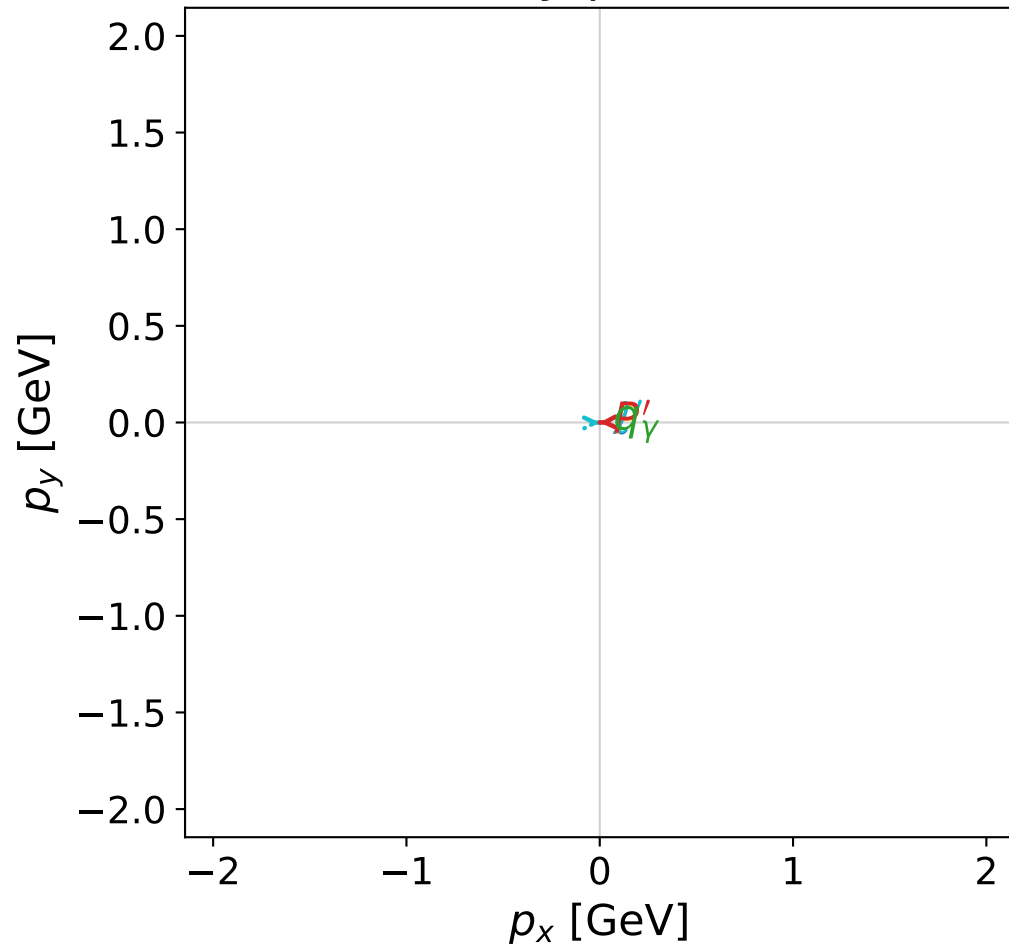


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



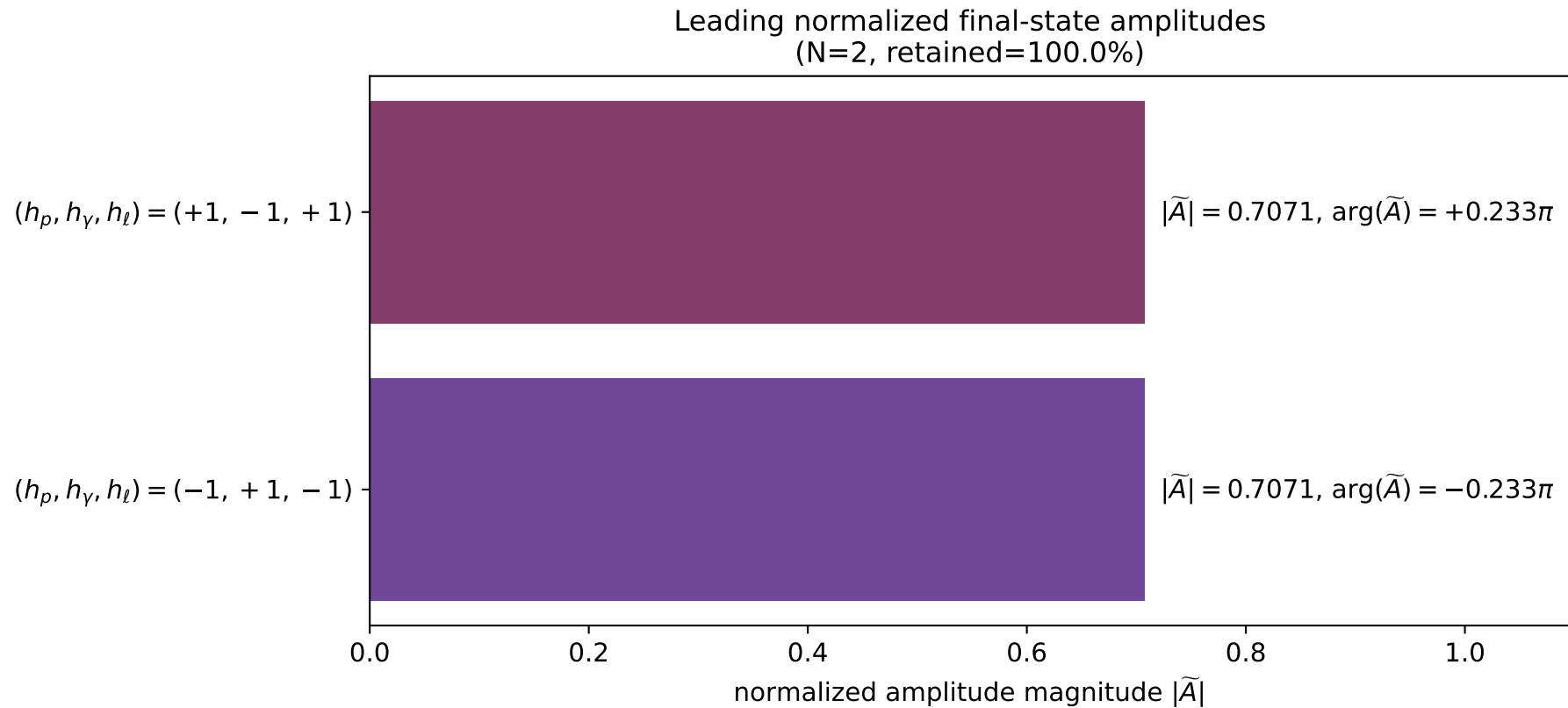
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_124
 $d_{\text{GHZ}}=4.80438\text{e-}08$
 region: polarization_cluster_03_local_minimum_124
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0124
 lepton mixing angle: $\alpha_e=2.3690498$ rad
 incoming lepton: $-0.716138|+\rangle +0.697959|-\rangle$
 proton mixing angle: $\alpha_p=0.77252117$ rad
 incoming proton: $0.716153|+\rangle +0.697943|-\rangle$

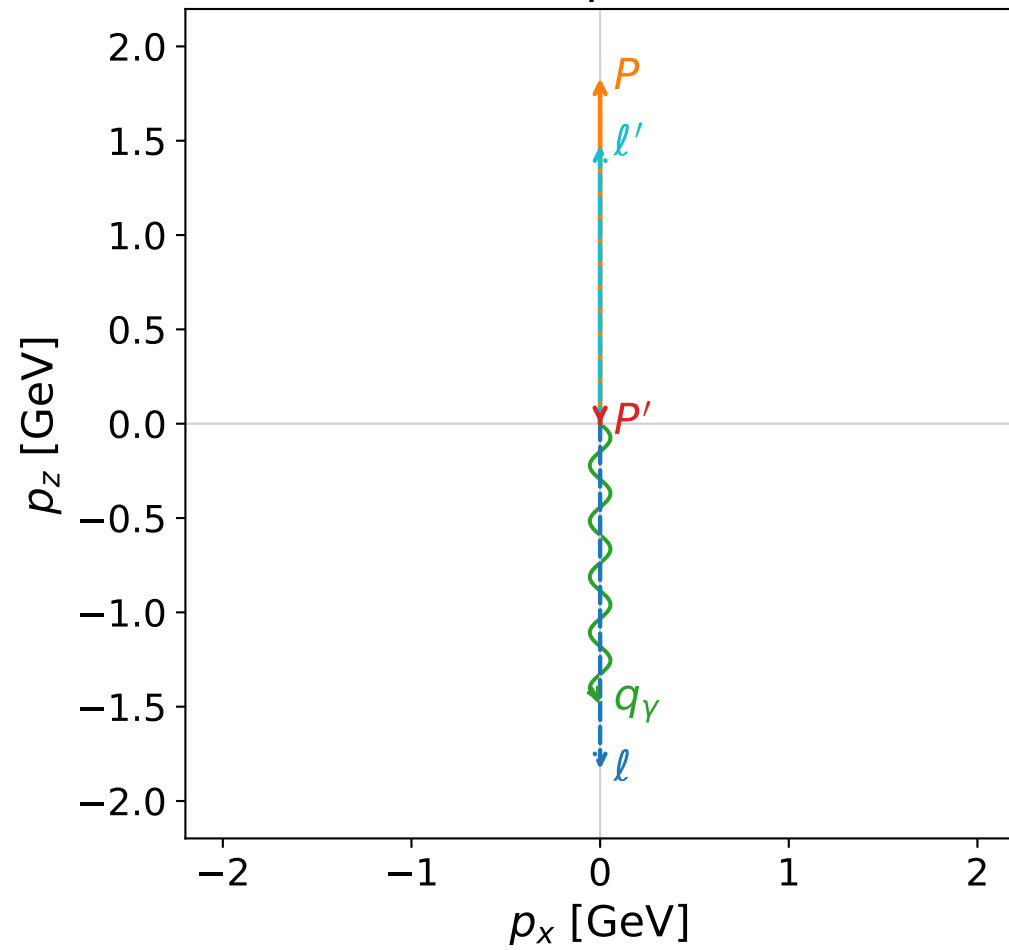
$s=14.4913$, $\sqrt{s}=3.80674$, $|\vec{P}|=1.78781$, $|\vec{P}'|=0.198316$
 $\theta_{p'}=3.14137$, $\theta_\gamma=3.14159$, $\phi_{p'}=3.17027$, $\phi_\gamma=0.760181$
 $E_\gamma=1.32485$, $\theta_{\ell'}=2.88699\text{e-}05$, $\phi_{\ell'}=0.0286774$
 $Q^2=10.8925$, $x_B=0.843897$, $t=-2.82066$, $W^2=2.89472$, $y=0.948272$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.7878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.7878$
 P $E= 2.0189$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.7878$
 ℓ' $E= 1.5232$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5232$
 P' $E= 0.9587$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.1983$
 q_γ $E= 1.3248$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3248$

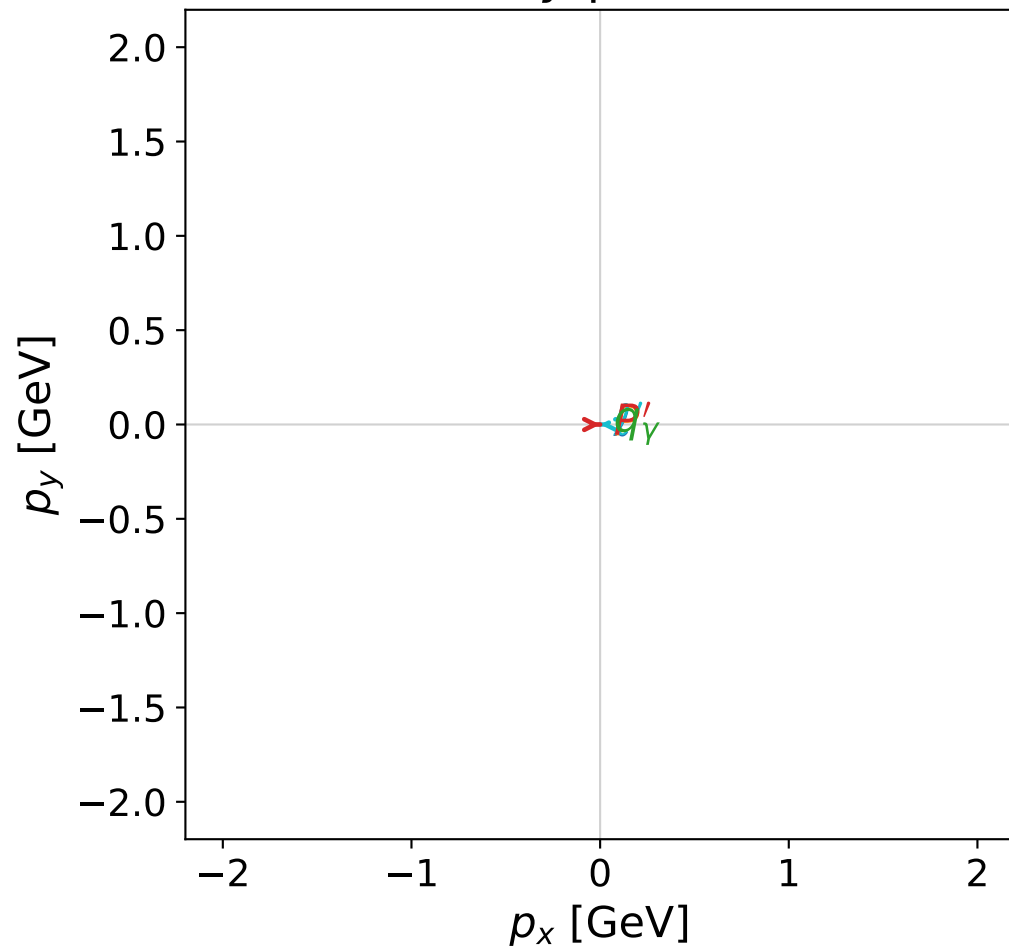


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



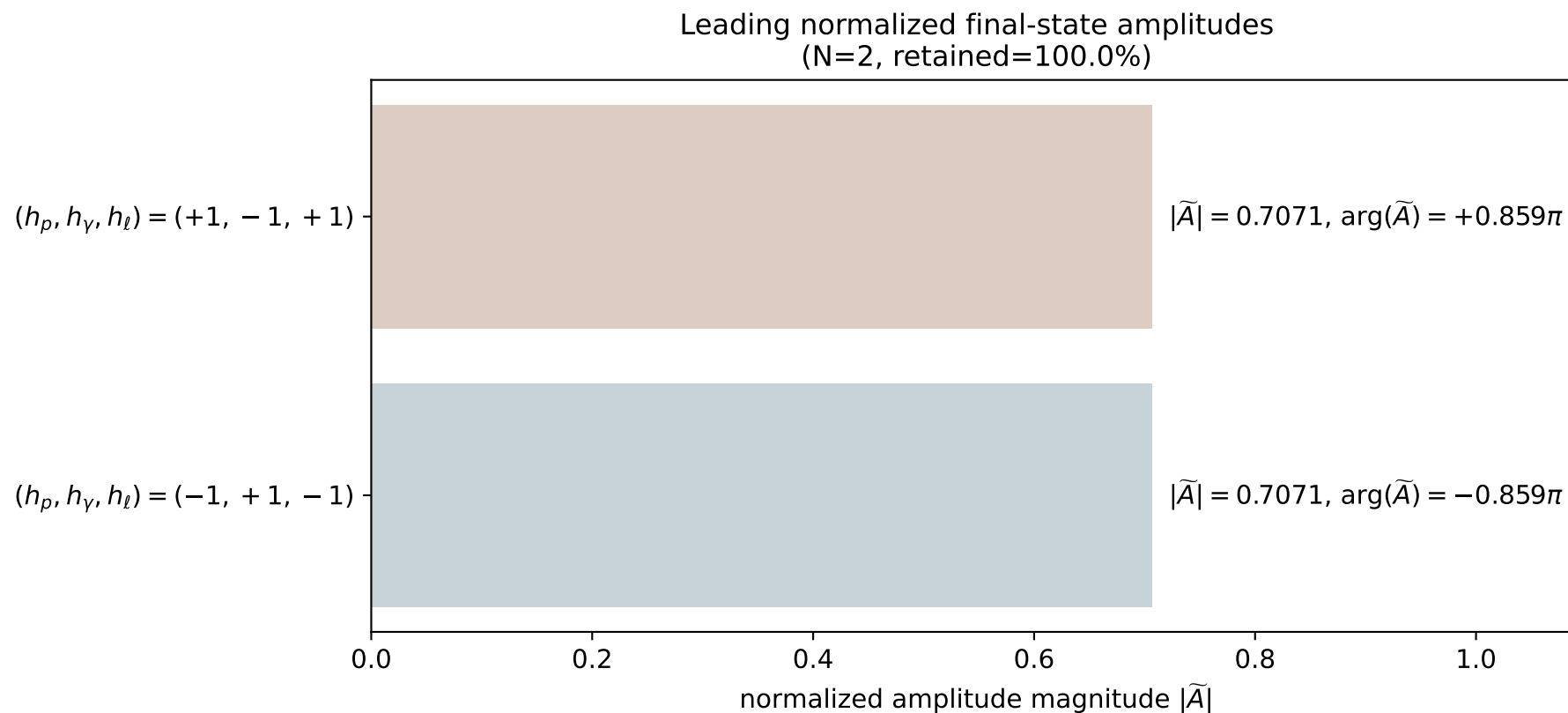
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_128
 $d_{\text{GHZ}}=5.13615\text{e-}08$
 region: polarization_cluster_03_local_minimum_128
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0128
 lepton mixing angle: $\alpha_e=2.5301823$ rad
 incoming lepton: $-0.818839|+\rangle +0.574023|-\rangle$
 proton mixing angle: $\alpha_p=0.61140686$ rad
 incoming proton: $0.818841|+\rangle +0.57402|-\rangle$

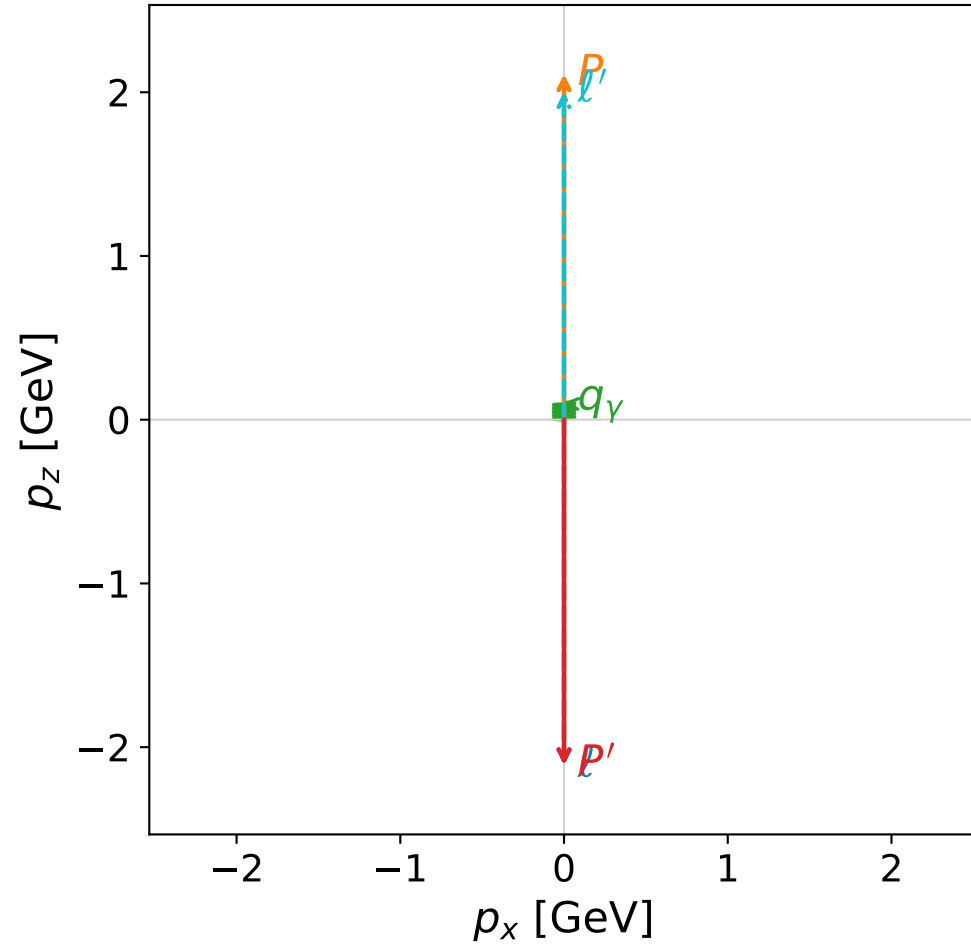
$s=15.1318$, $\sqrt{s}=3.88997$, $|\vec{P}|=1.83189$, $|\vec{P}'|=0.000225325$
 $\theta_{P'}=3.14145$, $\theta_\gamma=3.14159$, $\phi_{P'}=0.00233824$, $\phi_\gamma=5.84287$
 $E_\gamma=1.47587$, $\theta_{\ell'}=1.49012\text{e-}08$, $\phi_{\ell'}=3.14393$
 $Q^2=10.8162$, $x_B=0.79623$, $t=-2.10208$, $W^2=3.64791$, $y=0.953148$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8319$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8319$
 P $E= 2.0581$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8319$
 ℓ' $E= 1.4761$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4761$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0002$
 q_γ $E= 1.4759$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.4759$

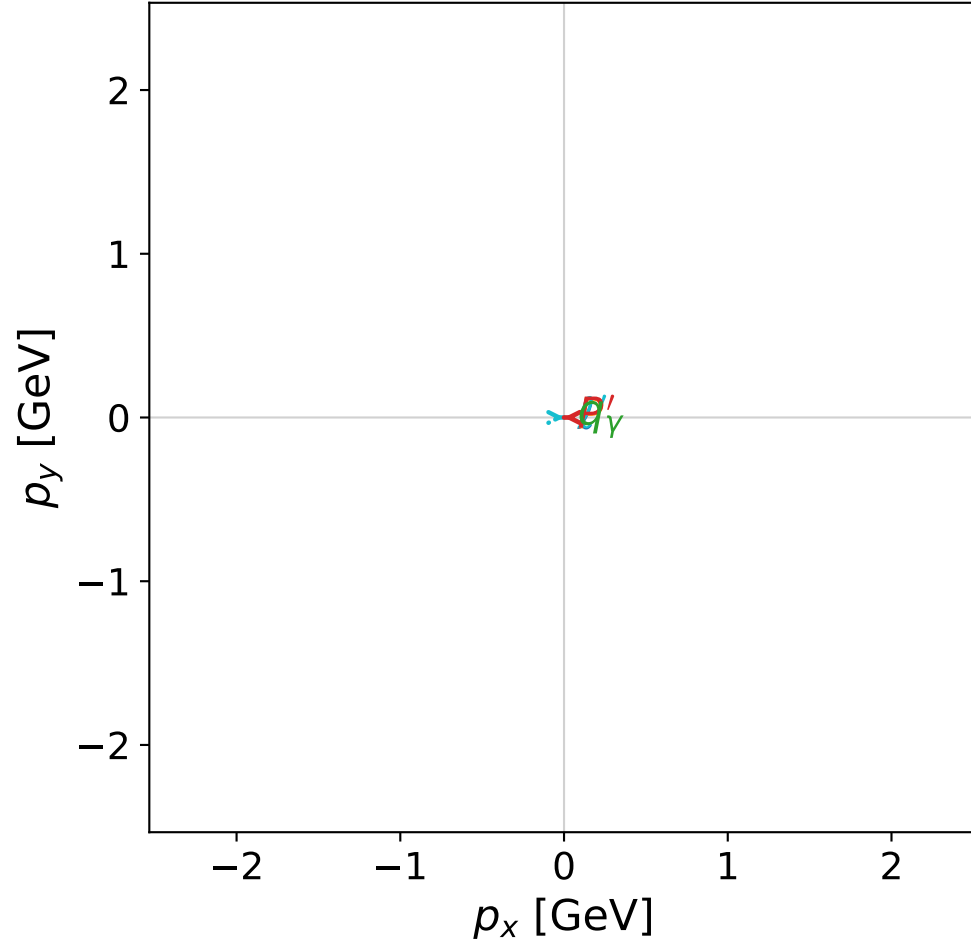


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



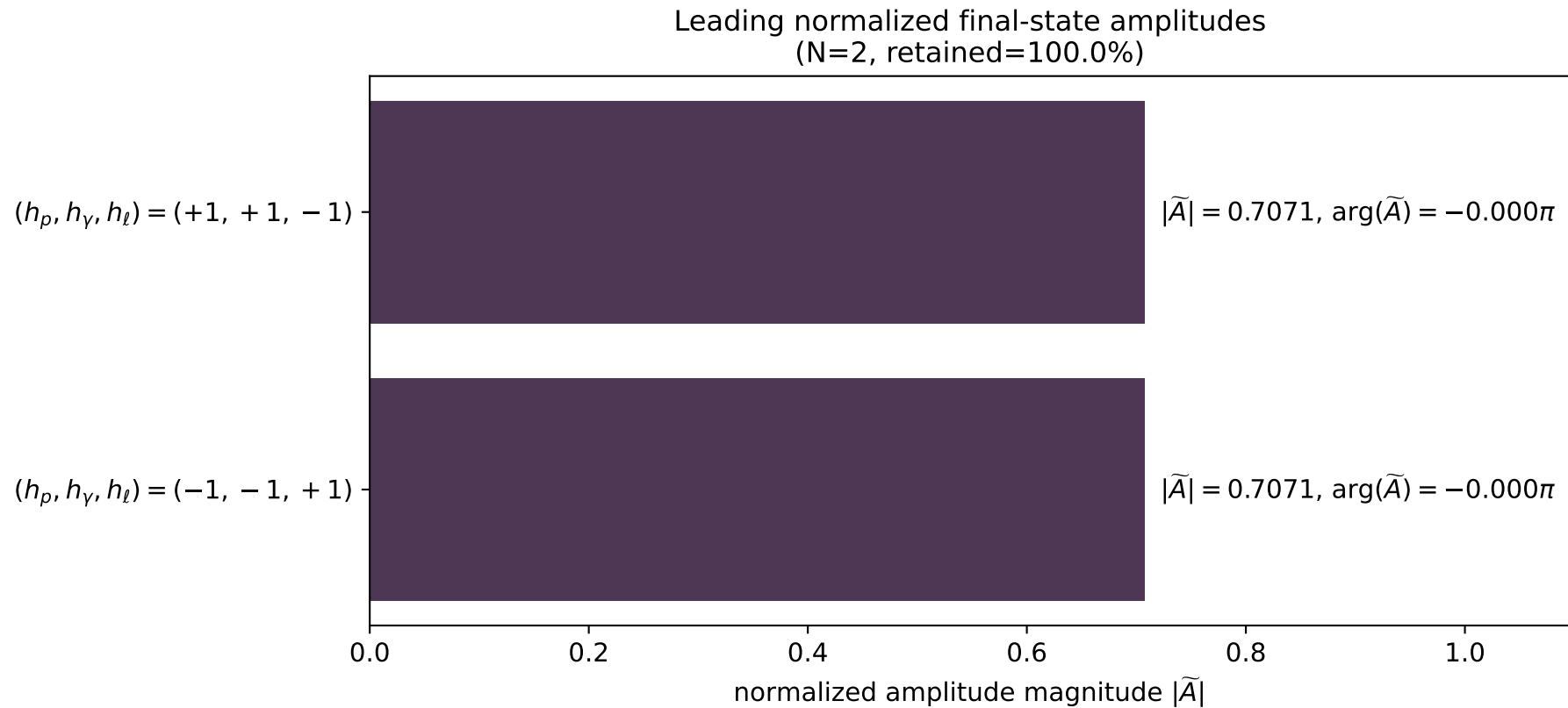
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_131
 $d_{\{\rm GHZ\}}=5.31748\text{e-}08$
 region: polarization_cluster_03_local_minimum_131
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0131
 lepton mixing angle: $\alpha_e=1.8802171$ rad
 incoming lepton: $-0.304507|+\rangle +0.95251|-\rangle$
 proton mixing angle: $\alpha_p=0.30941263$ rad
 incoming proton: $0.952513|+\rangle +0.304499|-\rangle$

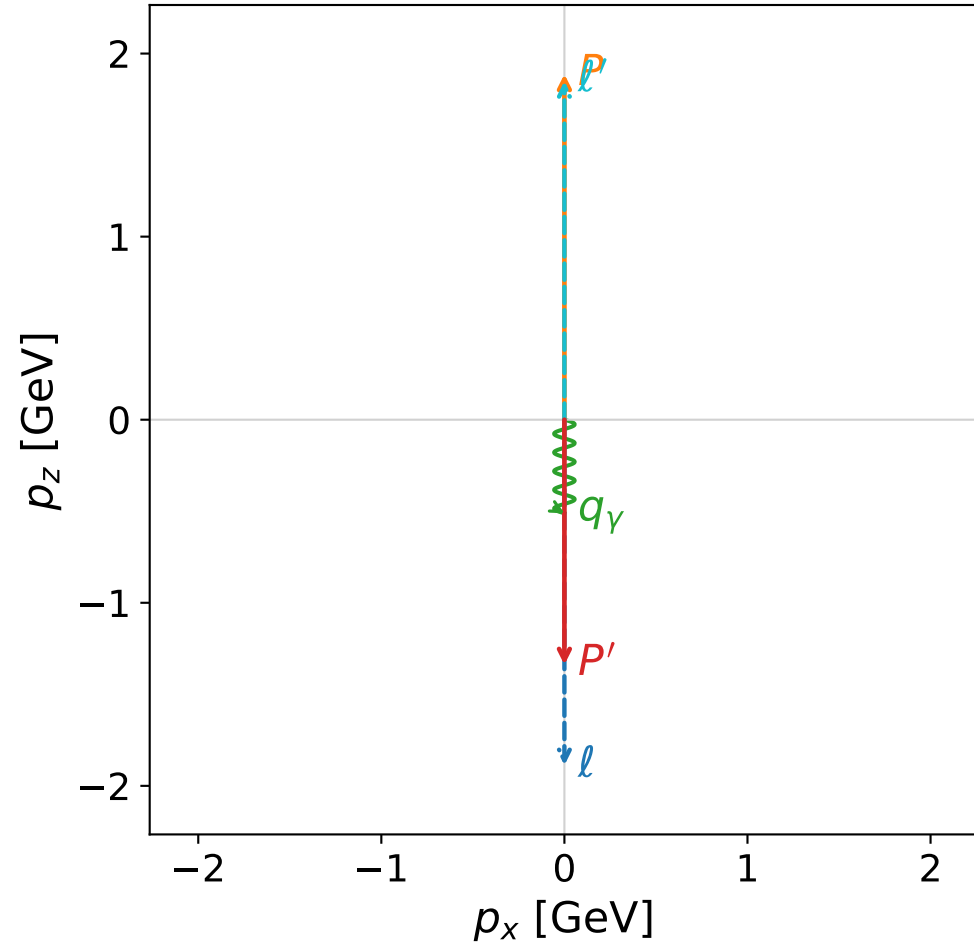
$s=19.5413$, $\sqrt{s}=4.42055$, $|\vec{P}|=2.11076$, $|\vec{P}'|=2.11076$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.40678$, $\phi_\gamma=1.45431$
 $E_\gamma=0.105533$, $\theta_{l'}=0$, $\phi_{l'}=0.265183$
 $Q^2=16.9302$, $x_B=0.947768$, $t=-17.8212$, $W^2=1.81287$, $y=0.957226$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.1108$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1108$
 P $E= 2.3098$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1108$
 l' $E= 2.0052$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0052$
 P' $E= 2.3098$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.1108$
 q_γ $E= 0.1055$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1055$

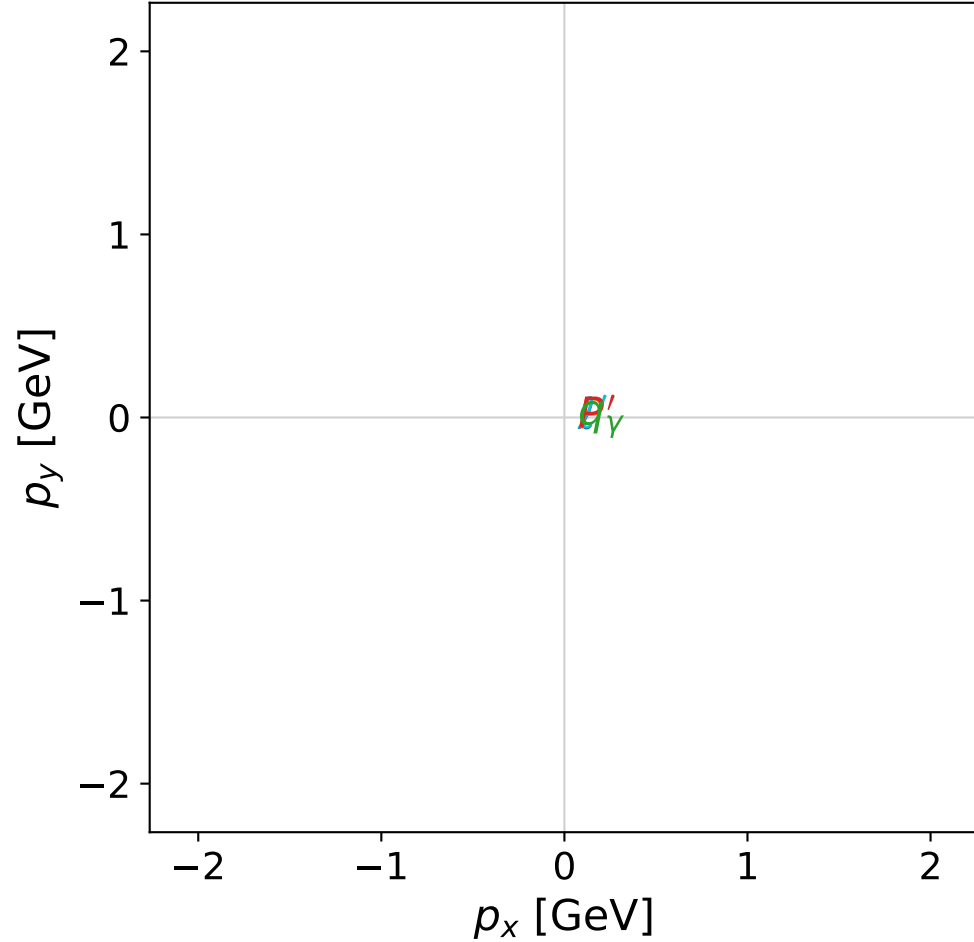


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



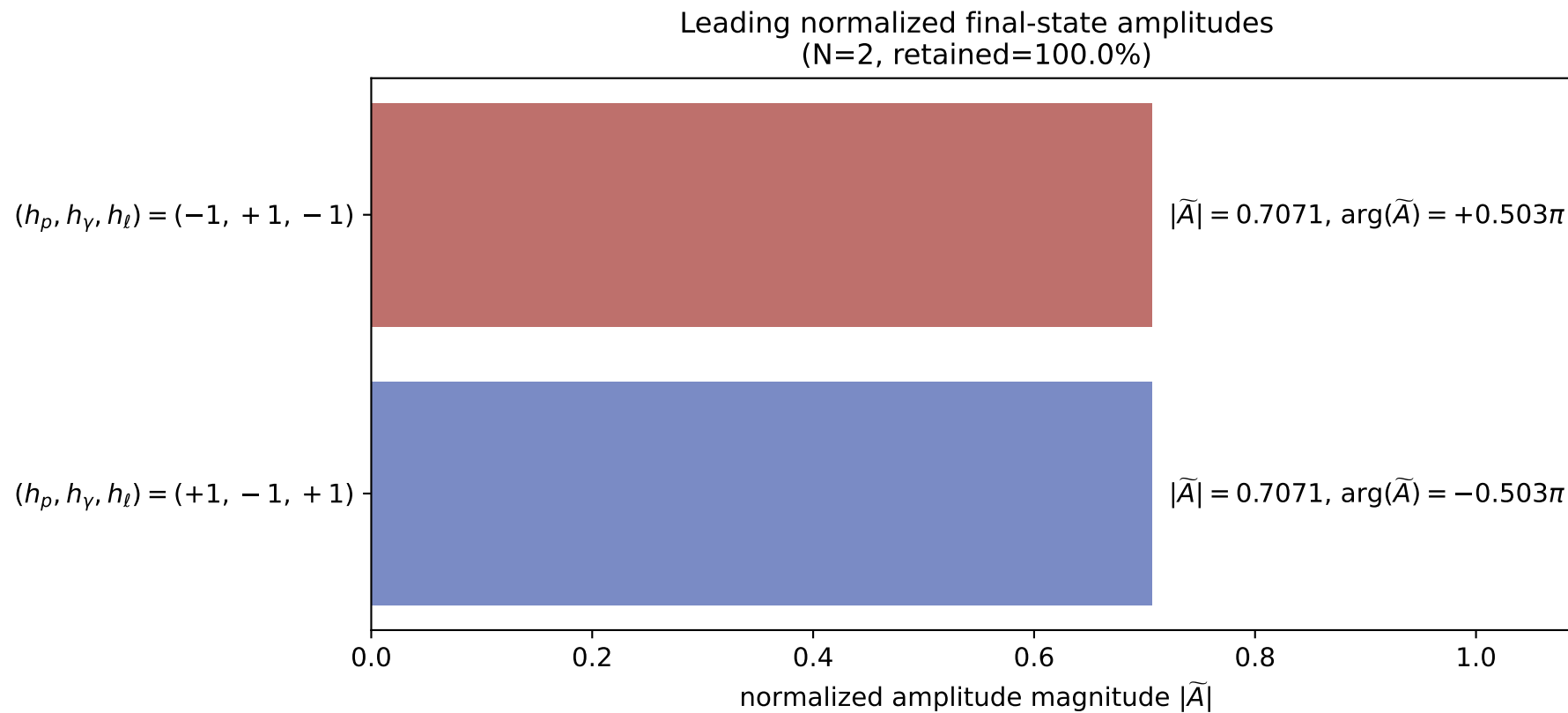
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_132
 $d_{\text{GHZ}}=5.32794\text{e-}08$
 region: polarization_cluster_03_local_minimum_132
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0132
 lepton mixing angle: $\alpha_e=2.7526505$ rad
 incoming lepton: $-0.925311|+\rangle +0.37921|-\rangle$
 proton mixing angle: $\alpha_p=0.38894583$ rad
 incoming proton: $0.925309|+\rangle +0.379213|-\rangle$

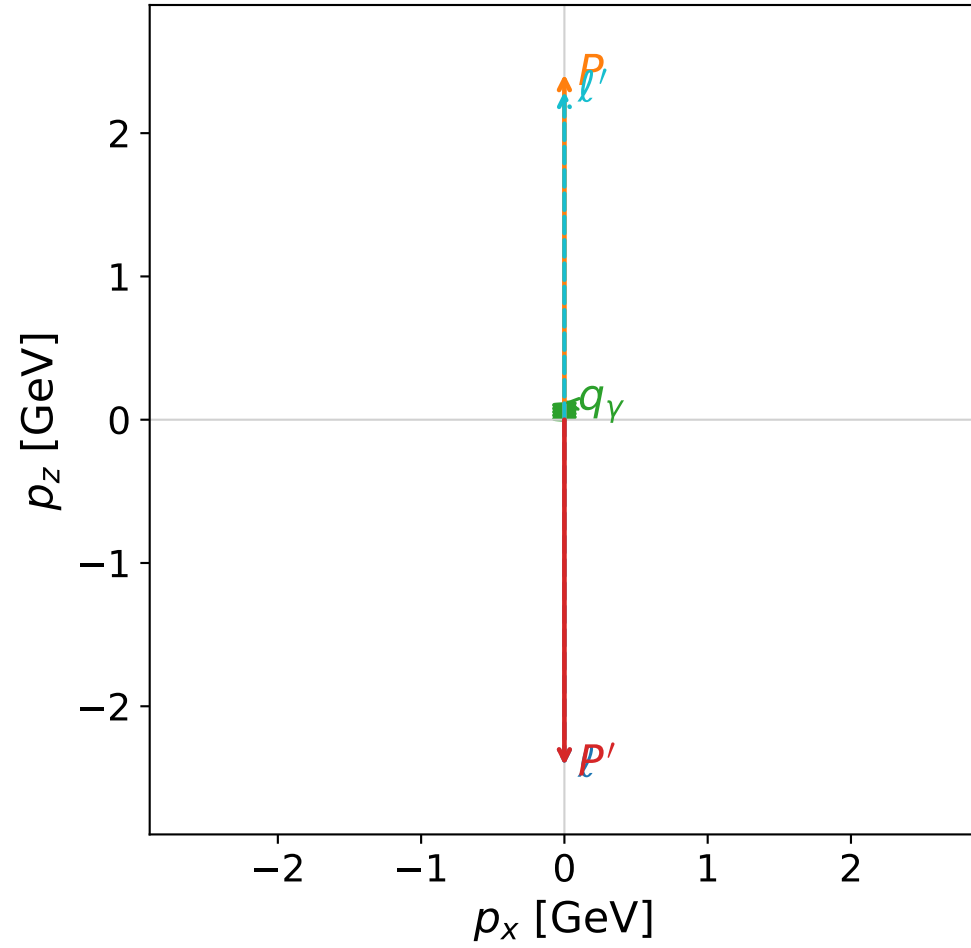
$s=15.965$, $\sqrt{s}=3.99562$, $|\vec{P}|=1.88771$, $|\vec{P}'|=1.33879$
 $\theta_{p'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{p'}=5.83465$, $\phi_\gamma=1.56232$
 $E_\gamma=0.511073$, $\theta_{\ell'}=0$, $\phi_{\ell'}=3.08424$
 $Q^2=13.968$, $x_B=0.978806$, $t=-10.1864$, $W^2=1.18229$, $y=0.945994$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.8877$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.8877$
 P $E= 2.1079$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.8877$
 ℓ' $E= 1.8499$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8499$
 P' $E= 1.6347$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -1.3388$
 q_γ $E= 0.5111$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.5111$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

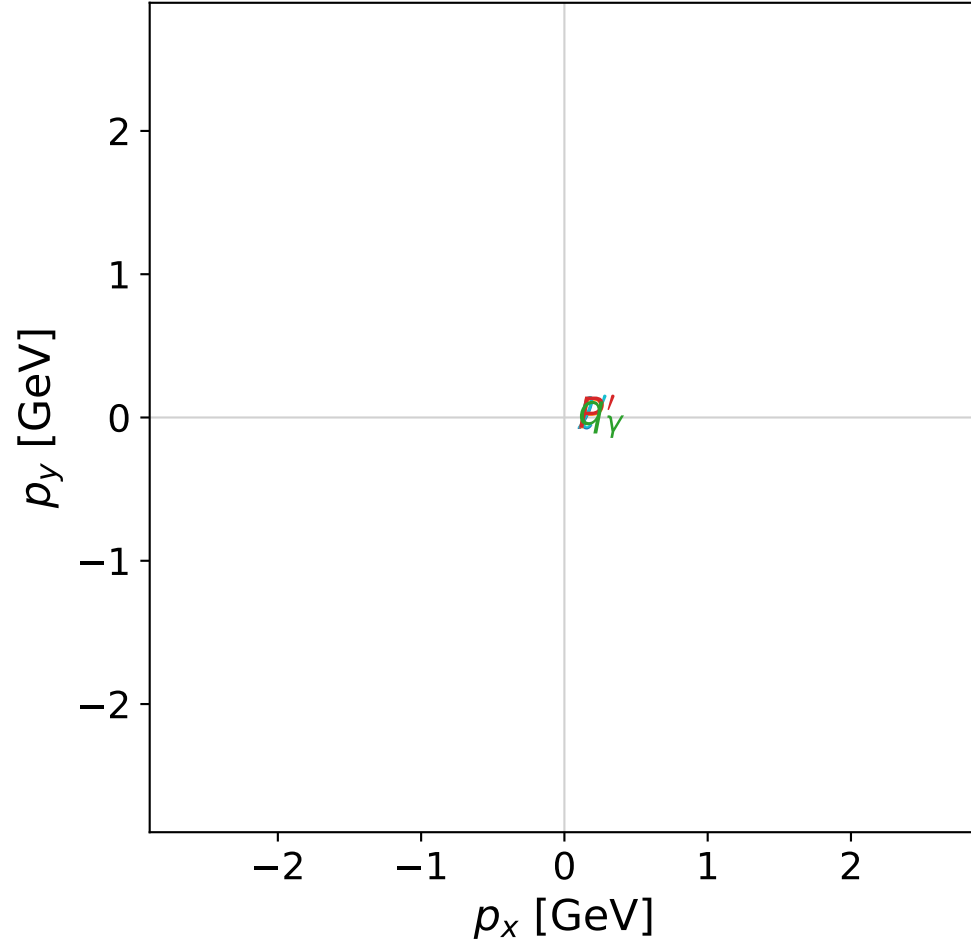


dghz_mixing_angles_polarization_03_local_minimum_134
 $d_{\text{GHZ}}=5.41283\text{e-}08$
 region: polarization_cluster_03_local_minimum_134
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0134
 lepton mixing angle: $\alpha_e=2.9154679$ rad
 incoming lepton: $-0.974543|+\rangle +0.224203|-\rangle$
 proton mixing angle: $\alpha_p=1.3446609$ rad
 incoming proton: $0.224213|+\rangle +0.97454|-\rangle$

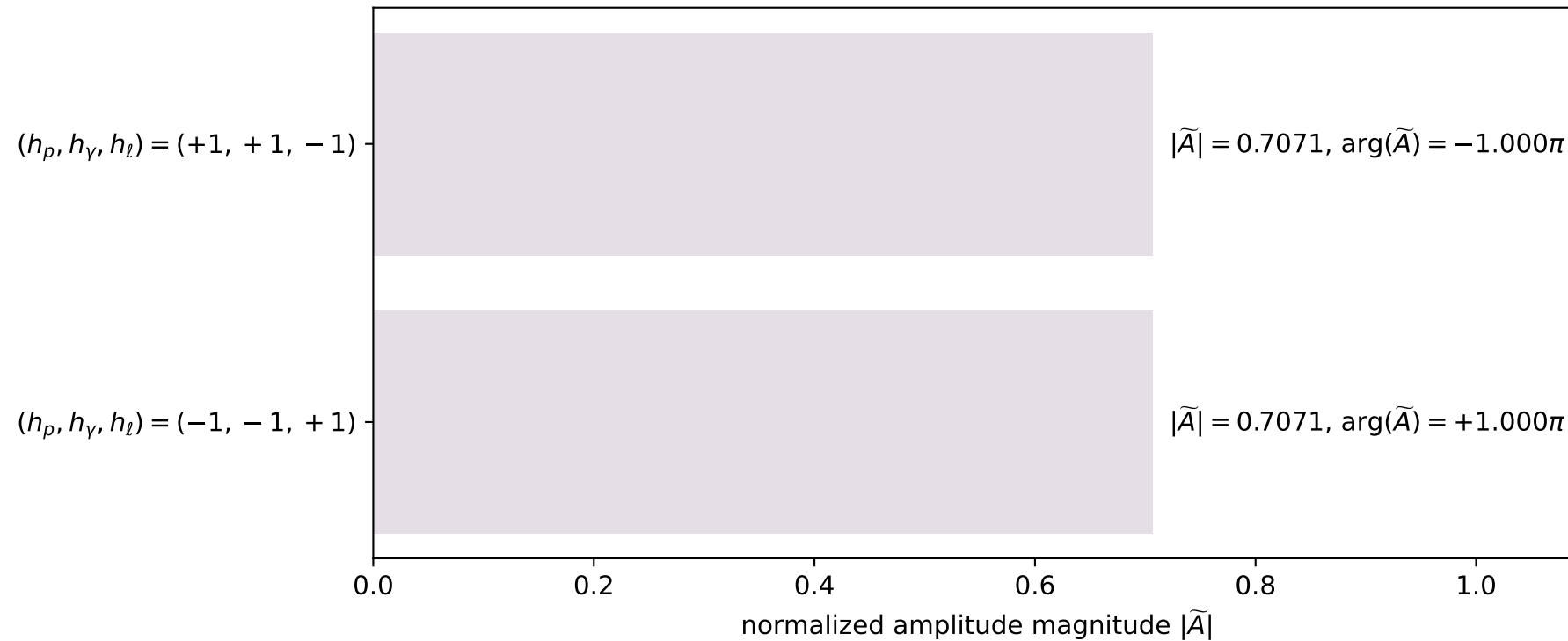
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=2.5701$, $\phi_\gamma=3.68393$
 $E_\gamma=0.120596$, $\theta_{l'}=0$, $\phi_{l'}=5.71169$
 $Q^2=22.1078$, $x_B=0.948273$, $t=-23.2713$, $W^2=2.0858$, $y=0.966566$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.2914$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.2914$
 P' $E= 2.5880$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

x--y plane

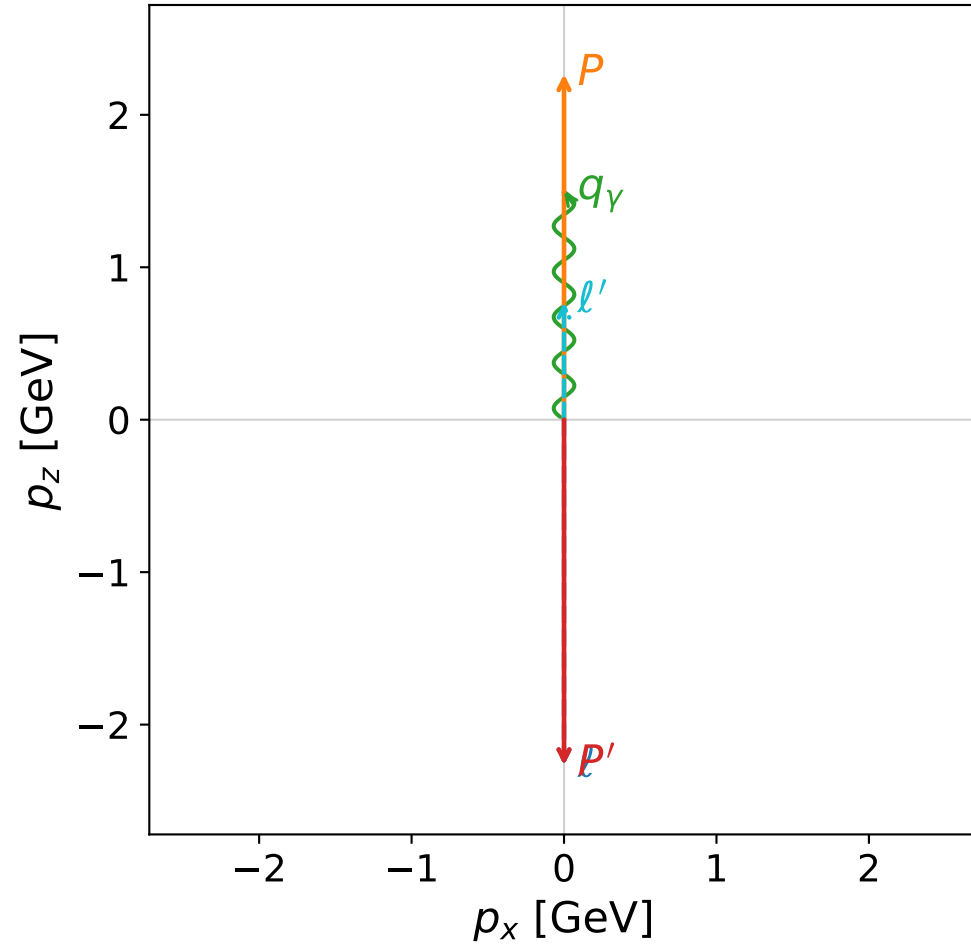


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

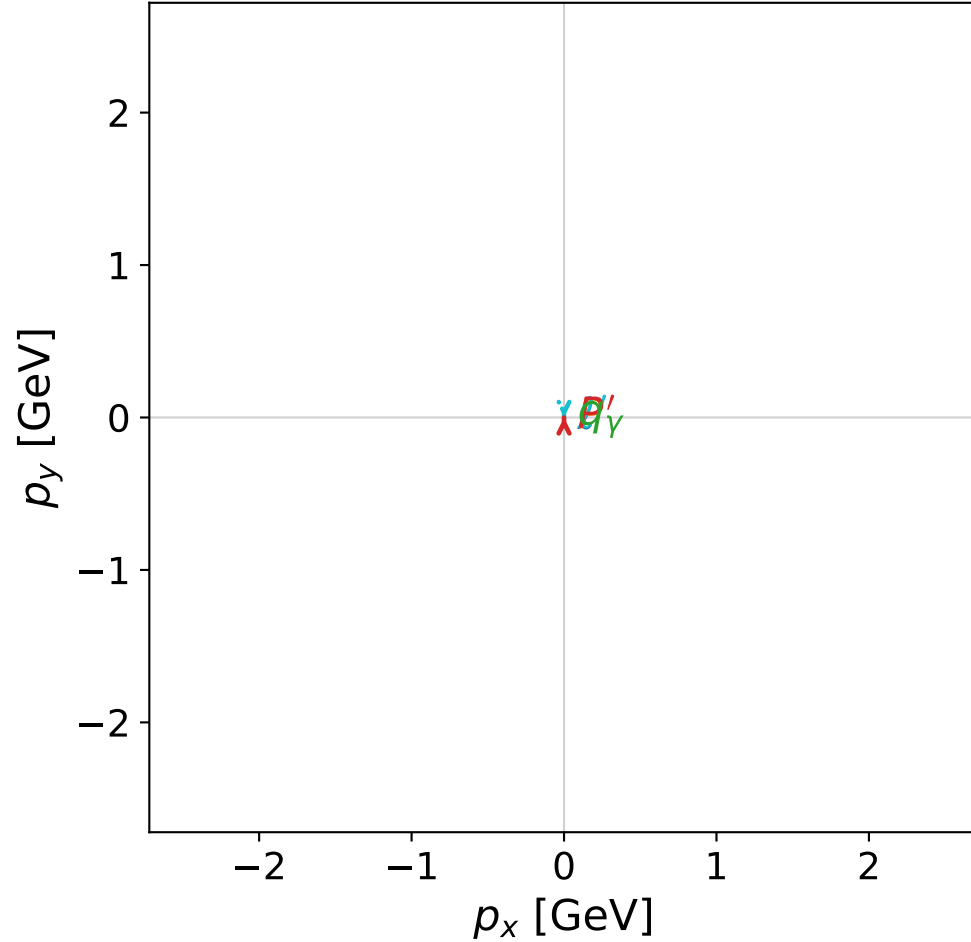


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



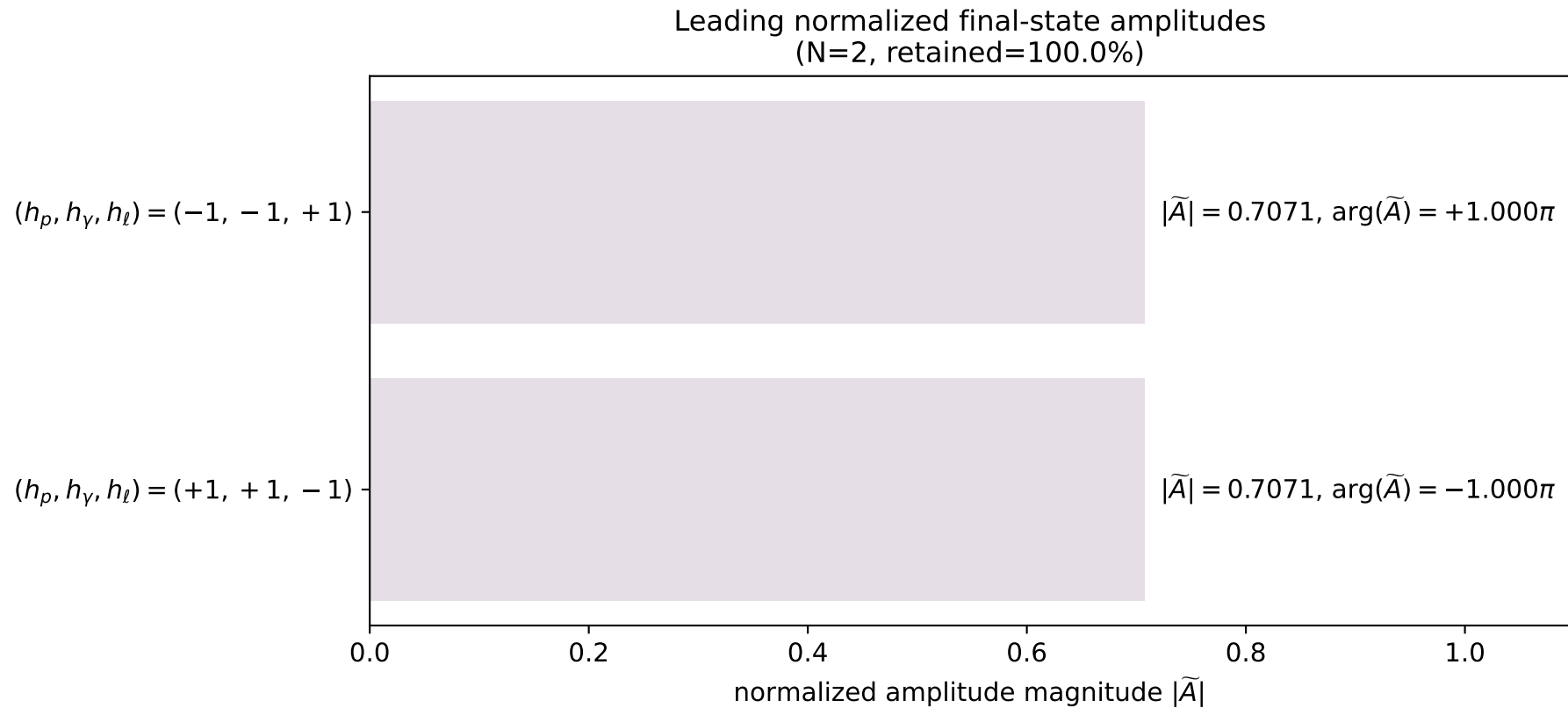
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_143
 $d_{\{\rm GHZ\}}=7.74877\text{e-}08$
 region: polarization_cluster_03_local_minimum_143
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0143
 lepton mixing angle: $\alpha_e=2.2187554$ rad
 incoming lepton: $-0.60356|+\rangle +0.797317|-\rangle$
 proton mixing angle: $\alpha_p=0.64796548$ rad
 incoming proton: $0.797313|+\rangle +0.603566|-\rangle$

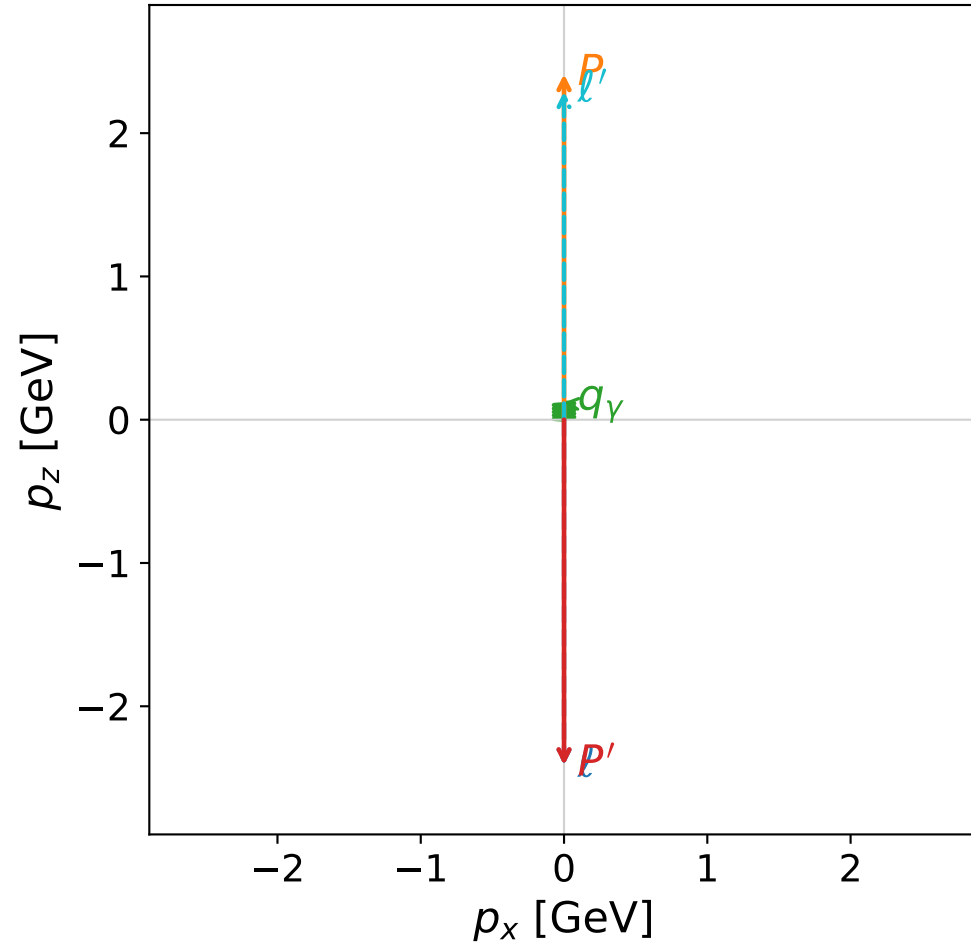
$s=22.2813$, $\sqrt{s}=4.7203$, $|\vec{P}|=2.26695$, $|\vec{P}'|=2.26695$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=1.65437$, $\phi_\gamma=3.92312$
 $E_\gamma=1.49471$, $\theta_{\ell'}=0$, $\phi_{\ell'}=4.79596$
 $Q^2=7.00254$, $x_B=0.331661$, $t=-20.5563$, $W^2=14.9908$, $y=0.986548$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2670$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 P $E= 2.4533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2670$
 ℓ' $E= 0.7722$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.7722$
 P' $E= 2.4533$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -2.2670$
 q_γ $E= 1.4947$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.4947$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

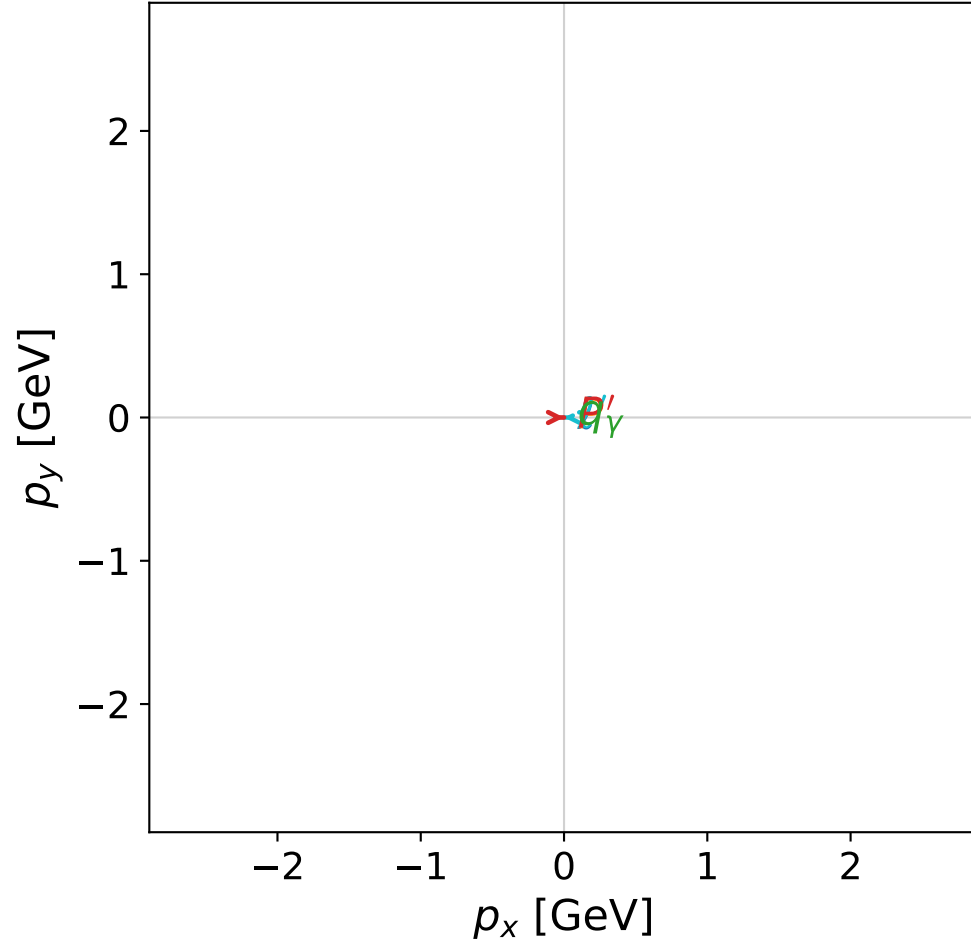


dghz_mixing_angles_polarization_03_local_minimum_154
 $d_{\{\text{rm GHZ}\}}=9.73504\text{e-}08$
 region: polarization_cluster_03_local_minimum_154
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0154
 lepton mixing angle: $\alpha_e=3.0190559$ rad
 incoming lepton: $-0.992502|+\rangle +0.12223|-\rangle$
 proton mixing angle: $\alpha_p=1.4482597$ rad
 incoming proton: $0.12223|+\rangle +0.992502|-\rangle$

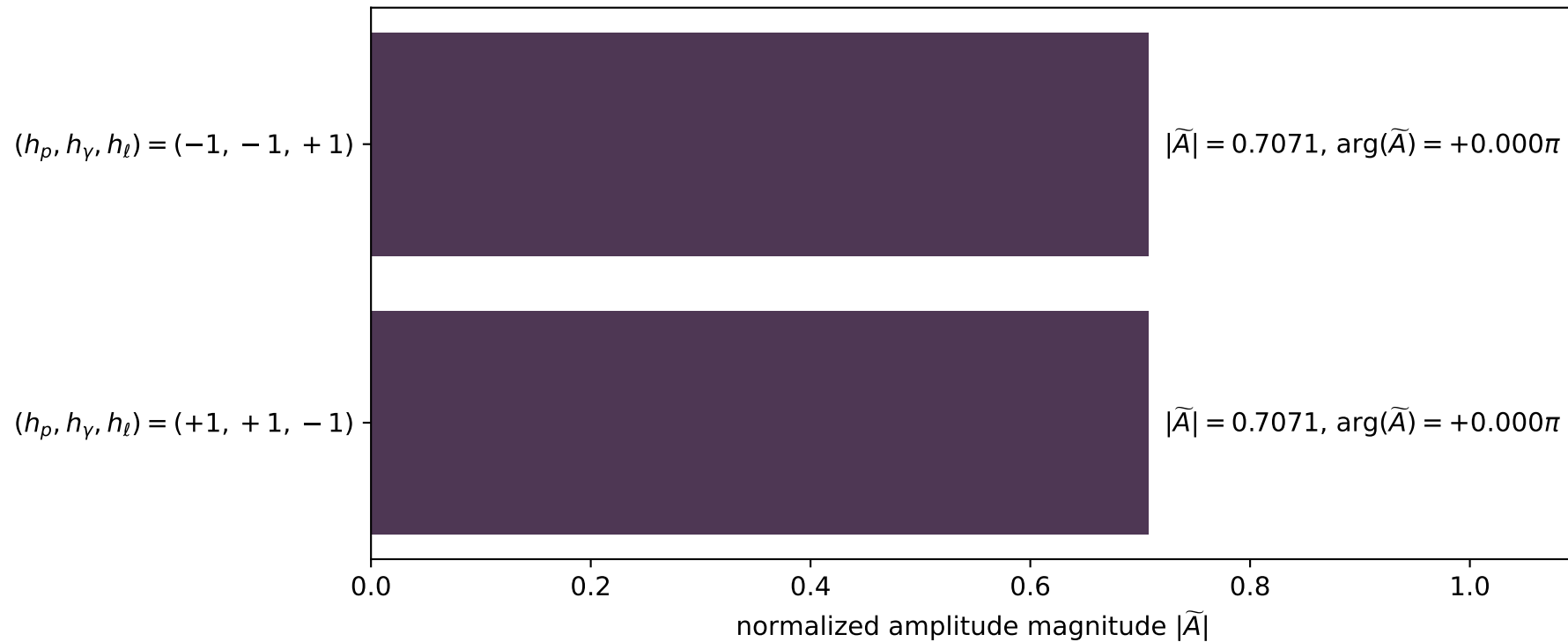
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=2.41202$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=0.116076$, $\phi_\gamma=6.10382$
 $E_\gamma=0.120666$, $\theta_{l'}=0$, $\phi_{l'}=3.25767$
 $Q^2=22.1071$, $x_B=0.948242$, $t=-23.2713$, $W^2=2.08651$, $y=0.966567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 2.2913$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 2.2913$
 P' $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 q_γ $E= 0.1207$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.1207$

x--y plane

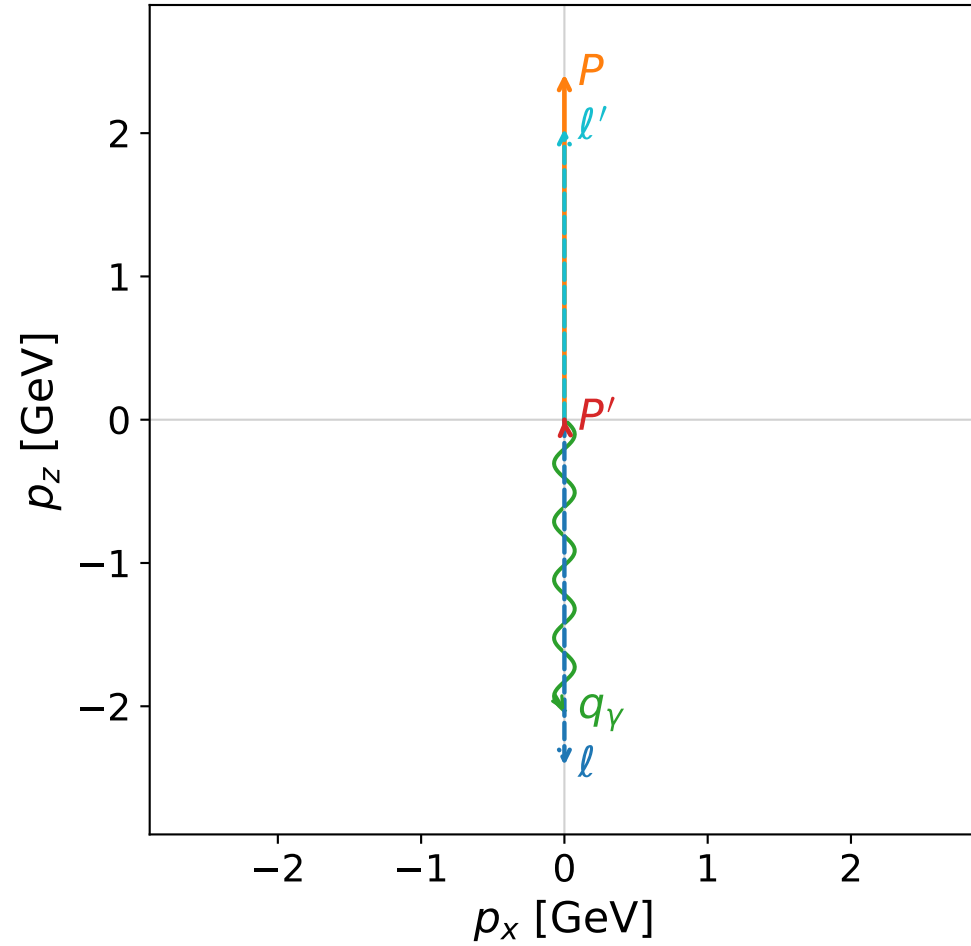


Leading normalized final-state amplitudes
(N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

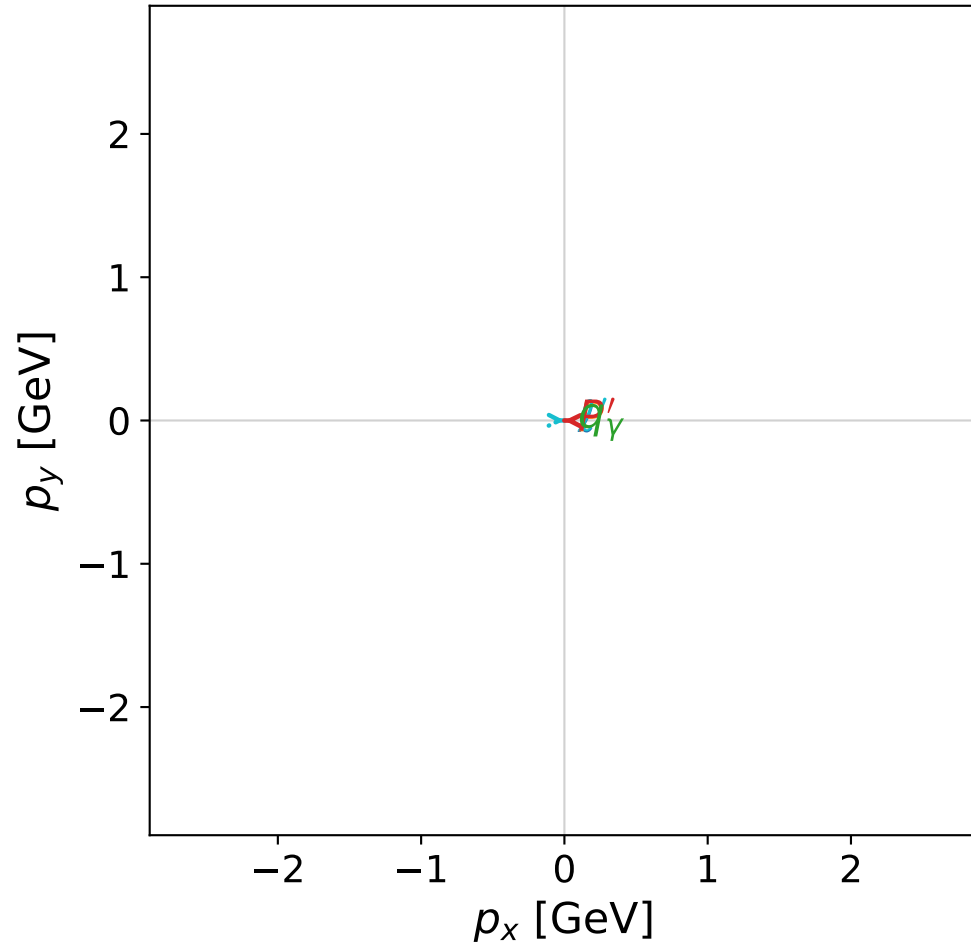


dghz_mixing_angles_polarization_03_local_minimum_156
 $d_{\text{GHZ}}=9.98211\text{e-}08$
 region: polarization_cluster_03_local_minimum_156
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0156
 lepton mixing angle: $\alpha_e=3.0075642$ rad
 incoming lepton: $-0.991032|+\rangle +0.133628|-\rangle$
 proton mixing angle: $\alpha_p=0.13401798$ rad
 incoming proton: $0.991033|+\rangle +0.133617|-\rangle$

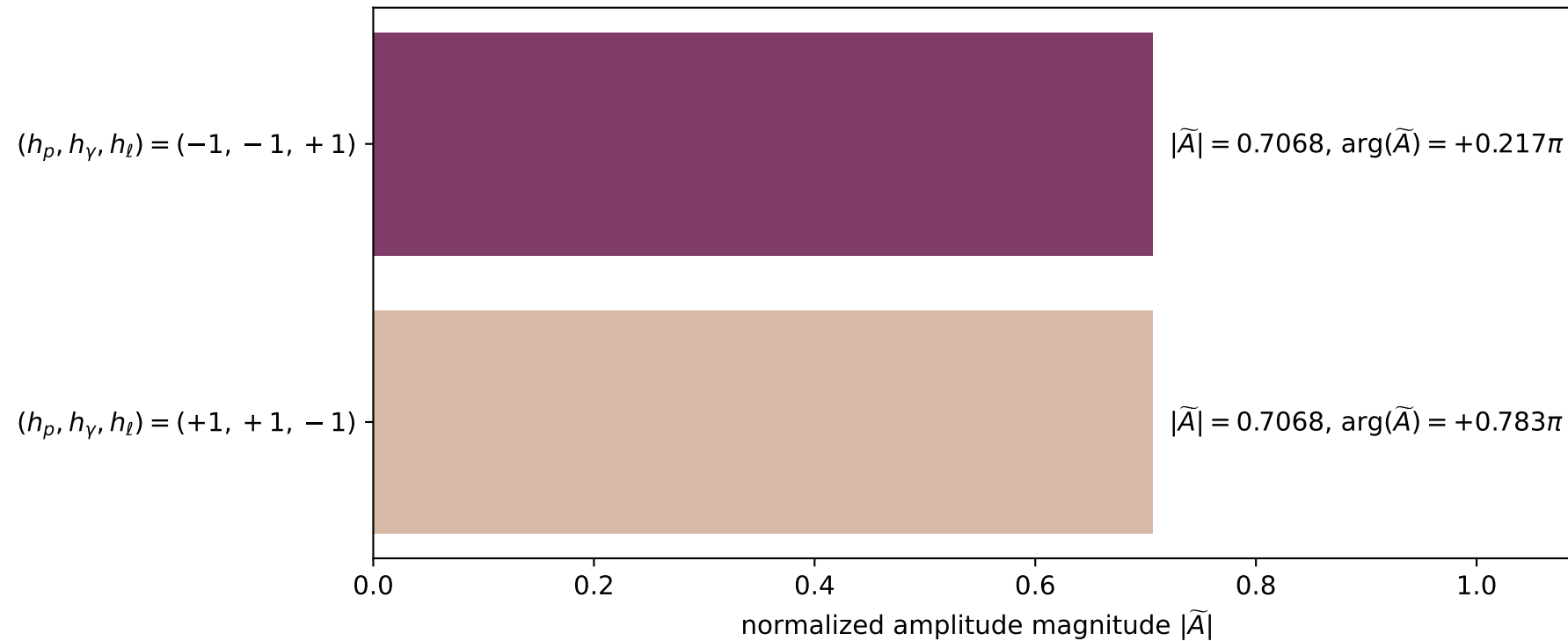
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.000632829$
 $\theta_{P'}=0.0587284$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.12895$, $\phi_\gamma=3.82447$
 $E_\gamma=2.03132$, $\theta_{\ell'}=1.82912\text{e-}05$, $\phi_{\ell'}=6.27054$
 $Q^2=19.5922$, $x_B=0.837076$, $t=-3.09232$, $W^2=4.69316$, $y=0.97037$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.0307$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 2.0307$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0006$
 q_γ $E= 2.0313$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.0313$

x--y plane

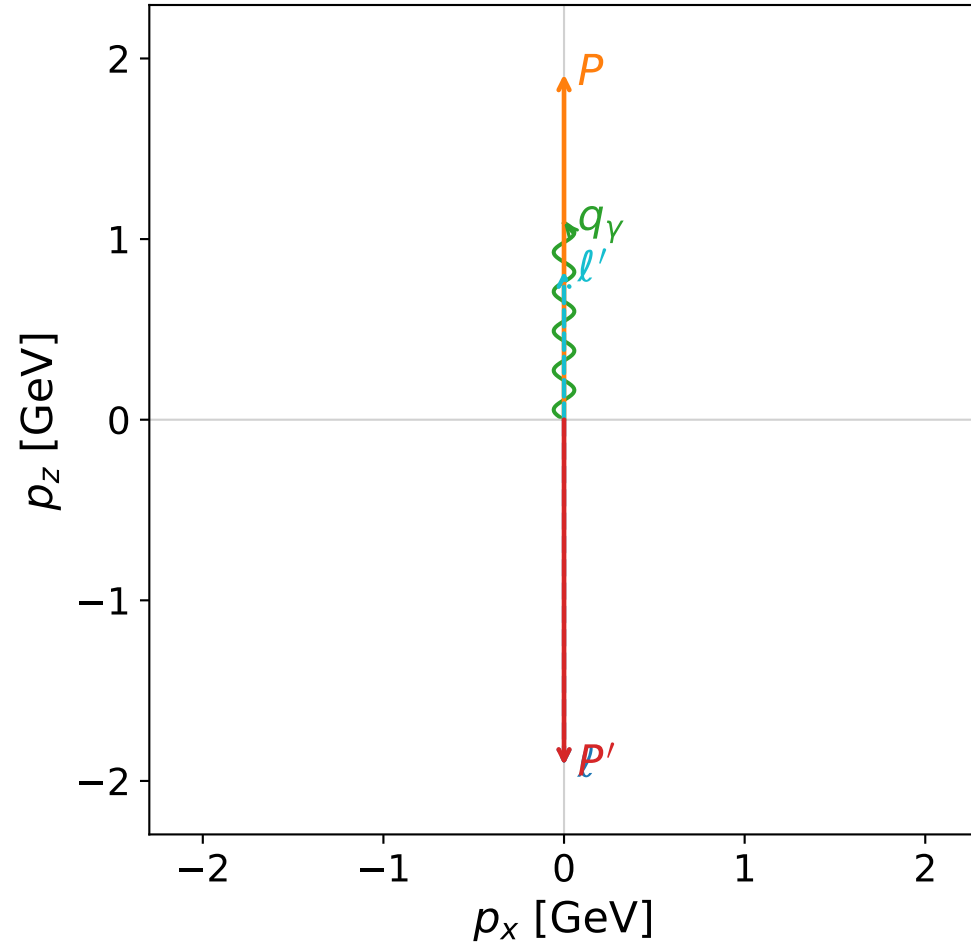


Leading normalized final-state amplitudes
(N=2, retained=99.9%)

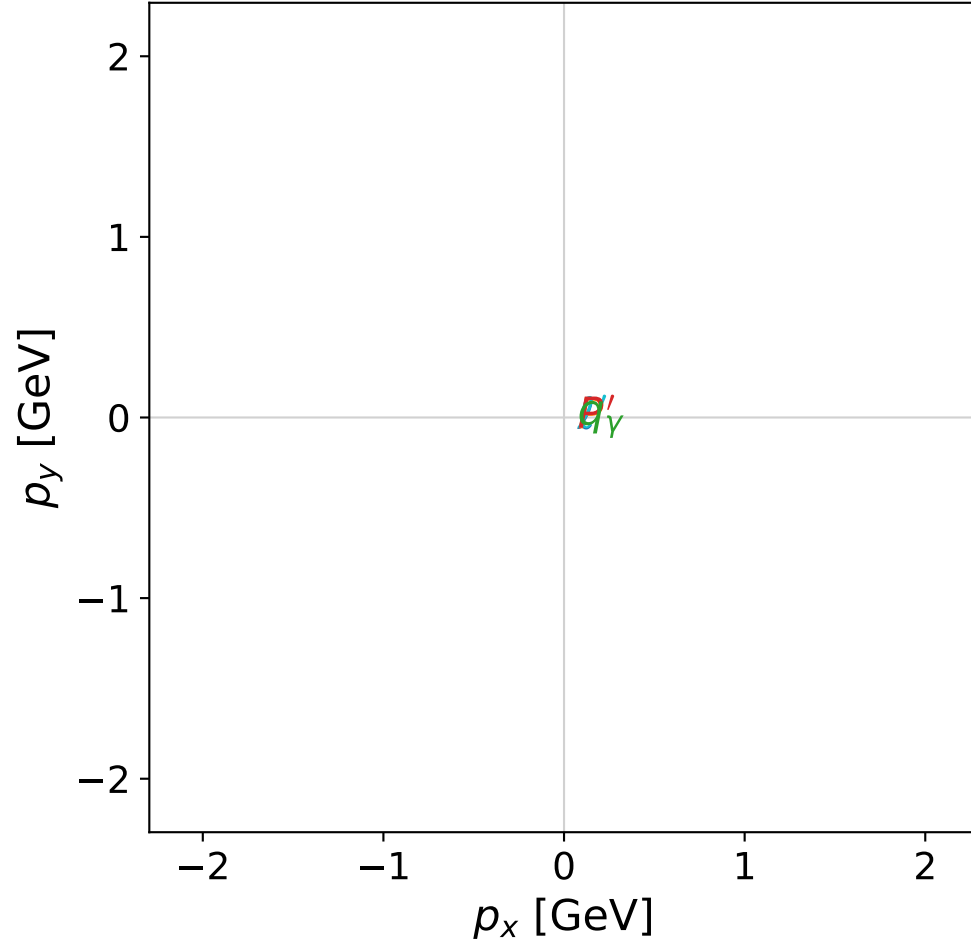


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



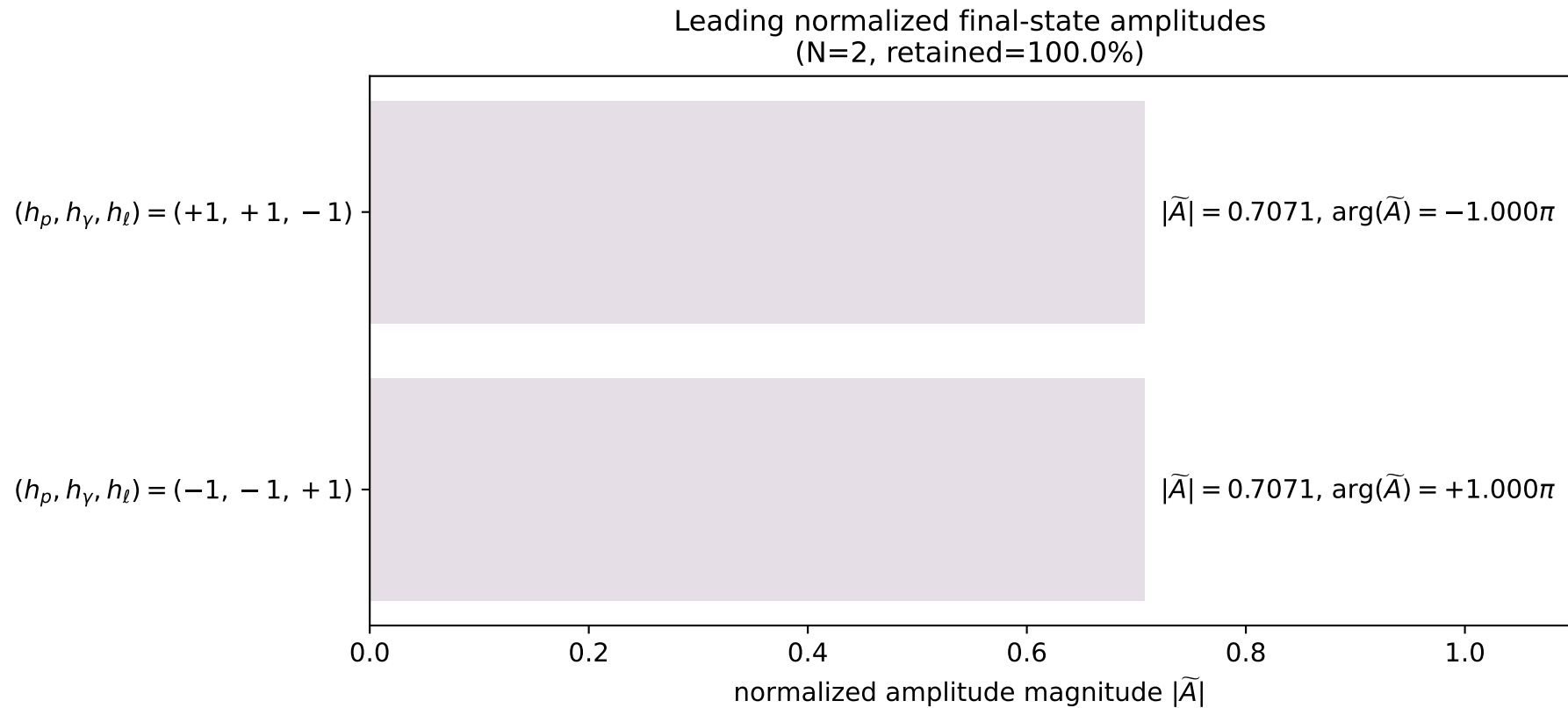
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_158
 $d_{\{\rm GHZ\}}=1.03904\text{e-}07$
 region: polarization_cluster_03_local_minimum_158
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0158
 lepton mixing angle: $\alpha_e=2.0331734$ rad
 incoming lepton: $-0.446077|+\rangle +0.894995|-\rangle$
 proton mixing angle: $\alpha_p=0.46236846$ rad
 incoming proton: $0.894999|+\rangle +0.446069|-\rangle$

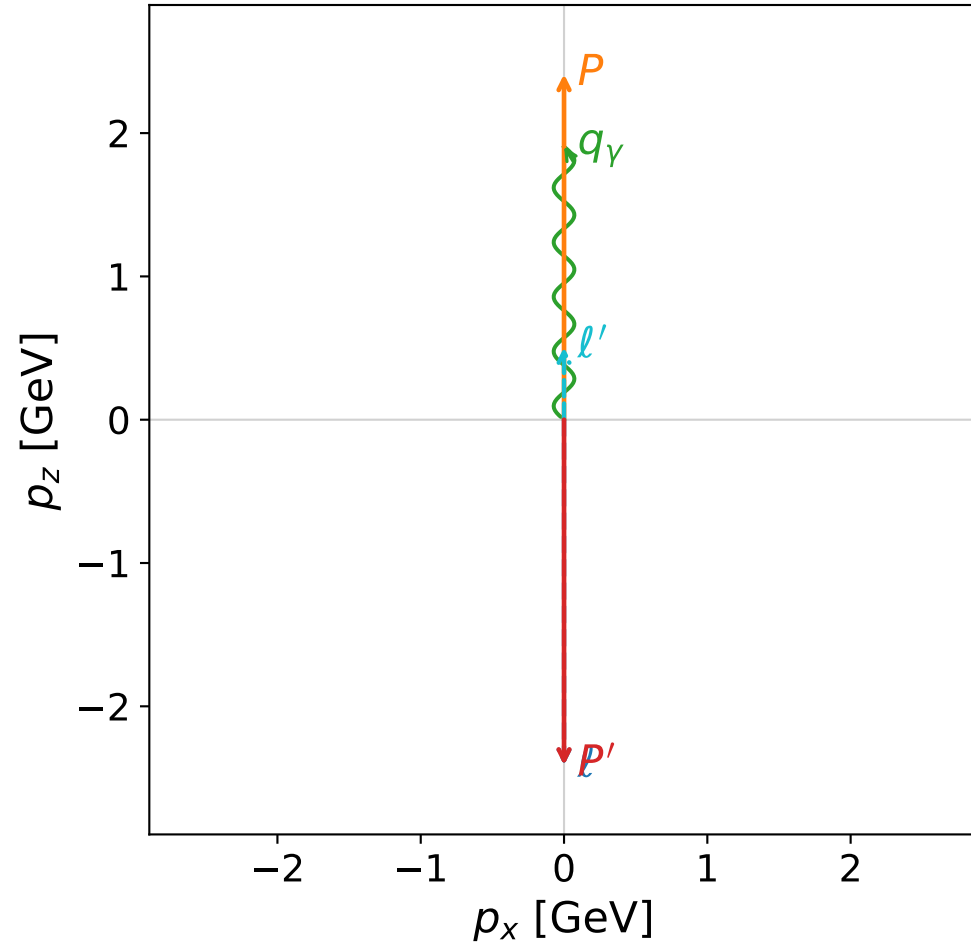
$s=16.3566$, $\sqrt{s}=4.04434$, $|\vec{P}|=1.91339$, $|\vec{P}'|=1.91339$
 $\theta_{P'}=3.14159$, $\theta_Y=0$, $\phi_{P'}=2.40943$, $\phi_Y=3.29633$
 $E_Y=1.09092$, $\theta_{\ell'}=0$, $\phi_{\ell'}=5.55102$
 $Q^2=6.29483$, $x_B=0.416353$, $t=-14.6443$, $W^2=9.70397$, $y=0.976878$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.9134$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 P $E= 2.1309$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.9134$
 ℓ' $E= 0.8225$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.8225$
 P' $E= 2.1309$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -1.9134$
 q_Y $E= 1.0909$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.0909$

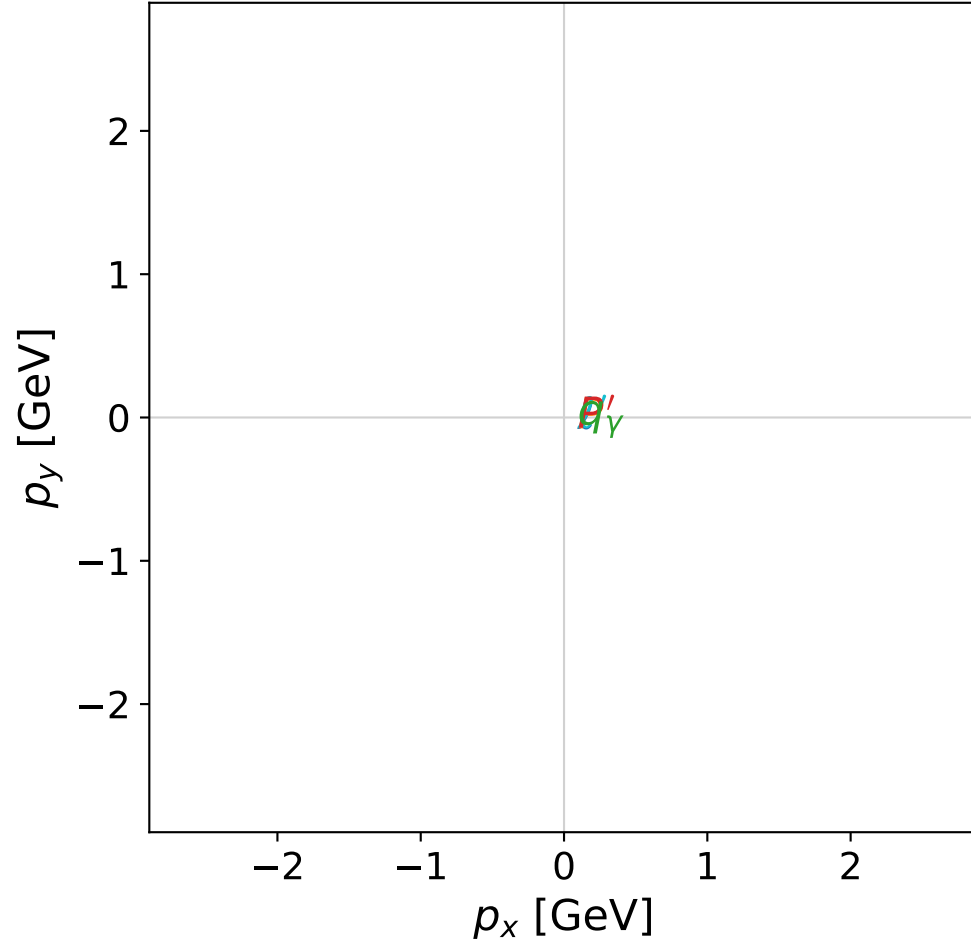


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



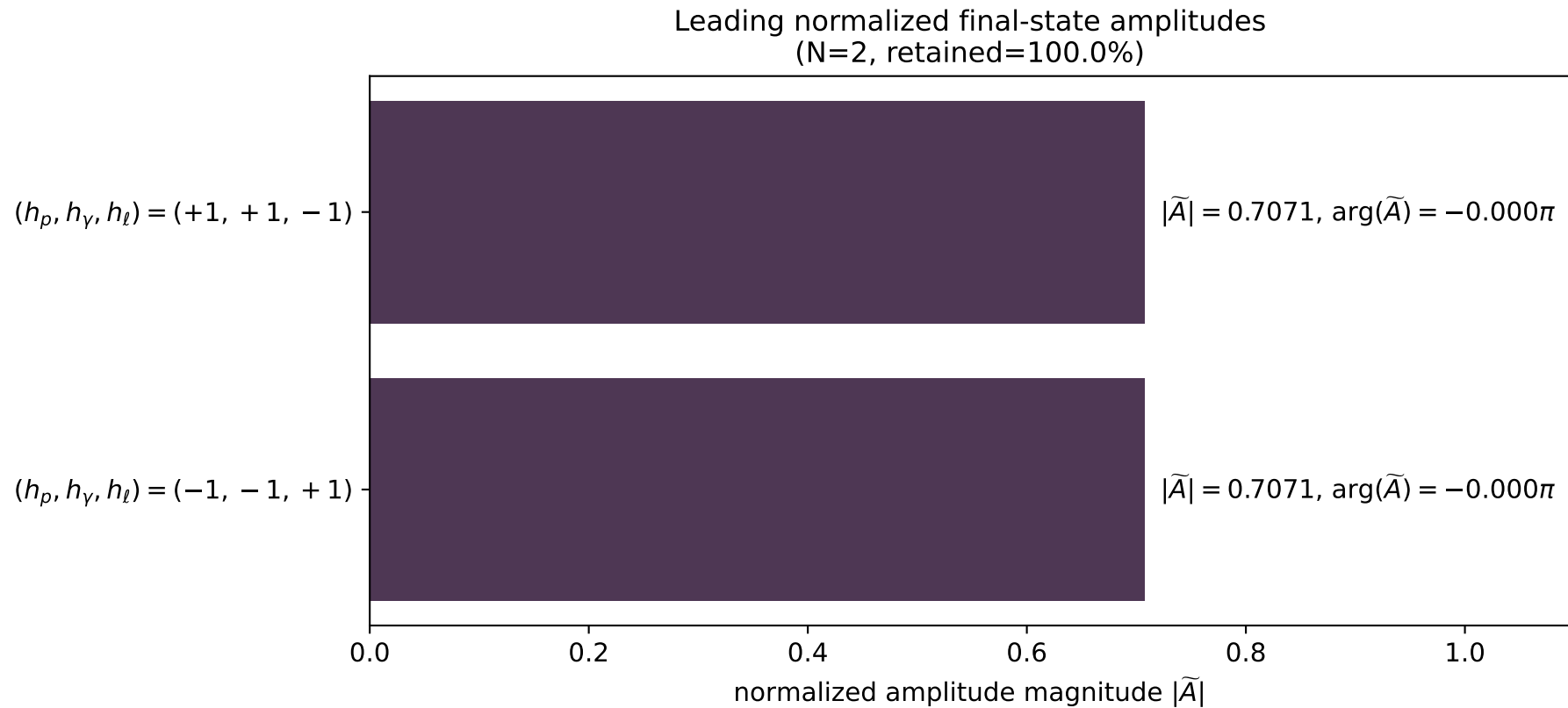
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_161
 $d_{\{\rm GHZ\}}=1.13122\text{e-}07$
 region: polarization_cluster_03_local_minimum_161
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0161
 lepton mixing angle: $\alpha_e=2.5224319$ rad
 incoming lepton: $-0.814366|+\rangle +0.580352|-\rangle$
 proton mixing angle: $\alpha_p=0.95162257$ rad
 incoming proton: $0.580362|+\rangle +0.814358|-\rangle$

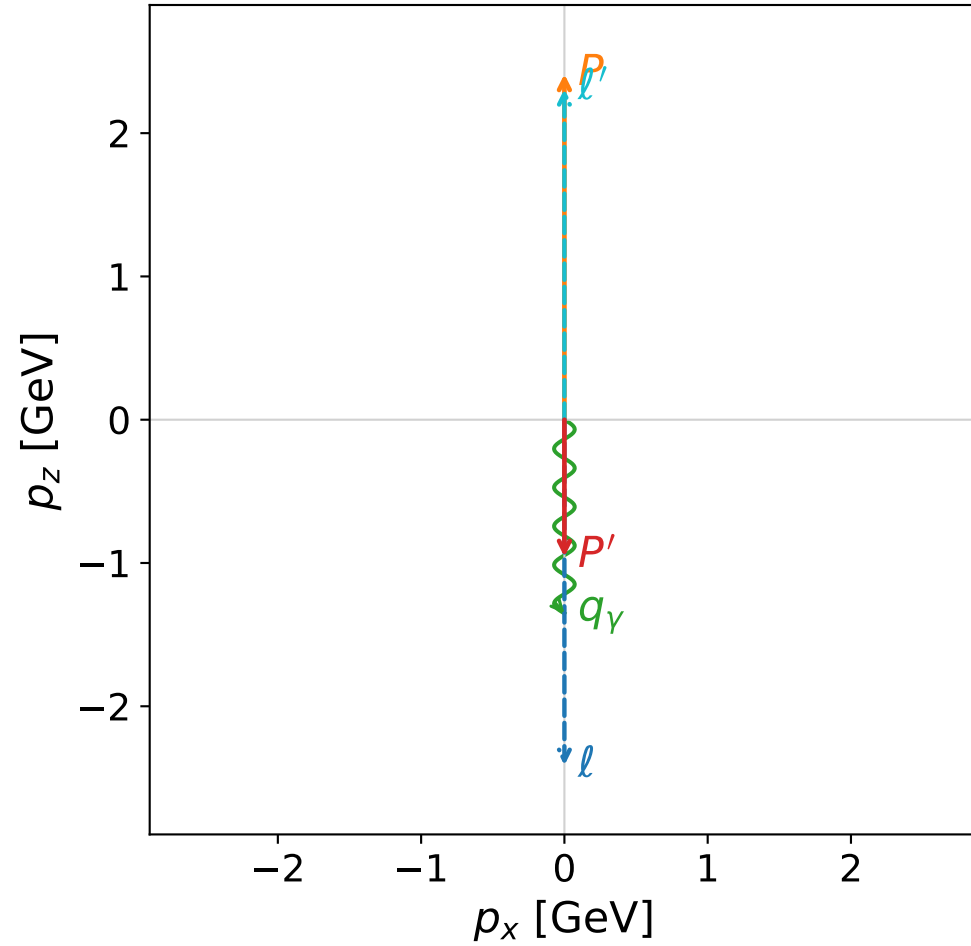
$s=24.9928$, $\sqrt{s}=4.99928$, $|\vec{P}|=2.41164$, $|\vec{P}'|=2.41164$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=5.60969$, $\phi_\gamma=5.70666$
 $E_\gamma=1.90533$, $\theta_{l'}=0$, $\phi_{l'}=2.4681$
 $Q^2=4.88418$, $x_B=0.204062$, $t=-23.2641$, $W^2=19.9304$, $y=0.992609$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4116$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4116$
 P $E= 2.5876$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4116$
 l' $E= 0.5063$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.5063$
 P' $E= 2.5876$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -2.4116$
 q_γ $E= 1.9053$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 1.9053$



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

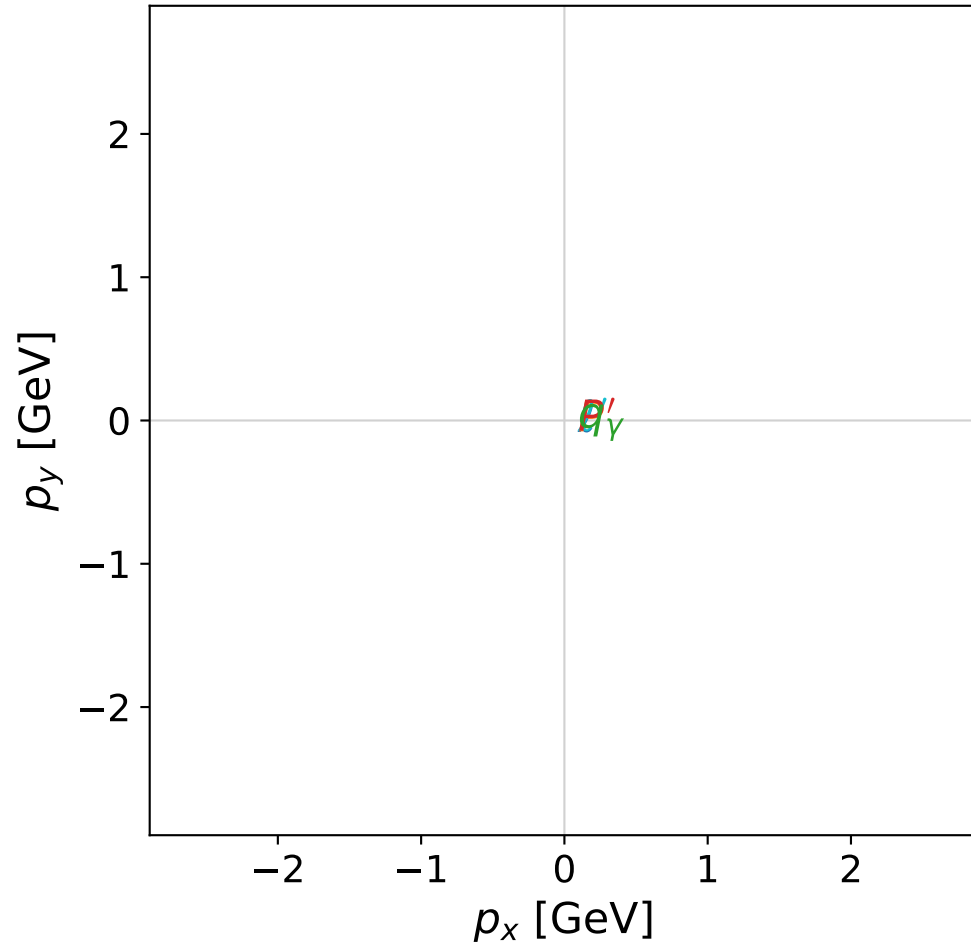


dghz_mixing_angles_polarization_03_local_minimum_174
 $d_{\text{GHZ}}=2.67083\text{e-}07$
 region: polarization_cluster_03_local_minimum_174
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0174
 lepton mixing angle: $\alpha_e=3.0976361$ rad
 incoming lepton: $-0.999034|+\rangle +0.0439424|-\rangle$
 proton mixing angle: $\alpha_p=0.043954198$ rad
 incoming proton: $0.999034|+\rangle +0.04394|-\rangle$

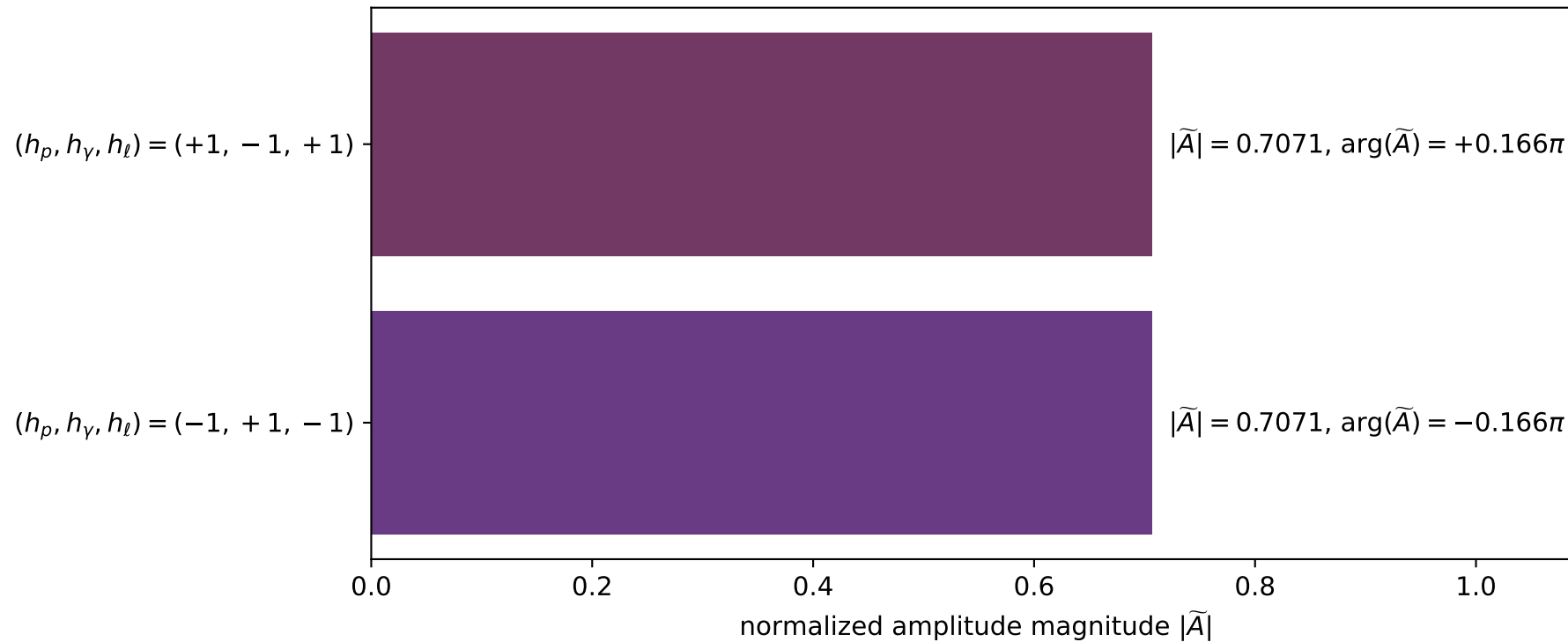
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=0.956774$
 $\theta_{P'}=3.14159$, $\theta_\gamma=3.14159$, $\phi_{P'}=3.14456$, $\phi_\gamma=3.66323$
 $E_\gamma=1.35168$, $\theta_{\ell'}=0$, $\phi_{\ell'}=0.307656$
 $Q^2=22.2721$, $x_B=0.955566$, $t=-9.79096$, $W^2=1.9155$, $y=0.966317$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 ℓ' $E= 2.3085$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3085$
 P' $E= 1.3399$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.9568$
 q_γ $E= 1.3517$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -1.3517$

x--y plane

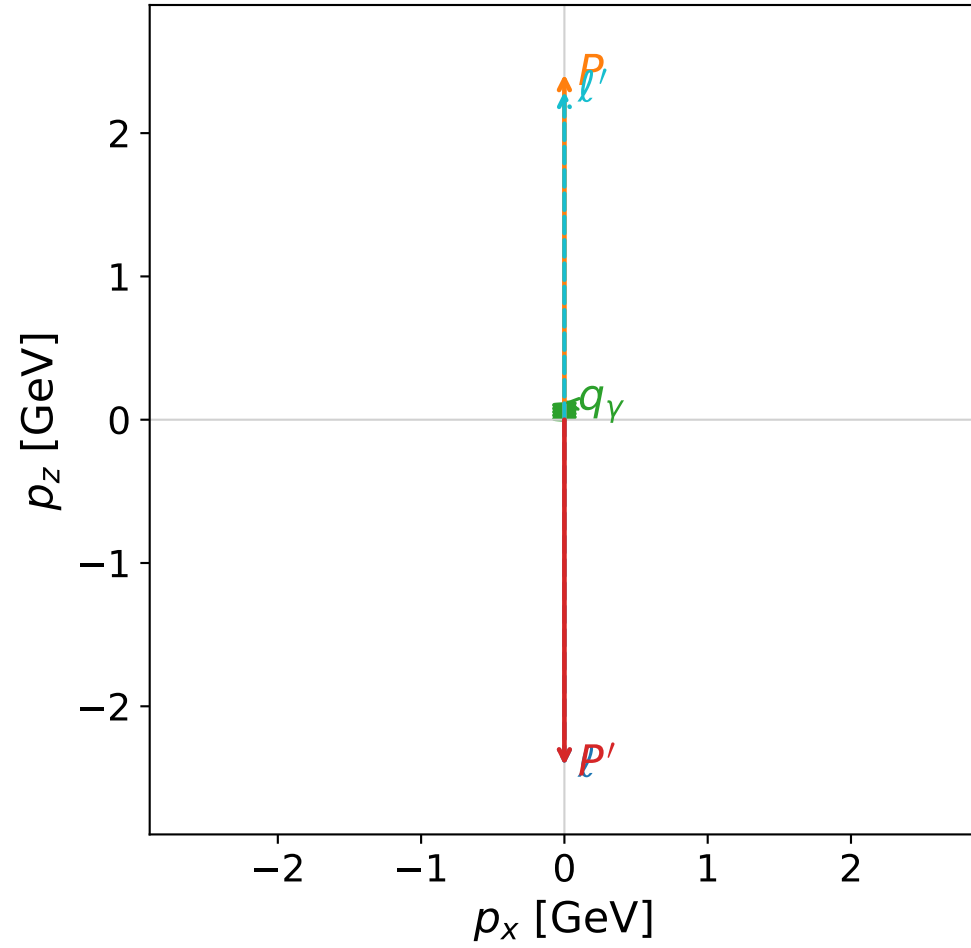


Leading normalized final-state amplitudes
 (N=2, retained=100.0%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

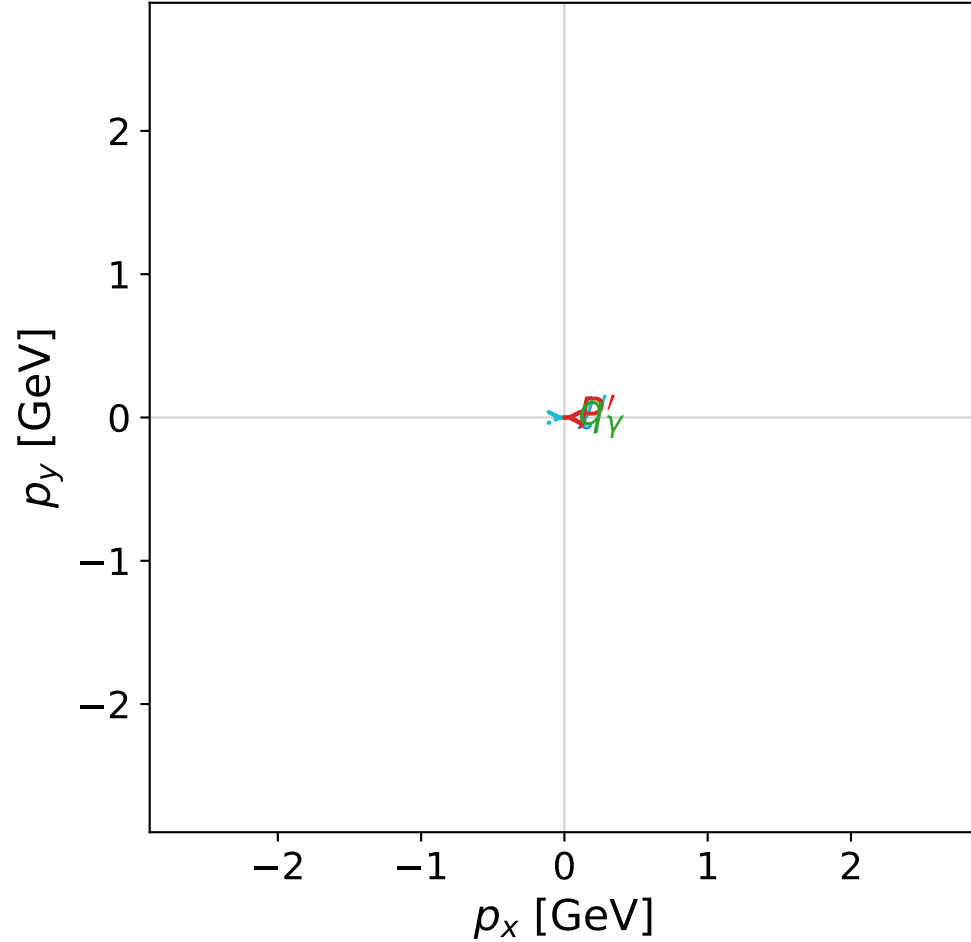


dghz_mixing_angles_polarization_03_local_minimum_178
 $d_{\text{GHZ}}=4.23054\text{e-}07$
 region: polarization_cluster_03_local_minimum_178
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0178
 lepton mixing angle: $\alpha_e=3.1136594$ rad
 incoming lepton: $-0.99961|+\rangle + 0.0279296|-\rangle$
 proton mixing angle: $\alpha_p=1.5428613$ rad
 incoming proton: $0.0279314|+\rangle + 0.99961|-\rangle$

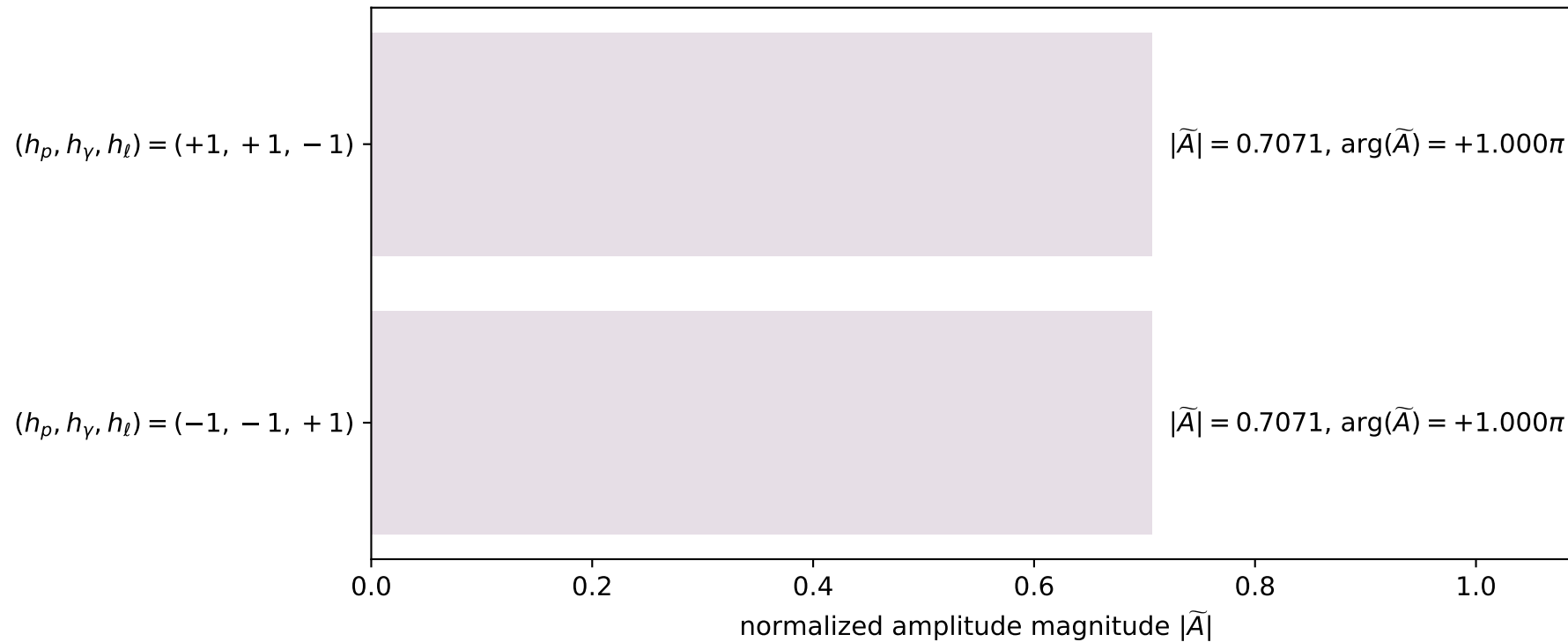
$s=24.9961$, $\sqrt{s}=4.99961$, $|\vec{P}|=2.41181$, $|\vec{P}'|=2.41181$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=3.46255$, $\phi_\gamma=3.37801$
 $E_\gamma=0.120586$, $\theta_{l'}=0$, $\phi_{l'}=0.320954$
 $Q^2=22.104$, $x_B=0.948272$, $t=-23.2674$, $W^2=2.08561$, $y=0.966561$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.4118$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4118$
 P $E= 2.5878$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4118$
 ℓ' $E= 2.2912$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.2912$
 P' $E= 2.5878$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.4118$
 q_γ $E= 0.1206$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.1206$

x--y plane

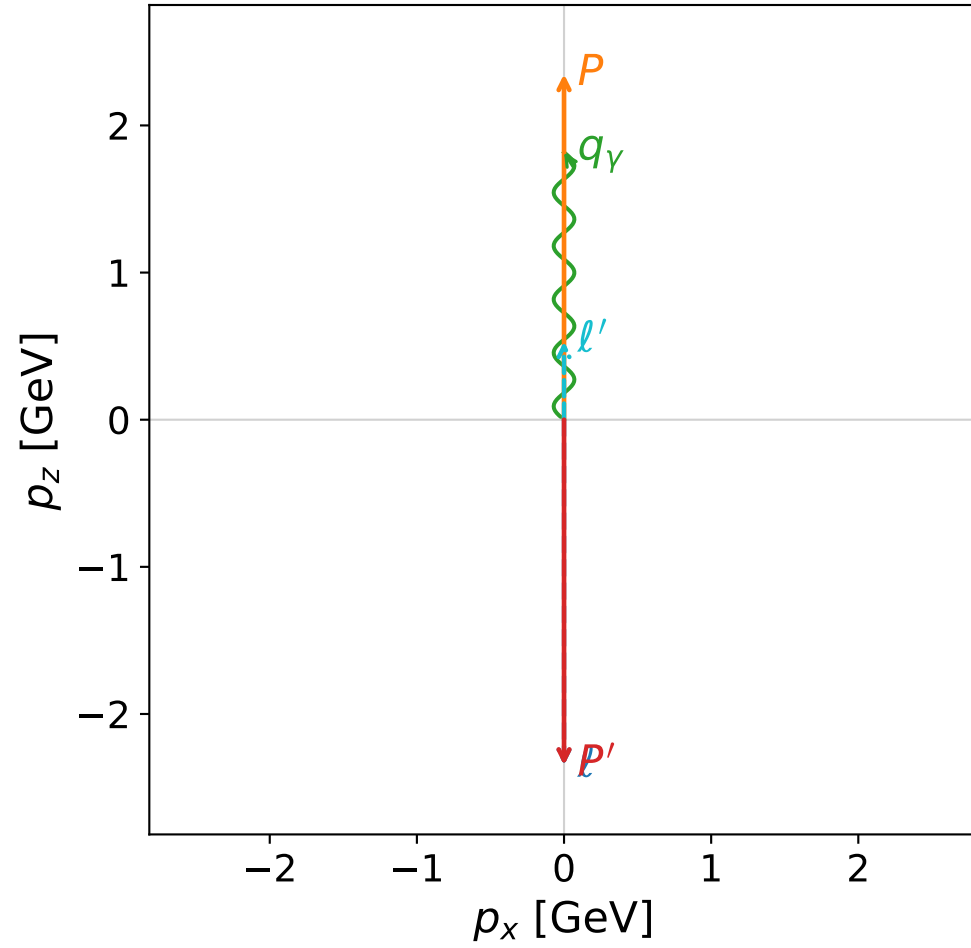


Leading normalized final-state amplitudes
(N=2, retained=100.0%)

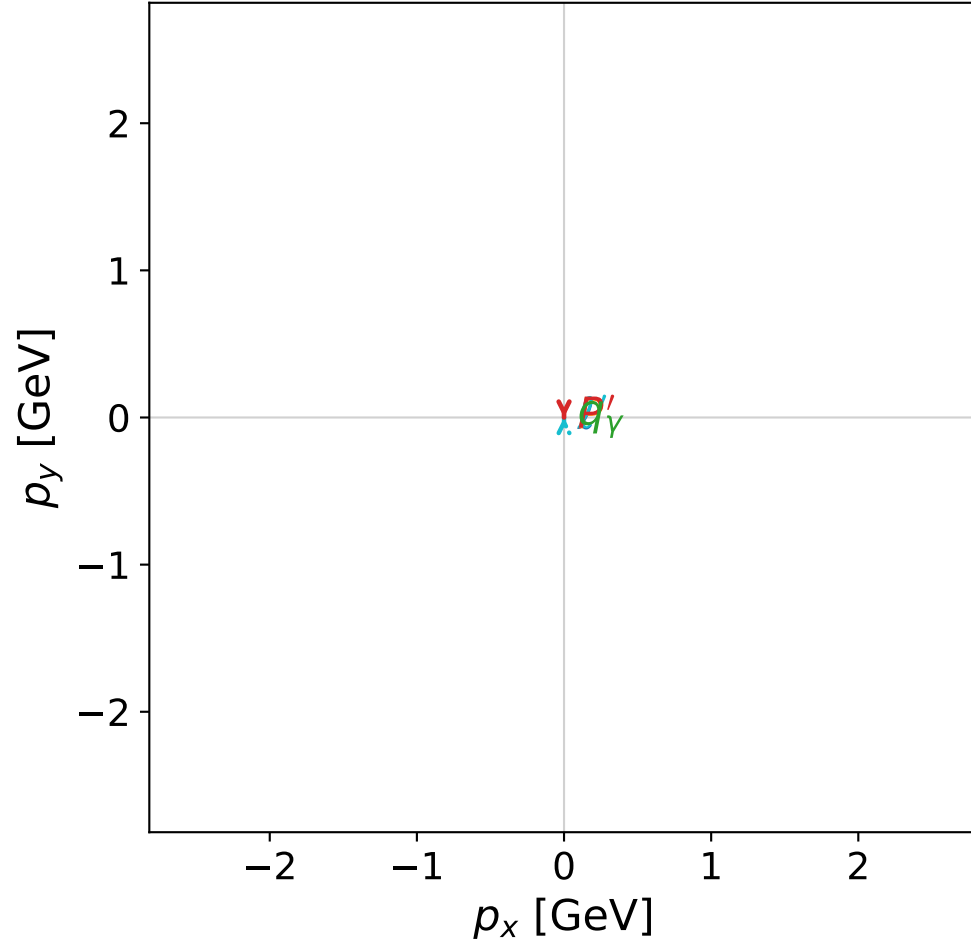


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

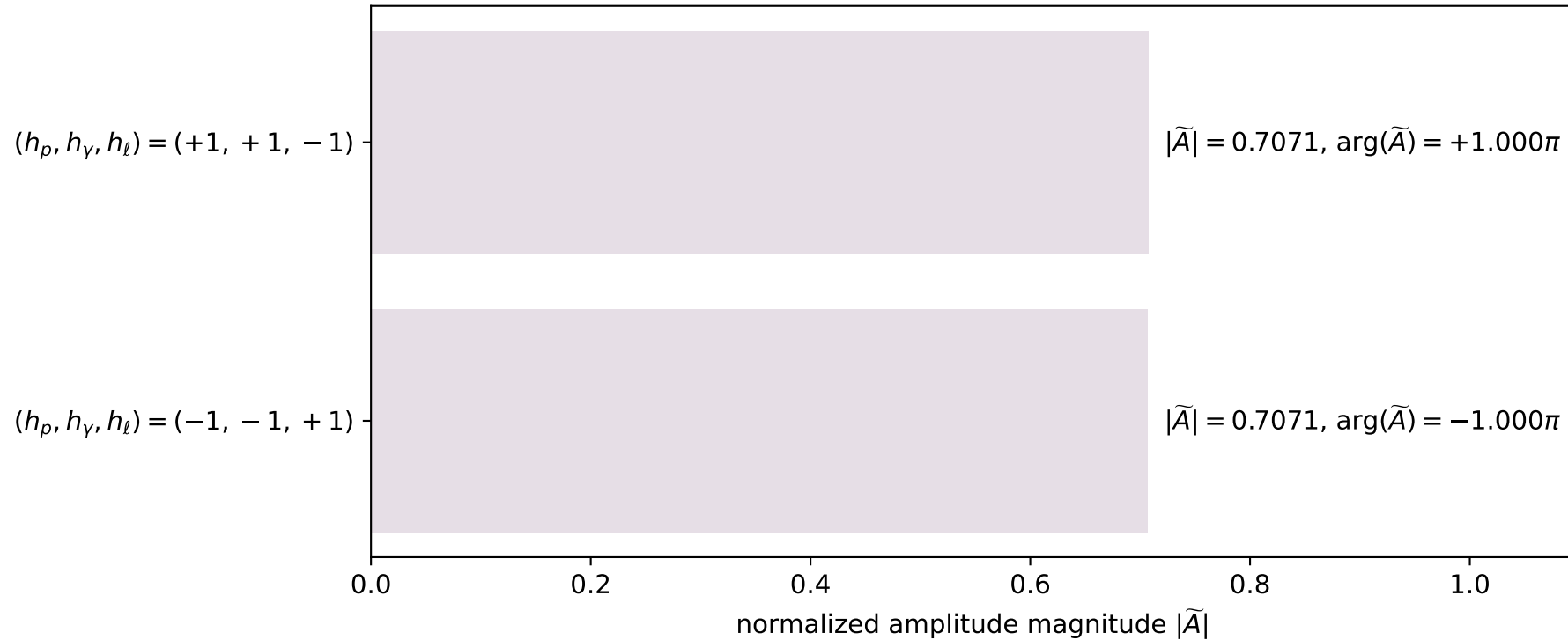


dghz_mixing_angles_polarization_03_local_minimum_181
 $d_{\{\rm GHZ\}}=5.4582\text{e-}07$
 region: polarization_cluster_03_local_minimum_181
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0181
 lepton mixing angle: $\alpha_e=3.0452158$ rad
 incoming lepton: $-0.995359|+\rangle +0.0962277|-\rangle$
 proton mixing angle: $\alpha_p=1.4744086$ rad
 incoming proton: $0.0962386|+\rangle +0.995358|-\rangle$

$s=23.7939$, $\sqrt{s}=4.8779$, $|\vec{P}|=2.34876$, $|\vec{P}'|=2.34876$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0$, $\phi_{p'}=4.56992$, $\phi_\gamma=2.75915$
 $E_\gamma=1.81698$, $\theta_{l'}=0$, $\phi_{l'}=1.42833$
 $Q^2=4.99613$, $x_B=0.219879$, $t=-22.0667$, $W^2=18.6059$, $y=0.991628$

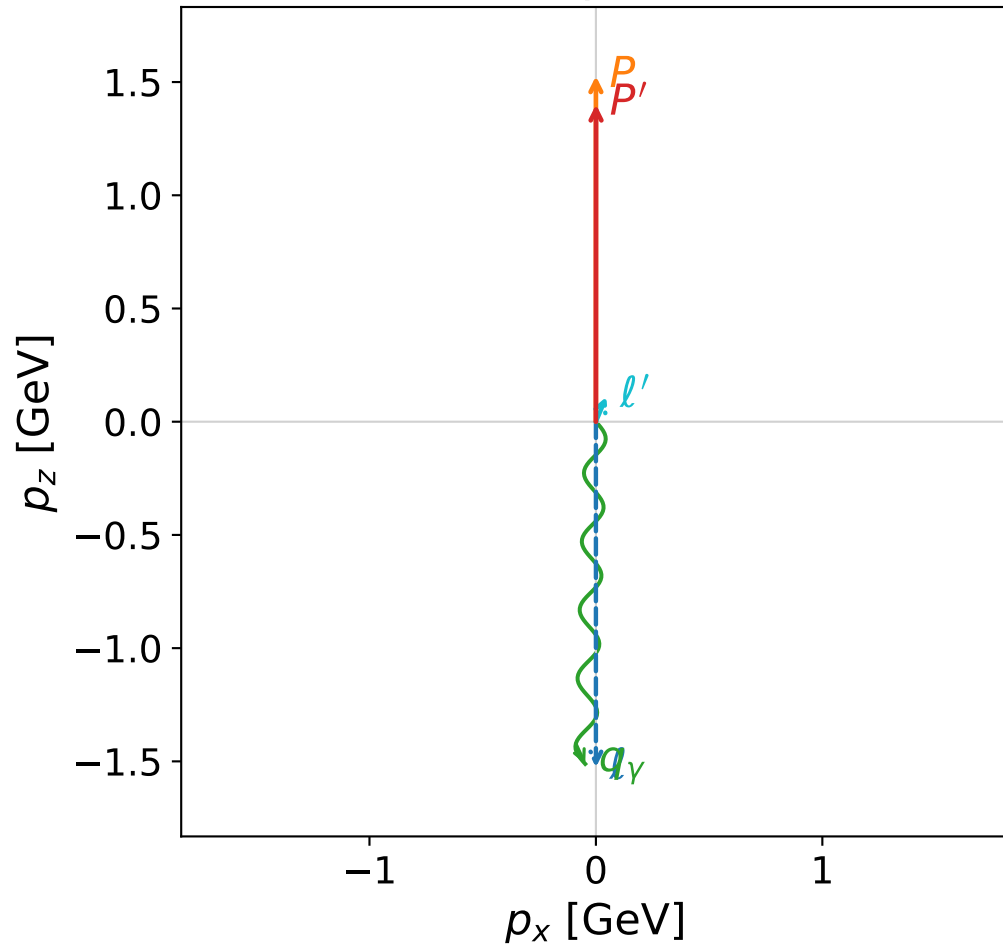
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3488$
 P $E= 2.5291$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3488$
 l' $E= 0.5318$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.5318$
 P' $E= 2.5291$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -2.3488$
 q_γ $E= 1.8170$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.8170$

Leading normalized final-state amplitudes (N=2, retained=100.0%)

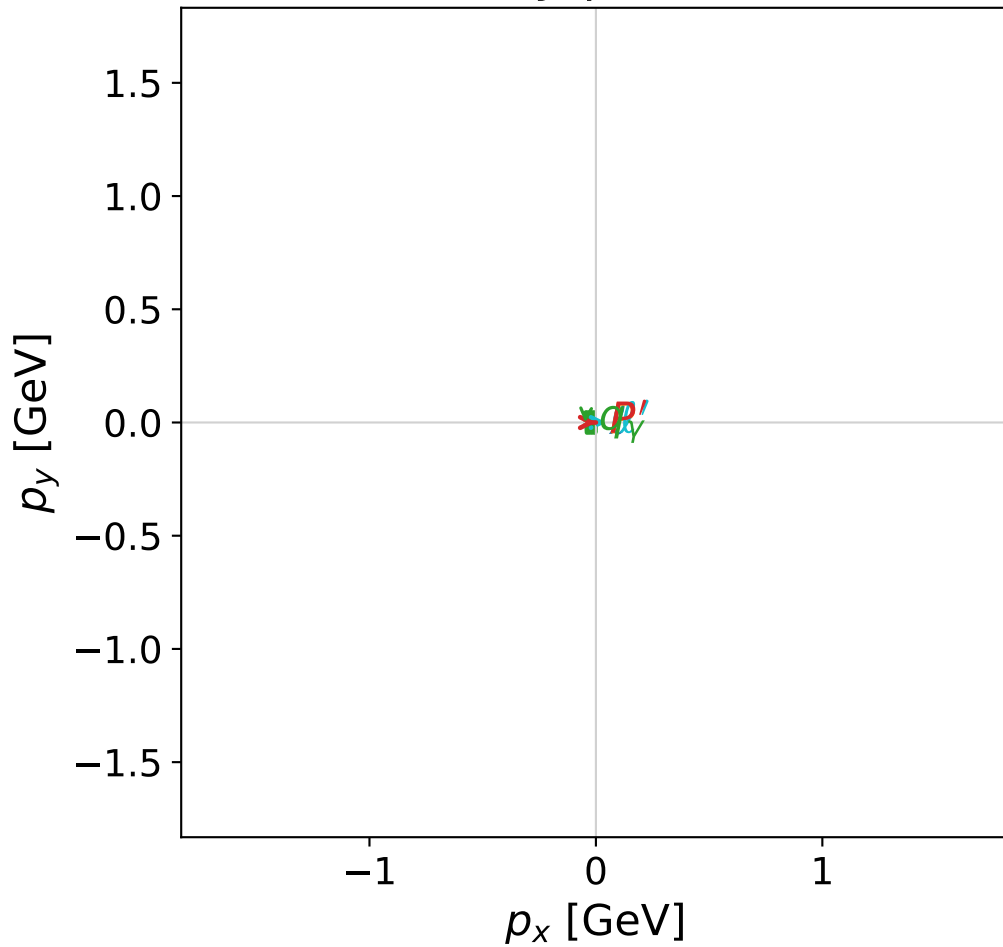


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

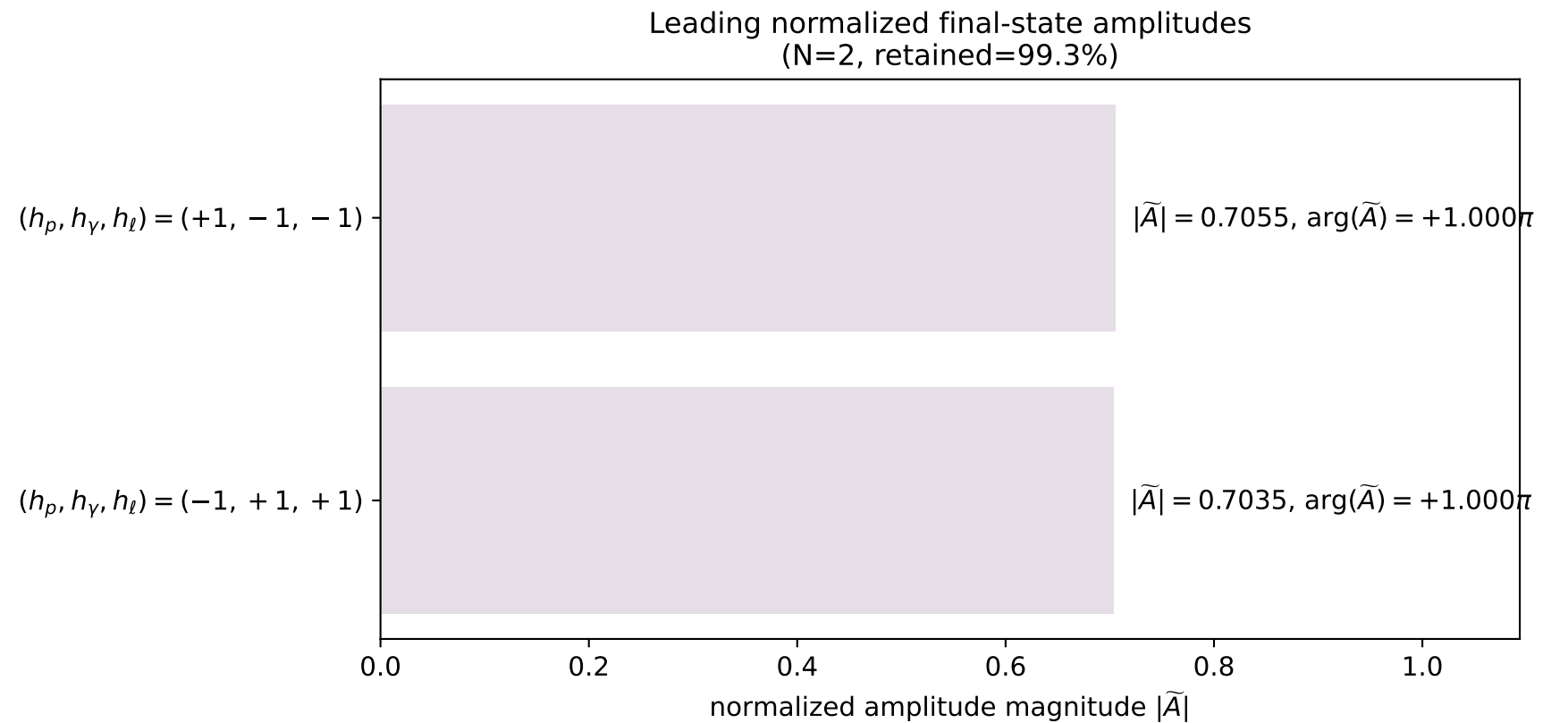


dghz_mixing_angles_polarization_03_local_minimum_188
 $d_{\{\text{rm GHZ}\}}=1.11719\text{e-}05$
 region: polarization_cluster_03_local_minimum_188
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0188
 lepton mixing angle: $\alpha_e=2.3764974$ rad
 incoming lepton: $-0.721316|+\rangle +0.692606|-\rangle$
 proton mixing angle: $\alpha_p=0.84717575$ rad
 incoming proton: $0.662102|+\rangle +0.749413|-\rangle$

$s=11.0066$, $\sqrt{s}=3.31762$, $|\vec{P}|=1.52621$, $|\vec{P}'|=1.40088$
 $\theta_{p'}=0.000417261$, $\theta_\gamma=3.11038$, $\phi_{p'}=6.35544\text{e-}05$, $\phi_\gamma=3.14165$
 $E_\gamma=1.51194$, $\theta_{\ell'}=0.399604$, $\phi_{\ell'}=6.18232\text{e-}05$
 $Q^2=0.702332$, $x_B=0.0699923$, $t=-0.00457805$, $W^2=10.2119$, $y=0.990883$

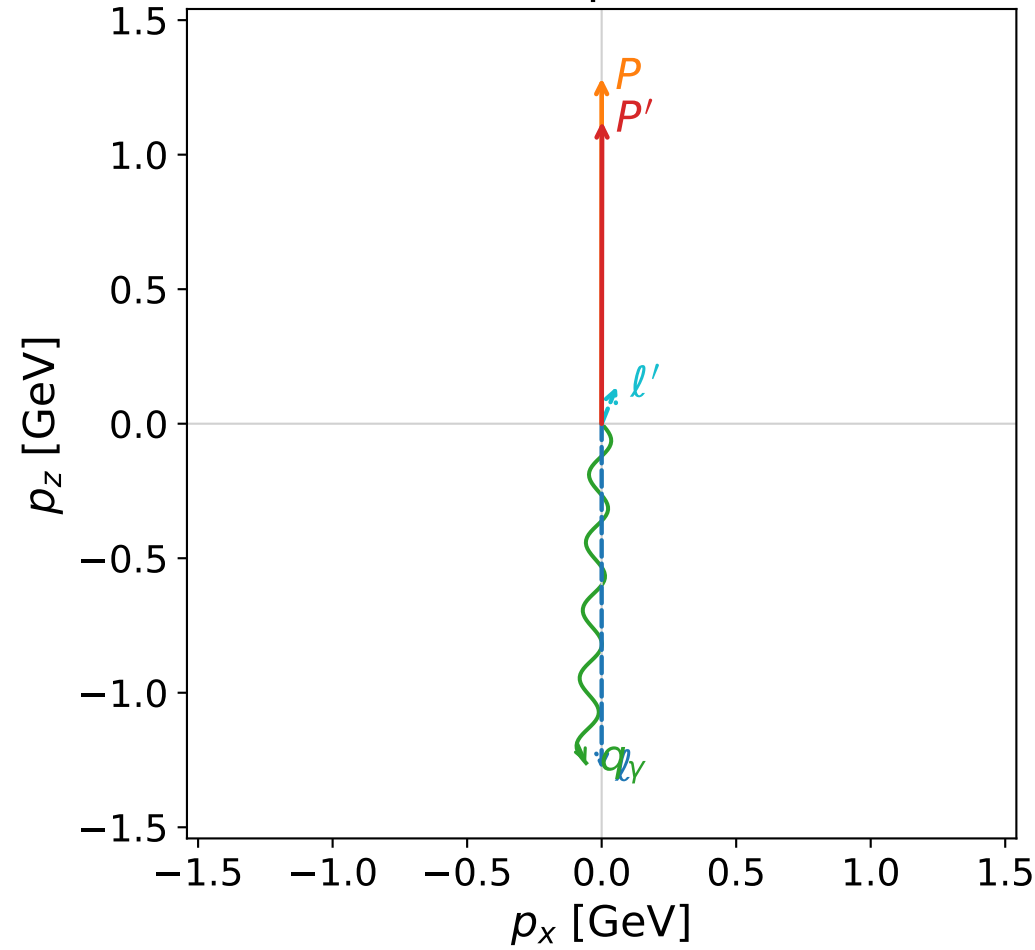
four-momenta $[E, p_x, p_y, p_z]$ GeV:

Particle	E	p_x	p_y	p_z
ℓ	1.5262	0.0000	0.0000	-1.5262
P	1.7914	0.0000	0.0000	1.5262
ℓ'	0.1198	0.0466	0.0000	0.1103
P'	1.6859	0.0006	0.0000	1.4009
q_γ	1.5119	-0.0472	-0.0000	-1.5112



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

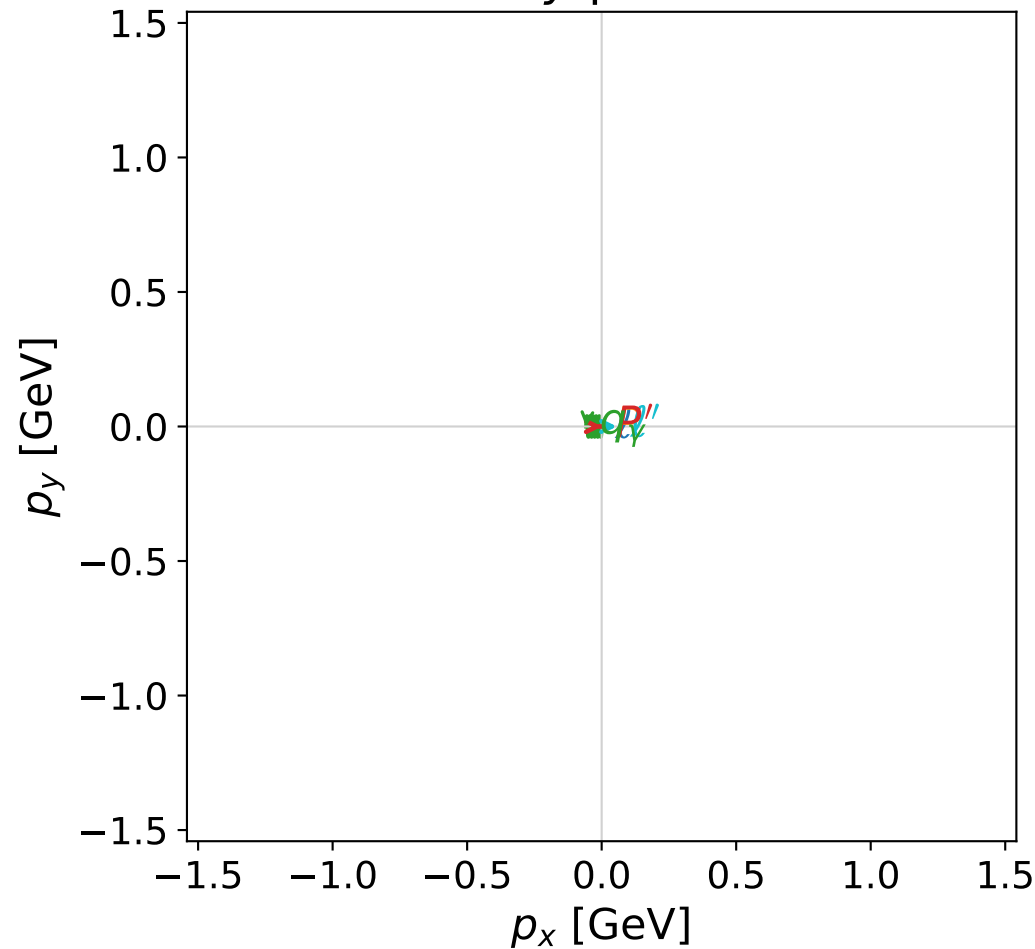


dghz_mixing_angles_polarization_03_local_minimum_189
 $d_{\text{GHZ}}=1.25241\text{e-}05$
 region: polarization_cluster_03_local_minimum_189
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0189
 lepton mixing angle: $\alpha_e=2.4349155$ rad
 incoming lepton: $-0.760524|+\rangle +0.64931|-\rangle$
 proton mixing angle: $\alpha_p=0.90852471$ rad
 incoming proton: $0.61491|+\rangle +0.788597|-\rangle$

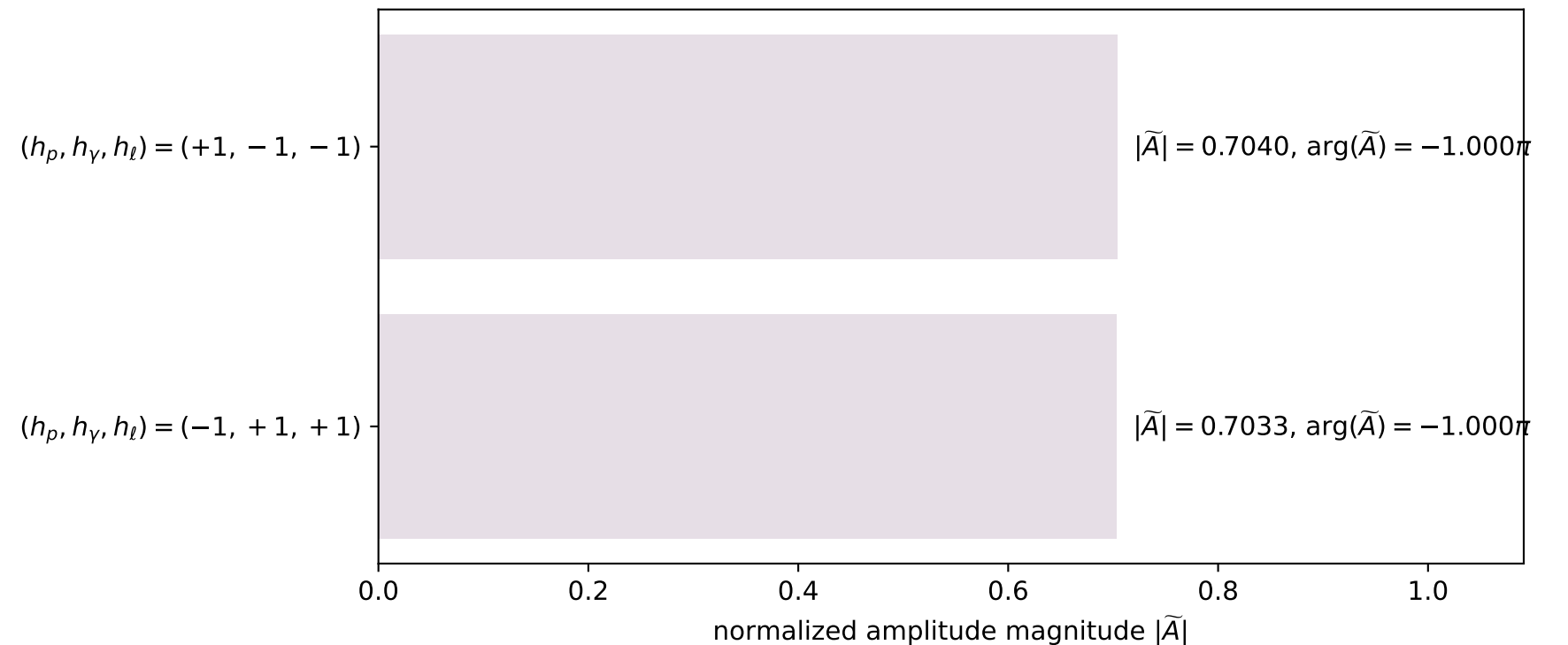
$s=8.26616$, $\sqrt{s}=2.87509$, $|\vec{P}|=1.28454$, $|\vec{P}'|=1.12322$
 $\theta_{p'}=0.00126895$, $\theta_\gamma=3.09582$, $\phi_{p'}=6.28313$, $\phi_\gamma=3.14157$
 $E_\gamma=1.2626$, $\theta_{\ell'}=0.38751$, $\phi_{\ell'}=6.28316$
 $Q^2=0.737755$, $x_B=0.101526$, $t=-0.00984887$, $W^2=7.40874$, $y=0.983798$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2845$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2845$
 P $E= 1.5906$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2845$
 ℓ' $E= 0.1491$ $p_x= 0.0563$ $p_y= -0.0000$ $p_z= 0.1381$
 P' $E= 1.4634$ $p_x= 0.0014$ $p_y= -0.0000$ $p_z= 1.1232$
 q_γ $E= 1.2626$ $p_x= -0.0578$ $p_y= 0.0000$ $p_z= -1.2613$

x--y plane

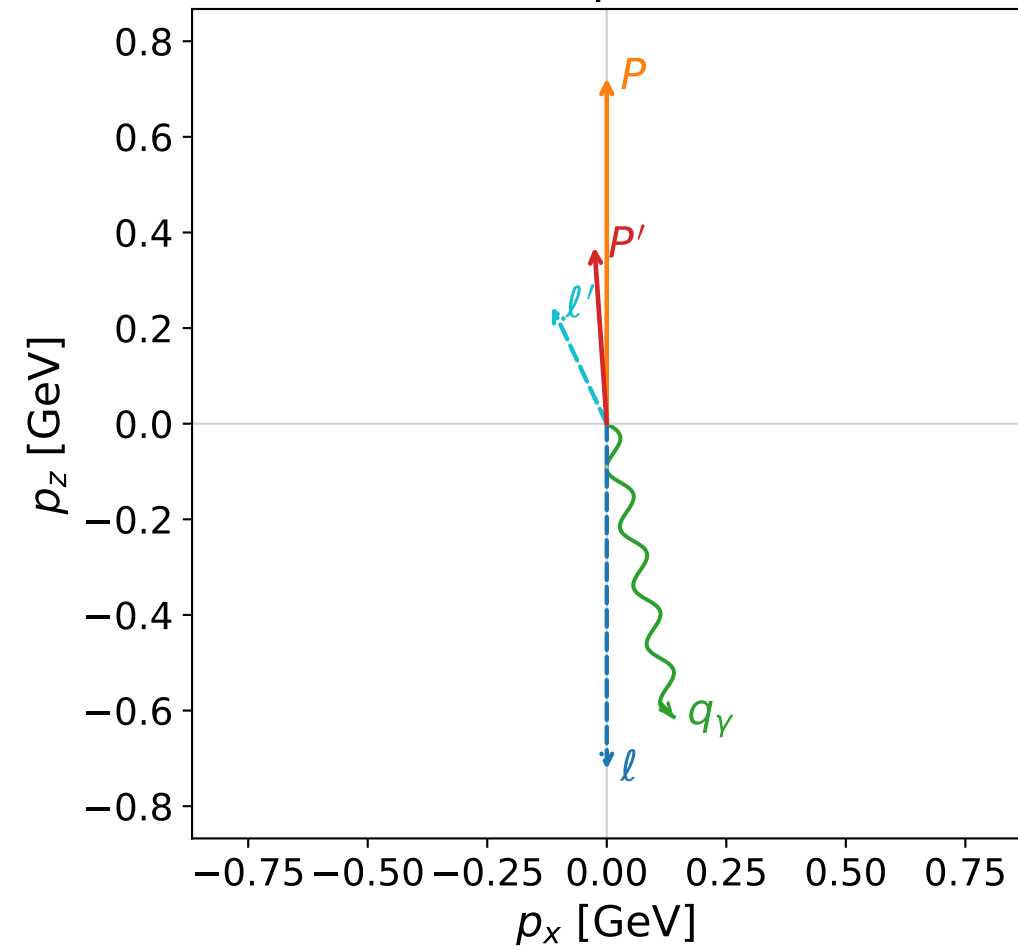


Leading normalized final-state amplitudes
(N=2, retained=99.0%)

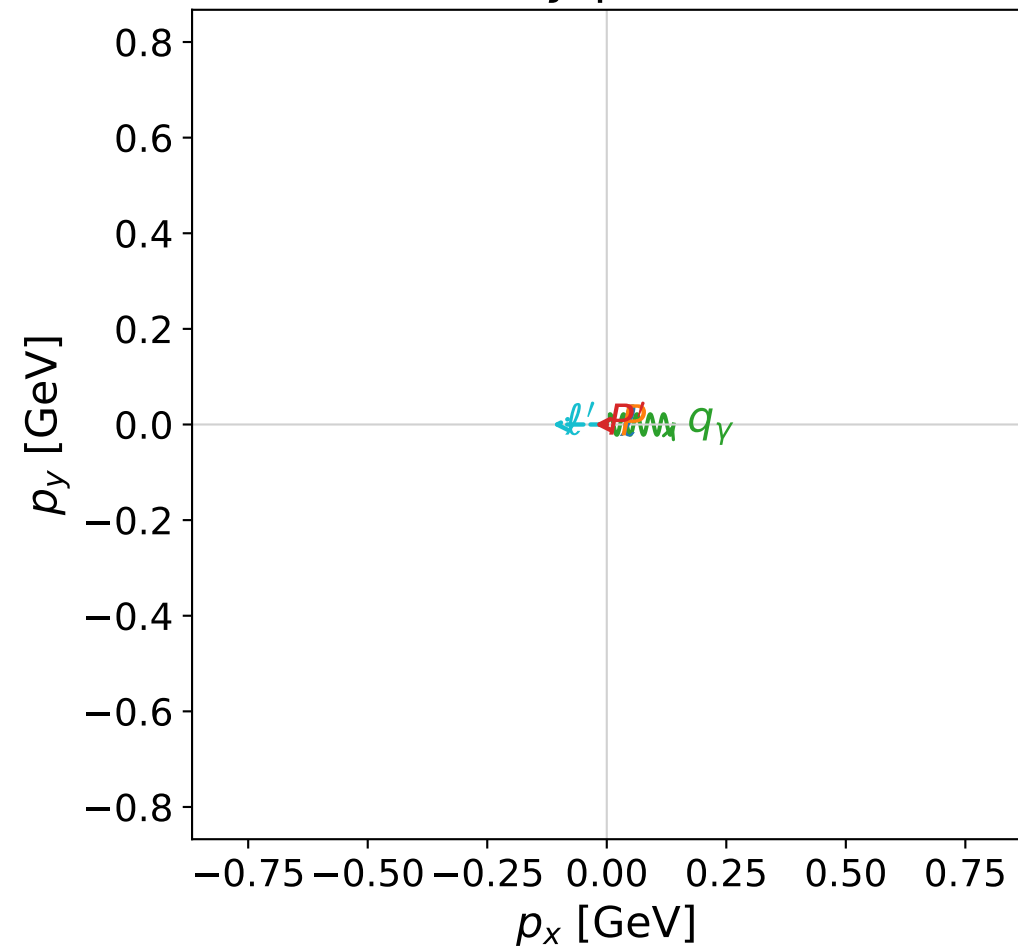


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_204

$d_{\{\text{rm GHZ}\}}=2.47799\text{e-}05$

region: polarization_cluster_03_local_minimum_204

outgoing-state purity: 1

kinematic point: gradient_local_minimum_0204

lepton mixing angle: $\alpha_e=2.0175873$ rad

incoming lepton: $-0.432074|+\rangle +0.901838|-\rangle$

proton mixing angle: $\alpha_p=0.38602922$ rad

incoming proton: $0.926411|+\rangle +0.376513|-\rangle$

$s=3.63688$, $\sqrt{s}=1.90706$, $|\vec{P}|=0.72285$, $|\vec{P}'|=0.370003$

$\theta_{P'}=0.0696716$, $\theta_\gamma=2.91723$, $\phi_{P'}=3.14156$, $\phi_\gamma=0$

$E_\gamma=0.629105$, $\theta_{\ell'}=0.437411$, $\phi_{\ell'}=3.1416$

$Q^2=0.742871$, $x_B=0.300568$, $t=-0.0948674$, $W^2=2.60853$, $y=0.896454$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.7228$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.7228$

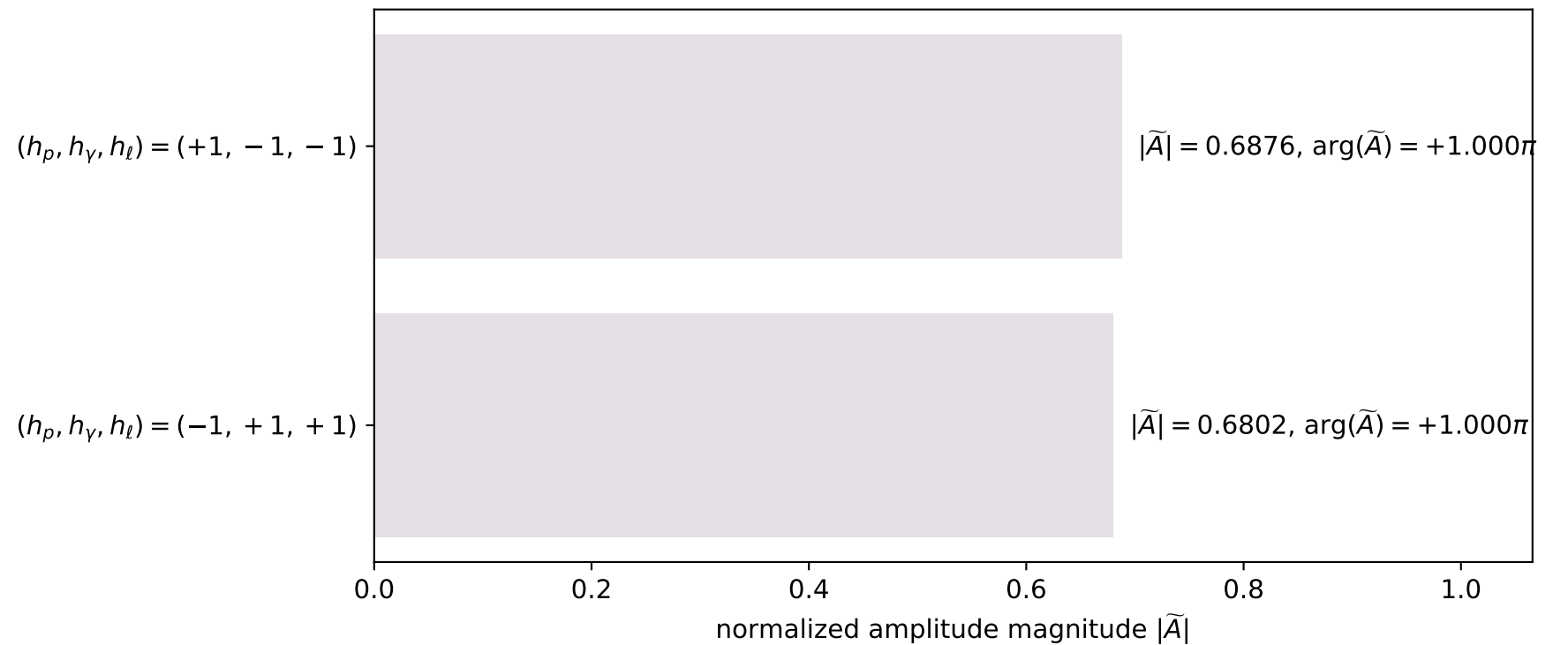
P $E= 1.1842$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.7228$

ℓ' $E= 0.2696$ $p_x= -0.1142$ $p_y= -0.0000$ $p_z= 0.2442$

P' $E= 1.0083$ $p_x= -0.0258$ $p_y= 0.0000$ $p_z= 0.3691$

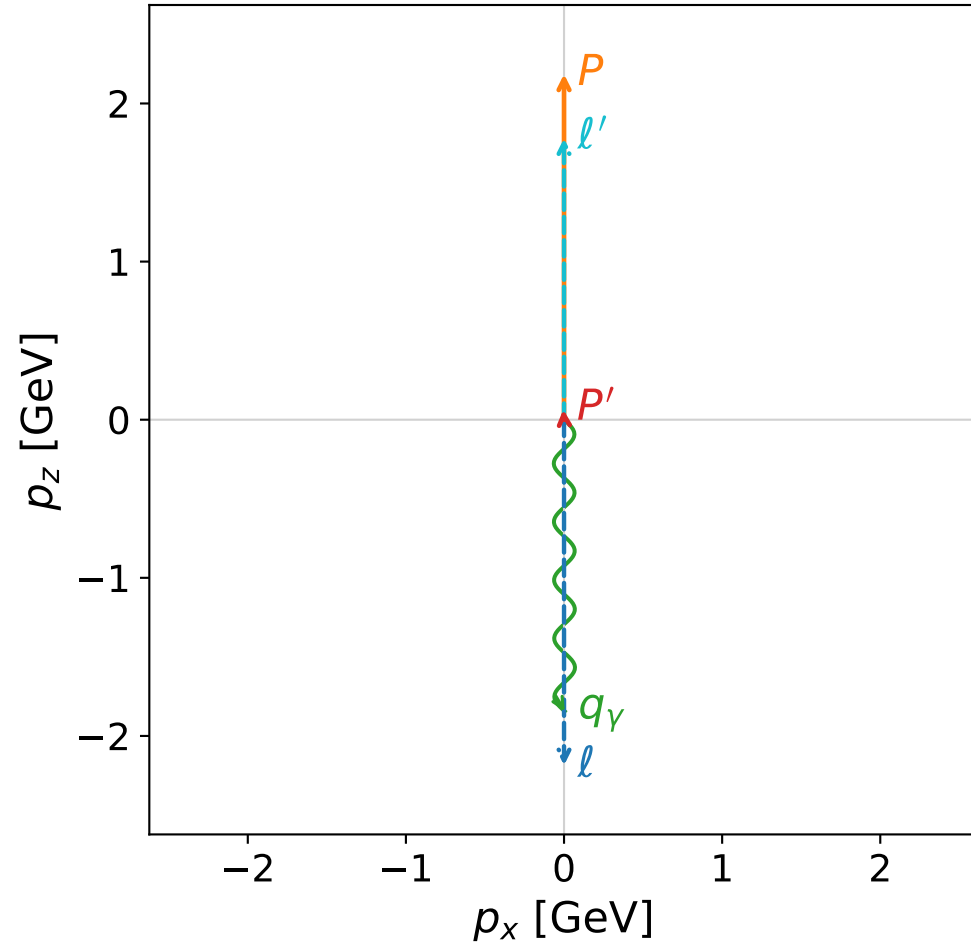
q_γ $E= 0.6291$ $p_x= 0.1400$ $p_y= 0.0000$ $p_z= -0.6133$

Leading normalized final-state amplitudes
(N=2, retained=93.6%)

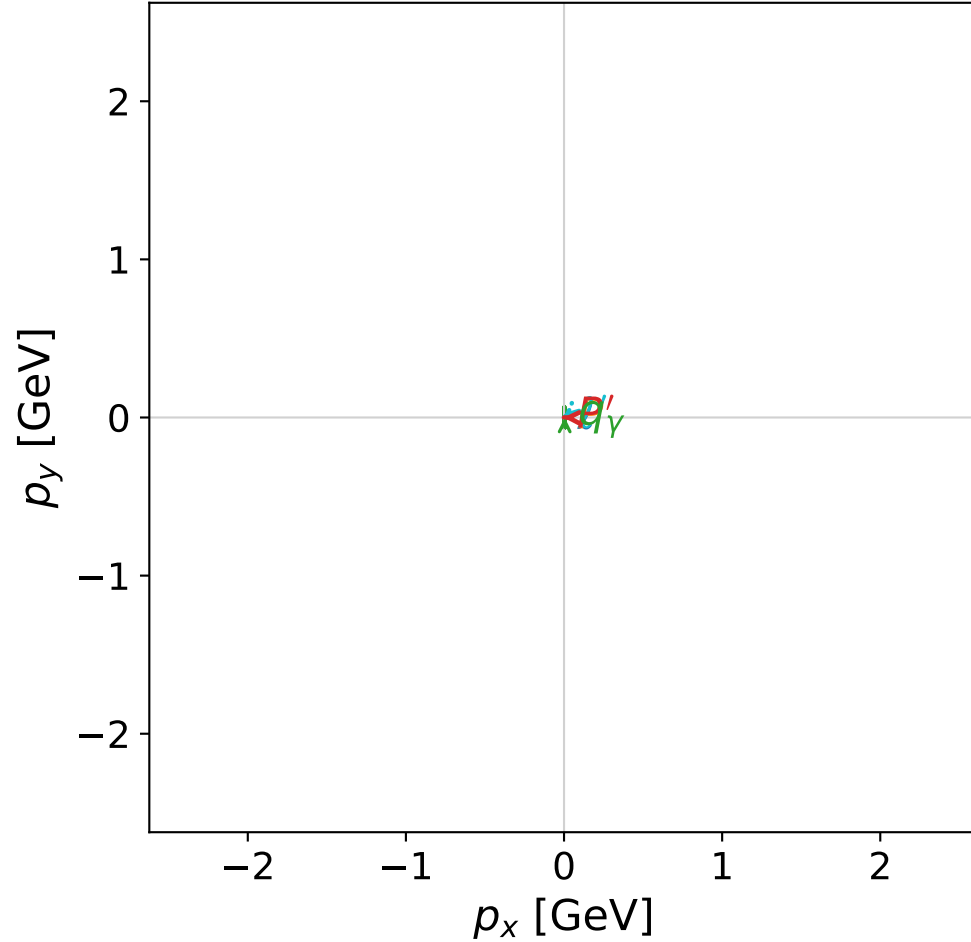


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



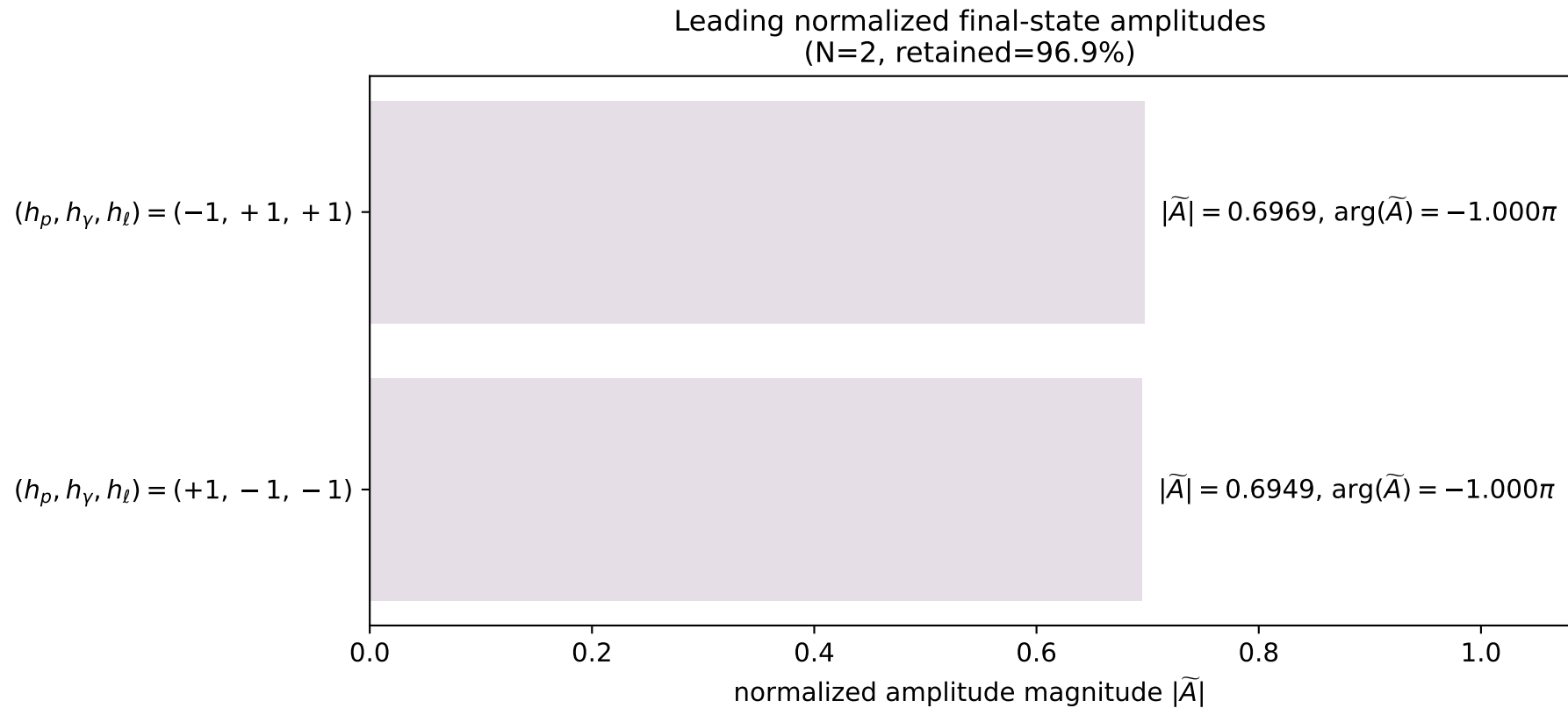
x--y plane



dghz_mixing_angles_polarization_03_local_minimum_248
 $d_{\{\text{rm GHZ}\}}=0.000127763$
 region: polarization_cluster_03_local_minimum_248
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0248
 lepton mixing angle: $\alpha_e=2.9049644$ rad
 incoming lepton: $-0.972134|+\rangle +0.234426|-\rangle$
 proton mixing angle: $\alpha_p=1.3339106$ rad
 incoming proton: $0.234676|+\rangle +0.972074|-\rangle$

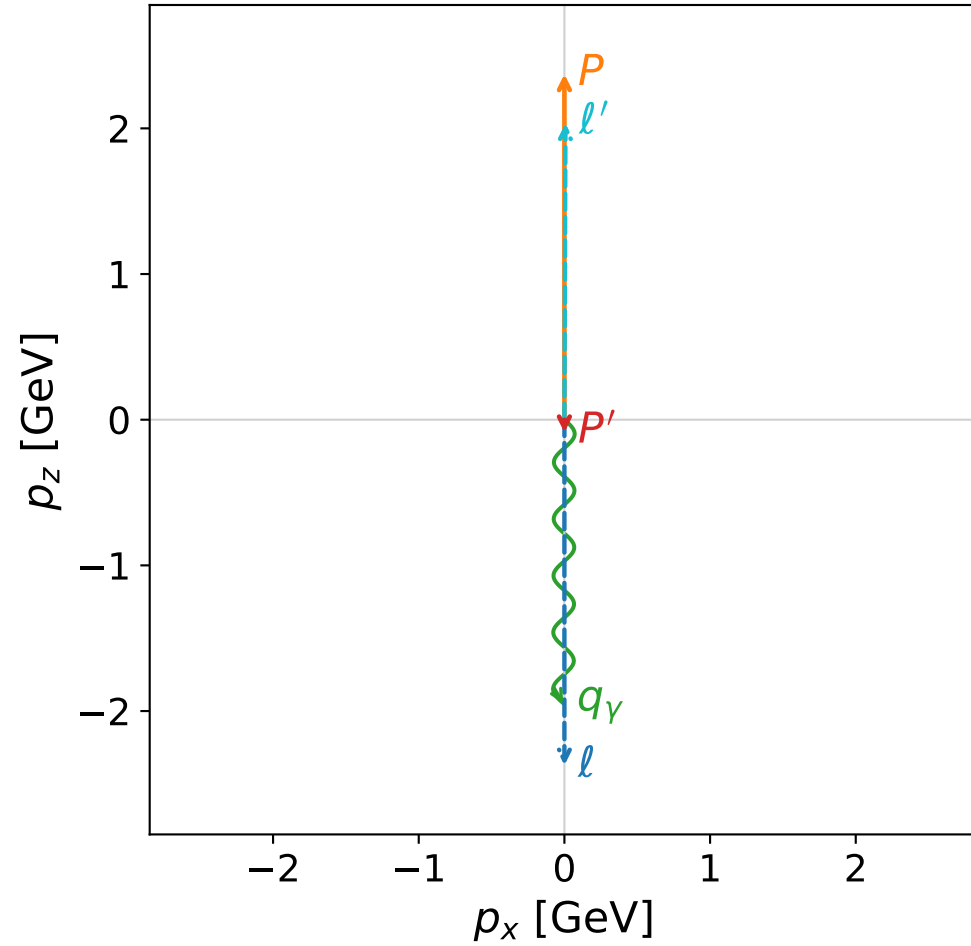
$s=20.8306$, $\sqrt{s}=4.56405$, $|\vec{P}|=2.18564$, $|\vec{P}'|=0.064417$
 $\theta_{P'}=0.0689908$, $\theta_\gamma=3.13913$, $\phi_{P'}=3.12116$, $\phi_\gamma=0$
 $E_\gamma=1.84406$, $\theta_{\ell'}=7.53344\text{e-}05$, $\phi_{\ell'}=3.88476$
 $Q^2=15.5599$, $x_B=0.807696$, $t=-2.43181$, $W^2=4.58449$, $y=0.965605$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.1856$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.1856$
 P $E= 2.3784$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.1856$
 ℓ' $E= 1.7798$ $p_x= -0.0001$ $p_y= -0.0001$ $p_z= 1.7798$
 P' $E= 0.9402$ $p_x= -0.0044$ $p_y= 0.0001$ $p_z= 0.0643$
 q_γ $E= 1.8441$ $p_x= 0.0045$ $p_y= 0.0000$ $p_z= -1.8441$

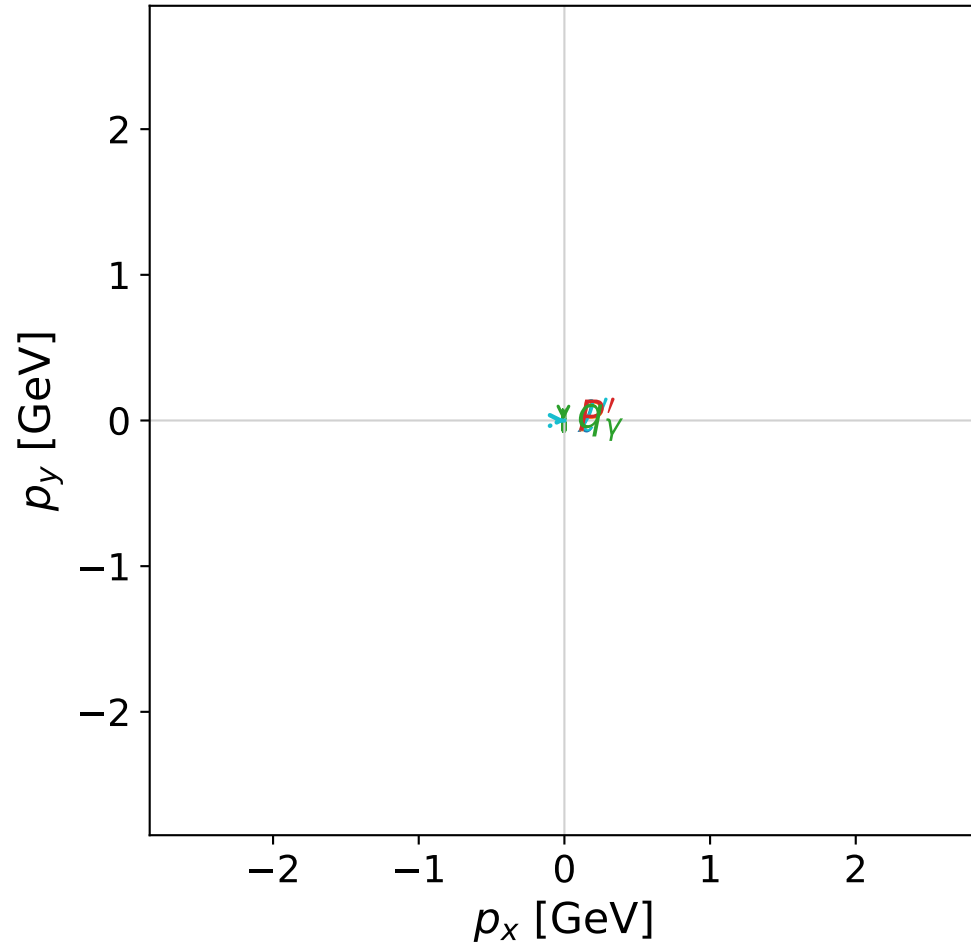


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane

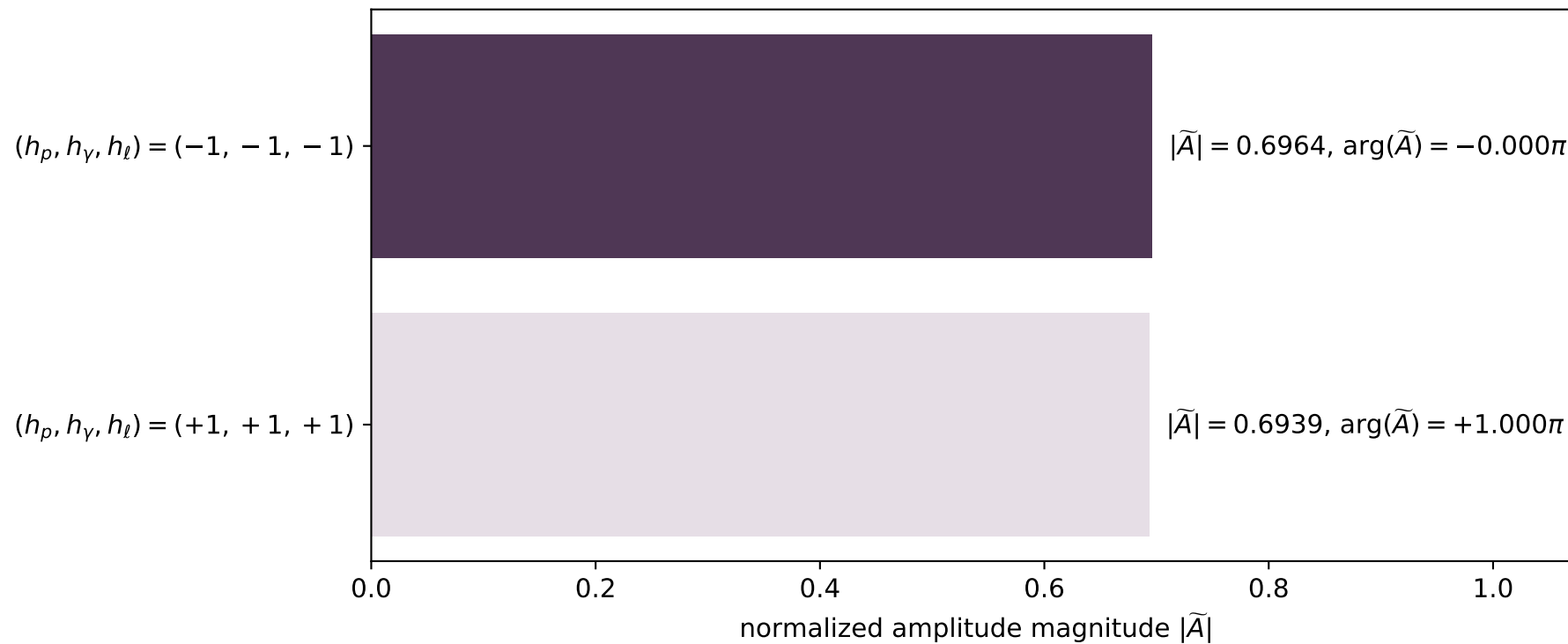


dghz_mixing_angles_polarization_03_local_minimum_262
 $d_{\text{GHZ}}=0.000177709$
 region: polarization_cluster_03_local_minimum_262
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0262
 lepton mixing angle: $\alpha_e=1.6990512$ rad
 incoming lepton: $-0.127904|+\rangle +0.991787|-\rangle$
 proton mixing angle: $\alpha_p=0.12870773$ rad
 incoming proton: $0.991729|+\rangle +0.128353|-\rangle$

$s=24.2413$, $\sqrt{s}=4.92355$, $|\vec{P}|=2.37242$, $|\vec{P}'|=0.0876034$
 $\theta_{P'}=3.14159$, $\theta_Y=3.13785$, $\phi_{P'}=3.00433$, $\phi_Y=3.14156$
 $E_Y=1.94693$, $\theta_{\ell'}=0.00358498$, $\phi_{\ell'}=6.28316$
 $Q^2=19.307$, $x_B=0.853001$, $t=-3.46271$, $W^2=4.20707$, $y=0.968871$

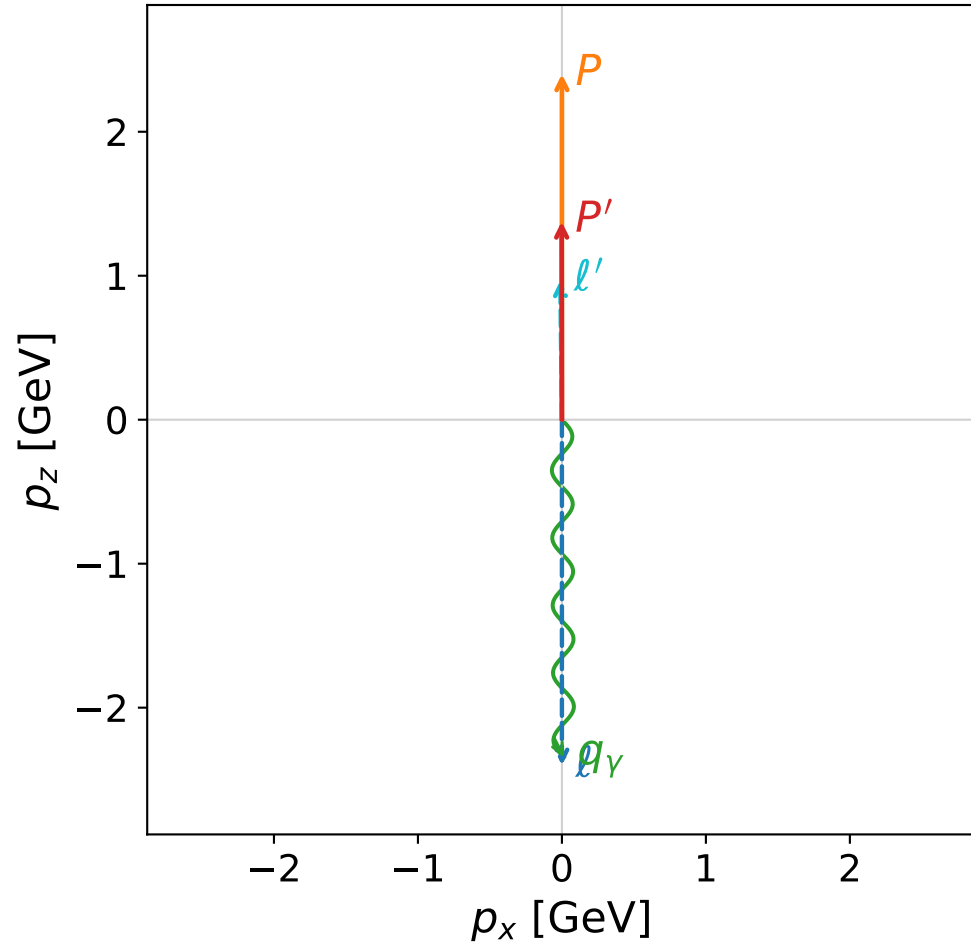
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3724$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.3724$
 P $E= 2.5511$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.3724$
 ℓ' $E= 2.0345$ $p_x= 0.0073$ $p_y= -0.0000$ $p_z= 2.0345$
 P' $E= 0.9421$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0876$
 q_Y $E= 1.9469$ $p_x= -0.0073$ $p_y= 0.0000$ $p_z= -1.9469$

Leading normalized final-state amplitudes
 (N=2, retained=96.6%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

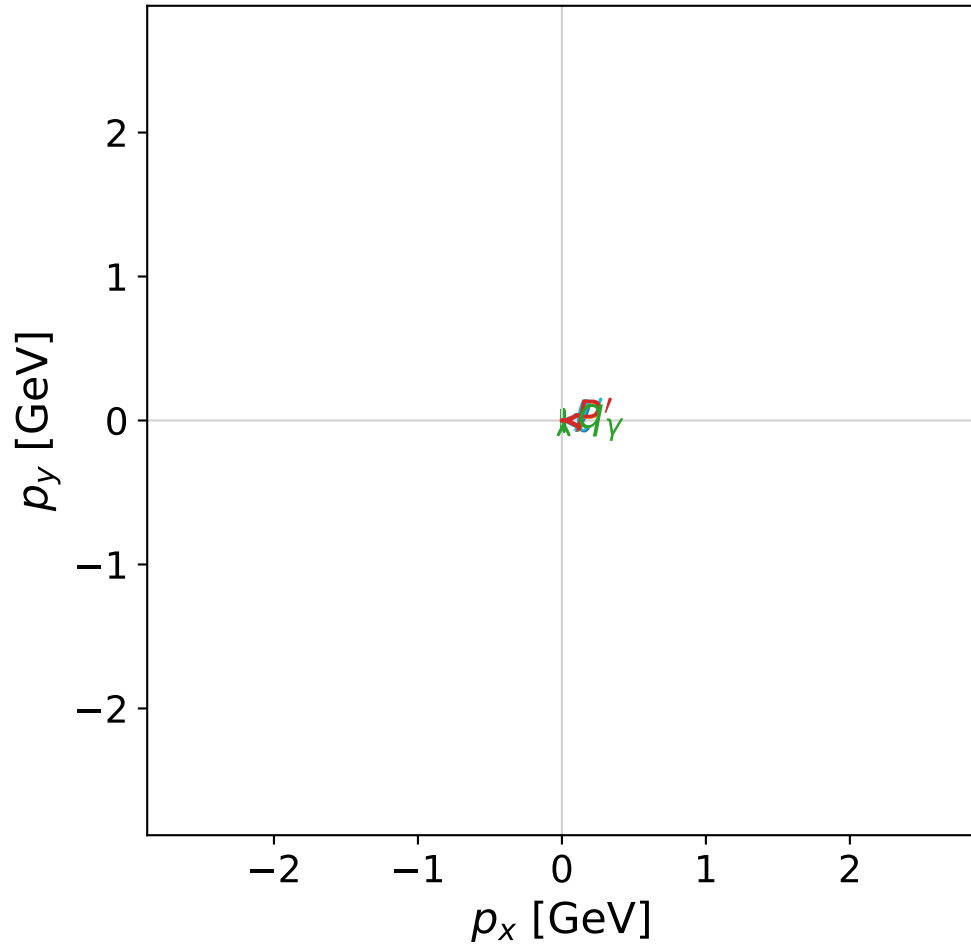


dghz_mixing_angles_polarization_03_local_minimum_267
 $d_{\{\rm GHZ\}}=0.000191782$
 region: polarization_cluster_03_local_minimum_267
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0267
 lepton mixing angle: $\alpha_e=2.647731$ rad
 incoming lepton: $-0.880509|+\rangle +0.47403|-\rangle$
 proton mixing angle: $\alpha_p=1.0728653$ rad
 incoming proton: $0.477609|+\rangle +0.878573|-\rangle$

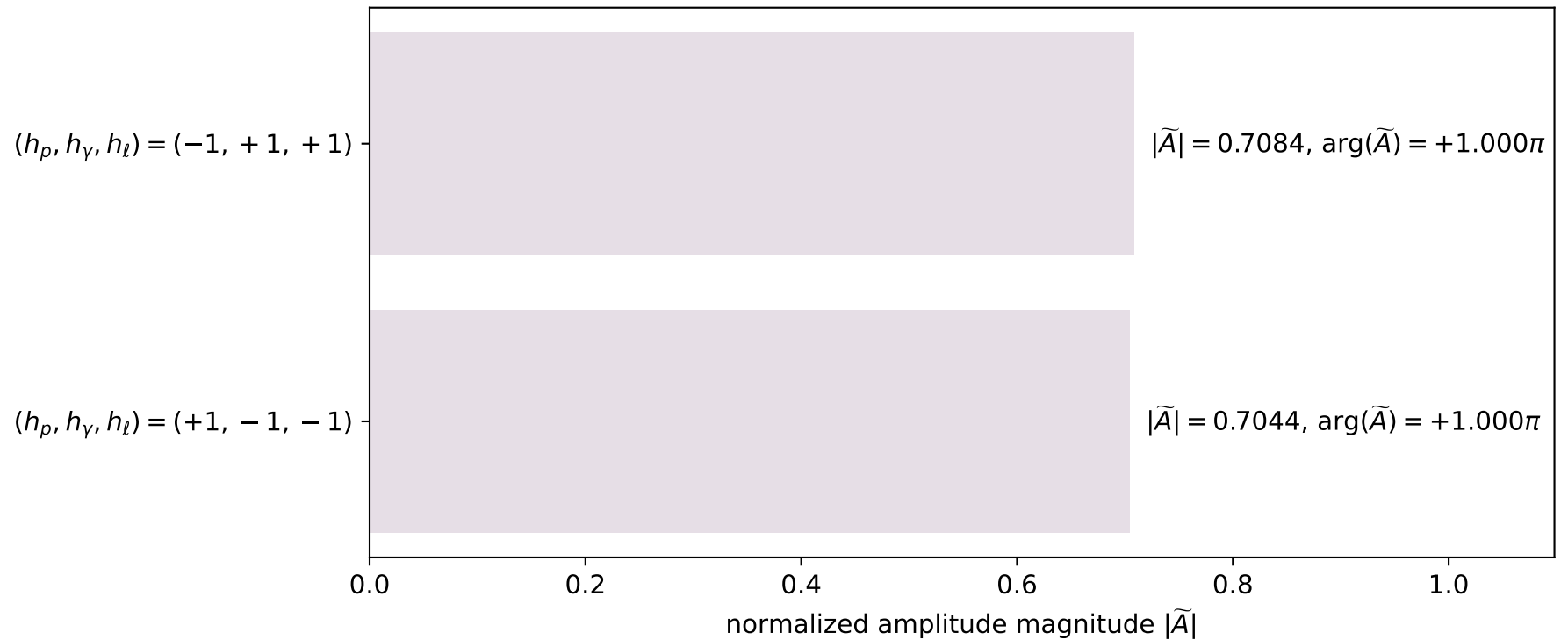
$s=24.7744$, $\sqrt{s}=4.97739$, $|\vec{P}|=2.40031$, $|\vec{P}'|=1.37624$
 $\theta_{p'}=0.00145035$, $\theta_\gamma=3.13587$, $\phi_{p'}=3.13934$, $\phi_\gamma=0$
 $E_\gamma=2.34405$, $\theta_{l'}=0.0117929$, $\phi_{l'}=3.14199$
 $Q^2=9.29214$, $x_B=0.394536$, $t=-0.217746$, $W^2=15.1398$, $y=0.985667$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4003$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4003$
 P $E= 2.5771$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4003$
 l' $E= 0.9678$ $p_x= -0.0114$ $p_y= -0.0000$ $p_z= 0.9678$
 P' $E= 1.6655$ $p_x= -0.0020$ $p_y= 0.0000$ $p_z= 1.3762$
 q_γ $E= 2.3441$ $p_x= 0.0134$ $p_y= 0.0000$ $p_z= -2.3440$

x--y plane

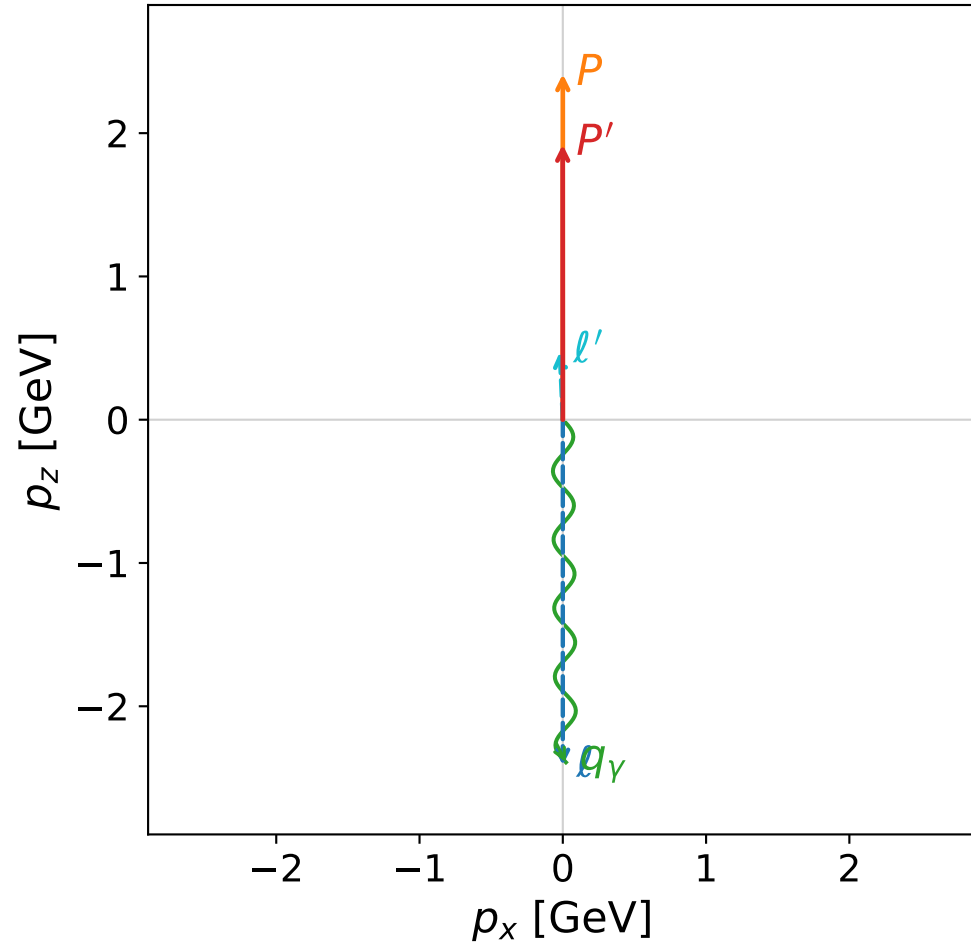


Leading normalized final-state amplitudes
(N=2, retained=99.8%)



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane

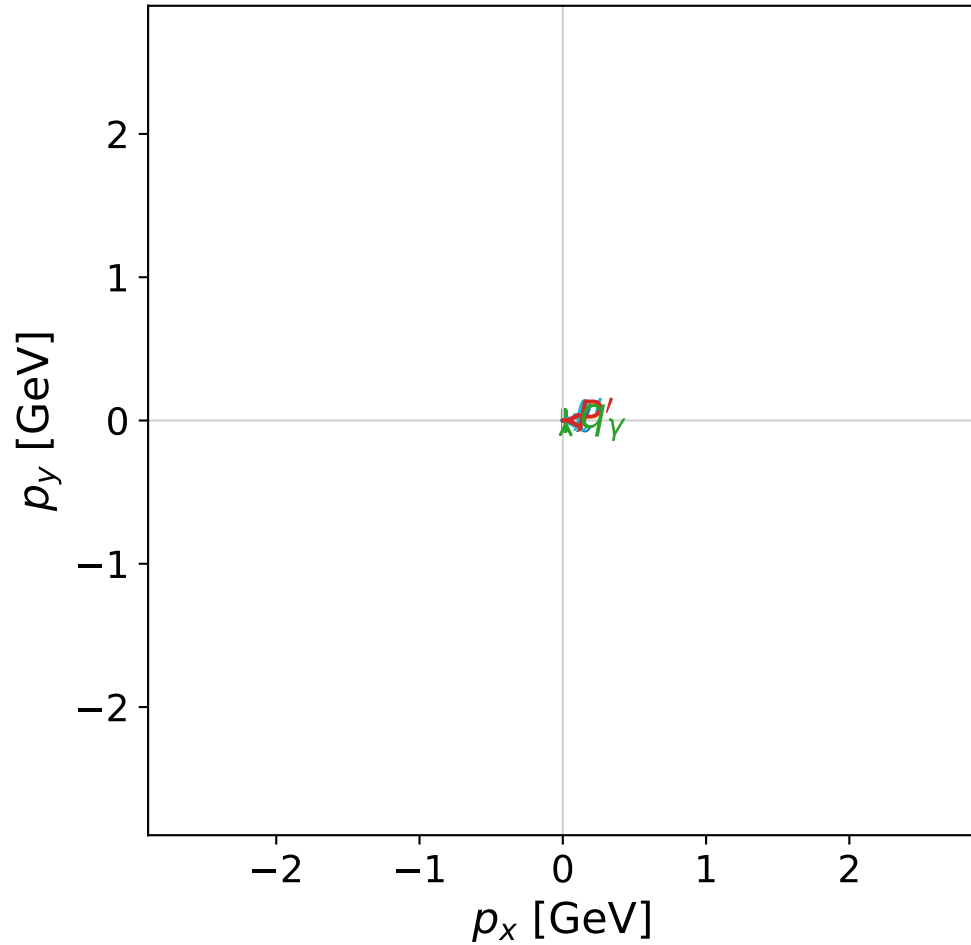


dghz_mixing_angles_polarization_03_local_minimum_316
 $d_{\text{GHZ}}=0.00040247$
 region: polarization_cluster_03_local_minimum_316
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0316
 lepton mixing angle: $\alpha_e=2.555944$ rad
 incoming lepton: $-0.833354|+\rangle + 0.55274|-\rangle$
 proton mixing angle: $\alpha_p=0.97424173$ rad
 incoming proton: $0.561796|+\rangle + 0.827276|-\rangle$

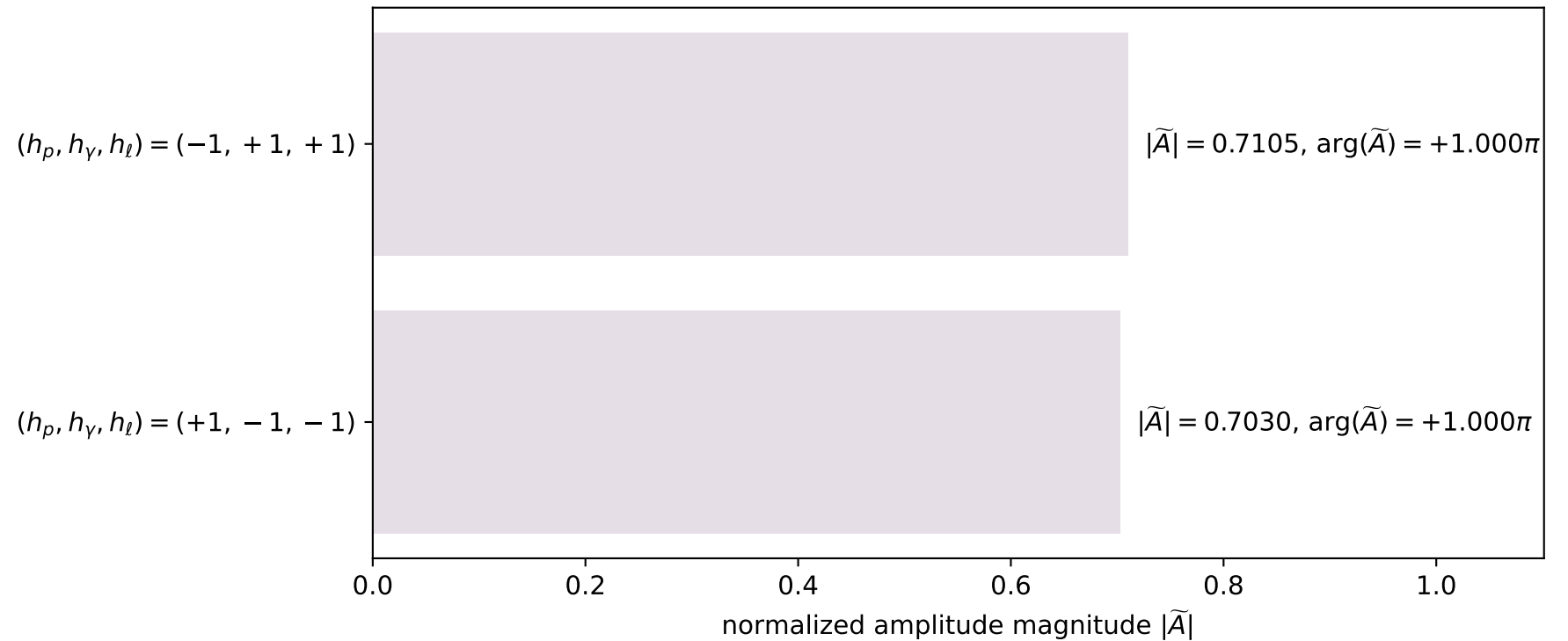
$s=25$, $\sqrt{s}=5$, $|\vec{P}|=2.41202$, $|\vec{P}'|=1.91934$
 $\theta_{p'}=0.000445521$, $\theta_\gamma=3.13237$, $\phi_{p'}=3.14125$, $\phi_\gamma=6.28316$
 $E_\gamma=2.39134$, $\theta_{l'}=0.0448925$, $\phi_{l'}=3.14158$
 $Q^2=4.55525$, $x_B=0.190186$, $t=-0.0386979$, $W^2=20.2762$, $y=0.993012$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.4120$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.4120$
 P $E= 2.5880$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.4120$
 l' $E= 0.4724$ $p_x= -0.0212$ $p_y= 0.0000$ $p_z= 0.4719$
 P' $E= 2.1363$ $p_x= -0.0009$ $p_y= 0.0000$ $p_z= 1.9193$
 q_γ $E= 2.3913$ $p_x= 0.0221$ $p_y= -0.0000$ $p_z= -2.3912$

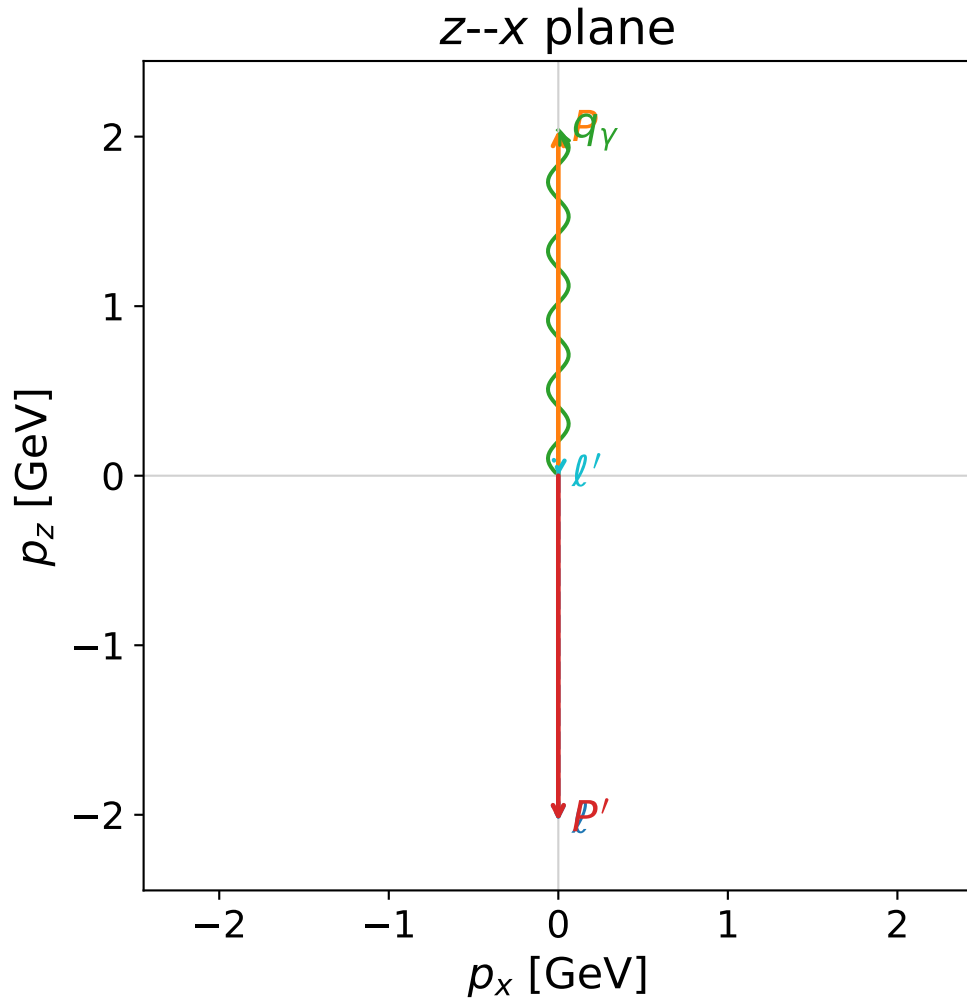
x--y plane



Leading normalized final-state amplitudes
 (N=2, retained=99.9%)



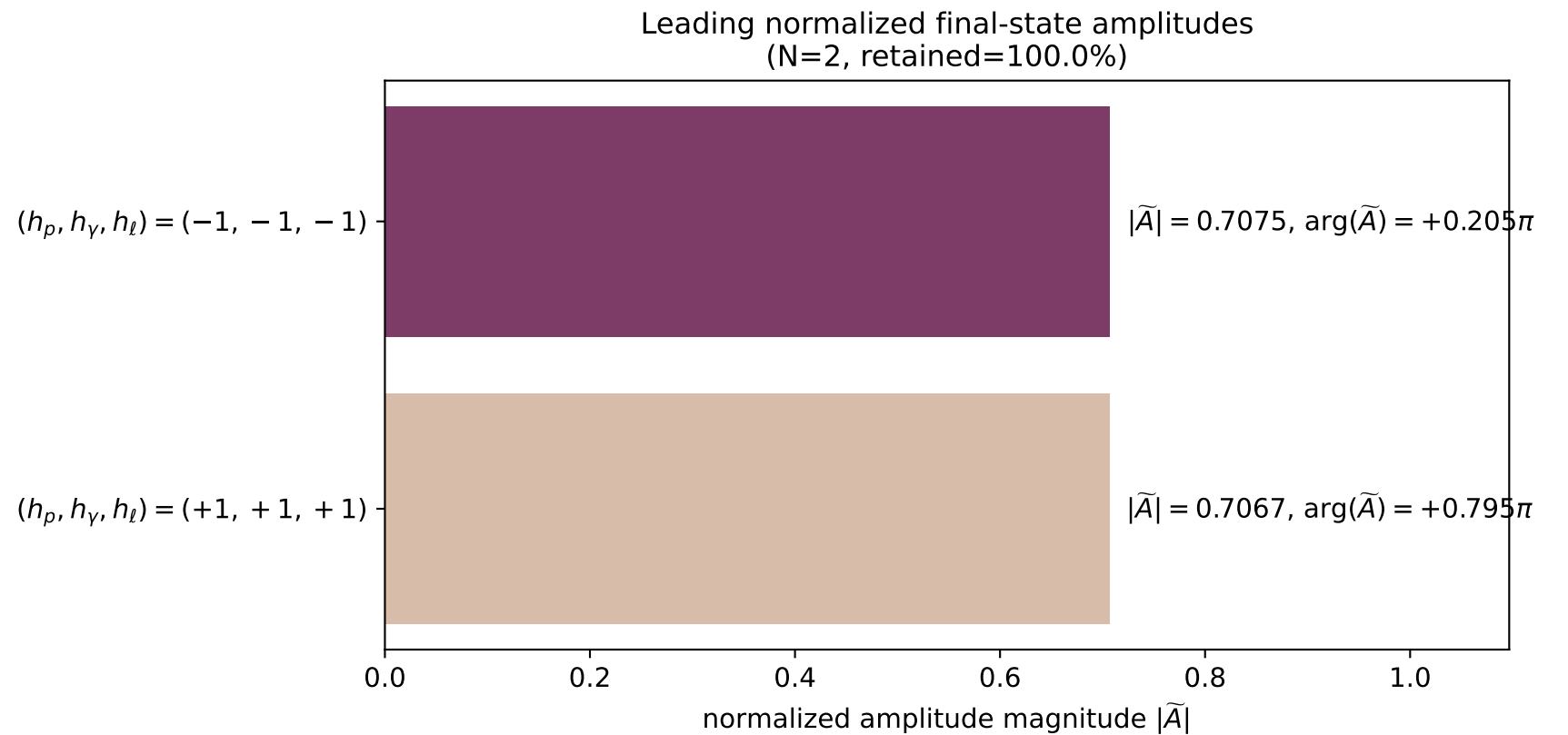
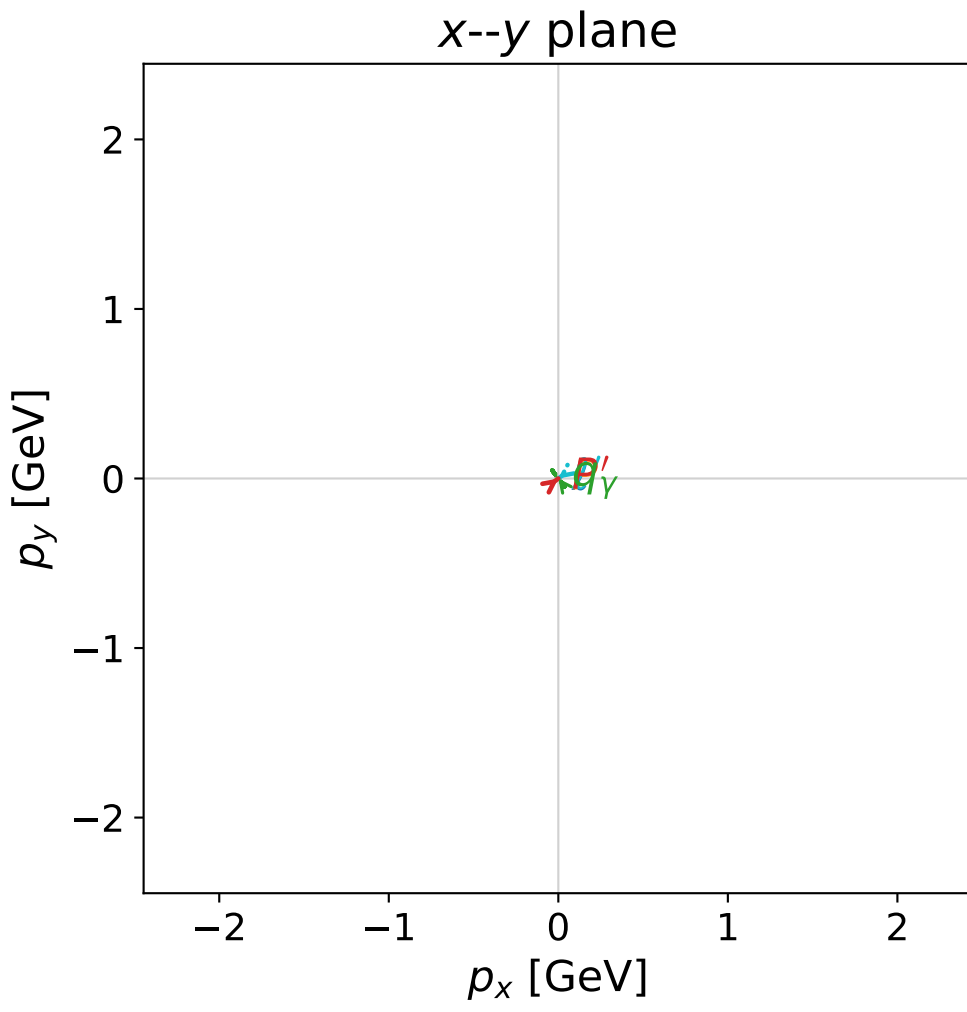
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

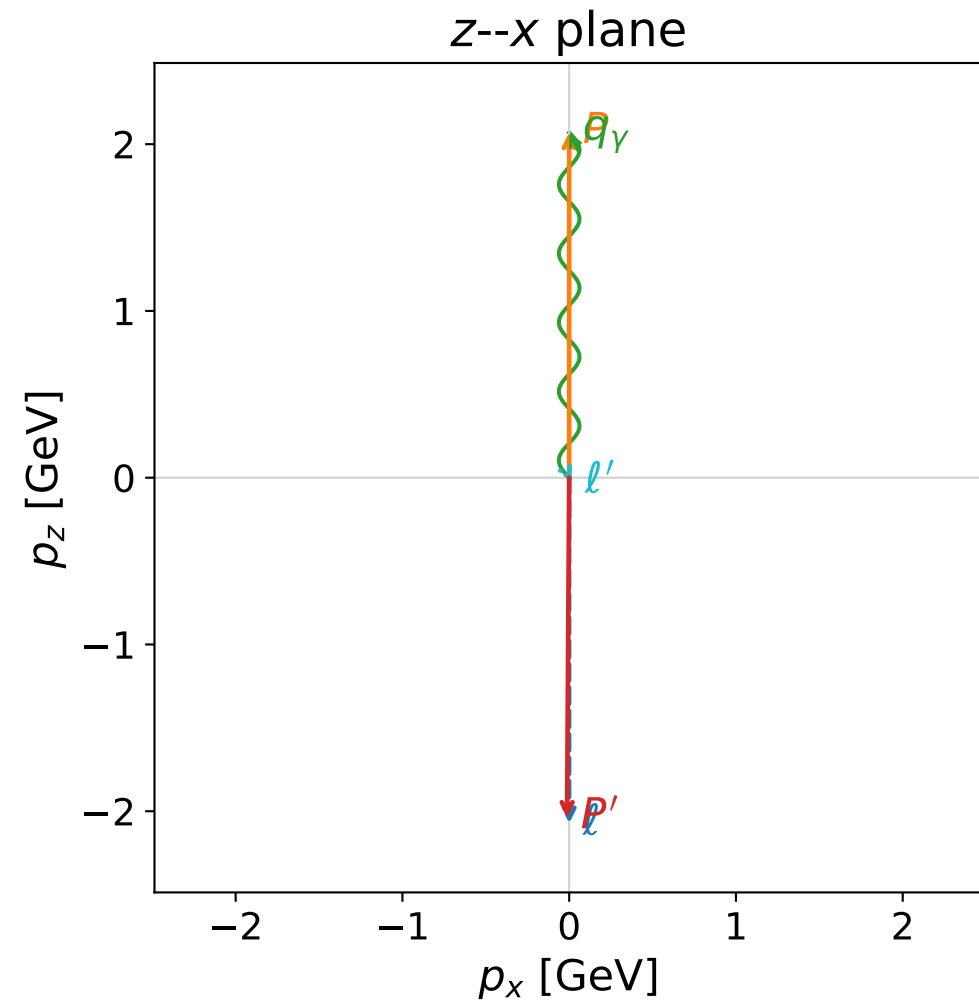


```
dghz_mixing_angles_polarization_03_local_minimum_357
d_{\rm GHZ}=0.00125916
region: polarization_cluster_03_local_minimum_357
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0357
lepton mixing angle: alpha_e=2.3565227 rad
incoming lepton: -0.707339|+> +0.706875|->
proton mixing angle: alpha_p=0.78632898 rad
incoming proton: 0.706448|+> +0.707765|->
```

```
s=18.3383, sqrt(s)=4.28232, |P|=2.03843, |P'|=2.03709
theta_p=3.14158, theta_gamma=2.79619e-05, phi_p=0.644969, phi_gamma=0.64563
E_gamma=2.03832, theta_l'=3.06993, phi_l'=3.78699
Q^2=0.000426632, x_B=2.44525e-05, t=-16.6098, W^2=18.3268, y=0.999369
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.0384 p_x=  0.0000 p_y=  0.0000 p_z= -2.0384
P      E=  2.2439 p_x=  0.0000 p_y=  0.0000 p_z=  2.0384
l'     E=  0.0013 p_x= -0.0001 p_y= -0.0001 p_z= -0.0012
P'     E=  2.2427 p_x=  0.0000 p_y=  0.0000 p_z= -2.0371
q_gamma E=  2.0383 p_x=  0.0000 p_y=  0.0000 p_z=  2.0383
```

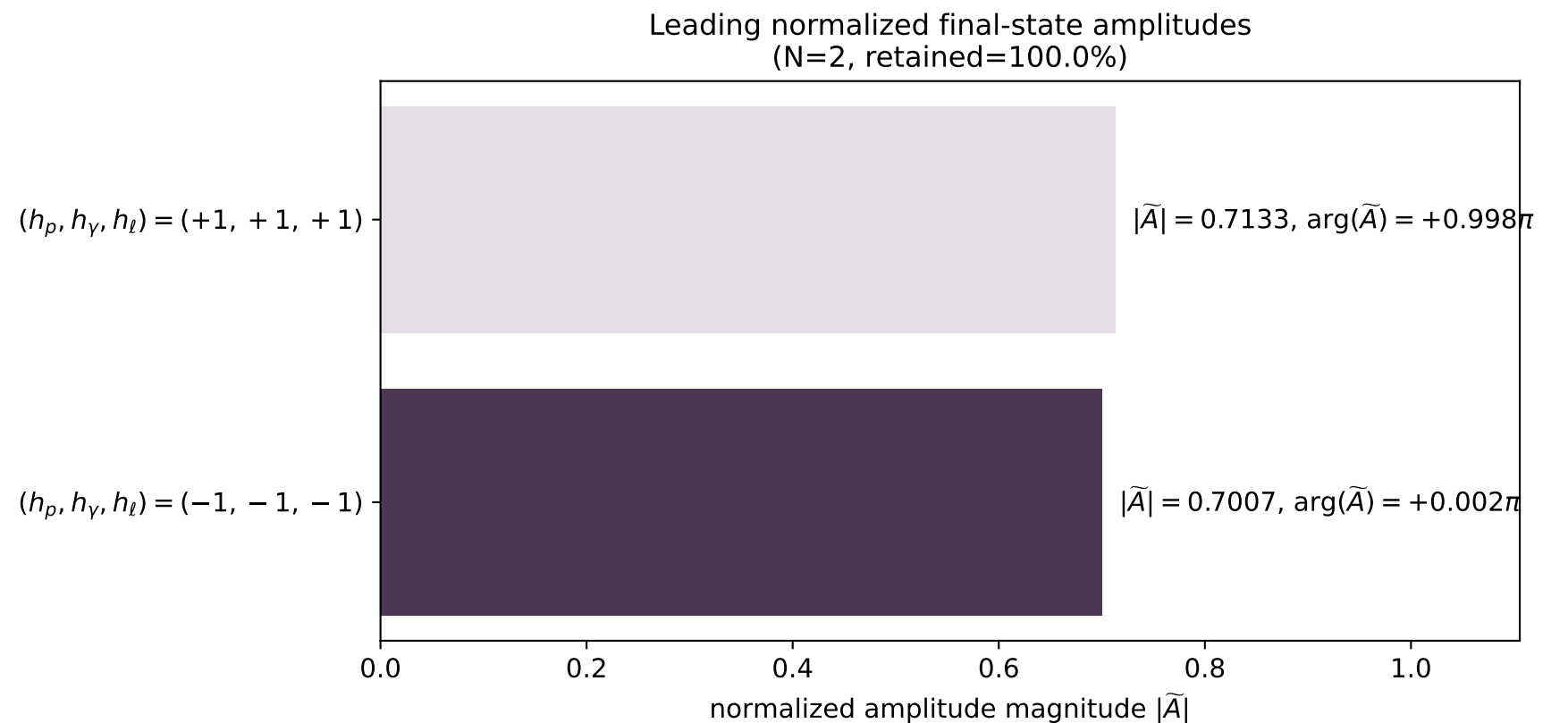
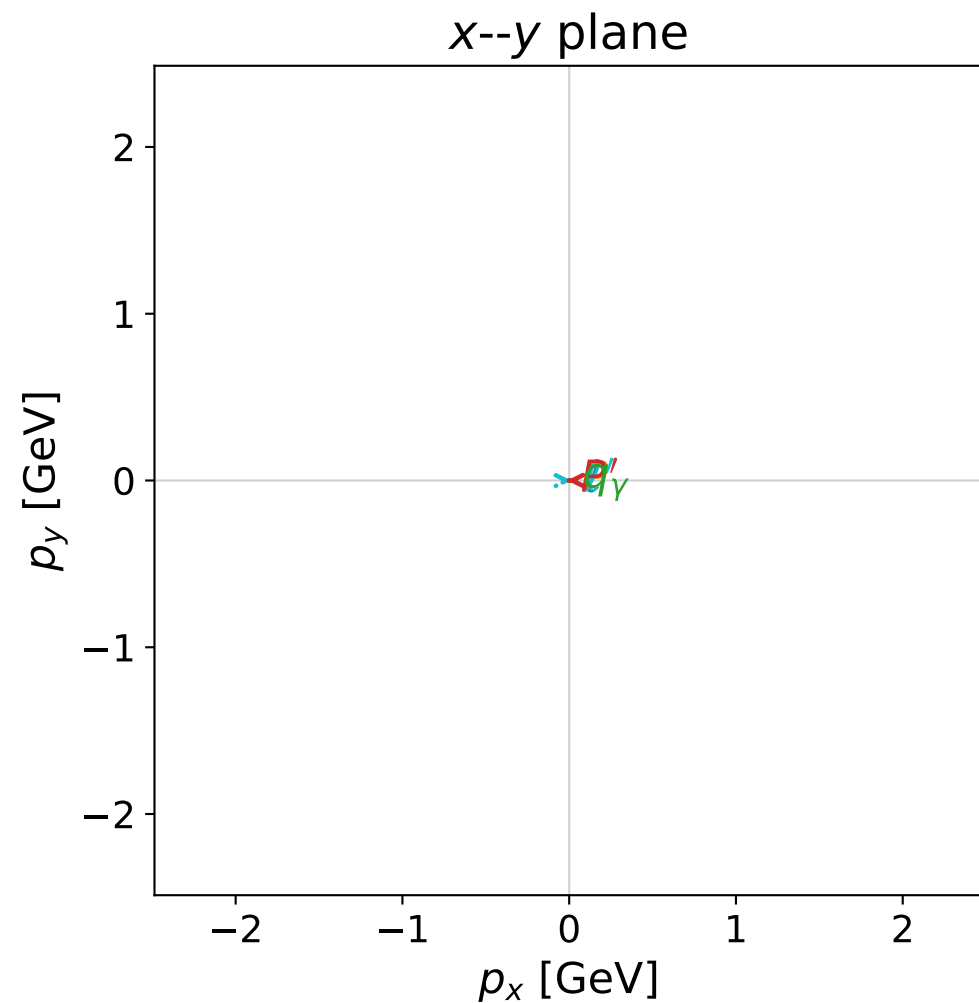


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

```
dghz_mixing_angles_polarization_03_local_minimum_372
d_{\rm GHZ}=0.0286865
region: polarization_cluster_03_local_minimum_372
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0372
lepton mixing angle: alpha_e=2.3560927 rad
incoming lepton: -0.707035|+> +0.707179|->
proton mixing angle: alpha_p=0.77718399 rad
incoming proton: 0.712891|+> +0.701275|->
```

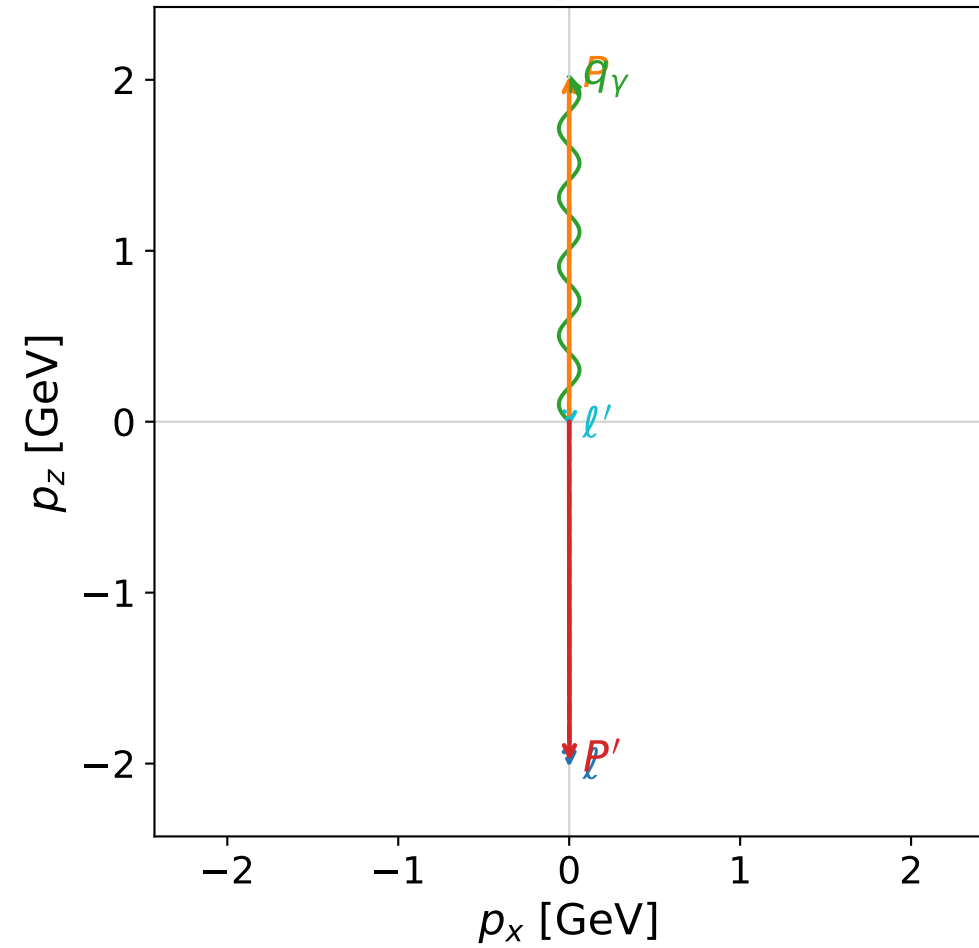
```
s=18.8971, sqrt(s)=4.34708, |P|=2.07234, |P'|=2.04126
theta_p=3.13533, theta_gamma=0, phi_p=3.14439, phi_gamma=1.40154
E_gamma=2.06954, theta_l'=2.71784, phi_l'=0.00279457
Q^2=0.0114083, x_B=0.000642414, t=-16.9207, W^2=18.6269, y=0.985637
```

```
four-momenta [E, p_x, p_y, p_z] GeV:
l      E=  2.0723 p_x=  0.0000 p_y=  0.0000 p_z= -2.0723
P      E=  2.2747 p_x=  0.0000 p_y=  0.0000 p_z=  2.0723
l'     E=  0.0311 p_x=  0.0128 p_y=  0.0000 p_z= -0.0283
P'     E=  2.2465 p_x= -0.0128 p_y= -0.0000 p_z= -2.0412
q_gamma E=  2.0695 p_x=  0.0000 p_y=  0.0000 p_z=  2.0695
```

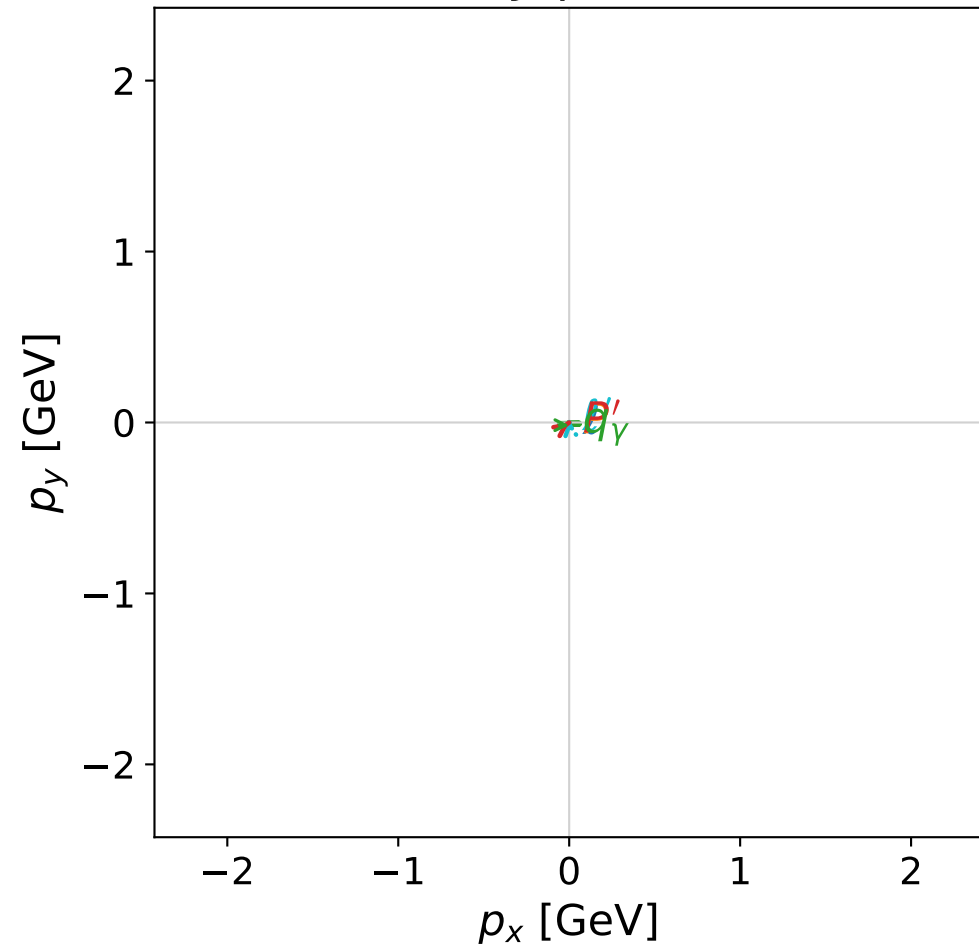


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

z--x plane



x--y plane



dghz_mixing_angles_polarization_03_local_minimum_374
 $d_{\text{GHZ}}=0.0335652$
 region: polarization_cluster_03_local_minimum_374
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum_0374
 lepton mixing angle: $\alpha_e=2.356188$ rad
 incoming lepton: $-0.707102|+\rangle +0.707111|-\rangle$
 proton mixing angle: $\alpha_p=0.79277071$ rad
 incoming proton: $0.701874|+\rangle +0.712301|-\rangle$

$s=18.0658$, $\sqrt{s}=4.25039$, $|\vec{P}|=2.02169$, $|\vec{P}'|=1.98608$
 $\theta_{p'}=3.14052$, $\theta_\gamma=0.00808957$, $\phi_{p'}=0.622432$, $\phi_\gamma=4.72104$
 $E_\gamma=2.01834$, $\theta_{l'}=2.70049$, $\phi_{l'}=1.69454$
 $Q^2=0.0137918$, $x_B=0.000816221$, $t=-16.0612$, $W^2=17.7632$, $y=0.983193$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.0217$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -2.0217$
 P $E= 2.2287$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 2.0217$
 l' $E= 0.0356$ $p_x= -0.0019$ $p_y= 0.0151$ $p_z= -0.0322$
 P' $E= 2.1964$ $p_x= 0.0017$ $p_y= 0.0012$ $p_z= -1.9861$
 q_γ $E= 2.0183$ $p_x= 0.0001$ $p_y= -0.0163$ $p_z= 2.0183$

